

THE WORKS OF HIPPOCRATES.

ON ANCIENT MEDICINE.

THE ARGUMENT.

Although, as stated in the second section of the Preliminary Discourse, the evidence in support of this treatise be unfortunately not such as clearly to establish its genuineness, all who read it with attention must admit that it is replete with important matters, and that if not the production of Hippocrates, it is not unworthy of his high reputation. Notwithstanding, then, that I am by no means so well convinced as M. Littré is, that the work is genuine, I have not hesitated to follow his example in placing it at the head of the list, as the nature of its contents is such as to form an excellent introduction to the study of the Hippocratic medicine.

It contains, as M. Littré remarks, a polemic, a method, and a system. The polemic is directed against those of his predecessors who had corrupted medicine by introducing hypotheses into it as the causes of disease, such as heat, cold, moisture, and dryness. These it will be seen that he combats with great force of argument and clearness of illustration. The philosophical dogmas to which he is supposed to refer in this place are those of the section of Pythagoreans, called the Eleatic, who would appear to have held nearly the same opinions as Pythagoras himself with regard to the elements.^[355] But, in fact, as I trust I have clearly made out in the third section of the Preliminary Discourse, all the ancient philosophers held substantially the same opinion regarding the elements, although they did not all express themselves in the same terms. It is of little consequence, then, to attempt to find out what particular class of philosophers our author directs his attack against, it being sufficient to say that he decidedly condemns the practice of founding the rules of medical practice on hypothesis.^[356] I may here remark, that the censure thus bestowed on hypothetical systems applies to modern times as well as to ancient, to those who proclaim theories by which, like Broussais, they account for all diseases upon figments which they call inflammations, and those who, like Cullen, attribute most diseases to spasms. We may rest assured, from the sensible observations which Hippocrates makes on this subject in the present work, that the causes of all diseases are realities, provided we could find them out, and that they are not vague abstractions, as the authors of these hypotheses suppose.

His method of cultivating medicine is founded on an attentive examination of all the circumstances connected with real life, and his system consists in studying the condition of the humors in the body, their origin, their coction, and their disappearance.

The most prominent feature, however, in the contents of this little treatise is the practical view which is here given of the origin of medicine, namely, from the necessities and weaknesses of the human race. The author clearly makes it out that Medicine is, as it were, a corollary to Dietetics. Nothing of the kind can well be imagined more ingenious and original than his observations and reasonings on this head in the introductory sections to this treatise. See in particular § 5.

The remarks in refutation of the hypothesis of cold, heat, moist, and dry, are very interesting. (§ 13.)

The reflections on the origin of fevers and inflammations are very just and original, but would appear not to have been properly appreciated by his successors; for among all the ancient authors who have treated of fevers, there is, perhaps, no one but himself who has stated in decided terms that there is something more in a fever than a mere increase of the innate (*or* animal) heat. See the Commentary on PAULUS ÆGINETA, B. II., 1.

The remarks on the effects of the cold bath at § 16 are much to the purpose, and deserve attention.

The observations on rheums *or* defluxions (§ 19) are also very striking, and even at the present day, after the many vicissitudes of medical theory which we have gone through, it would be difficult to deny that the opinions here advanced are well founded. At all events they must be allowed to be highly interesting, as containing the first germ of a theory which long flourished in the schools of medicine.

At § 20 the author seems to hold that philosophy is not so necessary to medicine as medicine is to philosophy. Schulze, with a considerable show of reason, argues that Celsus had this passage in view when he pronounced, concerning Hippocrates, that he was the first person who separated medicine from philosophy. (Hist. Med. I., 3, i., 26.) Schulze contends that what Celsus meant was, that Hippocrates discarded *à priori* arguments in medicine, and drew all his inferences from actual observation. This would appear to me the most plausible interpretation which has ever been given to this celebrated passage in the preface of Celsus. Philosophy, then, it would appear, freed medicine from the delusions of superstition, by substituting the errors of hypothesis in their place, and the important office which he who was called the Father of Medicine conferred upon the art was by discarding both superstition and hypothesis, and substituting the results of actual observation in the room of both.

From § 22 to the end of the work the author gives important observations on the modifications which diseases undergo in connection with the peculiar organization of the part in which they are situated. It may well be doubted whether the remarks

and reflections herein contained have ever obtained all the attention which they merit.

The style of this piece is certainly elegant and beautiful; and it is proper to mention that the text is remarkably improved in M. Littré's edition. In all the previous editions it was more corrupt than that of almost any other of the Hippocratic treatises.

The following remarks of M. Littré on the present work appear to me so just, and are so elegantly expressed, that I cannot deny myself the pleasure of introducing them here in the original:

“En résumé, le livre de *l'Ancienne Médecine* donne une idée des problèmes agités du temps d'Hippocrate, et de la manière dont ils étaient débattus. Il s'agissait, dans la plus grande généralité de la pathologie de déterminer la cause des maladies ou, en d'autres termes, de poser les bases d'un système de médecine. Certains médecins disaient que cette cause, étant une, résidait dans une propriété unique du corps, propriété qu'ils spécifiaient. Hippocrate répétait qu'en fait, cela était en contradiction avec l'expérience, qu'en principe une hypothèse était suspecte et stérile, et qu'il n'y avait de sûreté que dans l'étude des faits et dans la tradition de la science qui y ramène. Ainsi, quatre cents ans avant J. C., on essayait de rattacher toute la médecine à une seule propriété hypothétique, comme on l'a essayé de nos jours; mais cette propriété était ou le chaud, ou le froid, ou l'humide, ou le sec. Quatre cents ans avant J. C., un esprit sévère et éclairé combattait de telles opinions au nom de l'expérience, montrait que les causes des maladies ne pouvant pas se ramener à une seule, le champ de la pathologie générale était bien plus vaste qu'on ne croyait; et formulait ce que l'observation lui avait permis de conclure; mais sa conclusion n'embrasse guère que la trouble dans le mélange des humeurs, que leur coction et leurs crises. Depuis lors, la méthode de ceux qu'Hippocrate avait combattus, et la méthode d'Hippocrate, l'hypothèse et l'observation se sont perpétuées, ainsi que le témoigne l'histoire de la médecine, mais ce ne sont plus ni l'ancienne hypothèse, ni l'ancienne observation.

“Il est certainement instructif d'étudier, dans le cours du temps, les problèmes tels qu'ils ont été posés, et les discussions qu'ils ont soulevées. On le voit, la science antique a de grandes ressemblances avec la science moderne; dès l'époque que nous sommes forcés de regarder comme l'aurore de la médecine, dès les premiers monuments que nous possédons, les questions fondamentales sont débattues, et les limites de l'esprit humain sont touchées. Mais en dedans de ces limites, la science trouve, dans une immensité inépuisable de combinaisons, les matériaux qui la font grandir; et il est impossible de ne pas reconnaître que, sur un sol et avec les aliments que lui fournissent les choses et l'expérience, elle se développe en vertu d'un principe interne de vie, qui réside dans l'enchaînement nécessaire de son développement successif.”^[357]

ON ANCIENT MEDICINE.

1. Whoever having undertaken to speak or write on Medicine, have first laid down for themselves some hypothesis to their argument, such as hot, or cold, or moist, or dry, or whatever else they choose (thus reducing their subject within a narrow compass, and supposing only one or two original causes of diseases or of death among mankind), are all clearly mistaken in much that they say; and this is the more reprehensible as relating to an art which all men avail themselves of on the most important occasions, and the good operators and practitioners in which they hold in especial honor. For there are practitioners, some bad and some far otherwise, which, if there had been no such thing as Medicine, and if nothing had been investigated or found out in it, would not have been the case, but all would have been equally unskilled and ignorant of it, and everything concerning the sick would have been directed by chance. But now it is not so; for, as in all the other arts, those who practise them differ much from one another in dexterity and knowledge, so is it in like manner with Medicine. Wherefore I have not thought that it stood in need of an empty hypothesis, like those subjects which are occult and dubious, in attempting to handle which it is necessary to use some hypothesis; as, for example, with regard to things above us and things below the earth;^[358] if any one should treat of these and undertake to declare how they are constituted, the reader or hearer could not find out, whether what is delivered be true or false; for there is nothing which can be referred to in order to discover the truth.

2. But all these requisites belong of old to Medicine, and an origin and way have been found out, by which many and elegant discoveries have been made, during a length of time, and others will yet be found out, if a person possessed of the proper ability, and knowing those discoveries which have been made, should proceed from them to prosecute his investigations. But whoever, rejecting and despising all these, attempts to pursue another course and form of inquiry, and says he has discovered anything, is deceived himself and deceives others, for the thing is impossible. And for what reason it is impossible, I will now endeavor to explain, by stating and showing what the art really is. From this it will be manifest that discoveries cannot possibly be made in any other way. And most especially, it appears to me, that whoever treats of this art should treat of things which are familiar to the common people. For of nothing else will such a one have to inquire or treat, but of the diseases under which the common people have labored, which diseases and the causes of their origin and departure, their increase and decline, illiterate persons cannot easily find out themselves, but still it is easy for them to understand these things when discovered and expounded by others. For it is nothing more than that every one is put in mind of what had occurred to himself. But whoever does not reach the capacity of the illiterate vulgar and fails to make them listen to him, misses his mark. Wherefore, then, there is no necessity for any hypothesis.

3. For the art of Medicine would not have been invented at first, nor would it have been made a subject of investigation (for there would have been no need of it), if when men are indisposed, the same food and other articles of regimen which they eat and drink when in good health were proper for them, and if no others were preferable to these. But now necessity itself made medicine to be sought out and discovered by men, since the same things when administered to the sick, which agreed with them when in good health, neither did nor do agree with them. But to go still further back, I hold that the diet and food which people in health now use would not have been discovered, provided it had suited with man to eat and drink in like manner as the ox, the horse, and all other animals, except man, do of the productions of the earth, such as fruits, weeds, and grass; for from such things these animals grow, live free of disease, and require no other kind of food. And, at first, I am of opinion that man used the same sort of food, and that the present articles of diet had been discovered and invented only after a long lapse of time. For when they suffered much and severely from this strong and brutish diet, swallowing things which were raw, unmixed, and possessing great strength, they became exposed to strong pains and diseases, and to early deaths. It is likely, indeed, that from habit they would suffer less from these things than we would now, but still they would suffer severely even then; and it is likely that the greater number, and those who had weaker constitutions, would all perish; whereas the stronger would hold out for a longer time, as even nowadays some, in consequence of using strong articles of food, get off with little trouble, but others with much pain and suffering. From this necessity it appears to me that they would search out the food befitting their nature, and thus discover that which we now use: and that from wheat, by macerating it, stripping it of its hull, grinding it all down, sifting, toasting, and baking it, they formed bread,^[359] and from barley they formed cake (maza),^[360] performing many operations in regard to it; they boiled, they roasted, they mixed, they diluted those things which are strong and of intense qualities with weaker things, fashioning them to the nature and powers of man, and considering that the stronger things Nature would not be able to manage if administered, and that from such things pains, diseases, and death would arise, but such as Nature could manage, that from them food, growth, and health, would arise. To such a discovery and investigation what more suitable name could one give than that of Medicine? since it was discovered for the health of man, for his nourishment and safety, as a substitute for that kind of diet by which pains, diseases, and deaths were occasioned.

4. And if this is not held to be an art, I do not object. For it is not suitable to call any one an artist of that which no one is ignorant of, but which all know from usage and necessity. But still the discovery is a great one, and requiring much art and investigation. Wherefore those who devote themselves to gymnastics and training, are always making some new discovery, by pursuing the same line of inquiry, where, by eating and drinking certain things, they are improved and grow stronger than they were.^[361]

5. Let us inquire then regarding what is admitted to be Medicine; namely, that which was invented for the sake of the sick, which possesses a name and practitioners, whether it also seeks to accomplish the same objects, and whence it derived its origin. To me, then, it appears, as I said at the commencement, that nobody would have sought for medicine at all, provided the same kinds of diet had suited with men in sickness as in good health. Wherefore, even yet, such races of men as make no use of medicine, namely, barbarians, and even certain of the Greeks, live in the same way when sick as when in health; that it to say, they take what suits their appetite, and neither abstain from, nor restrict themselves in anything for which they have a desire. But those who have cultivated and invented medicine, having the same object in view as those of whom I formerly spoke, in the first place, I suppose, diminished the quantity of the articles of food which they used, and this alone would be sufficient for certain of the sick, and be manifestly beneficial to them, although not to all, for there would be some so affected as not to be able to manage even small quantities of their usual food, and as such persons would seem to require something weaker, they invented soups, by mixing a few strong things with much water, and thus abstracting that which was strong in them by dilution and boiling. But such as could not manage even soups, laid them aside, and had recourse to drinks, and so regulated them as to mixture and quantity, that they were administered neither stronger nor weaker than what was required.

6. But this ought to be well known, that soups do not agree with certain persons in their diseases, but, on the contrary, when administered both the fevers and the pains are exacerbated, and it becomes obvious that what was given has proved food and increase to the disease, but a wasting and weakness to the body. But whatever persons so affected partook of solid food, or cake, or bread, even in small quantity, would be ten times and more decidedly injured than those who had taken soups, for no other reason than from the strength of the food in reference to the affection: and to whomsoever it is proper to take soups and not eat solid food, such a one will be much more injured if he eat much than if he eat little, but even little food will be injurious to him. But all the causes of the sufferance refer themselves to this rule, that the strongest things most especially and decidedly hurt man, whether in health or in disease.

7. What other object, then, had he in view who is called a physician, and is admitted to be a practitioner of the art, who found out the regimen and diet befitting the sick, than he who originally found out and prepared for all mankind that kind of food which we all now use, in place of the former savage and brutish mode of living? To me it appears that the mode is the same, and the discovery of a similar nature. The one sought to abstract those things which the constitution of man cannot digest, because of their wildness and intemperance, and the other those things which are beyond the powers of the affection in

which any one may happen to be laid up. Now, how does the one differ from the other, except that the latter admits of greater variety, and requires more application, whereas the former was the commencement of the process?

8. And if one would compare the diet of sick persons with that of persons in health, he will find it not more injurious than that of healthy persons in comparison with that of wild beasts and of other animals. For, suppose a man laboring under one of those diseases which are neither serious and unsupportable, nor yet altogether mild, but such as that, upon making any mistake in diet, it will become apparent, as if he should eat bread and flesh, or any other of those articles which prove beneficial to healthy persons, and that, too, not in great quantity, but much less than he could have taken when in good health; and that another man in good health, having a constitution neither very feeble, nor yet strong, eats of those things which are wholesome and strengthening to an ox or a horse, such as vetches, barley, and the like, and that, too, not in great quantity, but much less than he could take; the healthy person who did so would be subjected to no less disturbance and danger than the sick person who took bread or cake unseasonably. All these things are proofs that Medicine is to be prosecuted and discovered by the same method as the other.

9. And if it were simply, as is laid down, that such things as are stronger prove injurious, but such as are weaker prove beneficial and nourishing, both to sick and healthy persons, it were an easy matter, for then the safest rule would be to circumscribe the diet to the lowest point. But then it is no less mistake, nor one that injures a man less, provided a deficient diet, or one consisting of weaker things than what are proper, be administered. For, in the constitution of man, abstinence may enervate, weaken, and kill. And there are many other ills, different from those of repletion, but no less dreadful, arising from deficiency of food; wherefore the practice in those cases is more varied, and requires greater accuracy. For one must aim at attaining a certain measure, and yet this measure admits neither weight nor calculation of any kind, by which it may be accurately determined, unless it be the sensation of the body; wherefore it is a task to learn this accurately, so as not to commit small blunders either on the one side or the other, and in fact I would give great praise to the physician whose mistakes are small, for perfect accuracy is seldom to be seen, since many physicians seem to me to be in the same plight as bad pilots, who, if they commit mistakes while conducting the ship in a calm do not expose themselves, but when a storm and violent hurricane overtake them, they then, from their ignorance and mistakes, are discovered to be what they are, by all men, namely, in losing their ship. And thus bad and commonplace physicians, when they treat men who have no serious illness, in which case one may commit great mistakes without producing any formidable mischief, (and such complaints occur much more frequently to men than dangerous ones); under these circumstances, when they commit mistakes, they do not expose themselves to ordinary men; but when they fall in with a great, a strong, and a dangerous disease, then their mistakes and want of skill are made apparent to all. Their punishment is not far off, but is swift in overtaking both the one and the other.^[362]

10. And that no less mischief happens to a man from unseasonable depletion than from repletion, may be clearly seen upon reverting to the consideration of persons in health. For, to some, with whom it agrees to take only one meal in the day, and they have arranged it so accordingly; whilst others, for the same reason, also take dinner, and this they do because they find it good for them, and not like those persons who, for pleasure or from any casual circumstance, adopt the one or the other custom: and to the bulk of mankind it is of little consequence which of these rules they observe, that is to say, whether they make it a practice to take one or two meals. But there are certain persons who cannot readily change their diet with impunity; and if they make any alteration in it for one day, or even for a part of a day, are greatly injured thereby. Such persons, provided they take dinner when it is not their wont, immediately become heavy and inactive, both in body and mind, and are weighed down with yawning, slumbering, and thirst; and if they take supper in addition, they are seized with flatulence, tormina, and diarrhoea, and to many this has been the commencement of a serious disease, when they have merely taken twice in a day the same food which they have been in the custom of taking once. And thus, also, if one who has been accustomed to dine, and this rule agrees with him, should not dine at the accustomed hour, he will straightway feel great loss of strength, trembling, and want of spirits, the eyes of such a person will become more pallid, his urine thick and hot, his mouth bitter; his bowels will seem, as it were, to hang loose; he will suffer from vertigo, lowness of spirit, and inactivity,—such are the effects; and if he should attempt to take at supper the same food which he was wont to partake of at dinner, it will appear insipid, and he will not be able to take it off; and these things, passing downwards with tormina and rumbling, burn up his bowels; he experiences insomnolency or troubled and disturbed dreams; and to many of them these symptoms are the commencement of some disease.

11. But let us inquire what are the causes of these things which happened to them. To him, then, who was accustomed to take only one meal in the day, they happened because he did not wait the proper time, until his bowels had completely derived benefit from and had digested the articles taken at the preceding meal, and until his belly had become soft, and got into a state of rest, but he gave it a new supply while in a state of heat and fermentation, for such bellies digest much more slowly, and require more rest and ease. And as to him who had been accustomed to dinner, since, as soon as the body required food, and when the former meal was consumed, and he wanted refreshment, no new supply was furnished to it, he wastes and is consumed from want of food. For all the symptoms which I describe as befalling to this man I refer to want of food. And I also

say that all men who, when in a state of health, remain for two or three days without food, experience the same unpleasant symptoms as those which I described in the case of him who had omitted to take dinner.

12. Wherefore, I say, that such constitutions as suffer quickly and strongly from errors in diet, are weaker than others that do not; and that a weak person is in a state very nearly approaching to one in disease; but a person in disease is the weaker, and it is, therefore, more likely that he should suffer if he encounters anything that is unseasonable. It is difficult, seeing that there is no such accuracy in the Art, to hit always upon what is most expedient, and yet many cases occur in medicine which would require this accuracy, as we shall explain. But on that account, I say, we ought not to reject the ancient Art, as if it were not, and had not been properly founded, because it did not attain accuracy in all things, but rather, since it is capable of reaching to the greatest exactitude by reasoning, to receive it and admire its discoveries, made from a state of great ignorance, and as having been well and properly made, and not from chance.

13. But I wish the discourse to revert to the new method of those who prosecute their inquiries in the Art by hypothesis. For if hot, or cold, or moist, or dry, be that which proves injurious to man, and if the person who would treat him properly must apply cold to the hot, hot to the cold, moist to the dry, and dry to the moist—let me be presented with a man, not indeed one of a strong constitution, but one of the weaker, and let him eat wheat, such as it is supplied from the thrashing-floor, raw and unprepared, with raw meat, and let him drink water. By using such a diet I know that he will suffer much and severely, for he will experience pains, his body will become weak, and his bowels deranged, and he will not subsist long. What remedy, then, is to be provided for one so situated? Hot? or cold? or moist? or dry? For it is clear that it must be one or other of these. For, according to this principle, if it is one of these which is injuring the patient, it is to be removed by its contrary. But the surest and most obvious remedy is to change the diet which the person used, and instead of wheat to give bread, and instead of raw flesh, boiled, and to drink wine in addition to these: for by making these changes it is impossible but that he must get better, unless completely disorganized by time and diet. What, then, shall we say? whether that, as he suffered from cold, these hot things being applied were of use to him, or the contrary? I should think this question must prove a puzzler to whomsoever it is put. For whether did he who prepared bread out of wheat remove the hot, the cold, the moist, or the dry principle in it?—for the bread is consigned both to fire and to water, and is wrought with many things, each of which has its peculiar property and nature, some of which it loses, and with others it is diluted and mixed.

14. And this I know, moreover, that to the human body it makes a great difference whether the bread be fine or coarse,^[363] of wheat with or without the hull, whether mixed with much or little water, strongly wrought or scarcely at all, baked or raw—and a multitude of similar differences; and so, in like manner, with the cake (maza); the powers of each, too, are great, and the one nowise like the other. Whoever pays no attention to these things, or, paying attention, does not comprehend them, how can he understand the diseases which befall a man? For, by every one of these things, a man is affected and changed this way or that, and the whole of his life is subjected to them, whether in health, convalescence, or disease. Nothing else, then, can be more important or more necessary to know than these things. So that the first inventors, pursuing their investigations properly, and by a suitable train of reasoning, according to the nature of man, made their discoveries, and thought the Art worthy of being ascribed to a god, as is the established belief. For they did not suppose that the dry or the moist, the hot or the cold, or any of these, are either injurious to man, or that man stands in need of them; but whatever in each was strong, and more than a match for a man's constitution, whatever he could not manage, that they held to be hurtful, and sought to remove. Now, of the sweet, the strongest is that which is intensely sweet; of the bitter, that which is intensely bitter; of the acid, that which is intensely acid; and of all things that which is extreme, for these things they saw both existing in man, and proving injurious to him. For there is in man the bitter and the salt, the sweet and the acid, the sour and the insipid,^[364] and a multitude of other things having all sorts of powers both as regards quantity and strength. These, when all mixed and mingled up with one another, are not apparent, neither do they hurt a man; but when any of them is separate, and stands by itself, then it becomes perceptible, and hurts a man. And thus, of articles of food, those which are unsuitable and hurtful to man when administered, every one is either bitter, or intensely so, or saltish or acid, or something else intense and strong, and therefore we are disordered by them in like manner as we are by the secretions in the body. But all those things of which a man eats and drinks are devoid of any such intense and well-marked quality, such as bread, cake, and many other things of a similar nature which man is accustomed to use for food, with the exception of condiments and confectionaries, which are made to gratify the palate and for luxury. And from those things, when received into the body abundantly, there is no disorder nor dissolution of the powers belonging to the body; but strength, growth, and nourishment result from them, and this for no other reason than because they are well mixed, have nothing in them of an immoderate character, nor anything strong, but the whole forms one simple and not strong substance.

15. I cannot think in what manner they who advance this doctrine, and transfer the Art from the cause I have described to hypothesis, will cure men according to the principle which they have laid down. For, as far as I know, neither the hot nor the cold, nor the dry, nor the moist, has ever been found unmixed with any other quality; but I suppose they use the same articles of meat and drink as all we other men do. But to this substance they give the attribute of being hot, to that cold, to that dry, and to that moist. Since it would be absurd to advise the patient to take something hot, for he would straightway ask what it is? so

that he must either play the fool, or have recourse to some one of the well-known substances: and if this hot thing happen to be sour, and that hot thing insipid, and this hot thing has the power of raising a disturbance in the body (and there are many other kinds of heat, possessing many opposite powers), he will be obliged to administer some one of them, either the hot and the sour, or the hot and the insipid, or that which, at the same time, is cold and sour (for there is such a substance), or the cold and the insipid. For, as I think, the very opposite effects will result from either of these, not only in man, but also in a bladder, a vessel of wood, and in many other things possessed of far less sensibility than man; for it is not the heat which is possessed of great efficacy, but the sour and the insipid, and other qualities as described by me, both in man and out of man, and that whether eaten or drunk, rubbed in externally, and otherwise applied.

16. But I think that of all the qualities heat and cold exercise the least operation in the body, for these reasons: as long time as hot and cold are mixed up with one another they do not give trouble, for the cold is tempered and rendered more moderate by the hot, and the hot by the cold; but when the one is wholly separate from the other, then it gives pain; and at that season when cold is applied it creates some pain to a man, but quickly, for that very reason, heat spontaneously arises in him without requiring any aid or preparation. And these things operate thus both upon men in health and in disease. For example, if a person in health wishes to cool his body during winter, and bathes either in cold water or in any other way, the more he does this, unless his body be fairly congealed, when he resumes his clothes and comes into a place of shelter, his body becomes more heated than before. And thus, too, if a person wish to be warmed thoroughly either by means of a hot bath or strong fire, and straightway having the same clothing on, takes up his abode again in the place he was in when he became congealed, he will appear much colder, and more disposed to chills than before. And if a person fan himself on account of a suffocating heat, and having procured refrigeration for himself in this manner, cease doing so, the heat and suffocation will be ten times greater in his case than in that of a person who does nothing of the kind. And, to give a more striking example, persons travelling in the snow, or otherwise in rigorous weather, and contracting great cold in their feet, their hands, or their head, what do they not suffer from inflammation and tingling when they put on warm clothing and get into a hot place? In some instances, blisters arise as if from burning with fire, and they do not suffer from any of those unpleasant symptoms until they become heated. So readily does either of these pass into the other; and I could mention many other examples. And with regard to the sick, is it not in those who experience a rigor that the most acute fever is apt to break out? And yet not so strongly neither, but that it ceases in a short time, and, for the most part, without having occasioned much mischief; and while it remains, it is hot, and passing over the whole body, ends for the most part in the feet, where the chills and cold were most intense and lasted longest: and, when sweat supervenes, and the fever passes off, the patient is much colder than if he had not taken the fever at all. Why then should that which so quickly passes into the opposite extreme, and loses its own powers spontaneously, be reckoned a mighty and serious affair? And what necessity is there for any great remedy for it?

17. One might here say—but persons in ardent fevers, pneumonia, and other formidable diseases, do not quickly get rid of the heat, nor experience these rapid alterations of heat and cold. And I reckon this very circumstance the strongest proof that it is not from heat simply that men get into the febrile state, neither is it the sole cause of the mischief, but that this species of heat is bitter, and that acid, and the other saltish, and many other varieties; and again there is cold combined with other qualities. These are what proves injurious: heat, it is true, is present also, possessed of strength as being that which conducts, is exacerbated and increased along with the other, but has no power greater than what is peculiar to itself.

18. With regard to these symptoms, in the first place those are most obvious of which we have all often had experience. Thus, then, in such of us as have a coryza and defluxion from the nostrils, this discharge is much more acrid than that which formerly was formed in and ran from them daily; and it occasions swelling of the nose, and it inflames, being of a hot and extremely ardent nature, as you may know, if you apply your hand to the place; and, if the disease remains long, the part becomes ulcerated although destitute of flesh and hard; and the heat in the nose ceases, not when the defluxion takes place and the inflammation is present, but when the running becomes thicker and less acrid, and more mixed with the former secretion, then it is that the heat ceases. But in all those cases in which this decidedly proceeds from cold alone, without the concurrence of any other quality, there is a change from cold to hot, and from hot to cold, and these quickly supervene, and require no coction. But all the others being connected, as I have said, with acrimony and intemperance of humors, pass off in this way by being mixed and concocted.

19. But such defluxions as are determined to the eyes being possessed of strong and varied acrimonies, ulcerate the eyelids, and in some cases corrode the cheeks and parts below the eyes upon which the flow, and even occasion rupture and erosion of the tunic which surrounds the eyeball. But pain, heat, and extreme burning prevail until the defluxions are concocted and become thicker, and concretions form about the eyes, and the coction takes place from the fluids being mixed up, diluted, and digested together. And in defluxions upon the throat, from which are formed hoarseness, cynanche, erysipelas, and pneumonia, all these have at first saltish, watery, and acrid discharges, and with these the diseases gain strength. But when the discharges become thicker, more concocted, and are freed from all acrimony, then, indeed, the fevers pass away, and the other symptoms which annoyed the patient; for we must account those things the cause of each complaint, which, being present in a certain fashion, the complaint exists, but it ceases when they change to another combination. But those which originate from pure heat

or cold, and do not participate in any other quality, will then cease when they undergo a change from cold to hot, and from hot to cold; and they change in the manner I have described before. Wherefore, all the other complaints to which man is subject arise from powers (qualities?). Thus, when there is an overflow of the bitter principle, which we call yellow bile, what anxiety, burning heat, and loss of strength prevail! but if relieved from it, either by being purged spontaneously, or by means of a medicine seasonably administered, the patient is decidedly relieved of the pains and heat; but while these things float on the stomach, unconcocted and undigested, no contrivance could make the pains and fever cease; and when there are acidities of an acrid and æruginous character, what varieties of frenzy, gnawing pains in the bowels and chest, and inquietude, prevail! and these do not cease until the acidities be purged away, or are calmed down and mixed with other fluids. The coction, change, attenuation, and thickening into the form of humors, take place through many and various forms; therefore the crises and calculations of time are of great importance in such matters; but to all such changes hot and cold are but little exposed, for these are neither liable to putrefaction nor thickening. What then shall we say of the change? that it is a combination (crasis) of these humors having different powers toward one another. But the hot does not lose its heat when mixed with any other thing except the cold; nor again, the cold, except when mixed with the hot. But all other things connected with man become the more mild and better in proportion as they are mixed with the more things besides. But a man is in the best possible state when they are concocted and at rest, exhibiting no one peculiar quality; but I think I have said enough in explanation of them.

20. Certain sophists and physicians say that it is not possible for any one to know medicine who does not know what man is [and how he was made and how constructed], and that whoever would cure men properly, must learn this in the first place. But this saying rather appertains to philosophy, as Empedocles and certain others have described what man in his origin is, and how he first was made and constructed.^[365] But I think whatever such has been said or written by sophist or physician concerning nature has less connection with the art of medicine than with the art of painting. And I think that one cannot know anything certain respecting nature from any other quarter than from medicine; and that this knowledge is to be attained when one comprehends the whole subject of medicine properly, but not until then; and I say that this history shows what man is, by what causes he was made, and other things accurately. Wherefore it appears to me necessary to every physician to be skilled in nature, and strive to know, if he would wish to perform his duties, what man is in relation to the articles of food and drink, and to his other occupations, and what are the effects of each of them to every one. And it is not enough to know simply that cheese is a bad article of food, as disagreeing with whoever eats of it to satiety, but what sort of disturbance it creates, and wherefore, and with what principle in man it disagrees; for there are many other articles of food and drink naturally bad which affect man in a different manner. Thus, to illustrate my meaning by an example, undiluted wine drunk in large quantity renders a man feeble; and everybody seeing this knows that such is the power of wine, and the cause thereof; and we know, moreover, on what parts of a man's body it principally exerts its action; and I wish the same certainty to appear in other cases. For cheese (since we used it as an example) does not prove equally injurious to all men, for there are some who can take it to satiety without being hurt by it in the least, but, on the contrary, it is wonderful what strength it imparts to those it agrees with; but there are some who do not bear it well, their constitutions are different, and they differ in this respect, that what in their body is incompatible with cheese, is roused and put in commotion by such a thing; and those in whose bodies such a humor happens to prevail in greater quantity and intensity, are likely to suffer the more from it. But if the thing had been pernicious to the whole nature of man, it would have hurt all. Whoever knows these things will not suffer from it.

21. During convalescence from diseases, and also in protracted diseases, many disorders occur, some spontaneously, and some from certain things accidentally administered. I know that the common herd of physicians, like the vulgar, if there happen to have been any innovation made about that day, such as the bath being used, a walk taken, or any unusual food eaten, all which were better done than otherwise, attribute notwithstanding the cause of these disorders, to some of these things, being ignorant of the true cause but proscribing what may have been very proper. Now this ought not to be so; but one should know the effects of a bath or a walk unseasonably applied; for thus there will never be any mischief from these things, nor from any other thing, nor from repletion, nor from such and such an article of food. Whoever does not know what effect these things produce upon a man, cannot know the consequences which result from them, nor how to apply them.

22. And it appears to me that one ought also to know what diseases arise in man from the powers, and what from the structures. What do I mean by this? By powers, I mean intense and strong juices; and by structures, whatever conformations there are in man. For some are hollow, and from broad contracted into narrow; some expanded, some hard and round, some broad and suspended,^[366] some stretched, some long, some dense, some rare and succulent,^[367] some spongy and of loose texture.^[368] Now, then, which of these figures is the best calculated to suck to itself and attract humidity from another body? Whether what is hollow and expanded, or what is solid and round, or what is hollow, and from broad, gradually turning narrow? I think such as from hollow and broad are contracted into narrow: this may be ascertained otherwise from obvious facts: thus, if you gape wide with the mouth you cannot draw in any liquid; but by protruding, contracting, and compressing the lips, and still more by using a tube, you can readily draw in whatever you wish. And thus, too, the instruments which are used for cupping are broad below and gradually become narrow, and are so constructed in order to suck and draw in from the fleshy parts. The nature and construction of the parts within a man are of a like nature; the bladder, the head, the uterus in

woman; these parts clearly attract, and are always filled with a juice which is foreign to them. Those parts which are hollow and expanded are most likely to receive any humidity flowing into them, but cannot attract it in like manner. Those parts which are solid and round could not attract a humidity, nor receive it when it flows to them, for it would glide past, and find no place of rest on them. But spongy and rare parts, such as the spleen, the lungs, and the breasts, drink up especially the juices around them, and become hardened and enlarged by the accession of juices. Such things happen to these organs especially. For it is not with the spleen as with the stomach, in which there is a liquid, which it contains and evacuates every day; but when it (the spleen) drinks up and receives a fluid into itself, the hollow and lax parts of it are filled, even the small interstices; and, instead of being rare and soft, it becomes hard and dense, and it can neither digest nor discharge its contents: these things it suffers, owing to the nature of its structure. Those things which engender flatulence or tormina in the body, naturally do so in the hollow and broad parts of the body, such as the stomach and chest, where they produce rumbling noises; for when they do not fill the parts so as to be stationary, but have changes of place and movements, there must necessarily be noise and apparent movements from them. But such parts as are fleshy and soft, in these there occur torpor and obstructions, such as happen in apoplexy. But when it (the flatus?) encounters a broad and resisting structure, and rushes against such a part, and this happens when it is by nature not strong so as to be able to withstand it without suffering injury; nor soft and rare, so as to receive or yield to it, but tender, juicy, full of blood, and dense, like the liver, owing to its density and broadness, it resists and does not yield. But flatus, when it obtains admission, increases and becomes stronger, and rushes toward any resisting object; but owing to its tenderness, and the quantity of blood which it (the liver) contains, it cannot be without uneasiness; and for these reasons the most acute and frequent pains occur in the region of it, along with suppurations and chronic tumors (phymata). These symptoms also occur in the site of the diaphragm, but much less frequently; for the diaphragm is a broad, expanded, and resisting substance, of a nervous (tendinous?) and strong nature, and therefore less susceptible of pain; and yet pains and chronic abscesses do occur about it.

23. There are both within and without the body many other kinds of structure, which differ much from one another as to sufferings both in health and disease; such as whether the head be small or large; the neck slender or thick, long or short; the belly long or round; the chest and ribs broad or narrow; and many others besides, all which you ought to be acquainted with, and their differences; so that knowing the causes of each, you may make the more accurate observations.

24. And, as has been formerly stated, one ought to be acquainted with the powers of juices, and what action each of them has upon man, and their alliances towards one another. What I say is this: if a sweet juice change to another kind, not from any admixture, but because it has undergone a mutation within itself; what does it first become?—bitter? salt? austere? or acid? I think acid. And hence, an acid juice is the most improper of all things that can be administered in cases in which a sweet juice is the most proper. Thus, if one should succeed in his investigations of external things, he would be the better able always to select the best; for that is best which is farthest removed from that which is unwholesome.

ON AIRS, WATERS, AND PLACES.

THE ARGUMENT.

Dr. Coray, in his excellent edition of this treatise, divides it into six chapters, as follows: first, the Introduction (from § 1–3) comprehends some general observations on the importance of cultivating a knowledge of the effects which the different seasons, the winds, the various kinds of water, the situation of cities, the nature of soils and the modes of life, exercise upon the health, and the necessity of a physician's making himself well acquainted with all these matters, if he would wish to practise his profession successfully. The author insists, with particular earnestness, on the utility of studying the constitution of the year and the nature of the seasons, and refutes the opinions of those persons, in his days, who held that a knowledge of all these things belongs to meteorology rather than to medicine. The second chapter (§ 3–7) treats of climate, and the diseases prevalent in localities characterized by their exposure to particular winds. Those winds being peculiar to Greece, their names occasion some trouble in order to understand them correctly, and we shall give below a summary of what the modern Greek Coray says in illustration of them. This part of the present treatise appears to have been highly elaborated, and contains much important information. The third chapter (§ 7–10) treats of the various kinds of water, and their effects in different states of the human constitution. The remarks contained here are of an eminently practical nature, and evidently must have been the results of patient observation and experiment, so that, even at the present day, it would be difficult to detect our author in a single error of judgment. In this place he has occasion to deliver his opinions on the formation of urinary calculi, which he does at considerable length; and I may be permitted to remark, whatever may be thought of his etiology of the disease, it will be admitted that his theory is plausible, and the best that could well have been framed in the state of knowledge which then prevailed on that subject. Indeed, even at the present day, it must be allowed that this is a dark subject; we have acquired, it is true, many new and curious facts connected with the minute structure of these concretions, but it can hardly be affirmed that we have been able to evolve from them any general principles, or certain rules of practice. In the fourth chapter (§ 10–12), the nature of the seasons is treated of, and their influence on the health circumstantially stated. Some of the observations contained in this part of the work are remarkable for their acuteness and originality, such as the following, that, in estimating the effects of a season on the health, we ought to take into account the seasons which preceded it. This is well expressed by Celsus, as follows: “neque solum interest quales dies sint sed etiam quales præcesserint.” (Præfat.) See also Hippocrates (de Humoribus, § 8); and Coray (ad h. l. § cix.) It will be seen in our annotations that a considerable number of the Aphorisms are abstracted from this part of the present treatise. In the fifth chapter (§ 12–17), the effects of climate and the institutions of society on the inhabitants of Asia are treated of at considerable length. Our author, in this place, evinces a great acquaintance with human life, and a most philosophical spirit in contemplating the subject which he is handling. Indeed few works in any language display so much accurate observation and originality of thought. The varieties of disposition, and of intellectual and moral development among mankind, are set down as being derived, in a great measure, from differences of climate and modes of government. Thus the Asiatics are of an effeminate and slavish disposition, because they live in a soft climate, on a rich soil, where they are little exposed to hardships or labor, and under a despotic form of government, which arrests the development of their mental energies.^[369] This part also contains some interesting observations on the Macrocephali and the inhabitants of Phasis. In the sixth and last chapter (§ 17 to the end), the peculiar traits of the European character, as connected with climate and institutions, are described in a very interesting manner. Here the observations on the Amazons, Sauromatæ, and Scythians are well deserving of an attentive perusal, and more particularly the description of the disease induced by continual riding on horseback, the probable nature of which we shall consider presently. Here, too, are given our author's remarks on diseases supposed to be divine, which, as we have stated in the Preliminary section on his life, evince a wonderful exemption from the superstitious belief of his age, and indicate an extraordinary depth of thought.

This is a general outline of the contents of this treatise, which is one of the most celebrated in the whole Collection. From what we have stated, it will at once be seen that it relates to a subject of commanding interest, and deserves to be carefully studied, as containing the oldest exposition which we possess of the opinions entertained by an original and enlightened mind on many important questions connected with Public Hygiene and Political Economy, two sciences which, of late years, have commanded a large amount of professional attention. Whether or not modern experience may confirm our author's judgment in every particular case, it surely can neither be unprofitable nor uninteresting to ascertain what his opinions on these subjects actually were. Let us be thankful, then, that the destroying hand of time has spared us so valuable a relic of antiquity; and, instead of undervaluing our ancient instructor because he shows himself ignorant of many truths which we are now familiar with, let us be grateful to him for the amount of information which he has supplied to us, and for setting us an example which it must be both safe and profitable for us to follow. Surely great praise is due to the man who first mooted so many important questions, and stated their bearings in distinct terms, although he did not always succeed in solving them.^[370]

I may take the present opportunity of mentioning that M. Littré, with some appearance of truth, blames Hippocrates for having rather overrated the influence of climate and institutions, in producing military valor, which, as he justly remarks, has

been proved by modern examples to be most intimately connected with discipline, and a knowledge of the arts of war. But if Hippocrates was wrong on this point, it was because he did not avail himself properly of the lights of his own age; for he might have learned from his contemporary, Socrates, the very doctrine which M. Littré here inculcates. “The question being put to him,” says Xenophon, “whether valor was a thing that could be taught, or was natural? I am of opinion, he said, that as one body is born with greater powers than another for enduring labor, so is one soul produced by nature stronger than another for enduring dangers. For I see persons brought up under the same institutions and habits differing much from one another in courage. But I think that every nature may be improved in valour by learning and discipline. For it is obvious that the Scythians and Thracians would not dare to contend with the Lacedemonians with bucklers and spears; and it is clear that, the Lacedemonians would not be willing to contend with the Thracians with small targets and javelins, or with the Scythians with bows and arrows.” (*Memorab.* iii., 9.) The same doctrine is taught with remarkable subtlety of argument and originality of thought in the “*Protagoras*” of Plato, (see § 97). If, then, Hippocrates was wrong on this head, (which, however, may be doubted), it is clear that he is not to be screened by the alleged ignorance of his age, and that he might have put himself right by attending to the instructions of a contemporary with whom he, in all probability, was familiar, and who undoubtedly was the greatest master of human nature that ever existed.

As there are certain matters connected with this treatise which will require a more lengthened discussion than can well suit with foot notes, I think it advisable to treat of them in this place:—

1. With regard to the seasons of the year, as indicated by the risings and settings of the stars. the following observations, taken in a great measure from Clifton’s Preface will supply, in as brief a space as possible, all the information that will be required; “As the reader will find frequent mention of seasons, equinoxes, solstices, risings and settings of the sun and stars (particularly Arcturus, the Dog-star, and the Pleiades), it may not be amiss to premise, in the first place, that a year was divided by the ancients into four parts, every one of these was distinguished astronomically.

“Thus, for instance, the winter began at the setting of the Pleiades, and continued to the vernal equinox.

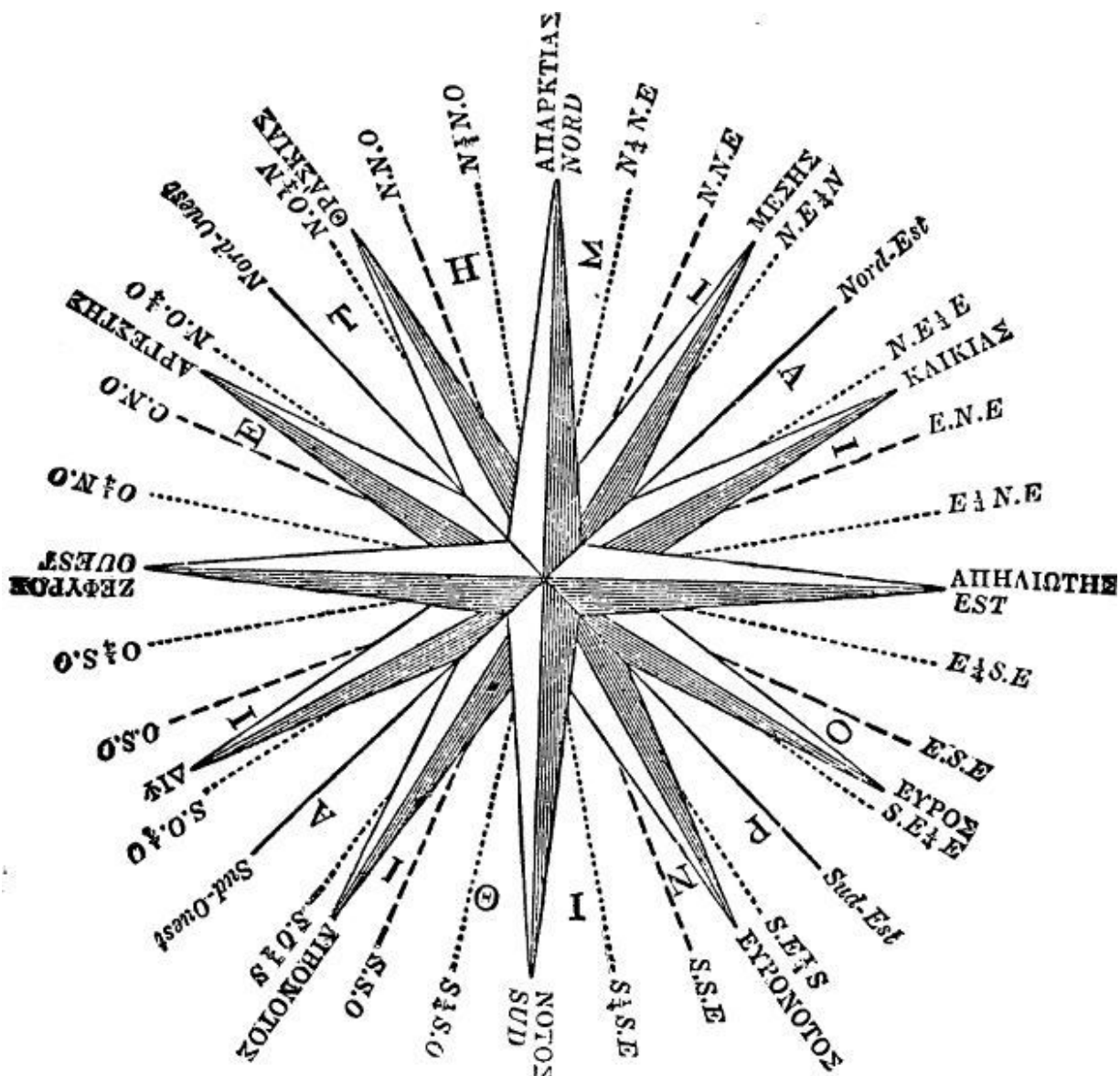
“The spring began at the vernal equinox, and ended at the rising of the Pleiades.

“The summer began at the rising of the Pleiades, and ended at the rising of Arcturus.

“The autumn began at the rising of Arcturus, and ended at the setting of the Pleiades.

“The rising and setting of the stars is always to be understood of what astronomers call the heliacal rising or setting, i. e. when a star rises or sets with the sun.

“The rising and setting of the sun in summer or winter (an expression which often occurs in this treatise), implies those points of the compass the sun rises and sets at.”^[371]



II. On the winds, of which frequent mention is made by our author, Coray has treated with a degree of prolixity and earnestness for which it is difficult to recognize the necessity. The figure given above, if properly studied and understood, will supply the professional reader with all the information he will require on this head.

III. One of the most singular diseases noticed in this work is the effeminacy with which the Scythians are said to have been attacked in consequence of spending the greater part of their time on horseback. (See § 22.) As the subject has attracted a good deal of attention lately, I will give a summary of the information which has been collected respecting it. See Coray, etc., t. ii., p. 331; Littré, t. ii., p. 5, 6; and Avert., xxxix., p. 47; t. iv., p. 9.

In the first place, then, it can scarcely admit of doubt that the disease is the same as that which Herodotus describes in the following passage: "Venus inflicted upon the Scythians, who pillaged her temple at Ascalon, and on their descendants, *the feminine disease*; at least it is to this cause that they attribute their disease; and travellers that go to the land of Scythia see how these persons are affected whom the Scythians called *accursed* (ἐναρετες)." ^[372]

All the opinions which have been entertained respecting this affection are referred by M. Littré to the three following categories:

I. A vice, namely (A), Pederasty, which, he says is the most ancient opinion we have respecting it, as indicated by Longinus ^[373] (on the Sublime, 25), and defended by his commentators, Toll and Pearce, and by Casaubon and Coster. ^[374] (B),

Onanism, the opinion to which Sprengel inclines in his work on Hippocrates.

2. A bodily disease, to wit: (A), Hemorrhoids, as maintained by Paul Thomas de Girac,^[375] by Valkenæer, by Bayer,^[376] and by the Compilers of the 'Universal History.'^[377] (B), A true menstruation, as appears to be maintained by Lefevre and Dacier,^[378] and by others. (C), Blenorragia, as Guy Patin^[379] and others suppose. (D), A true impotence, as held by Mercuriali and others.

3. A mental disease, as maintained by Sauvages,^[380] Heyne,^[381] Coray,^[382] and others.

M. Rosenbaum is at great pains to make out that the affection in question was pederasty, and that the *accursed* (ἐνάρπες) of Herodotus were the same as the *pathici* of the Romans. I must say, that in my opinion Rosenbaum makes out a strong case in support of this opinion. In particular it will be remarked, that Herodotus says, the descendants of these Scythians were also afflicted with this complaint. Now Celsus Aurelianus says expressly, that the affection of the *pathici* was hereditary.^[383] Taking everything into account, I must say that my own opinion has always been that the disease in question must have been some variety of *spermatorrhœa*. I need scarcely remark that this affection induces a state, both of body and mind, analogous to that of the *pathici*, as described by ancient authors.

Before leaving this subject, however, I should mention that M. Littré, in the fourth volume of his Hippocrates (p. xi.), brings into view a thesis by M. Graff, the object of which is to prove that the disease of the Scythians was a true sort of impotence; and in illustration of it, he cites a passage from the memoirs of M. Larrey, containing a description of a species of impotence, attended with wasting of the testicle, which attacked the French army in Egypt. But, as far as I can see, this disease described by Larrey had nothing to do with riding on horseback, and I cannot see any relation between it and the diseases described by Herodotus and Hippocrates.

IV. Of all the legendary tales of antiquity, there is probably no one which was so long and so generally credited by the best informed historians, critics, geographers, poets, and philosophers, as the story of the Amazons. They are noticed historically by Homer (*Iliad*, iii., 186; vi., 152); Apollonius Rhodius (ii., 196); Pindar (*Olymp.* xiii., 84); Herodotus (ix., 27); Lysias (*Epitaph.* 3); Plato (*Menex.*); Isocrates (*Panegy.*); Ctesias (*Persic.*); Plutarch (*Theseus*); Strabo (*Geogr.* ix.); Pausanias (iv., 31, 6; vii., 2, 4); Arrian (*Exped. Alexand.*); Quintus Curtius (vi., 4). Now it is singular that in all this list of authorities, which, it will be remarked, comprehends the *élite* of ancient scholars, no one, with the exception of Strabo, ventures to express the slightest doubt respecting the actual existence of the Amazons. Some of them, indeed, admit that the race had become extinct in their time; but they all seem satisfied that the Amazons had truly existed in a bygone age, and consequently they acknowledge them as real historical personages. See, in particular, Arrian, who, although compelled by his respect for truth to acknowledge that they did not exist in the days of Alexander the Great, still does not hesitate to declare that it appeared incredible that this race of women, celebrated as they were by the most eminent authors, should never have existed at all. Yet, notwithstanding the mass of evidence in support of their actual existence, I suppose few scholars nowadays will hesitate to agree with Heyne (*Apollodor.* ii., 5, 9), and with Grote (*Hist. of Greece*, i., 2), in setting down the whole story as mere myth. But, considering how generally it had been believed, we need not wonder that Hippocrates in this treatise should appear to entertain no doubt of their actual existence. The reader will remark that he makes the locality of the Amazons to be in Europe, among the Sarmatians, on the north side of the Euxine. It is generally taken for granted, however, in the ancient myths, that their place of residence was on the banks of the Thermodon, in Cappadocia, and they are described as having afterwards crossed to the opposite side of the Euxine, when expelled from this locality. But, in fact, they are remarkable so much for nothing as their ubiquity, being sometimes located in Asia, sometimes in Africa, and at other times in Athens. I may remark, before concluding, that Mr. Payne Knight (*Symbolical Language*, etc., *Classical Journal*, 23), and Creuzer (*Symbolik*, etc.), give a symbolical interpretation to the story of the Amazons; but this mode of explaining the myths of antiquity is altogether fanciful and unsatisfactory. It seems safer and more judicious to deal with them as Mr. Grote has done,^[384] that is to say, to receive them as tales in which the ancients believed, without having any rational foundation for their faith. That there may have been a certain basis of truth in the story of the Amazons need not be denied; but in this, as in all the ancient myths, it is a hopeless task to attempt to separate truth from fiction.

ON AIRS, WATERS, AND PLACES.

1. Whoever wishes to investigate medicine properly, should proceed thus: in the first place to consider the seasons of the year, and what effects each of them produces (for they are not at all alike, but differ much from themselves in regard to their changes).^[385] Then the winds, the hot and the cold, especially such as are common to all countries, and then such as are peculiar to each locality. We must also consider the qualities of the waters, for as they differ from one another in taste and weight, so also do they differ much in their qualities. In the same manner, when one comes into a city to which he is a stranger, he ought to consider its situation, how it lies as to the winds and the rising of the sun; for its influence is not the same whether it lies to the north or the south, to the rising or to the setting sun. These things one ought to consider most attentively, and concerning the waters which the inhabitants use, whether they be marshy and soft, or hard, and running from elevated and

rocky situations, and then if saltish and unfit for cooking; and the ground, whether it be naked and deficient in water, or wooded and well watered, and whether it lies in a hollow, confined situation, or is elevated and cold; and the mode in which the inhabitants live, and what are their pursuits, whether they are fond of drinking and eating to excess, and given to indolence, or are fond of exercise and labor, and not given to excess in eating and drinking.^[386]

2. From these things he must proceed to investigate everything else. For if one knows all these things well, or at least the greater part of them, he cannot miss knowing, when he comes into a strange city, either the diseases peculiar to the place, or the particular nature of common diseases, so that he will not be in doubt as to the treatment of the diseases, or commit mistakes, as is likely to be the case provided one had not previously considered these matters. And in particular, as the season and the year advances, he can tell what epidemic diseases will attack the city, either in summer or in winter, and what each individual will be in danger of experiencing from the change of regimen. For knowing the changes of the seasons, the risings and settings of the stars, how each of them takes place, he will be able to know beforehand what sort of a year is going to ensue. Having made these investigations, and knowing beforehand the seasons, such a one must be acquainted with each particular, and must succeed in the preservation of health, and be by no means unsuccessful in the practice of his art. And if it shall be thought that these things belong rather to meteorology,^[387] it will be admitted, on second thoughts, that astronomy contributes not a little, but a very great deal, indeed, to medicine. For with the seasons the digestive organs of men undergo a change.

3. But how each of the afore-mentioned things should be investigated and explained, I will now declare in a clear manner. A city that is exposed to hot winds (these are between the wintry rising, and the wintry setting of the sun), and to which these are peculiar, but which is sheltered from the north winds; in such a city the waters will be plenteous and saltish, and as they run from an elevated source, they are necessarily hot in summer, and cold in winter;^[388] the heads of the inhabitants are of a humid and pituitous constitution, and their bellies subject to frequent disorders, owing to the phlegm running down from the head; the forms of their bodies, for the most part, are rather flabby; they do not eat nor drink much; drinking wine in particular, and more especially if carried to intoxication, is oppressive to them; and the following diseases are peculiar to the district: in the first place, the women are sickly and subject to excessive menstruation; then many are unfruitful from disease, and not from nature, and they have frequent miscarriages; infants are subject to attacks of convulsions and asthma, which they consider to be connected with infancy,^[389] and hold to be a sacred disease (epilepsy). The men are subject to attacks of dysentery, diarrhœa, hepialus,^[390] chronic fevers in winter, of epinyctis,^[391] frequently, and of hemorrhoids about the anus. Pleurisies, peripneumonies, ardent fevers, and whatever diseases are reckoned acute, do not often occur, for such diseases are not apt to prevail where the bowels are loose. Ophthalmies occur of a humid character, but not of a serious nature, and of short duration, unless they attack epidemically from the change of the seasons. And when they pass their fiftieth year, defluxions supervening from the brain, render them paralytic when exposed suddenly to strokes of the sun,^[392] or to cold. These diseases are endemic to them, and, moreover, if any epidemic disease connected with the change of the seasons, prevail, they are also liable to it.

4. But the following is the condition of cities which have the opposite exposure, namely, to cold winds, between the summer settings and the summer risings of the sun, and to which these winds are peculiar, and which are sheltered from the south and the hot breezes. In the first place the waters are, for the most part, hard and cold. The men must necessarily be well braced and slender, and they must have the discharges downwards of the alimentary canal hard, and of difficult evacuation, while those upwards are more fluid, and rather bilious than pituitous. Their heads are sound and hard, and they are liable to burstings (of vessels?) for the most part. The diseases which prevail epidemically with them, are pleurisies, and those which are called acute diseases. This must be the case when the bowels are bound; and from any causes, many become affected with suppurations in the lungs, the cause of which is the tension of the body, and hardness of the bowels; for their dryness and the coldness of the water dispose them to ruptures (of vessels?). Such constitutions must be given to excess of eating, but not of drinking; for it is not possible to be gourmands and drunkards at the same time. Ophthalmies, too, at length supervene; these being of a hard and violent nature, and soon ending in rupture of the eyes; persons under thirty years of age are liable to severe bleedings at the nose in summer; attacks of epilepsy are rare but severe. Such people are likely to be rather long-lived; their ulcers are not attended with serous discharges, nor of a malignant character; in disposition they are rather ferocious than gentle. The diseases I have mentioned are peculiar to the men, and besides they are liable to any common complaint which may be prevailing from the changes of the seasons. But the women, in the first place, are of a hard constitution, from the waters being hard, indigestible, and cold; and their menstrual discharges are not regular, but in small quantity, and painful. Then they have difficult parturition, but are not very subject to abortions. And when they do bring forth children, they are unable to nurse them; for the hardness and indigestible nature of the water puts away their milk. Phthisis frequently supervenes after childbirth, for the efforts of it frequently bring on ruptures and strains.^[393] Children while still little are subject to dropsies in the testicle, which disappear as they grow older; in such a town they are late in attaining manhood. It is, as I have now stated, with regard to hot and cold winds and cities thus exposed.

5. Cities that are exposed to winds between the summer and the winter risings of the sun, and those the opposite to them, have the following characters:—Those which lie to the rising of the sun are all likely to be more healthy than such as are

turned to the North, or those exposed to the hot winds, even if there should not be a furlong between them.^[394] In the first place, both the heat and cold are more moderate. Then such waters as flow to the rising sun, must necessarily be clear, fragrant, soft, and delightful to drink, in such a city. For the sun in rising and shining upon them purifies them, by dispelling the vapors which generally prevail in the morning. The persons of the inhabitants are, for the most part, well colored and blooming, unless some disease counteract. The inhabitants have clear voices, and in temper and intellect are superior to those, which are exposed to the north, and all the productions of the country in like manner are better. A city so situated resembles the spring as to moderation between heat and cold, and the diseases are few in number, and of a feeble kind, and bear a resemblance to the diseases which prevail in regions exposed to hot winds. The women there are very prolific, and have easy deliveries. Thus it is with regard to them.

6. But such cities as lie to the west, and which are sheltered from winds blowing from the east, and which the hot winds and the cold winds of the north scarcely touch, must necessarily be in a very unhealthy situation: in the first place the waters are not clear, the cause of which is, because the mist prevails commonly in the morning, and it is mixed up with the water and destroys its clearness, for the sun does not shine upon the water until he be considerably raised above the horizon. And in summer, cold breezes from the east blow and dews fall; and in the latter part of the day the setting sun particularly scorches the inhabitants, and therefore they are pale and enfeebled, and are partly subject to all the aforesaid diseases, but no one is peculiar to them. Their voices are rough and hoarse owing to the state of the air, which in such a situation is generally impure and unwholesome, for they have not the northern winds to purify it; and these winds they have are of a very humid character, such being the nature of the evening breezes. Such a situation of a city bears a great resemblance to autumn as regards the changes of the day, inasmuch as the difference between morning and evening is great. So it is with regard to the winds that are conducive to health, or the contrary.

7. And I wish to give an account of the other kinds of waters, namely, of such as are wholesome and such as are unwholesome, and what bad and what good effects may be derived from water; for water contributes much towards health.^[395] Such waters then as are marshy, stagnant, and belong to lakes, are necessarily hot in summer, thick, and have a strong smell, since they have no current; but being constantly supplied by rain-water, and the sun heating them, they necessarily want their proper color, are unwholesome and form bile; in winter, they become congealed, cold, and muddy with the snow and ice, so that they are most apt to engender phlegm, and bring on hoarseness; those who drink them have large and obstructed spleens, their bellies are hard, emaciated, and hot; and their shoulders, collar-bones, and faces are emaciated; for their flesh is melted down and taken up by the spleen, and hence they are slender; such persons then are voracious and thirsty; their bellies are very dry both above and below, so that they require the strongest medicines.^[396] This disease is habitual to them both in summer and in winter, and in addition they are very subject to dropsies of a most fatal character; and in summer dysenteries, diarrhoeas, and protracted quartan fevers frequently seize them, and these diseases when prolonged dispose such constitutions to dropsies, and thus prove fatal. These are the diseases which attack them in summer; but in winter younger persons are liable to pneumonia, and maniacal affections; and older persons to ardent fevers, from hardness of the belly. Women are subject to oedema and leucophlegmasiæ;^[397] when pregnant then during nursing they become wasted and sickly, and the lochial discharge after parturition does not proceed properly with the women. The children are particularly subject to hernia, and adults to varices and ulcers on their legs, so that persons with such constitutions cannot be long-lived, but before the usual period they fall into a state of premature old age. And further, the women appear to be with child, and when the time of parturition arrives, the fulness of the belly disappears, and this happens from dropsy of the uterus.^[398] Such waters then I reckon bad for every purpose. The next to them in badness are those which have their fountains in rocks, so that they must necessarily be hard, or come from a soil which produces thermal waters, such as those having iron, copper, silver, gold, sulphur, alum, bitumen, or nitre (soda) in them: for all these are formed by the force of heat.^[399] Good waters cannot proceed from such a soil, but those that are hard and of a heating nature, difficult to pass by urine, and of difficult evacuation by the bowels. The best are those which flow from elevated grounds, and hills of earth; these are sweet, clear, and can bear a little wine; they are hot in summer and cold in winter, for such necessarily must be the waters from deep wells. But those are most to be commended which run to the rising of the sun, and especially to the summer sun; for such are necessarily more clear, fragrant, and light. But all such as are saltish, crude, and hard, are not good for drink. But there are certain constitutions and diseases with which such waters agree when drunk, as I will explain presently. Their characters are as follows: the best are such as have their fountains to the east; the next, those between the summer risings and settings of the sun, and especially those to the risings; and third, those between the summer and winter settings; but the worst are those to the south, and the parts between the winter rising and setting, and those to the south are very bad, but those to the north are better. They are to be used as follows: whoever is in good health and strength need not mind, but may always drink whatever is at hand. But whoever wishes to drink the most suitable for any disease, may accomplish his purpose by attending to the following directions: To persons whose bellies are hard and easily burnt up, the sweetest, the lightest, and the most limpid waters will be proper; but those persons whose bellies are soft, loose, and pituitous, should choose the hardest, those kinds that are most crude, and the saltest, for thus will they be most readily dried up; for such waters as are adapted for boiling, and are of a very solvent nature, naturally loosen readily and melt

down the bowels; but such as are intractable, hard, and by no means proper for boiling, these rather bind and dry up the bowels. People have deceived themselves with regard to salt waters, from inexperience, for they think these waters purgative, whereas they are the very reverse; for such waters are crude, and ill adapted for boiling, so that the belly is more likely to be bound up than loosened by them.^[400] And thus it is with regard to the waters of springs.

8. I will now tell how it is with respect to rain-water, and water from snow. Rain waters, then, are the lightest, the sweetest, the thinnest, and the clearest; for originally the sun raises and attracts the thinnest and lightest part of the water, as is obvious from the nature of salts; for the saltish part is left behind owing to its thickness and weight, and forms salts; but the sun attracts the thinnest part, owing to its lightness, and he abstracts this not only from the lakes, but also from the sea, and from all things which contain humidity, and there is humidity in everything; and from man himself the sun draws off the thinnest and lightest part of the juices. As a strong proof of this, when a man walks in the sun, or sits down having a garment on, whatever parts of the body the sun shines upon do not sweat, for the sun carries off whatever sweat makes its appearance; but those parts which are covered by the garment, or anything else, sweat, for the particles of sweat are drawn and forced out by the sun, and are preserved by the cover so as not to be dissipated by the sun; but when the person comes into the shade the whole body equally perspires, because the sun no longer shines upon it.^[401] Wherefore, of all kinds of water, these spoil the soonest; and rain water has a bad smell, because its particles are collected and mixed together from most objects, so as to spoil the soonest. And in addition to this, when attracted and raised up, being carried about and mixed with the air, whatever part of it is turbid and darkish is separated and removed from the other, and becomes cloud and mist, but the most attenuated and lightest part is left, and becomes sweet, being heated and concocted by the sun, for all other things when concocted become sweet. While dissipated then and not in a state of consistence it is carried aloft. But when collected and condensed by contrary winds, it falls down wherever it happens to be most condensed. For this is likely to happen when the clouds being carried along and moving with a wind which does not allow them to rest, suddenly encounters another wind and other clouds from the opposite direction: there it is first condensed, and what is behind is carried up to the spot, and thus it thickens, blackens, and is conglomerated, and by its weight it falls down and becomes rain. Such, to all appearance, are the best of waters, but they require to be boiled and strained;^[402] for otherwise they have a bad smell, and occasion hoarseness and thickness of the voice to those who drink them.^[403] Those from snow and ice are all bad, for when once congealed, they never again recover their former nature; for whatever is clear, light, and sweet in them, is separated and disappears; but the most turbid and weightiest part is left behind.^[404] You may ascertain this in the following manner: If in winter you will pour water by measure into a vessel and expose it to the open air until it is all frozen, and then on the following day bring it into a warm situation where the ice will thaw, if you will measure the water again when dissolved you will find it much less in quantity. This is a proof that the lightest and thinnest part is dissipated and dried up by the congelation, and not the heaviest and thickest, for that is impossible:^[405] wherefore I hold that waters from snow and ice, and those allied to them, are the worst of any for all purposes whatever. Such are the characters of rain-water, and those from ice and snow.

9.^[406] Men become affected with the stone, and are seized with diseases of the kidneys, strangury, sciatica, and become ruptured, when they drink all sorts of waters, and those from great rivers into which other rivulets run, or from a lake into which many streams of all sorts flow, and such as are brought from a considerable distance. For it is impossible that such waters can resemble one another, but one kind is sweet, another saltish and aluminous, and some flow from thermal springs; and these being all mixed up together disagree, and the strongest part always prevails; but the same kind is not always the strongest, but sometimes one and sometimes another, according to the winds, for the north wind imparts strength to this water, and the south to that, and so also with regard to the others. There must be deposits of mud and sand in the vessels from such waters, and the aforesaid diseases must be engendered by them when drunk, but why not to all I will now explain. When the bowels are loose and in a healthy state,^[407] and when the bladder is not hot, nor the neck of the bladder very contracted, all such persons pass water freely, and no concretion forms in the bladder; but those in whom the belly is hot, the bladder must be in the same condition; and when preternaturally heated, its neck becomes inflamed; and when these things happen, the bladder does not expel the urine, but raises its heat excessively. And the thinnest part of it is secreted, and the purest part is passed off in the form of urine, but the thickest and most turbid part is condensed and concreted, at first in small quantity, but afterwards in greater; for being rolled about in the urine, whatever is of a thick consistence it assimilates to itself, and thus it increases and becomes indurated. And when such persons make water, the stone forced down by the urine falls into the neck of the bladder and stops the urine, and occasions intense pain; so that calculous children rub their privy parts and tear at them, as supposing that the obstruction to the urine is situated there. As a proof that it is as I say, persons affected with calculus have very limpid urine, because the thickest and foulest part remains and is concreted.^[408] Thus it generally is in cases of calculus. It forms also in children from milk, when it is not wholesome, but very hot and bilious, for it heats the bowels and bladder, so that the urine being also heated undergoes the same change. And I hold that it is better to give children only the most diluted wine, for such will least burn up and dry the veins. Calculi do not form so readily in women, for in them the urethra is short and wide, so that in them the urine is easily expelled; neither do they rub the pudendum with their hands, nor handle the passage like males;^[409]

for the urethra in women opens direct into the pudendum, which is not the case with men, neither in them is the urethra so wide, and they drink more than children do.^[410] Thus, or nearly so, is it with reward to them.

10. And respecting the seasons, one may judge whether the year will prove sickly or healthy from the following observations:^[411]—If the appearances connected with the rising and setting stars be as they should be; if there be rains in autumn; if the winter be mild, neither very tepid nor unseasonably cold, and if in spring the rains be seasonable, and so also in summer, the year is likely to prove healthy. But if the winter be dry and northerly, and the spring showery and southerly, the summer will necessarily be of a febrile character, and give rise to ophthalmies and dysenteries.^[412] For when suffocating heat sets in all of a sudden, while the earth is moistened by the vernal showers, and by the south wind, the heat is necessarily doubled from the earth, which is thus soaked by rain and heated by a burning sun, while, at the same time, men's bellies are not in an orderly state, nor the brain properly dried; for it is impossible, after such a spring, but that the body and its flesh must be loaded with humors, so that very acute fevers will attack all, but especially those of a phlegmatic constitution. Dysenteries are also likely to occur to women and those of a very humid temperament. And if at the rising of the Dog-star rain and wintry storms supervene, and if the etesian winds blow, there is reason to hope that these diseases will cease, and that the autumn will be healthy; but if not, it is likely to be a fatal season to children and women, but least of all to old men; and that convalescents will pass into quartans, and from quartans into dropsies; but if the winter be southerly, showery and mild, but the spring northerly, dry, and of a wintry character, in the first place women who happen to be with child, and whose accouchement should take place in spring, are apt to miscarry; and such as bring forth, have feeble and sickly children, so that they either die presently or are tender, feeble, and sickly, if they live. Such is the case with the women. The others are subject to dysenteries^[413] and dry ophthalmies, and some have catarrhs beginning in the head and descending to the lungs. Men of a phlegmatic temperament are likely to have dysenteries; and women, also, from the humidity of their nature, the phlegm descending downwards from the brain; those who are bilious, too, have dry ophthalmies from the heat and dryness of their flesh; the aged, too, have catarrhs from their flabbiness and melting of the veins, so that some of them die suddenly and some become paralytic on the right side or the left.^[414] For when, the winter being southerly and the body hot, the blood and veins are not properly constricted; a spring that is northerly, dry, and cold, having come on, the brain when it should have been expanded and purged, by the coryza and hoarseness is then constricted and contracted, so that the summer and the heat occurring suddenly, and a change supervening, these diseases fall out. And such cities as lie well to the sun and winds, and use good waters, feel these changes less, but such as use marshy and poorly waters, and lie well both as regards the winds and the sun, these all feel it more. And if the summer be dry, those diseases soon cease, but if rainy, they are protracted; and there is danger of any sore that there is becoming phagedenic from any cause; and lenteries and dropsies supervene at the conclusion of diseases; for the bowels are not readily dried up. And if the summer be rainy and southerly, and next the autumn, the winter must, of necessity, be sickly, and ardent fevers are likely to attack those that are phlegmatic, and more elderly than forty years, and pleurisies and peripneumonies^[415] those that are bilious. But if the summer is parched and northerly, but the autumn rainy and southerly, headache and sphaelus of the brain^[416] are likely to occur; and in addition hoarseness, coryza, coughs, and in some cases, consumption.^[417] But if the season is northerly and without water, there being no rain, neither after the Dog-star nor Arcturus; this state agrees best with those who are naturally phlegmatic, with those who are of a humid temperament, and with women: but it is most inimical to the bilious; for they become much parched up, and ophthalmies of a dry nature supervene, fevers both acute and chronic, and in some cases melancholy;^[418] for the most humid and watery part of the bile being consumed, the thickest and most acrid portion is left, and of the blood likewise, whence these diseases come upon them. But all these are beneficial to the phlegmatic, for they are thereby dried up, and reach winter not oppressed with humors, but with them dried up.

11. Whoever studies and observes these things may be able to foresee most of the effects which will result from the changes of the seasons: and one ought to be particularly guarded during the greatest changes of the seasons, and neither willingly give medicines, nor apply the cautery to the belly, nor make incisions there until ten or more days be past. Now, the greatest and most dangerous are the two solstices, and especially the summer, and also the two equinoxes, but especially the autumnal.^[419] One ought also to be guarded about the rising of the stars, especially of the Dog-star, then of Arcturus, and then the setting of the Pleiades; for diseases are especially apt to prove critical in those days, and some prove fatal, some pass off, and all others change to another form and another constitution. So it is with regard to them.

12. I wish to show, respecting Asia and Europe, how, in all respects, they differ from one another, and concerning the figure of the inhabitants, for they are different, and do not at all resemble one another. To treat of all would be a long story, but I will tell you how I think it is with regard to the greatest and most marked differences. I say, then, that Asia differs very much from Europe as to the nature of all things, both with regard to the productions of the earth and the inhabitants, for everything is produced much more beautiful and large in Asia; the country is milder, and the dispositions of the inhabitants also are more gentle and affectionate.^[420] The cause of this is the temperature of the seasons, because it lies in the middle of the risings of the sun^[421] towards the east, and removed from the cold (and heat),^[422] for nothing tends to growth and mildness so much as

when the climate has no predominant quality, but a general equality of temperature prevails. It is not everywhere the same with regard to Asia, but such parts of the country as lie intermediate between the heat and the cold, are the best supplied with fruits and trees, and have the most genial climate, and enjoy the purest waters, both celestial and terrestrial. For neither are they much burnt up by the heat, nor dried up by the drought and want of rain, nor do they suffer from the cold; since they are well watered from abundant showers and snow, and the fruits of the season,^[423] as might be supposed, grow in abundance, both such as are raised from seed that has been sown, and such plants as the earth produces of its own accord, the fruits of which the inhabitants make use of, training them from their wild state and transplanting them to a suitable soil; the cattle also which are reared there are vigorous, particularly prolific, and bring up young of the fairest description; the inhabitants too, are well fed, most beautiful in shape, of large stature, and differ little from one another either as to figure or size; and the country itself, both as regards its constitution and mildness of the seasons, may be said to bear a close resemblance to the spring. Manly courage, endurance of suffering, laborious enterprise, and high spirit, could not be produced in such a state of things either among the native inhabitants or those of a different country, for there pleasure necessarily reigns. For this reason, also, the forms of wild beasts there are much varied.^[424] Thus it is, as I think, with the Egyptians and Libyans.

13. But concerning those on the right hand of the summer risings of the sun as far as the Palus Mæotis^[425] (for this is the boundary of Europe and Asia), it is with them as follows: the inhabitants there differ far more from one another than those I have treated of above, owing to the differences of the seasons and the nature of the soil. But with regard to the country itself, matters are the same there as among all other men; for where the seasons undergo the greatest and most rapid changes, there the country is the wildest and most unequal; and you will find the greatest variety of mountains, forests, plains, and meadows; but where the seasons do not change much there the country is the most even; and, if one will consider it, so is it also with regard to the inhabitants; for the nature of some is like to a country covered with trees and well watered; of some, to a thin soil deficient in water; of others, to fenny and marshy places; and of some again, to a plain of bare and parched land.^[426] For the seasons which modify their natural frame of body are varied, and the greater the varieties of them the greater also will be the differences of their shapes.

14. I will pass over the smaller differences among the nations, but will now treat of such as are great either from nature, or custom; and, first, concerning the Macrocephali.^[427] There is no other race of men which have heads in the least resembling theirs. At first, usage was the principal cause of the length of their head, but now nature cooperates with usage. They think those the most noble who have the longest heads. It is thus with regard to the usage: immediately after the child is born, and while its head is still tender, they fashion it with their hands, and constrain it to assume a lengthened shape by applying bandages and other suitable contrivances whereby the spherical form of the head is destroyed, and it is made to increase in length. Thus, at first, usage operated, so that this constitution was the result of force: but, in the course of time, it was formed naturally; so that usage had nothing to do with it; for the semen comes from all parts of the body, sound from the sound parts, and unhealthy from the unhealthy parts. If, then, children with bald heads are born to parents with bald heads; and children with blue eyes to parents who have blue eyes; and if the children of parents having distorted eyes squint also for the most part; and if the same may be said of other forms of the body, what is to prevent it from happening that a child with a long head should be produced by a parent having a long head?^[428] But now these things do not happen as they did formerly, for the custom no longer prevails owing to their intercourse with other men. Thus it appears to me to be with regard to them.

15. As to the inhabitants of Phasis,^[429] their country is fenny, warm, humid, and wooded; copious and severe rains occur there at all seasons; and the life of the inhabitants is spent among the fens; for their dwellings are constructed of wood and reeds, and are erected amidst the waters; they seldom practise walking either to the city or the market, but sail about, up and down, in canoes constructed out of single trees, for there are many canals there.^[430] They drink the hot and stagnant waters, both when rendered putrid by the sun, and when swollen with rains. The Phasis itself is the most stagnant of all rivers, and runs the smoothest;^[431] all the fruits which spring there are unwholesome, of feeble and imperfect growth, owing to the redundance of water, and on this account they do not ripen, for much vapor from the waters overspreads the country. For these reasons the Phasians have shapes different from those of all other men; for they are large in stature, and of a very gross habit of body, so that not a joint nor vein is visible; in color they are sallow, as if affected with jaundice. Of all men they have the roughest voices, from their breathing an atmosphere which is not clear, but misty and humid; they are naturally rather languid in supporting bodily fatigue. The seasons undergo but little change either as to heat or cold; their winds for the most part are southerly, with the exception of one peculiar to the country, which sometimes blows strong, is violent and hot, and is called by them the wind *cenchron*. The north wind scarcely reaches them, and when it does blow it is weak and gentle. Thus it is with regard to the different nature and shape of the inhabitants of Asia and Europe.

16. And with regard to the pusillanimity and cowardice of the inhabitants, the principal reason why the Asiatics are more unwarlike and of more gentle disposition than the Europeans is, the nature of the seasons, which do not undergo any great changes either to heat or cold, or the like; for there is neither excitement of the understanding nor any strong change of the body by which the temper might be ruffled, and they are roused to inconsiderate emotion and passion, rather than living as they

do always in the same state. It is changes of all kinds which arouse the understanding of mankind, and do not allow them to get into a torpid condition. For these reasons, it appears to me, the Asiatic race is feeble, and further, owing to their laws; for monarchy prevails in the greater part of Asia, and where men are not their own masters nor independent, but are the slaves of others, it is not a matter of consideration with them how they may acquire military discipline, but how they may seem not to be warlike, for the dangers are not equally shared, since they must serve as soldiers, perhaps endure fatigue, and die for their masters, far from their children, their wives, and other friends; and whatever noble and manly actions they may perform lead only to the aggrandizement of their masters, whilst the fruits which they reap are dangers and death; and, in addition to all this, the lands of such persons must be laid waste by the enemy and want of culture.^[432] Thus, then, if any one be naturally warlike and courageous, his disposition will be changed by the institutions. As a strong proof of all this, such Greeks or barbarians in Asia as are not under a despotic form of government, but are independent, and enjoy the fruits of their own labors, are of all others the most warlike; for these encounter dangers on their own account, bear the prizes of their own valor, and in like manner endure the punishment of their own cowardice. And you will find the Asiatics differing from one another, for some are better and others more dastardly; of these differences, as I stated before, the changes of the seasons are the cause. Thus it is with Asia.

17. In Europe there is a Scythian race, called Sauromatæ, which inhabits the confines of the Palus Mæotis, and is different from all other races.^[433] Their women mount on horseback, use the bow, and throw the javelin from their horses, and fight with their enemies as long as they are virgins; and they do not lay aside their virginity until they kill three of their enemies, nor have any connection with men until they perform the sacrifices according to law. Whoever takes to herself a husband, gives up riding on horseback unless the necessity of a general expedition obliges her. They have no right breast; for while still of a tender age their mothers heat strongly a copper instrument constructed for this very purpose, and apply it to the right breast, which is burnt up, and its development being arrested, all the strength and fullness are determined to the right shoulder and arm.

18. As the other Scythians have a peculiarity of shape, and do not resemble any other, the same observation applies to the Egyptians, only that the latter are oppressed by heat and the former by cold.^[434] What is called the Scythian desert is a prairie, abounding in meadows, high-lying, and well watered; for the rivers which carry off the water from the plains are large. There live those Scythians which are called Nomades, because they have no houses, but live in wagons. The smallest of these wagons have four wheels, but some have six; they are covered in with felt, and they are constructed in the manner of houses, some having but a single apartment, and some three; they are proof against rain, snow, and winds. The wagons are drawn by yokes of oxen, some of two and others of three, and all without horns, for they have no horns, owing to the cold.^[435] In these wagons the women live, but the men are carried about on horses, and the sheep, oxen, and horses accompany them; and they remain on any spot as long as there is provender for their cattle, and when that fails they migrate to some other place. They eat boiled meat, and drink the milk of mares, and also eat *hippace*, which is cheese prepared from the milk of the mare. Such is their mode of life and their customs.^[436]

19. In respect of the seasons and figure of body, the Scythian race, like the Egyptian, have a uniformity of resemblance, different from all other nations; they are by no means prolific, and the wild beasts which are indigenous there are small in size and few in number, for the country lies under the Northern Bears, and the Rhiphæan mountains, whence the north wind blows; the sun comes very near to them only when in the summer solstice, and warms them but for a short period, and not strongly; and the winds blowing from the hot regions of the earth do not reach them, or but seldom, and with little force; but the winds from the north always blow, congealed, as they are, by the snow, the ice, and much water, for these never leave the mountains, which are thereby rendered uninhabitable. A thick fog covers the plains during the day, and amidst it they live, so that winter may be said to be always present with them; or, if they have summer, it is only for a few days, and the heat is not very strong. Their plains are high-lying and naked, not crowned with mountains, but extending upwards under the Northern Bears.^[437] The wild beasts there are not large, but such as can be sheltered under-ground; for the cold of winter and the barrenness of the country prevent their growth, and because they have no covert nor shelter.^[438] The changes of the seasons, too, are not great nor violent, for, in fact, they change gradually; and therefore their figures resemble one another, as they all equally use the same food, and the same clothing summer and winter, respiring a humid and dense atmosphere, and drinking water from snow and ice; neither do they make any laborious exertions, for neither body nor mind is capable of enduring fatigue when the changes of the seasons are not great.^[439] For these reasons their shapes are gross and fleshy, with ill-marked joints, of a humid temperament, and deficient in tone: the internal cavities, and especially those of the intestines, are full of humors; for the belly cannot possibly be dry in such a country, with such a constitution and in such a climate; but owing to their fat, and the absence of hairs from their bodies, their shapes resemble one another, the males being all alike, and so also with the women: for the seasons being of an uniform temperature, no corruption or deterioration takes place in the concretion of the semen, unless from some violent cause, or from disease.^[440]

20. I will give you a strong proof of the humidity (laxity?) of their constitutions.^[441] You will find the greater part of the Scythians, and all the Nomades, with marks of the cautery on their shoulders, arms, wrists, breasts, hip-joints, and loins, and that for no other reason but the humidity and flabbiness of their constitution, for they can neither strain with their bows, nor launch the javelin from their shoulder owing to their humidity and atony: but when they are burnt, much of the humidity in their joints is dried up, and they become better braced, better fed, and their joints get into a more suitable condition.^[442] They are flabby and squat at first, because, as in Egypt, they are not swathed(?),^[443] and then they pay no attention to horsemanship, so that they may be adepts at it; and because of their sedentary mode of life; for the males, when they cannot be carried about on horseback, sit the most of their time in the wagon, and rarely practise walking, because of their frequent migrations and shiftings of situation; and as to the women, it is amazing how flabby and sluggish they are. The Scythian race are tawny from the cold, and not from the intense heat of the sun, for the whiteness of the skin is parched by the cold, and becomes tawny.

21. It is impossible that persons of such a constitution could be prolific, for, with the man, the sexual desires are not strong, owing to the laxity of his constitution, the softness and coldness of his belly, from all which causes it is little likely that a man should be given to venery; and besides, from being jaded by exercise on horseback, the men become weak in their desires. On the part of the men these are the causes; but on that of the women, they are embonpoint and humidity; for the womb cannot take in the semen, nor is the menstrual discharge such as it should be, but scanty and at too long intervals; and the mouth of the womb is shut up by fat and does not admit the semen; and, moreover, they themselves are indolent and fat, and their bellies cold and soft.^[444] From these causes the Scythian race is not prolific. Their female servants furnish a strong proof of this; for they no sooner have connection with a man than they prove with child, owing to their active course of life and the slenderness of body.

22. And, in addition to these, there are many eunuchs among the Scythians, who perform female work, and speak like women. Such persons are called effeminates.^[445] The inhabitants of the country attribute the cause of their impotence to a god, and venerate and worship such persons, every one dreading that the like might befall himself; but to me it appears that such affections are just as much divine as all others are, and that no one disease is either more divine or more human than another, but that all are alike divine, for that each has its own nature, and that no one arises without a natural cause.^[446] But I will explain how I think that the affection takes its rise. From continued exercise on horseback they are seized with chronic defluxions in their joints (*kedmata*^[447]) owing to their legs always hanging down below their horses; they afterwards become lame and stiff at the hip-joint, such of them, at least, as are severely attacked with it. They treat themselves in this way: when the disease is commencing, they open the vein behind either ear, and when the blood flows, sleep, from feebleness, seizes them, and afterwards they awaken, some in good health and others not. To me it appears that the semen is altered by this treatment, for there are veins behind the ears which, if cut, induce impotence; now, these veins would appear to me to be cut.^[448] Such persons afterwards, when they go in to women and cannot have connection with them, at first do not think much about it, but remain quiet; but when, after making the attempt two, three, or more times, they succeed no better, fancying they have committed some offence against the god whom they blame for the affection, they put on female attire, reproach themselves for effeminacy, play the part of women, and perform the same work as women do. This the rich among the Scythians endure, not the basest, but the most noble and powerful, owing to their riding on horseback; for the poor are less affected, as they do not ride on horses. And yet, if this disease had been more divine than the others, it ought not to have befallen the most noble and the richest of the Scythians alone, but all alike, or rather those who have little, as not being able to pay honors to the gods, if, indeed, they delight in being thus rewarded by men, and grant favors in return; for it is likely that the rich sacrifice more to the gods, and dedicate more votive offerings, inasmuch as they have wealth, and worship the gods; whereas the poor, from want, do less in this way, and, moreover, upbraid the gods for not giving them wealth, so that those who have few possessions were more likely to bear the punishments of these offences than the rich. But, as I formerly said, these affections are divine just as much as others, for each springs from a natural cause, and this disease arises among the Scythians from such a cause as I have stated. But it attacks other men in like manner, for whenever men ride much and very frequently on horseback, then many are affected with rheums in the joints, sciatica, and gout, and they are inept at venery. But these complaints befall the Scythian, and they are the most impotent of men for the aforesaid causes, and because they always wear breeches, and spend the most of their time on horseback,^[449] so as not to touch their privy parts with the hand, and from the cold and fatigue they forget the sexual desire, and do not make the attempt until after they have lost their virility.^[450] Thus it is with the race of the Scythians.

23. The other races in Europe differ from one another, both as to stature and shape, owing to the changes of the seasons, which are very great and frequent, and because the heat is strong, the winters severe, and there are frequent rains, and again protracted droughts, and winds, from which many and diversified changes are induced. These changes are likely to have an effect upon generation in the coagulation of the semen, as this process cannot be the same in summer as in winter, nor in rainy as in dry weather; wherefore, I think, that the figures of Europeans differ more than those of Asiatics; and they differ very much from one another as to stature in the same city; for vitiations of the semen occur in its coagulation more frequently during frequent changes of the seasons, than where they are alike and equable. And the same may be said of their dispositions,

for the wild, and unsociable, and the passionate occur in such a constitution; for frequent excitement of the mind induces wildness, and extinguishes sociableness and mildness of disposition, and therefore I think the inhabitants of Europe more courageous than those of Asia; for a climate which is always the same induces indolence, but a changeable climate, laborious exertions both of body and mind; and from rest and indolence cowardice is engendered, and from laborious exertions and pains, courage. On this account the inhabitants of Europe are more warlike than the Asiatics, and also owing to their institutions, because they are not governed by kings like the latter, for where men are governed by kings there they must be very cowardly, as I have stated before; for their souls are enslaved, and they will not willingly, or readily undergo dangers in order to promote the power of another; but those that are free undertake dangers on their own account, and not for the sake of others; they court hazard and go out to meet it, for they themselves bear off the rewards of victory, and thus their institutions contribute not a little to their courage.^[451]

Such is the general character of Europe and Asia.^[452]

24. And there are in Europe other tribes, differing from one another in stature, shape, and courage: the differences are those I formerly mentioned, and will now explain more clearly. Such as inhabit a country which is mountainous, rugged, elevated, and well watered, and where the changes of the seasons are very great, are likely to have great variety of shapes among them, and to be naturally of an enterprising and warlike disposition;^[453] and such persons are apt to have no little of the savage and ferocious in their nature; but such as dwell in places which are low-lying, abounding in meadows and ill ventilated, and who have a larger proportion of hot than of cold winds, and who make use of warm waters—these are not likely to be of large stature nor well proportioned, but are of a broad make, fleshy, and have black hair; and they are rather of a dark than of a light complexion, and are less likely to be phlegmatic than bilious; courage and laborious enterprise are not naturally in them, but may be engendered in them by means of their institutions. And if there be rivers in the country which carry off the stagnant and rain water from it, these may be wholesome and clear; but if there be no rivers, but the inhabitants drink the waters of fountains, and such as are stagnant and marshy, they must necessarily have prominent bellies and enlarged spleens. But such as inhabit a high country, and one that is level, windy, and well-watered, will be large of stature, and like to one another; but their minds will be rather unmanly and gentle. Those who live on thin, ill-watered, and bare soils, and not well attuned to the changes of the seasons, in such a country they are likely to be in their persons rather hard and well braced, rather of a blond than a dark complexion, and in disposition and passions haughty and self-willed. For, where the changes of the seasons are most frequent, and where they differ most from one another, there you will find their forms, dispositions, and nature the most varied. These are the strongest of the natural causes of difference, and next the country in which one lives, and the waters; for, in general, you will find the forms and dispositions of mankind to correspond with the nature of the country; for where the land is fertile, soft, and well-watered, and supplied with waters from very elevated situations, so as to be hot in summer and cold in winter, and where the seasons are fine, there the men are fleshy, have ill-formed joints,^[454] and are of a humid temperament; they are not disposed to endure labor, and, for the most part, are base in spirit; indolence and sluggishness are visible in them, and to the arts they are dull, and not clever nor acute. When the country is bare, not fenced, and rugged, blasted by the winter and scorched by the sun, there you may see the men hardy, slender, with well-shaped joints,^[454] well-braced, and shaggy; sharp industry and vigilance accompany such a constitution; in morals and passions they are haughty and opinionative, inclining rather to the fierce than to the mild; and you will find them acute and ingenious as regards the arts, and excelling in military affairs; and likewise all the other productions of the earth corresponding to the earth itself.^[455] Thus it is with regard to the most opposite natures and shapes; drawing conclusions from them, you may judge of the rest without any risk of error.

ON THE PROGNOSTICS.

THE ARGUMENT.

Of the genuineness of this work I have treated in the Preliminary Discourse, and have also briefly touched upon its relation to two other important treatises in the Hippocratic collection, the “Prorrhethics” and the “Coacæ Prænotiones.” The latter subject I am now to resume, and in doing so I mean to avail myself of the talented dissertation of Dr. Ermerins, to which also I have already made allusion. Indeed I am persuaded that I cannot do a more acceptable service to my profession in Britain than by laying before them a brief exposition of the important views brought forward in this “Dissertatio Inauguralis.”^[456]

After some preliminary observations on the ancient Temples of Health, which are mainly derived from Sprengel’s “History of Medicine”^[457] he passes on to consider the opinion started[*? typo for stated, but started also in googlebooks] by this author and others before his time, that the first book of the “Prorrhethics” and the “Coacæ Prænotiones” are the results of isolated observations made upon the sick in the Asclepion of Cos. The probability of this opinion being well founded he shows to be very great; and he next endeavors to solve the question whether the first book of the “Prorrhethics” be derived from the “Coacæ Prænotiones,” or whether the latter be the more modern work of the two. He comes to the conclusion that the “Prorrhethics” is the more ancient work, for the following reasons: 1st. Because in it the names of the patients are frequently given, which is rarely the case in the “Coacæ Prænotiones.” 2d. Because queries and doubts are oftener found in this book than in the other, when one takes into account the number of presages. 3d. Because the number of observations which this book contains is much smaller than those which the “Coacæ” embrace. 4th. This is confirmed by the circumstance that the enunciations of the prognoses are far less extended in the “Prorrhethics,” whence it is clearly proved that they are not derived from so great a field of observations as those we meet with in the other work. He then gives a most lucid view of the parallelism which subsists between the “Prorrhethica” and the

“Coacæ,” and, as the results of his observations upon them, he draws the following most important conclusions:

1. “By a most fortunate occurrence certain monuments of the medical art, as cultivated by the Asclepiadæ, are preserved to us in the first “Prorrhethics” and the “Prænotiones Coacæ” which books appear to be fragments and excerpts from the histories of diseases and cures which were formerly found on the votive tablets of the Coan temple.

2. This sacerdotal medicine was at first a certain medical divination, which, as it was the offspring of pure observation, so the system of prognostics of the Coans was altogether aloof from the theories and systems of the philosophers, and is therefore to be reckoned most worthy of our attention, both from the great love of observation which we admire in it, and from the exquisite and beautiful sense of the simple truth which it evinces.

3. We must keep in view the origin of these presages from individual observations gradually collected, in order that we may have a knowledge of this system of prognostic semeiology. Hence we comprehend how we meet with so many doubtful propositions, and so many uncertain and vague remarks, and that imperfect etiology which confounded causes with their effects, and again, the latter with the former.

4. The readers must particularly keep before their eyes this origin, and the antiquity of those writings, if they would pass a correct judgment on the merits of the Asclepiadæ towards the art of medicine. Whatever in their works we have the pleasure of possessing, all attest the infancy of the art; many things are imperfect, and not unfrequently do we see them, while in the pursuit of truth, groping, as it were, and proceeding with uncertain steps, like men wandering about in darkness; but yet the method which they applied, and to which they would seem to have betaken themselves of their own accord, was so excellent, that nothing could surpass it. It was the same method which Hippocrates himself always adopted, and which, in fine, Lord Bacon, many ages afterwards, commended as the only instrument by which truth in medicine can be found out.

5. As this method is founded on true induction, so are its dicta to be held the more worthy of admiration, the more they possess a universal signification. To give an example; what assiduous observation, and what abundance of rational experience, must have been required for enunciating the following admirable truth, and, as it were, law of nature: “Those things which bring alleviation with bad signs, and do not remit with good, are troublesome and difficult.”

6. Many passages bear reference to the condition of the vital powers, which they took into account at all times, both in making presages and in exercising the art. For, although they had not our theories of the vital force, they perceived its effects very well by observation; and for this very reason, that they did not search for the art in theories, but in observation alone, we owe so many excellent things to them, since they did not adapt their observation to theories, but related a trustworthy and faithful history of the operations of nature.

7. They sought after many things from a comparison of health with disease, in which also they rightly calculated the manners and customs of men. Thus they call that, in the first place, the best mode of reclining, which is adopted by the patient

when in good health, and hence they estimate the other modes as being less good, or altogether unfavorable. Nor did they only compare health with disease, but they compared also the symptoms of diseases with one another, and interpreted the one from the other. Thus they first depict and pronounce a favorable opinion on the best kind of excretions, and then they described the other abnormal kinds, and pass an unfavorable judgment on them.

8. They particularly relate the operations of a *natura medicatrix*, which, in a region such as Greece is, and in athletic, strong bodies, on which they appear to have practised the art, and for the most part in acute diseases, and the few chronic ones derived from them which they have left described, might especially be looked for. Hence that doctrine of crises most deserving of attention, the rudiments, indeed, of which we only have here preserved, but a just notion of which we may easily draw from these fragments.

9. The Asclepiadæ would appear to have accommodated and directed their art to this natural Therapia. Hence the advice that convulsions arising from a great hemorrhage, forcibly stopped, should be cured by the abstraction of blood. It is to be regretted that but a few monuments of their practice remain; but these embrace admirable imitations of nature, and the most prudent caution in administering remedies.

10. Neither did they neglect surgery, but deliver many excellent remarks on things pertaining to wounds, ulcers, and fractures.

11. Although it cannot be made out for certain that everything which is preserved in these writings existed before Hippocrates, there can be no doubt that many of them are more ancient than he. And although we may attribute some things rather to Hippocrates himself, it is nevertheless certain that the method of deducing the art from observation and comparison had existed before him. Some may, perhaps, object that these books are to be attributed to the youth of Hippocrates, and that the others, more elaborate and perfect, had proceeded from the same person in his old age; but this supposition we may refute by a single argument, namely, that it would be absurd to ascribe so many observations about so many diseases to one man.

12. From the whole Coan system of cultivating medicine, the best hopes might justly have been expected; and from what follows it will be seen that the result did not disappoint this expectation.”

These deductions, I must say, appear to be most legitimately drawn; and having thus satisfactorily made out that the “Coacæ Prænotiones” are founded on the “Prorrhethics,” Dr. Ermerins proceeds to make an interesting comparison between the former and the book of “Prognostics.” Here again we can only find room for the general conclusions.

1. “We have compared together two monuments of antiquity embracing entirely the same doctrine, so that we may hold it as put out of all doubt that they must have derived their origin from the same school, only the one yields to the other in antiquity, as its more expanded mode of expression shows.

2. The more recent work is attributed to Hippocrates by all the critics and interpreters; the most ancient authors have made mention of it, and all the characteristic marks by which the genuine works of Hippocrates are distinguished from the spurious, without doubt, are found in it; for whether you look to the brevity and gravity of the language, or the paucity of the reasonings, the correctness of the observations, or the dialect in which they are expressed, or, in fine, its agreement with the whole Hippocratic doctrine,—all these attest that “the divine old man” is the author of this work.

3. From a comparison of the “Coacæ Prænotiones” with the “Prognostics,” it is as clear as the light of day that Hippocrates composed this work from them, in such a manner that he circumscribed many of the symptoms, limited the enunciations, and amplified them all by his own experience in the medical Art. Hence the Prognostics may not inaptly be called the Commentary of Hippocrates on the “Coacæ Prænotiones.”

4. With regard to the exquisite and artificial order, in which we see many things proposed in his book, we agree entirely with Sprengel, who thinks that they have proceeded from a more recent describer. This is confirmed by our comparison of both works.

5. This work exhibits the fundamental principles and originals of the Hippocratic doctrine, and although we hardly know anything as to the manner in which Hippocrates composed his writings, and of the form which he gave them, it does not seem at all out of the way to hold this book to be the oldest of all the works which “the Father of Medicine” has left to us.

6. Inasmuch as this work is entitled the Book of Prognostics, so it turns on the *prescience* πρόνοια, that is to say, the foreknowledge of the physician, which Hippocrates recommends to physicians for three reasons: first, for the confidence of mankind, which it will conciliate to the physician; then because it will free the practitioner from all blame, if he has announced beforehand the fatal result of diseases; and further, as being a very great instrument in effecting the cure.

7. Like the Coan priests, Hippocrates drew his Prognostics from a comparison of disease with health. This he held to be of so great importance, that he first delivers physiological semeiotics, and then adds pathological.

8. In calculating and judging of signs he neglected neither age nor sex, and, in the first place, directed his mind to the power of habit on the human body.

9. Nor did Hippocrates stop here, but directed care to be had of the attack of epidemics, and the condition of the season.

10. The Prognostics of Hippocrates are not of one time or place, but extend through every age, and through the whole world; inasmuch as the prognostic signs have been proved to be true in Libya, in Delos, and in Scythia, and it should be well known that every year, and at every season of the year, bad symptoms bode ill, and good symptoms good.

11. But he who would wish to know properly beforehand those who will recover from a disease, and those who will die, and those in whom the disease will persevere for many days, and those in whom it will last for a few, should be able to comprehend and estimate the doctrine of all the signs, and weigh in his mind and compare together their strength. The Hippocratic foreknowledge rests not only on the observation of the signs, but also on the understanding of them.

12. The Book of Prognostics exhibits observations of acute diseases, and of chronic arising from them, in which Hippocrates has diligently noted the times and modes of the crises.

13. Such is the authority of critical days and signs, that in those fevers which cease without the symptoms of resolution, and not upon critical days, a relapse is to be expected.

14. The series of critical days which Hippocrates delivers, proceeds solely upon the observation of nature. Yet neither can any of them be exactly numbered by entire days, since neither the year nor the months are usually numbered by entire days."

Dr. Ermerins, in the remaining part of his Essay, shows, in a very lucid manner, that the rules of Prognosis laid down in this treatise by Hippocrates, are manifestly those by which he is regulated in his other works, and more especially in the Epidemics and Aphorisms. We must not, however, occupy room with any further exposition of the contents of this important treatise, which does equal credit to the author himself, and to the medical system of education pursued in the learned university from which it emanated.

I will now give some remarks and reflections of my own on the treatise under consideration.

In this work, then, Hippocrates appears to have had for his object, to give such a general description of the phenomena of disease as would apply to all the disorders of the animal frame. With this intention he brings into review the state of the countenance, the position of the patient in bed, the movements of the hands, the respiration, the sweats, the state of the hypochondria, dropsies which are the consequences of acute diseases, the sleep, the urine, the alvine dejections, the vomitings, and the sputa. In doing this, his uniform practice is to contrast the healthy with the morbid appearances. Although M. Littré regards it as a treatise on special Pathology, it appears to me to be decidedly a general work on Semeiology. Certain it is that all the best commentators, such as Erotian and Stephanus,^[458] decidedly regard it as a semeiological work. The class of ancient writings with which it admits of being most closely compared, are the works on the prognostics of the weather. On this subject Greek literature contains several works of a very philosophical nature, such as the *Phænomena* of Aratus, and several of the minor tracts of Theophrastus. Now as the object of these authors was to connect the most striking phenomena in the sky, the earth, and the sea, with the changes in the weather, of which they are the precursors, so the intention of the medical writer of Prognostics was to point out the alterations in the animal frame, which certain preternatural symptoms usually indicate. And as the utility of an acquaintance with prognostics of the weather to the husbandman and sailor is sufficiently obvious, the benefit to be derived from a knowledge of medical prognostics by the physician is equally so. Our author, it will be seen in the Preface to this work, enumerates three objects to be attained by cultivating an acquaintance with prognostics; first, to attract the confidence of one's patients; second, to free the physician from blame by enabling him to announce beforehand the issue of the disorder about which he is consulted; and, third, to give him a decided advantage in conducting the treatment by preparing him for remarkable changes in the diseases before they occur. And, in like manner, I may be allowed to remark, the master of a ship who shows himself prepared for all changes of the weather, will naturally attract the confidence of those intrusted to his charge; and whatever may be the result, he will be freed from blame if his ship should be damaged in a storm which he had previously predicted; and surely his knowledge of impending commotions in the sea and sky, will be of advantage to him by enabling him to make preparations for them.

Looking then to the importance of general Prognostics, I have often wondered why this branch of Semeiology is no longer cultivated by the profession. Did not the ancient physicians follow the best possible plan when they first described the general phenomena of diseased action, and then applied them to particular cases? Surely they did right in first taking a comprehensive view of the whole subject of disease before attempting to examine the different parts of it in detail. This, in fact, constitutes the great superiority of the ancient *savans* over the modern, that the former possessed a much greater talent for apprehending general truths than the latter, who confine their attention to particular facts, and too much neglect the observation of general

appearances. I trust no one will be offended if I venture to pronounce regarding the present condition of our professional literature, that (to borrow an illustration from the Logic of Kant) it is altogether Cyclopic,—that is to say, it wants the eye of Philosophy, for, although we have learned to examine particular objects with greater accuracy than our forefathers did, the sphere of our mental vision, so to speak, is more confined than theirs, and cannot embrace the same enlarged views of general subjects. Surely then we might gain a useful lesson by endeavoring to combine their more comprehensive views with our own more accurate and minute observation.

Some people may be inclined to think that we have greatly detracted from the credit which Hippocrates has long enjoyed as being the undoubted author of this work, by showing that in composing it he was so much indebted to the labors of his predecessors. But I have long been impressed with the conviction that in compositions even of the highest order, there is much less originality than is generally supposed, and that true genius frequently is displayed more in its own felicitous way of dealing with materials formerly prepared and collected for its use than in searching out new matter to work upon,^[459] and hence it will be found upon examination that many of the most distinguished efforts of human intellect have consisted in the successful performance of tasks which had been frequently attempted by previous laborers in the same line. Many artists, before the time of Phidias, had acquired reputation by their attempts at making the statue of Jupiter,^[460] but this did not deter him from undertaking the same task: and we may well believe that he would avail himself of every practical lesson which he could draw from the success or failure of his predecessors, in perfecting that matchless performance which completely cast all others into the background. The sad misfortunes of Œdipus had been often represented on the Athenian stage before Sophocles made them the subject of those inimitable dramas, which still enjoy an unrivalled reputation, nor will it be often considered how much assistance he may have derived from the labors of those who had gone before him. It is well known that of all the literary performances of Aristotle, there is no one which gained him so enduring a reputation as his Categories, and yet it is admitted that his division of the subject into the ten Predicaments, was taken from the Pythagorean philosopher Archytas,^[461] in short, the great merit of Aristotle on this as on many other occasions, consisted in defining and arranging a subject on which much had been previously effected by the labors of his predecessors. And, to give one example more, long before the time of Galen, the temperaments, and the facts in physiology and pathology bearing upon Hygiene, had been frequently and successfully investigated, but he, by recasting all these subject-matters into his *Ars Medica*, composed a work which posterity regarded as his master-performance, and every word and tittle of which, for a succession ages, were commented upon and admired in the Schools of Medicine. And of all our Author's admired performances, there is perhaps no one which has exerted so great an influence upon the literature of the profession as the present work, for all the Greek, Roman, and Arabian writers on medicine, subsequent to him, make use of his terms, and copy his descriptions of morbid phenomena.

THE BOOK OF PROGNOSTICS.

1. It appears to me a most excellent thing for the physician to cultivate Prognosis; for by foreseeing and foretelling, in the presence of the sick, the present, the past, and the future, and explaining the omissions which patients have been guilty of,^[462] he will be the more readily believed to be acquainted with the circumstances of the sick; so that men will have confidence to intrust themselves to such a physician. And he will manage the cure best who has foreseen what is to happen from the present state of matters. For it is impossible to make all the sick well; this, indeed, would have been better than to be able to foretell what is going to happen; but since men die, some even before calling the physician, from the violence of the disease, and some die immediately after calling him, having lived, perhaps, only one day or a little longer, and before the physician could bring his art to counteract the disease; it therefore becomes necessary to know the nature of such affections, how far they are above the powers of the constitution; and, moreover, if there be anything divine in the diseases,^[463] and to learn a foreknowledge of this also. Thus a man will be the more esteemed to be a good physician, for he will be the better able to treat those aright who can be saved, from having long anticipated everything; and by seeing and announcing beforehand those who will live and those who will die, he will thus escape censure.^[464]

2. He should observe thus in acute diseases: first, the countenance of the patient, if it be like those of persons in health, and more so, if like itself, for this is the best of all; whereas the most opposite to it is the worst, such as the following; *a sharp nose, hollow eyes, collapsed temples; the ears cold, contracted, and their lobes turned out: the skin about the forehead being rough, distended, and parched; the color of the whole face being green, black, livid, or lead-colored.*^[465] If the countenance be such at the commencement of the disease, and if this cannot be accounted for from the other symptoms, inquiry must be made whether the patient has long wanted sleep; whether his bowels have been very loose; and whether he has suffered from want of food; and if any of these causes be confessed to, the danger is to be reckoned so far less; and it becomes obvious, in the course of a day and a night, whether or not the appearance of the countenance proceeded from these causes.^[466] But if none of these be said to exist, and if the symptoms do not subside in the aforesaid time, it is to be known for certain that death is at hand. And, also, if the disease be in a more advanced stage either on the third or fourth day, and the countenance be such, the same inquiries as formerly directed are to be made, and the other symptoms are to be noted, those in the whole countenance, those on the body, and those in the eyes; for if they shun the light, or weep involuntarily, or squint, or if the one be less than the

other, or if the white of them be red, livid, or has black veins in it; if there be a gum upon the eyes, if they are restless, protruding, or are become very hollow; and if the countenance be squalid and dark, or the color of the whole face be changed—all these are to be reckoned bad and fatal symptoms. The physician should also observe the appearance of the eyes from below the eyelids in sleep; for when a portion of the white appears, owing to the eyelids not being closed together, and when this is not connected with diarrhoea or purgation from medicine, or when the patient does not sleep thus from habit, it is to be reckoned an unfavorable and very deadly symptom; but if the eyelid be contracted, livid, or pale, or also the lip, or nose, along with some of the other symptoms, one may know for certain that death is close at hand. It is a mortal symptom, also, when the lips are relaxed, pendent, cold, and blanched.

3.^[467] It is well when the patient is found by his physician reclining upon either his right or his left side, having his hands, neck, and legs slightly bent, and the whole body lying in a relaxed state, for thus the most of persons in health recline, and these are the best of postures which most resemble those of healthy persons. But to lie upon one's back, with the hands, neck, and the legs extended, is far less favorable. And if the patient incline forward, and sink down to the foot of the bed, it is a still more dangerous symptom; but if he be found with his feet naked and not sufficiently warm, and the hands, neck, and legs tossed about in a disorderly manner and naked, it is bad, for it indicates aberration of intellect. It is a deadly symptom, also, when the patient sleeps constantly with his mouth open, having his legs strongly bent and plaited together, while he lies upon his back; and to lie upon one's belly, when not habitual to the patient to sleep thus while in good health, indicates delirium, or pain in the abdominal regions. And for the patient to wish to sit erect at the acme of a disease is a bad symptom in all acute diseases, but particularly so in pneumonia.^[468] To grind the teeth in fevers, when such has not been the custom of the patient from childhood, indicates madness and death, both which dangers are to be announced beforehand as likely to happen; and if a person in delirium do this it is a very deadly symptom. And if the patient had an ulcer previously, or if one has occurred in the course of the disease, it is to be observed; for if the man be about to die the sore will become livid and dry, or yellow and dry before death.^[469]

4. Respecting the movement of the hands I have these observations to make: When in acute fevers, pneumonia, phrenitis, or headache, the hands are waved before the face, hunting through empty space, as if gathering bits of straw, picking the nap from the coverlet, or tearing chaff from the wall—all such symptoms are bad and deadly.^[470]

5. Respiration, when frequent, indicates pain or inflammation in the parts above the diaphragm: a large respiration performed at a great interval announces delirium; but a cold respiration at nose or mouth is a very fatal symptom. Free respiration is to be looked upon as contributing much to the safety of the patient in all acute diseases, such as fevers, and those complaints which come to a crisis in forty days.^[471]

6. Those sweats are the best in all acute diseases which occur on the critical days, and completely carry off the fever. Those are favorable, too, which taking place over the whole body, show that the man is bearing the disease better. But those that do not produce this effect are not beneficial. The worst are cold sweats, confined to the head, face, and neck; these in an acute fever prognosticate death, or in a milder one, a prolongation of the disease; and sweats which occur over the whole body, with the characters of those confined to the neck, are in like manner bad. Sweats attended with a miliary eruption, and taking place about the neck, are bad; sweats in the form of drops and of vapour are good. One ought to know the entire character of sweats, for some are connected with prostration of strength in the body, and some with intensity of the inflammation.^[472]

7.^[473] That state of the hypochondrium is best when it is free from pain, soft, and of equal size on the right side and the left. But if inflamed, or painful, or distended; or when the right and left sides are of disproportionate sizes;—all these appearances are to be dreaded. And if there be also pulsation in the hypochondrium, it indicates perturbation or delirium; and the physician should examine the eyes of such persons; for if their pupils be in rapid motion, such persons may be expected to go mad. A swelling in the hypochondrium, that is hard and painful, is very bad, provided it occupy the whole hypochondrium; but if it be on either side, it is less dangerous when on the left. Such swellings at the commencement of the disease prognosticate speedy death; but if the fever has passed twenty days, and the swelling has not subsided, it turns to a suppuration.^[474] A discharge of blood from the nose occurs to such in the first period, and proves very useful; but inquiry should be made if they have headache or indistinct vision; for if there be such, the disease will be determined thither. The discharge of blood is rather to be expected in those who are younger than thirty-five years. Such swellings as are soft, free from pain, and yield to the finger, occasion more protracted crises, and are less dangerous than the others. But if the fever continue beyond sixty days, without any subsidence of the swelling, it indicates that empyema is about to take place; and a swelling in any other part of the cavity will terminate in like manner. Such, then, as are painful, hard, and large, indicate danger of speedy death; but such as are soft, free of pain, and yield when pressed with the finger, are more chronic than these. Swellings in the belly less frequently form abscesses than those in the hypochondrium; and seldomest of all, those below the navel are converted into suppuration; but you may rather expect a hemorrhage from the upper parts. But the suppuration of all protracted swellings about these parts is to be anticipated. The collections of matter there are to be thus judged of: such as are determined outwards are the best when they are small, when they protrude very much, and swell to a point; such as are large

and broad, and which do not swell out to a sharp point, are the worst. Of such as break internally, the best are those which have no external communication, but are covered and indolent; and when the whole place is free from discoloration. That pus is best which is white, homogeneous, smooth, and not at all fetid; the contrary to this is the worst.

8.^[475] All dropsies arising from acute diseases are bad; for they do not remove the fever, and are very painful and fatal. The most of them commence from the flanks and loins, but some from the liver; in those which derive their origin from the flanks and loins the feet swell, protracted diarrhœas supervene, which neither remove the pains in the flanks and loins, nor soften the belly;^[476] but in dropsies which are connected with the liver there is a tickling cough, with scarcely any perceptible expectoration, and the feet swell; there are no evacuations from the bowels, unless such as are hard and forced; and there are swellings about the belly, sometimes on the one side and sometimes on the other, and these increase and diminish by turns.^[477]

9. It is a bad symptom when the head, hands, and feet are cold, while the belly and sides are hot; but it is a very good symptom when the whole body is equally hot.^[478] The patient ought to be able to turn round easily, and to be agile when raised up; but if he appear heavy in the rest of his body as well as in his hands and feet, it is more dangerous; and if, in addition to the weight, his nails and fingers become livid, immediate death may be anticipated; and if the hands and feet be black it is less dangerous than if they be livid, but the other symptoms must be attended to; for if he appear to bear the illness well, and if certain of the salutary symptoms appear along with these there may be hope that the disease will turn to a deposition, so that the man may recover; but the blackened parts of the body will drop off. When the testicles and members are retracted upwards, they indicate strong pains and danger of death.^[479]

10. With regard to sleep—as is usual with us in health, the patient should wake during the day and sleep during the night. If this rule be anywise altered it is so far worse: but there will be little harm provided he sleep in the morning for the third part of the day; such sleep as takes place after this time is more unfavorable; but the worst of all is to get no sleep either night or day; for it follows from this symptom that the insomnolency is connected with sorrow and pains, or that he is about to become delirious.^[480]

11. The excrement is best which is soft and consistent, is passed at the hour which was customary to the patient when in health, in quantity proportionate to the ingesta; for when the passages are such, the lower belly is in a healthy state.^[481] But if the discharges be fluid, it is favorable that they are not accompanied with a noise, nor are frequent, nor in great quantity; for the man being oppressed by frequently getting up, must be deprived of sleep; and if the evacuations be both frequent and large, there is danger of his falling into deliquium animi.^[482] But in proportion to the ingesta he should have evacuations twice or thrice in the day, once at night and more copiously in the morning, as is customary with a person in health. The fœces should become thicker when the disease is tending to a crisis; they ought to be yellowish and not very fetid. It is favorable that round worms be passed with the discharges when the disease is tending to a crisis.^[483] The belly, too, through the whole disease, should be soft and moderately distended; but excrements that are very watery, or white, or green, or very red, or frothy, are all bad. It is also bad when the discharge is small, and viscid, and white, and greenish, and smooth; but still more deadly appearances are the black, or fatty, or livid, or verdigris-green, or fetid. Such as are of varied characters indicate greater duration of the complaint, but are no less dangerous; such as those which resemble scrapings,^[484] those which are bilious, those resembling leeks, and the black; these being sometimes passed together, and sometimes singly.^[485] It is best when wind passes without noise, but it is better that flatulence should pass even thus than that it should be retained; and when it does pass thus, it indicates either that the man is in pain or in delirium, unless he gives vent to the wind spontaneously.^[486] Pains in the hypochondria, and swellings, if recent, and not accompanied with inflammation, are relieved by borborygmi supervening in the hypochondrium, more especially if it pass off with fœces, urine, and wind; but even although not, it will do good by passing along, and it also does good by descending to the lower part of the belly.^[487]

12. The urine is best when the sediment is white, smooth, and consistent during the whole time, until the disease come to a crisis, for it indicates freedom from danger, and an illness of short duration; but if deficient, and if it be sometimes passed clear, and sometimes with a white and smooth sediment, the disease will be more protracted, and not so void of danger. But if the urine be reddish, and the sediment consistent and smooth, the affection, in this case, will be more protracted than the former, but still not fatal.^[488] But farinaceous sediments in the urine are bad, and still worse are the leafy;^[489] the white and thin are very bad, but the furfuraceous are still worse than these. Clouds carried about in the urine are good when white, but bad if black. When the urine is yellow and thin, it indicates that the disease is unconcocted; and if it (the disease) should be protracted, there may be danger lest the patient should not hold out until the urine be concocted.^[490] But the most deadly of all kinds of urine are the fetid, watery, black, and thick; in adult men and women the black is of all kinds of urine the worst, but in children, the watery.^[491] In those who pass thin and crude urine for a length of time, if they have otherwise symptoms of convalescence, an abscess may be expected to form in the parts below the diaphragm.^[492] And fatty substances floating on the surface are to be dreaded, for they are indications of melting. And one should consider respecting the kinds of urine, which have clouds, whether they tend upwards or downwards, and the colors which they have and such as fall downwards, with the

colors as described, are to be reckoned good and commended; but such as are carried upwards, with the colors as described, are to be held as bad, and are to be distrusted.^[493] But you must not allow yourself to be deceived if such urine be passed while the bladder is diseased; for then it is a symptom of the state, not of the general system, but of a particular viscus.^[494]

13. That vomiting is of most service which consists of phlegm and bile mixed together, and neither very thick nor in great quantity; but those vomitings which are more unmixed are worse. But if that which is vomited be of the color of leeks or livid, or black, whatever of these colors it be, it is to be reckoned bad; but if the same man vomit all these colors, it is to be reckoned a very fatal symptom. But of all the vomitings, the livid indicates the most imminent danger of death, provided it be of a fetid smell. But all the smells which are somewhat putrid and fetid, are bad in all vomitings.^[495]

14. The expectoration in all pains about the lungs and sides, should be quickly and easily brought up, and a certain degree of yellowness should appear strongly mixed up with the sputum. But if brought up long after the commencement of the pain, and of a yellow or ruddy color, or if it occasions much cough, or be not strongly mixed, it is worse; for that which is intensely yellow is dangerous, but the white, and viscid, and round, do no good. But that which is very green and frothy is bad; but if so intense as to appear black, it is still more dangerous than these; it is bad if nothing is expectorated, and the lungs discharge nothing, but are gorged with matters which boil (as it were) in the air-passages. It is bad when coryza and sneezing either precede or follow affections of the lungs, but in all other affections, even the most deadly, sneezing is a salutary symptom.^[496] A yellow spittle mixed up with not much blood in cases of pneumonia, is salutary and very beneficial if spit up at the commencement of the disease, but if on the seventh day, or still later, it is less favorable. And all sputa are bad which do not remove the pain. But the worst is the black, as has been described. Of all others the sputa which remove the pain are the best.^[497]

15. When the pains in these regions do not cease, either with the discharge of the sputa, nor with alvine evacuations, nor from venesection, purging with medicine, nor a suitable regimen, it is to be held that they will terminate in suppurations.^[498] Of empyemata such as are spit up while the sputum is still bilious, are very fatal, whether the bilious portion be expectorated separate, or along with the other; but more especially if the empyema begin to advance after this sputum on the seventh day of the disease. It is to be expected that a person with such an expectoration shall die on the fourteenth day, unless something favorable supervene. The following are favorable symptoms: to support the disease easily, to have free respiration, to be free from pain, to have the sputa readily brought up, the whole body to appear equally warm and soft, to have no thirst, the urine, and fæces, sleep, and sweats to be all favorable, as described before; when all these symptoms concur, the patient certainly will not die; but if some of these be present and some not, he will not survive longer than the fourteenth day. The bad symptoms are the opposite of these, namely, to bear the disease with difficulty, respiration large and dense, the pain not ceasing, the sputum scarcely coughed up, strong thirst, to have the body unequally affected by the febrile heat, the belly and sides intensely hot, the forehead, hands, and feet cold; the urine, and excrements, the sleep, and sweats, all bad, agreeably to the characters described above; if such a combination of symptoms accompany and either on the ninth or eleventh. Thus then one may conclude regarding this expectoration, that it is very deadly, and that the patient will not survive until the fourteenth day. It is by balancing the concomitant symptoms whether good or bad, that one is to form a prognosis; for thus it will most probably prove to be a true one. Most other suppurations burst, some on the twentieth, some on the thirtieth, some on the fortieth, and some as late as the sixtieth day.^[499]

16. One should estimate when the commencement of the suppuration will take place, by calculating from the day on which the patient was first seized with fever, or if he had a rigor, and if he says, that there is a weight in the place where he had pain formerly, for these symptoms occur in the commencement of suppurations. One then may expect the rupture of the abscesses to take place from these times according to the periods formerly stated. But if the empyema be only on either side, one should turn him and inquire if he has pain on the other side; and if the one side be hotter than the other, and when laid upon the sound side, one should inquire if he has the feeling of a weight hanging from above, for if so, the empyema will be upon the opposite side to that on which the weight was felt.^[500]

17. Empyema may be recognized in all cases by the following symptoms: In the first place, the fever does not go off, but is slight during the day, and increases at night, and copious sweats supervene, there is a desire to cough, and the patients expectorate nothing worth mentioning, the eyes become hollow, the cheeks have red spots on them, the nails of the hands are bent, the fingers are hot especially their extremities, there are swellings in the feet, they have no desire of food, and small blisters (phlyctænæ) occur over the body. These symptoms attend chronic empyemata, and may be much trusted to; and such as are of short standing are indicated by the same, provided they be accompanied by those signs which occur at the commencement, and if at the same time the patient has some difficulty of breathing. Whether they will break earlier or later may be determined by these symptoms; if there be pain at the commencement, and if the dyspnœa, cough, and ptalism be severe, the rupture may be expected in the course of twenty days or still earlier; but if the pain be more mild, and all the other symptoms in proportion, you may expect from these the rupture to be later; but pain, dyspnœa, and ptalism, must take place before the rupture of the abscess. Those patients recover most readily whom the fever leaves the same day that the abscess

bursts,—when they recover their appetite speedily, and are freed from the thirst,—when the alvine discharges are small and consistent, the matter white, smooth, uniform in color, and free of phlegm, and if brought up without pain or strong coughing. Those die whom the fever does not leave, or when appearing to leave them it returns with an exacerbation; when they have thirst, but no desire of food, and there are watery discharges from the bowels; when the expectoration is green or livid, or pituitous and frothy; if all these occur they die, but if certain of these symptoms supervene, and others not, some patients die and some recover, after a long interval. But from all the symptoms taken together one should form a judgment, and so in all other cases.

18. When abscesses form about the ears, after peripneumonic affections, or depositions of matter take place in the inferior extremities and end in fistula, such persons recover. The following observations are to be made upon them: if the fever persist, and the pain do not cease, if the expectoration be not normal, and if the alvine discharges be neither bilious, nor free and unmixed; and if the urine be neither copious nor have its proper sediment, but if, on the other hand, all the other salutary symptoms be present, in such cases abscesses may be expected to take place. They form in the inferior parts when there is a collection of phlegm about the hypochondria; and in the upper when the hypochondria continue soft and free of pain, and when dyspnœa having been present for a certain time, ceases without any obvious cause.^[501] All deposits which take place in the legs after severe and dangerous attacks of pneumonia, are salutary, but the best are those which occur at the time when the sputa undergo a change; for if the swelling and pain take place while the sputa are changing from yellow and becoming of a purulent character, and are expectorated freely, under these circumstances the man will recover most favorably and the abscess becoming free of pain, will soon cease; but if the expectoration is not free, and the urine does not appear to have the proper sediment, there is danger lest the limb should be maimed, or that the case otherwise should give trouble. But if the abscesses disappear and go back, while expectoration does not take place, and fever prevails, it is a bad symptom; for there is danger that the man may get into a state of delirium and die. Of persons having empyema after peripneumonic affections, those that are advanced in life run the greatest risk of dying; but in the other kinds of empyema younger persons rather die.^[502] In cases of empyema treated by the cautery or incision, when the matter is pure, white, and not fetid, the patient recovers; but if of a bloody and dirty character, he dies.^[503]

19. Pains accompanied with fever which occur about the loins and lower parts, if they attack the diaphragm, and leave the parts below, are very fatal. Wherefore one ought to pay attention to the other symptoms, since if any unfavorable one supervene, the case is hopeless; but if while the disease is determined to the diaphragm, the other symptoms are not bad, there is great reason to expect that it will end in empyema.^[504] When the bladder is hard and painful, it is an extremely bad and mortal symptom, more especially in cases attended with continued fever; for the pains proceeding from the bladder alone are enough to kill the patient; and at such a time the bowels are not moved, or the discharges are hard and forced. But urine of a purulent character, and having a white and smooth sediment, relieves the patient. But if no amendment takes place in the characters of the urine, nor the bladder become soft, and the fever is of the continual type, it may be expected that the patient will die in the first stages of the complaint. This form attacks children more especially, from their seventh to their fifteenth year.^[505]

20. Fevers come to a crisis on the same days as to number on which men recover and die. For the mildest class of fevers, and those originating with the most favorable symptoms, cease on the fourth day or earlier; and the most malignant, and those setting in with the most dangerous symptoms, prove fatal on the fourth day or earlier. The first class of them as to violence ends thus: the second is protracted to the seventh day, the third to the eleventh, the fourth to the fourteenth, the fifth to the seventeenth, and the sixth to the twentieth. Thus these periods from the most acute disease ascend by fours up to twenty. But none of these can be truly calculated by whole days, for neither the year nor the months can be numbered by entire days. After these in the same manner, according to the same progression, the first period is of thirty-four days, the second of forty days, and the third of sixty days. In the commencement of these it is very difficult to determine those which will come to a crisis after a long interval; for these beginnings are very similar, but one should pay attention from the first day, and observe further at every additional tetrad, and then one cannot miss seeing how the disease will terminate. The constitution of quartans is agreeable to the same order. Those which will come to a crisis in the shortest space of time, are the easiest to be judged of; for the differences of them are greatest from the commencement, thus those who are going to recover breathe freely, and do not suffer pain, they sleep during the night, and have the other salutary symptoms, whereas those that are to die have difficult respiration, are delirious, troubled with insomnolency, and have other bad symptoms. Matters being thus, one may conjecture, according to the time, and each additional period of the diseases, as they proceed to a crisis. And in women, after parturition, the crises proceed agreeably to the same ratio.^[506]

21. Strong and continued headaches with fever, if any of the deadly symptoms be joined to them, are very fatal. But if without such symptoms the pain be prolonged beyond twenty days, a discharge of blood from the nose or some abscess in the inferior parts may be anticipated; but while the pain is recent, we may expect in like manner a discharge of blood from the nose, or a suppuration, especially if the pain be seated above the temples and forehead; but the hemorrhage is rather to be looked for in persons younger than thirty years, and the suppuration in more elderly persons.^[507]

22. Acute pain of the ear, with continual and strong fever, is to be dreaded; for there is danger that the man may become delirious and die. Since, then, this is a hazardous spot, one ought to pay particular attention to all these symptoms from the commencement. Younger persons die of this disease on the seventh day, or still earlier, but old persons much later; for the fevers and delirium less frequently supervene upon them, and on that account the ears previously come to a suppuration, but at these periods of life, relapses of the disease coming on generally prove fatal. Younger persons die before the ear suppurates; only if white matter run from the ear, there may be hope that a younger person will recover, provided any other favorable symptom be combined.^[508]

23. Ulceration of the throat with fever, is a serious affection, and if any other of the symptoms formerly described as being bad, be present, the physician ought to announce that his patient is in danger.^[509] Those quinsies are most dangerous, and most quickly prove fatal, which make no appearance in the fauces, nor in the neck, but occasion very great pain and difficulty of breathing; these induce suffocation on the first day, or on the second, the third, or the fourth.^[510] Such as, in like manner, are attended with pain, are swelled up, and have redness (erythema) in the throat, are indeed very fatal, but more protracted than the former, provided the redness be great.^[511] Those cases in which both the throat and the neck are red, are more protracted, and certain persons recover from them, especially if the neck and breast be affected with erythema, and the erysipelas be not determined inwardly.^[512] If neither the erysipelas disappear on the critical day, nor any abscess form outwardly, nor any pus be spit up, and if the patient fancy himself well, and be free from pain, death, or a relapse of the erythema is to be apprehended. It is much less hazardous when the swelling and redness are determined outwardly; but if determined to the lungs, they superinduce delirium, and frequently some of these cases terminate in empyema.^[513] It is very dangerous to cut off or scarify enlarged uvulae while they are red and large, for inflammations and hemorrhages supervene; but one should try to reduce such swellings by some other means at this season. When the whole of it is converted into an abscess, which is called Uva, or when the extremity of the variety called Columella is larger and round, but the upper part thinner, at this time it will be safe to operate. But it will be better to open the bowels gently before proceeding to the operation, if time will permit, and the patient be not in danger of being suffocated.^[514]

24. When the fevers cease without any symptoms of resolution occurring, and not on the critical days, in such cases a relapse may be anticipated.^[515] When any of the fevers is protracted, although the man exhibits symptoms of recovery, and there is no longer pain from any inflammation, nor from any other visible cause, in such a case a deposit, with swelling and pain, may be expected in some one of the joints, and not improbably in those below. Such deposits occur more readily and in less time to persons under thirty years of age; and one should immediately suspect the formation of such a deposit, if the fever be protracted beyond twenty days; but to aged persons these less seldom happen, and not until the fever be much longer protracted. Such a deposit may be expected, when the fever is of a continual type, and that it will pass into a quartan, if it become intermittent, and its paroxysms come on in an irregular manner, and if in this form it approach autumn. As deposits form most readily in persons below thirty years of age, so quartans most commonly occur to persons beyond that age. It is proper to know that deposits occur most readily in winter, that then they are most protracted, but are less given to return.^[516] Whoever, in a fever that is not of a fatal character, says that he has pain in his head, and that something dark appears to be before his eyes, and that he has pain at the stomach, will be seized with vomiting of bile; but if rigor also attack him, and the inferior parts of the hypochondrium are cold, vomiting is still nearer at hand; and if he eat or drink anything at such a season, it will be quickly vomited. In these cases, when the pain commences on the first day, they are particularly oppressed on the fourth and the fifth; and they are relieved on the seventh, but the greater part of them begin to have pain on the third day, and are most especially tossed on the fifth, but are relieved on the ninth or eleventh; but in those who begin to have pains on the fifth day, and other matters proceed properly with them, the disease comes to a crisis on the fourteenth day. But when in such a fever persons affected with headache, instead of having a dark appearance before their eyes, have dimness of vision, or flashes of light appear before their eyes, and instead of pain at the pit of the stomach, they have in their hypochondrium a fulness stretching either to the right or left side, without either pain or inflammation, a hemorrhage from the nose is to be expected in such a case, rather than a vomiting. But it is in young persons particularly that the hemorrhage is to be expected, for in persons beyond the age of thirty-five, vomitings are rather to be anticipated. Convulsions occur to children if acute fever be present, and the belly be constipated, if they cannot sleep, are agitated, and moan, and change color, and become green, livid, or ruddy. These complaints occur most readily to children which are very young up to their seventh year; older children and adults are not equally liable to be seized with convulsions in fevers, unless some of the strongest and worst symptoms precede, such as those which occur in frenzy. One must judge of children as of others, which will die and which recover, from the whole of the symptoms, as they have been specially described.^[517] These things I say respecting acute diseases, and the affections which spring from them.

25. He who would know correctly beforehand those that will recover, and those that will die, and in what cases the disease will be protracted for many days, and in what cases for a shorter time, must be able to form a judgment from having made himself acquainted with all the symptoms, and estimating their powers in comparison with one another, as has been described, with regard to the others, and the urine and sputa, as when the patient coughs up pus and bile together. One ought also to

consider promptly the influx of epidemical diseases and the constitution of the season.^[518] One should likewise be well acquainted with the particular signs and the other symptoms, and not be ignorant how that, in every year, and at every season, bad symptoms prognosticate ill, and favorable symptoms good, since the aforesaid symptoms appear to have held true in Libya, in Delos, and in Scythia;^[519] from which it may be known that, in the same regions, there is no difficulty in attaining a knowledge of many more things than these; if having learned them, one knows also how to judge and reason correctly of them. But you should not complain because the name of any disease may happen not to be described here, for you may know all such as come to a crisis in the afore-mentioned times, by the same symptoms.^[520]

APPENDIX
TO
THE BOOK OF PROGNOSTICS.

As announced in the Preliminary Discourse (Sect. II., 18), I shall now proceed to give an abstract of the principal matters contained in the SECOND BOOK OF PRORRHETICS, which appear to me to be highly interesting, and as they relate to the subjects treated of in the Prognostics, they may be more suitably introduced here than in any other place.

The author commences the treatise with expressing his disapproval of certain modes of making prognostics which he had seen practiced. He says he had heard of many and famous predictions having been made by physicians, such as he himself did not pretend that he could make. Such, for example, as for a physician to call in upon a patient who was looked upon as being in a desperate condition by another physician, and predict that he would not die, but would lose his sight. Or to predict with regard to another patient supposed to be in a bad way, that he will recover, but will become lame of a hand. And of a third who, to all appearance, cannot recover, to predict that he will get well, but that his toes will blacken and putrefy. Similar predictions are related under this class. Another mode of prediction is to prophecy to buyers and traders, to one death, to another madness, and to the rest diseases, and that from what is now occurring, or has occurred before, and all the predictions to turn out true. Another kind of predictions relates to Athletæ, and those who practice gymnastic and laborious exercises for the cure of diseases, where the practitioner pretends to so much exactness, that if the patient is guilty of any act of omission or commission in regard to food, drink, or venery, the physician will detect it. He himself makes no pretensions to any such skill in divination, but announces it as his object to describe the symptoms by which it may be known whether a man will die or live, and whether his disease will be of short or of long duration. With regard to the predictions of abscesses, lameness, death, or madness, the author holds that they can only be made after the morbid conditions leading to them have fairly set in. He strongly disapproves of all ostentatious modes of making predictions, and gives it as his advice that in all such cases the greatest prudence and reserve should be observed, since if a man become an adept in this art of prognostications, he will gain great credit with his patient, whereas if he fall into mistakes, he will incur odium, and will be looked upon as being deranged.

With regard to the prognostics made by those who practice gymnastics, he recommends them not to be made in a charlatan manner, but with suitable caution, and directs minute attention to be paid to the circumstances of the patient, which one has superior advantages in observing under this system. He says, for example, that a physician who feels a patient's belly and pulse, pays attention to the breathing at the nostrils, and listens to the speech, and sound of the respiration, will be less likely to be deceived in forming a prognostic on his patients than he who neglects these things. He expresses himself, however, as being incredulous as to the possibility of detecting any little transgressions of orders which a patient may commit, although greater departures from instructions may be suspected. After some general observations in respect to diet, and other matters relating to it, he proceeds to a more circumstantial description of the symptoms upon which a prognosis is to be founded. And first, with regard to the alvine dejections, those of persons who live a laborious life, and use food and drink sparingly, are small and hard, and are passed every day, every third day, or every fourth day, but if they pass the last period there is danger of the man's being seized with fever or diarrhœa. When the stools are so liquid that they do not assume a shape, they are all of a worse character in these cases. The dejections of persons who lead an active life are less copious than those of the indolent, provided they use the same amount of food. Liquid dejections taking place on the seventh day, and quickly coming to a crisis, are beneficial, if they occur all at once, and are not repeated. But if accompanied with fever, or if the diarrhœa is prolonged, all such dejections are bad, whether bilious, pituitous, or of indigested matters, and require a particular regimen and mode of treatment.

With regard to the urine, it should be in proportion to the drink that is taken, and somewhat thicker than the fluid that is drunk. If it be more copious than natural, this indicates either that the patient has disobeyed orders as to the amount of his drink, or that his body is in a state of atrophy. If the urine is passed in deficient quantity, with a noise, it indicates either that the man stands in need of purging, or that the bladder is diseased. A small quantity of blood passed without fever and pain does not indicate anything bad, but proves a solution to a state of lassitude. But if in large quantity, with the addition of any of these symptoms, it is to be dreaded. But if the urine be passed with pain, and if pus be passed along with the urine in a fever, the physician should announce that the patient will thus be relieved of his complaints.^[521] Thick urine having a thin sediment indicates some pain and swelling about the joints. All the other sediments which occur in the urine of persons who practice exercises are connected with disease about the bladder; this will be clearly shown by the obstinate pains with which they are accompanied. The author, although he states that he had been conversant with the teachers of prognostics from urine, and their children and disciples, seems to express himself doubtful as to the possibility of acquiring a great degree of accuracy in regard to these matters.

Respecting dropsy, consumption, gout, and epilepsy, he states generally that if they are hereditary they are difficult to remove. A favorable prognosis is to be formed in dropsy when the patient's viscera are sound, when his strength is firm, the digestion and respiration natural, when he is free of pain, the temperature of the body moderate, and when there is no wasting of the extremities. It is favorable when there is no cough, thirst, nor dryness of the tongue, when the bowels are easily moved

by medicine, and when, at other times, the dejections are consistent. Dropsy, supervening, along with fever, upon a great discharge of blood, is of a most intractable nature, and the physician should intimate the danger to some other person beforehand. When great swellings suddenly subside and rise again, there is more hope in such a case than in dropsies connected with a discharge of blood. He concludes his observations on dropsies with the remark, that they are apt to deceive the patients, so that they desert their physicians and thus perish.

With regard to consumptive patients, he says, he has the same observations to make with regard to the sputa and cough as he had written with regard to empyema.^[522] If the patient is to recover, the sputa should be white, equable, of one color, without phlegm; the defluxion from the head should be determined to the nose; there should be no fever, nor anorexia, nor thirst; the alvine discharges firm, proportionate to the ingesta, and the patient should not get thin. The best form of the chest is when it is quadrangular and hairy, and when the cartilage is small, and covered with flesh. Young persons, who become affected with empyema from determination (metastasis?), or fistula, or from any other similar cause, or from the retrocession of an abscess, do not recover unless many of the favorable symptoms combine in the case. They die, most commonly, in autumn, which proves peculiarly fatal in protracted diseases. Of all others, virgins, and women suffering from amenorrhœa, seldome recover; and in their cases there is no hope unless menstruation be restored. All sexes, he seems to say (but the meaning appears to me rather ambiguous), have a better chance of recovery, when there is a discharge of blood, especially in those cases in which there are pains in the back and chest, connected with black bile; and if, after the evacuation, there be a remission of the pain; if the cough and fever do not set in; and if the thirst be tolerable. He seems to state (but the text is in an unsatisfactory condition), that relapses take place unless there be deposits in the place, the best of which are those which contain most blood; and that in those cases in which there are pains in the chest, if the patients get emaciated, and cough, and a dyspnoea supervenes, without fever or empyema, they should be asked whether, when they cough, and have difficulty of breathing, the sputa be compact, and attended with little smell.

With regard to persons affected with the gout, those who are aged, have tofi in their joints, who have led a hard life, and whose bowels are constipated, are beyond the power of medicine to cure. But, the best natural remedy for them is an attack of dysentery, or other determination to the bowels. Persons, under opposite circumstances, may be cured by a skilful physician.

The prognosis in epilepsy is unfavorable when the disease is congenital, and when it endures to manhood, and when it occurs to a grown person, and without any obvious cause. When connected with the head it is particularly to be apprehended, but least so when it seems to be derived from the hands or feet. The cure may be attempted in young persons, but not in old.

In the case of children, he mentions various complaints, such as distortion of the eyes, tubercles about the neck, pain in the bowels, omental hernia, etc., which, upon inquiry, will be found to be the consequences of an attack of epilepsy.

The judgment to be formed in the case of ulcers is to be founded on the age of the patient, the situation of the sore, and its appearance.

Strumous tubercles, which end in suppuration, occur most frequently in young persons. Adults are subject to bad favi, internal cancers, and herpetic sores, after epinyetis, until they pass sixty. Old persons are subject to cancers, both deep-seated and superficial, which never leave them. They are particularly intractable when seated in the armpits, the loins, and the thighs.

Of affections of the joints, the most dangerous are those seated in the thumb and great toe. When there is a chronic sore on the side of the tongue the surgeon should examine whether it be not occasioned by the sharp edge of a tooth.^[523]

The most dangerous wounds are those which implicate the large veins (blood-vessels), in the neck and groin; then those of the brain and liver; next, those of the bowels and bladder. These cases are all dangerous, but not uniformly fatal, as some suppose. Much depends upon constitution, as to liability to fever and inflammation after a wound. Sometimes, also, the wounds of smaller vessels prove fatal by inducing hemorrhage, fever, or delirium. In all recent wounds, however, the physician should endeavor to afford assistance.

Of spreading ulcers, the most fatal are such mortifications as are very deep, black, and dry; and those are bad and dangerous which are accompanied with a black ichorous discharge. Those which are white and mucous are less dangerous, but are more subject to relapse, and become inveterate. Herpes is the least dangerous of the spreading sores, but is most difficult to remove about deep-seated cancers.^[524] An ephemeral fever, with very white and thick pus, is beneficial in such a case; also, sphacelus of a nerve, of a bone, or of both, in deep-seated and black mortifications. For a free discharge of pus takes place and carries off the mortification.

The prognostics in wounds of the head are given in nearly the same terms as laid down in the treatise on that subject, and therefore I need not enter minutely into an exposition of what is stated regarding them here. Those in the upper part of the head, more especially if they implicate a suture, are said to be particularly dangerous. The author directs the surgeon to inquire whether, at the time of the accident, the patient fell down or became comatose, as in this case greater danger is to be apprehended.

Large wounds of the joints, if they involve the connecting nerves, necessarily leave the limb maimed. Several other observations connected with these injuries are added, of which one of the most important is the direction to practice flexion and extension of the limb, frequently, with the view, no doubt, of preventing rigidity of the joint.

Large excisions in the arm becoming inflamed end in suppurations, which require to be evacuated by the knife or cautery. Injuries of the spinal marrow, whether from disease or accident, are attended with loss of motion and sensibility, retention of the alvine and urinary discharges; but, after a time, involuntary evacuations take place, which are soon followed by death. When the throat is frequently filled with blood, and there is no headache or cough, nor any other morbid symptoms, the physician should examine whether there be not an ulcer or a leech in the part.

With regard to the eyes, the prognostics are given with so much prolixity of detail that I must be content with a brief abstract of them. Much attention is paid to the characters of the discharge from the eyes in diseases of them, namely, of the glutinous matter and tears; thus, if the gum be white and soft, the tears mixed with it not very hot, and the swelling light and loose, under these circumstances the eyelids are glued up during the night, so that the eye is free of pain, and thus the disease is without danger, and of short duration. The other appearances of the eye, and the discharges, are also minutely given. When the discharge is green or livid, the tears copious and hot, a burning heat in the head, and pains darting through the head to the eye, there must necessarily be ulceration in the eye; and there is much reason to apprehend that it will burst. If, when one can get a sight of the eye, it should be found burst, and the pupil projecting above the rupture, it is bad and difficult to restore; and, if there be sloughing, the eye will be wholly useless. According to the form and depth of the ulcers must be the subsequent cicatrices. These are minutely described according to their different varieties. Mention is also made of the prognostics from the eyes in fevers, as described by the author in another work. It is most likely that allusion is here made to the first book of "Prorrhethics." In conclusion, the surgeon is directed to pay great attention to the state of the urine in diseases of the eyes.

Dysenteries, when they set in with fever, alvine discharges of a mixed character, or with inflammation of the liver, or of the hypochondrium, or of the stomach, such as are painful, with retention of the food and thirst, all these are bad; and the more of these symptoms there are, the greater the danger; and the fewer, the more hope is there of recovery. Children from five to ten years of age are the most apt to die of this complaint; the other ages less so. Such dysenteries as are of a beneficial nature, and are attended with blood and scrapings of the bowels, cease on the seventh, or fourteenth, or twentieth, or thirtieth day, or within that period. In such cases even a pregnant woman may recover and not suffer abortion.

All cases of lenteria are said to be of a bad character when they are continued and protracted, both day and night, and when the dejections are either very crude, or black, soft, and fetid; for they occasion thirst and determine the fluids otherwise than to the bladder, give rise to ulcerations (aphthæ?) in the mouth, redness and ephelis^[525] of all colors, and at the same time the belly is in a state of ferment, and has a foul, wrinkled appearance externally. This disease is most to be dreaded by old persons; it is formidable to men of middle age, but less so in the other ages. The indications of cure, it is acutely stated, are to determine the fluids to the urine, to relieve the body from its atrophy, and change the color of the skin.

All the other varieties of diarrhœa without fever are of short duration and mild; for they will all cease when washed out, or of their own accord. The discharge may be predicted as about to cease when, upon touching the belly, there is no movement, and flatulence passes with the discharge. Eversion of the gut takes place in the case of middle-aged persons having piles, of children affected with the stone, and in protracted and intense discharges from the bowels, and of old persons having mucous concretions (scybalæ?).

Women may be judged of whether they are in a fit state for conception or not by attending to the following circumstances:—In the first place to their shapes. Women of smaller stature more readily conceive than taller persons; the thin than the fat; the white than the ruddy; the dark than the pale; those who have prominent veins than the contrary. In oldish women it is bad to have much flesh, but a good thing to have swelled and large breasts. In addition, inquiry should be made whether or not the menstruation be regular as to time and quantity. And it should be ascertained whether the uterus be healthy, of a dry temperament, and soft; neither in a state of retraction nor prolapsus; and its mouth neither turned aside, nor too close, nor too open. When any of these obstructions come in the way, it is impossible that conception can take place.

Such women as cannot conceive, but appear green, without fever, and the viscera are not in fault; these will say that the head is pained, and that the menstrual discharge is vitiated and irregular. But such of these as have the proper color, are of a fat habit of body, the veins are inconspicuous, they have no pains, and the menses either never appear at all, or are scanty and intense, and this is one of the most difficult states of sterility to remove. In other cases the health is not to blame, but the fault lies in the position of the womb. The other contingencies in this place are attended with pains, discoloration, and wasting.

Ulceration in the womb from parturition, an abscess of a chronic nature, or from any other cause, is necessarily accompanied with fevers, buboes, and pains in the place; and if the lochial discharge be also suppressed, all these evils are more intense and inveterate, along with pains of the hypochondrium and head. And when the ulcer heals, the part necessarily is smoother and harder, and the woman is less adapted for conception. If, however, the ulceration be in the right side only, the woman may conceive of a female child, or if in the left, of a male. When a woman cannot conceive, and fever comes on with a

slight cough, inquiry should be made whether she has any ulcer about the uterus, or any other of the complaints I have described; for if she has no complaint in that region to account for her loss of flesh and sterility, it may be expected that she will have vomiting of blood, and the catamenia will necessarily be suppressed. But if the fever be carried off by the evacuation of blood, and if the catamenia appear, she will then prove with child. But if looseness of the bowels having a bad character take place before there is an evacuation of blood, there is danger lest the woman perish before a vomiting of blood can take place.

In cases of false conception, in which women are deceived by the non-appearance of the menses, and by the increase of the belly and movement in it, they will be found to have had headache and pains about the neck and hypochondria, and there is no milk in the breasts except a little of a watery character. But when the swelling of the womb passes away they will conceive, unless there be any other impediment. For this affection is beneficial in so far as it produces a change in the uterus, so that afterwards the woman may prove with child. Women with child have not these pains unless the headache be habitual to them, and in addition they have milk in their breasts. Women affected with chronic discharges are to be asked whether they have pains in the head and loins, and in the lower part of the belly, and whether their teeth be set on edge, and if they have dimness of sight, and noises in their ears. Such women as vomit bile for several days while in a fasting state, though they are not with child nor have fever, are to be asked whether they have vomited up round lumbrici, and if they say not, they should be warned that this will happen to them. This affection happens principally to married women, then to virgins, and less seldom to other people.

Pains without fever are not deadly, but mostly prove protracted, and have many changes and relapses. Several varieties of headache are described, and the prognosis in each laid down. The natural cure of them is a coryza, a discharge of mucus from the nose, or sneezing. Pains spreading from the head to the neck and back, are relieved by abscesses, expectoration of pus, hemorrhoids, exanthemata on the body, or pityriasis on the head.

Heaviness and pruritus in the head, either in a part or through the whole of it, if, on inquiry, they extend to the tip of the tongue, indicate a confirmed disease, and one difficult to remove. They are best removed by the occurrence of an abscess. But those cases which are accompanied with vertigo are difficult to cure, and are apt to pass into mania. Other diseases in the head, of a very strong and protracted character, occur to both men and women, but especially to young persons, and virgins at the season of manhood, and especially at the catamenial period. Women, however, are less subject to pruritus and melancholic affections than the men, unless the menses have disappeared.

Both men and women who have long had a bad color, but not in the form of jaundice, are subject to headaches, eat stones and earth, and have piles. Those who have green colors, without decided jaundice, are affected in like manner, only instead of eating stones and earth, they are more subject to pains in the hypochondriac region. Persons who are pale for a length of time, and have the face tumid, will be found to have headache, or pains about the viscera, or some disease in the anus; and in most cases, not one, but many, or all these evils make their appearance.

Nyctalopia is most apt to attack young persons, either males or females, and to pass off spontaneously on the fortieth day or in seven months, and in some cases it endures for a whole year. Its duration may be estimated from the strength of the disease and the age of the patient. They are relieved by deposits which determine downwards, but these rarely occur in youth. Married women and virgins that have the menstrual discharge regular are not subject to the complaint. Persons having protracted defluxions of tears who are attacked with nyctalopia, are to be questioned whether they had any previous complaint in the head.

Such persons as have frequent pains in the vertex and temples, without fever or loss of color, unless they have some other obvious deposit in the face, or speak in a rough tone, or have pain in the teeth, may be expected to have a hemorrhage from the nose. Those who have bleeding at the nose, although they may appear to be otherwise in good health, will be found to have enlarged spleen, or pain in the head, or flashes of light before their eyes. Most of these patients have both headache and affection of the spleen.

The gums are diseased and the mouth fetid in persons who have enlarged spleens. But persons who, although they have enlarged spleens, are exempt from hemorrhages and fœtor of the mouth have malignant ulcers on the legs and black cicatrices. But if they have any obvious deposit in the countenance, or if their speech be rough, or if they have toothache, a hemorrhage from the nose may be expected. Those who have great swellings below the eyes will be found to have enlarged spleens. And if there come on swellings in the feet, and if they appear to be dropsical, the belly and loins must be attended to.

Distortions of the countenance, if not sympathetic with some other part of the body, quickly pass off either spontaneously or by remedial means. The others are of an apoplectic nature. In other cases, when the diseased part wastes from want of motion, there can be no relief afforded. But when wasting does not take place there may be recovery. With regard to the time when this may occur, it is to be prognosticated by attending to the severity of the disease, to its duration, to the age of the patient, and to the season, it being known that of all cases the inveterate, and such as are the consequences of repeated attacks,

are the worst, and the most difficult to remove, and those in aged persons. Autumn and winter are more unfavorable seasons for such complaints than spring and summer.

Pains in the shoulder, which, passing down the arms, occasion torpor and pains, do not usually terminate in deposits, but the patients get better by vomiting black bile. But when the pains remain in the shoulders, or extend to the back, the patients are relieved by vomiting pus or black bile. They are to be judged of thus:—If their breathing be free, and if they be slender, it is rather to be expected that they will vomit black bile. But if they have more difficulty of breathing, and if there is any unusual color on the countenance, whether reddish or black, it is to be expected that they will rather spit blood. It should also be attended to whether there be swellings on the feet. This disease attacks men most violently from forty to sixty years of age. At this period of life ischiatic diseases are most troublesome.

Ischiatic diseases are to be thus judged of:—In the case of old persons, when the torpor and coldness of the loins and legs are very strong, and when they lose the power of erections, and the bowels are not moved, or with difficulty, and the fæces are passed with much mucus, the disease will be very protracted, and it should be announced beforehand that the disease will not last shorter time than a year from its commencement; and amendment is to be looked for in spring and summer. Ischiatic diseases are no less painful in young men, but are of shorter duration, for they pass off in forty days; and neither is the torpor great, nor is there coldness of the legs and loins. In those cases in which the disease is seated in the loins and leg, but the patient does not suffer so much as to be confined to bed, examine whether any concretions have taken place in the hip-joint, and make inquiry whether the pain extends to the groin; for if both these symptoms be present, the disease will be of long duration. And the physician should also inquire whether there be torpor in the thigh, and if it extend to the ham; and if he says so, he is to be again asked if it spreads along the leg to the ankle of the foot. Those who confess to the most of these symptoms are to be told that the limb will be sometimes hot and sometimes cold; but those persons in whom the pain leaves the loins, and is turned downwards, are to be encouraged; but when the disease does not leave the hip and loins, such persons are to be warned that it is to be dreaded. In those cases in which there are pains and swellings about the joints, and they do not pass off, after the manner of gout, you will find the bowels enlarged, and a white sediment in the urine; and, if you inquire, the patient will admit that the temples are often pained, and he will say that he has nightly sweats. If the urine have not this sediment, nor do the sweats take place, there is danger either that the joints will become lame, or that the tumor called meliceris will form in it. This disease forms in those persons who in their youth had epistaxis, and in whom it had ceased afterwards. They are to be interrogated whether they had discharges of blood in their youth, and if they have pruritus in the breast and back. And the same thing happens to those who have severe pains in the bowels, without disorder of them, or who have hemorrhoids. This is the origin of these complaints. But if the patients have a bad color, they are to be interrogated whether their head be pained, for they will say that it is. In those cases in which the bowels are pained on the right side, the pains are stronger, and especially when the pain terminates in the hypochondrium at the liver. Such pains are immediately relieved if borborygmi take place in the belly. But when the pain ceases, they pass thick and green urine. The disease is not deadly, but very protracted. But when the disease is already of long standing, the patients have dimness of sight in consequence of it. But they are to be interrogated whether, when young, they had a flow of blood, and regarding the dimness of vision, the greenness of the urinary discharge, and regarding the borborygmi, if they took place and gave relief; for they will confess to all these symptoms.

Lichen, leprosy, and leucè, when they occur in young children and infants, or when they appear at first small, and gradually increase in the course of a long time—in these cases the eruption is not to be regarded as a deposit, but as a disease; but when they set in rank and suddenly, this case is a deposit. Leucè also arises from the most fatal diseases, such as the disease called phthisis;^[526] but leprosy and lichen are connected with black bile. These complaints are the more easily cured the more recent they are, and the younger the patients, and the more soft and fleshy the parts of the body in which they occur.

ON REGIMEN IN ACUTE DISEASES.

THE ARGUMENT.

In this treatise two very important questions are discussed: first, a nosological question, regarding the proper distinction of diseases from one another; and secondly, a therapeutical, respecting the rules by which the regimen in acute diseases ought to be regulated. The former of these is of a polemical nature, being an attack directed against the physicians of the Cnidian school of medicine, who distinguished diseases from one another in an arbitrary manner, from incidental varieties in their constitution, and without proper attention to their true constitution and identity. As will be seen in the annotations, the Cnidians pretended to recognize several varieties of disease connected with bile,—several fanciful divisions of diseases of the bladder, and so forth; to which mode of distinguishing diseases there would obviously be no end, since of incidental varieties in any case there can be no limit. The other question discussed in this treatise relates to what may justly be pronounced to be one of the most important points connected with the practice of medicine, namely, the proper regimen in acute diseases; that is to say, in idiopathic fevers and febrile diseases, comprising most of those diseases now classed under the head of *Zymotic*, and which constitute by far the highest item in our bills of mortality at the present day. Our author distinguishes them by the names of pleurisy, pneumonia, phrenitis, lethargy, causus, and their cognate diseases, including fever of the continual type. Now it is to be borne in mind, that the phrenitis,^[527] lethargy, and causus of Hippocrates, were all epidemic fevers, so that, with the exception of pleurisy and pneumonia, all the diseases here treated of are fevers of the country in which Hippocrates resided. One, then, cannot well imagine a question which from the commencement of the medical Art must have been felt of higher importance than this,—how so numerous and formidable a class of diseases ought to be treated. In the attempt to solve it, every imaginable mode of treatment, as might have been expected beforehand, was tried, and its effects determined by experience. Herodicus, the master of Hippocrates in gymnastics, applied his panacea in the treatment of febrile diseases, and, as we are informed, with the most disastrous results. “Herodicus,” says the author of the sixth Book of Epidemics, “killed persons in fever by promenading, much wrestling, and fomentations.” (§ iii., 18.) It may *now* appear wonderful that so extraordinary a mode of practice should have ever been attempted in this case; but while men of all ranks continue to resort for the cure of all sorts of diseases to any individual who has got a single hobby with which he constantly works to his own profit, whether it be *gymnastics*, or *shampooing*, or the *wet sheet*, we may expect to hear that such rash experiments have been repeated. Truly mankind pay as dearly for their tame submission to the insane practices of professional chiefs, as the Greeks are represented by the poet to have suffered from the follies of their princes:

“Quicquid delirant Reges, plectuntur Achivi.”^[528]

And surely it is much to be desired that men would learn a lesson from the Past, and not allow every new page in the history of society and of the profession to furnish a repetition of the oft-told tale of supine credulity on the one side, and of audacious folly on the other. From what has been stated, it will readily be understood that it was soon settled that active exercise is inadmissible in febrile diseases.^[529] It would next come to be determined, what rule was to be followed with regard to the administration of food in fevers. On this point, as will be seen below in our annotations, the most diametrically opposite plans of treatment were essayed. One authority administered the most highly nutritious articles of food, namely, fleshes, to his patients, while, on the other hand, some wasted them by enforcing a total abstinence for several days. Experience, we may be well assured, was not long in deciding against both the starving and the gluttoning system: the palled appetite would soon refuse to accept of solids, and the parched tongue would speedily crave some allowance of liquids. Even before the days of Hippocrates, there is every reason to suppose that these extreme modes of treatment had been abandoned; but still he complains that in his time many important points in the treatment of acute diseases were wholly undetermined, such as the following: whether plain drink, that is to say water, was to be administered;—or, water seasoned by the admixture of something farinaceous, such as the decoction of barley;—whether the same should be given so thick as to constitute a nutritious gruel, or strained so as to form merely a drink;—whether wine should be given in small quantity, or more copiously;—whether any of these things should be given from the commencement of the disease, or not until after an interval of certain days. Hippocrates informs us that the most discordant opinions prevailed upon these points, and his professed object, in this treatise, is to reduce the rules of practice to certain fixed principles. How our author performs this task, the reader is left to judge for himself; it may be interesting, however, to know, that Galen with all his devoted admiration of Hippocrates, is not disposed to admit that his solution of the question at issue is quite lucid and satisfactory. This opinion Galen pronounces on two separate occasions; in his commentary on this treatise, and in his great Work “On the Tenets of Hippocrates and Plato.” As I look upon his observations contained in the latter Work to be of great importance toward understanding the bearing of this treatise, I shall not scruple to introduce a translation of the greater part of them in this place.

The ninth book of the Work we have mentioned opens with an elaborate disquisition on the logical principles which ought to guide us in deciding with regard to identity and difference, both in Philosophy and Medicine: on the former of these subjects

he quotes freely from Plato, and on the other from Hippocrates. Coming, then, to the question in hand, he says:—"And thus Hippocrates proceeded in the work 'On the Regimen of Acute Diseases,' finding fault with the Cnidian physicians, as being ignorant of the differences of diseases with regard to genus and species; and he himself points out the definitions according to which that which appears to be one, being divided becomes many, not only in the case of diseases, but also in that of all other things; in which we find that many of the most celebrated physicians fall into mistakes, even with regard to the remedies. For some, coming to the particular use of them, have established a most immethodical method of instruction; whilst others, stating a very general precept, lay down a rule which at first sight appears very methodical, but in truth is very bad, and hence they disagree among themselves; some, as for example those treating of the remedy for a certain affection, such as pleurisy, declaring it to be venesection, others purging, some fomentations by means of sponges, and others of bags, or something of the like kind. And they differ, in the same manner, with regard to the use and disuse of the bath, of oxymel, of hydromel, and of water, of wine, and of ptisan, either giving of the strained juice only, or of the barley portion only; and some, with regard to food, giving discordant decisions as to the differences of the sick, and the indications which a pleuritic affection requires. And that he, as being the first discoverer, has handled these subjects in rather a confused manner, I have shown in my Commentary on the treatise which has been improperly entitled, 'Against the Cnidian Sentences,' and 'On the Ptisan.' But in order that those who are desirous of learning, may have a clear exposition of this question in a brief form, I shall not scruple to give here a summary of it. In the commencement of pleuritic attacks, when the side is just beginning to be pained, inasmuch as the nature of the disease is not yet obvious, he directs fomentations, otherwise called heating applications, to be tried, and he explains the materials of which they consist. And then, if the complaint is not removed, it is to be ascertained whether the patient took food recently, and whether the bowels have been moved, and he gives instructions what should be done in these cases. But if the disease does not yield to these means, he gives definitions of those cases which require venesection and purging, and those in which one should use hydromel for drink, or oxymel, or water until the crisis, without giving any food; and those in which the juice of ptisan is to be used, or the barley along with it, and when food is to be administered. In like manner, with regard to the administration of wine, it is determined in what cases it is to be given, and in what not, and when, and of what quality. And in like manner respecting baths, and other matters of the like kind. And as a twofold mistake is committed with regard to the divisions (of diseases), some doing it in a deficient manner, and others carrying this process to excess, Hippocrates, finding fault with both, expresses himself thus, in the beginning of the book: 'Some of them, indeed, were not ignorant of the many varieties of each complaint, and their manifold division, but when they wish to tell clearly the members (species?) of each disease, they do not write correctly; for the species would be almost innumerable if every symptom experienced by the patients were held to constitute a disease, and receive a different name.' And again, respecting the remedies, as being deficient, he writes thus: 'And not only do I not give them credit on this account, but also because those they use are few in number.' Afterwards, assuming what is of great importance to the question, he does not give a clear solution of it, and therefore the whole bearing of the question is misunderstood by many physicians. I have, therefore, given an exposition of the whole subject, in my first Commentary 'On the Regimen of Acute Diseases;' and it is necessary to show the import of it briefly. The question is given by Hippocrates in the following terms: 'But it appears to me that those things are more especially deserving of being consigned to writing, which are undetermined by physicians, notwithstanding that they are of vital importance, and either do much good or much harm. By undetermined, I mean such as those: wherefore certain physicians, during their whole lives, are constantly administering untrained ptisans, and fancy they thus accomplish the cure properly, whereas others take great pains that the patient may not swallow a particle of the barley (thinking it would do much harm), but strain the juice through a cloth before giving it: others, again, will neither give thick ptisan nor the juice, some until the seventh day of the disease, and some until after the crisis. Physicians are not in the practice of mooted such questions, nor perhaps, if mooted, would a solution of them be readily found, although the whole Art is thereby exposed to much censure from the vulgar, who fancy that really there is no such science as Medicine, since, in acute diseases, practitioners differ so much among themselves, that those things which one administers, as thinking it the best that can be given, another holds to be bad.' And a little afterwards: 'I say, then, that this question is a most excellent one, and allied to very many others, and some of the most vital importance in the Art: for, that it can contribute much to the recovery of the sick, and to the preservation of health in the case of those who use it well, and that it promotes the strength of those who take gymnastic exercises, and is useful to whatever one may wish to apply it.' The inquiry regarding the differences of opinion among practitioners, he says, is of the greatest consequence, not only to the sick, for the recovery of health, but also to those in health, for the preservation of it, and to those who practise it for the recovery and preservation of deportment. And he afterwards adds, 'to whatever one may wish;' as indicating that the solution of this inquiry is applicable not only to medicine but to all the other arts to which one may choose to apply it. For it is wonderful that physicians practising an art, in which the remedies applied may be determined by experience whether they are beneficial or hurtful, should yet make the most conflicting statements respecting those things which are beneficial and those which are prejudicial. For, in philosophy, it is not to be wondered at that there should be no end to most disputes, since these things cannot be clearly determined by experience; and therefore some hold that the world is uncreated, some that it was created, some that there is nothing beyond its boundary, some that there is, and some declaring what that which is contained is, and some pronouncing it to be a vacuum, having no substance in it, and some holding that worlds in inconceivable numbers, and infinite, exist. For such discrepancy of opinion cannot be set at rest by any clear appeal

to the senses. But it is not so with respect to the benefit or injury derived from remedies administered to the body, since the differences among physicians, in this case, may be decided by experience, as to which of them are beneficial and which injurious. Wherefore the solution of this question is not very clearly stated by Hippocrates, and on that account it has excited the observation of almost all the commentators on this book. It is this: some of the sick require abstinence from food, until the disease come to a crisis, and some require food, and of these some require the unstrained ptisan, and some the strained, as also some require still more substantial food, and, moreover, some require oxmyel, or hydromel, and some water, or wine. Wherefore to those physicians who have cultivated the Art upon experience alone, that only appears beneficial which perchance has seemed useful in most cases. Neither do they venture to try the opposite mode of regimen, for fear of failure. He alone, then, who knows the constitution of the sick, and the nature of the disease, and the powers of the remedy which is administered, and the time in which it ought to be used, will be able rationally to devise the remedy to be applied, and confirm his expectation of it by experience."

Galen gives other remarks, not devoid of interest, on the same subject, but these want of room obliges me to pass by. I may mention, however, that after giving, in the form of extract, the passage on wine (§ 12), he makes the remark, that if the question be put whether wine should be given to persons in fever, the proper answer to it would be, that it is to be given in some certain cases, and in others not. (See tom. v., p. 773, ed. Kühn.) Thus far Galen.

Before quitting this subject, I would beg leave to make a few remarks on some points of medical practice which are here treated of, and which appear to me to be either overlooked, or not satisfactorily determined at the present day; and also upon some modern innovations on the practice of the ancients. As far as I have observed, it is quite a common practice now to administer food, such as farinaceous gruels, or animal broths, without much reserve, after evacuation of the system either by purging or bleeding. Now it will be seen that Hippocrates forbids food to be administered at such a season, as the body, being weakened by the depletion, is unable to digest it properly, and consequently what is given as a support to the frame proves a load to it. To the reason here assigned for this practice, might be added that the vascular system, having been emptied, greedily absorbs the food before it is properly digested. I am not sure that this physiological principle is stated in any of the works of Hippocrates, but it is frequently to be met with in the works of Galen, and in those of the toxicologists, from Nicander to Actuarius. See PAULUS ÆGINETA, Book V., 2, Syd. Soc. edit.

I would beg leave to call the attention of my professional readers to the guarded and judicious manner in which pleurisy is treated by our author, beginning with hot fomentations to the side, and gradually advancing to the more active means, namely, purging and venesection. It will be remarked that Hippocrates holds depletion to be the only legitimate mode of removing the pain of the side, and that his commentator, in illustration of his meaning, pointedly condemns the use of narcotics in this case. Now this is a most important consideration, as bearing on a mode of practice which has obtained much favor of late years; I allude, of course, to the treatment by a combination of mercurials and opium. The experience of some thirty years would seem to decide in its favor, but how often have certain methods of treatment in other cases obtained the sanction of professional favor for a much longer period, and yet in the end been abandoned as positively prejudicial? In my younger days I knew old practitioners, of the highest reputation, who administered these medicines in scrofula,—in cancer,—in every case! One cannot think of the changes in professional opinions on the mercurial treatment of syphilis, since the days of John Hunter, without the most painful feeling of distrust in all modes of treatment where one cannot recognize some reasonable bond of connection between the remedy applied and the effects produced, or where long experience and analogy are in favor of them, and where the judgment runs no risk of being imposed upon by fallacious appearances and collateral circumstances. In a word, who does not feel disposed, in the practice of medicine, constantly to recur to the great truth proclaimed by our author in his first Aphorism? "Experience is fallacious, and judgment is difficult."

I am almost afraid further to put the question to the profession of the present day, whether or not the administration of antimonials in pleuro-pneumonia be an improvement on the ancient practice, or the reverse? Shall we say, then, that experience has decided that this substance (antimony), which, when applied to the cuticle, or to its prolongation, the epithelium of the stomach and bowels, occasions pain, heat, and vascular congestion, produces the very opposite effects on the lungs, when absorbed into the blood and conveyed to them? I dare not venture to answer these questions myself, but suggest them as deserving to be reconsidered, with serious impartiality, by the profession. I trust, however, it will not be supposed that I incline to stand up for ancient modes of practice, because they are old, or to condemn modern methods because they are new; I merely state the reflections which the comparison of ancient and modern usages, on this important subject, has suggested to me.

Our author, it will be seen, attaches much importance to the administration of the ptisan, or decoction of barley, in pleuro-pneumonia. Our modern Hippocrates, I mean, of course, Sydenham, was equally partial to this practice,^[530] which is still very much followed on the continent.

It will be remarked, that Hippocrates says nothing of counter-irritants to the skin, in the treatment of pleurisy, all his external applications being of the soothing kind. The stimulant treatment, however, is not altogether modern, having been

recommended in certain cases by the Arabians. (See PAULUS ÆEGINETA, Vol. I., p. 501.) Celsus also approves of sinapisms to the side. (iv., 6.)

The use of the bath and of the *douche*, or affusion of hot water in febrile diseases, is an important question, which well deserves to be reconsidered by the profession. (See the annotations on § 18.)

The reader will no doubt have been struck with the remark of Galen, in the extract given above, that our author's plan in the present work is deficient in method, because he himself was the discoverer of the subject-matters to which it relates. Galen then seems to have been of opinion that it was too much to expect from any individual, that he should produce a work which would be remarkable at the same time for the originality of its materials, and for the methodical arrangement of them. In confirmation of Galen's judgment in this case, I would direct attention to the difference that there is between this treatise and the "Prognostics;" for all must admit that the matters of which the latter work is composed are admirably methodized, and we have shown above that they were derived in a great measure from the previous labors of the Asclepiadæ.

ON REGIMEN IN ACUTE DISEASES.

Those who composed what are called "The Cnidian Sentences"^[531] have described accurately what symptoms the sick experience in every disease, and how certain of them terminate; and in so far a man, even who is not a physician, might describe them correctly, provided he put the proper inquiries to the sick themselves what their complaints are. But those symptoms which the physician ought to know beforehand without being informed of them by the patient, are, for the most part, omitted, some in one case and some in others, and certain symptoms of vital importance for a conjectural judgment.^[532] But when, in addition to the diagnosis, they describe how each complaint should be treated, in these cases I entertain a still greater difference of opinion with them respecting the rules they have laid down; and not only do I not agree with them on this account, but also because the remedies they use are few in number; for, with the exception of acute diseases, the only medicines which they give are drastic purgatives, with whey, and milk at certain times. If, indeed, these remedies had been good and suitable to the complaints in which they are recommended, they would have been still more deserving of recommendation, if, while few in number, they were sufficient; but this is by no means the case. Those, indeed, who have remodeled these "Sentences" have treated of the remedies applicable in each complaint more in a medical fashion. But neither have the ancients written anything worth mentioning respecting regimen, although this be a great omission. Some of them, indeed, were not ignorant of the many varieties of each complaint, and their manifold divisions, but when they wish to tell clearly the numbers (species?) of each disease they do not write correctly;^[533] for their species would be almost innumerable if every symptom experienced by the patients were held to constitute a disease, and receive a different name.^[534]

2. For my part, I approve of paying attention to everything relating to the art, and that those things which can be done well or properly should all be done properly; such as can be quickly done should be done quickly; such as can be neatly done should be done neatly; such operations as can be performed without pain should be done with the least possible pain; and that all other things of the like kind should be done better than they could be managed by the attendants. But I would more especially commend the physician who, in acute diseases, by which the bulk of mankind are cut off, conducts the treatment better than others. Acute diseases are those which the ancients named pleurisy, pneumonia, phrenitis, lethargy, causus, and the other diseases allied to these, including the continual fevers. For, unless when some general form of pestilential disease is epidemic, and diseases are sporadic and [not] of a similar character, there are more deaths from these diseases than from all the others taken together.^[535] The vulgar, indeed, do not recognize the difference between such physicians and their common attendants, and are rather disposed to commend and censure extraordinary remedies. This, then, is a great proof that the common people are most incompetent, of themselves, to form a judgment how such diseases should be treated: since persons who are not physicians pass for physicians owing most especially to these diseases, for it is an easy matter to learn the names of those things which are applicable to persons laboring under such complaints. For, if one names the juice of ptisan, and such and such a wine, and hydromel, the vulgar fancy that he prescribes exactly the same things as the physicians do, both the good and the bad, but in these matters there is a great difference between them.

3. But it appears to me that those things are more especially deserving of being consigned to writing which are undetermined by physicians, notwithstanding that they are of vital importance, and either do much good or much harm. By undetermined I mean such as these, wherefore certain physicians, during their whole lives, are constantly administering unstrained ptisans, and fancy they thus accomplish the cure properly, whereas others take great pains that the patient should not swallow a particle of the barley (thinking it would do much harm), but strain the juice through a cloth before giving it; others, again, will neither give thick ptisan nor the juice, some until the seventh day of the disease, and some until after the crisis.^[536] Physicians are not in the practice of mooted such questions; nor, perhaps, if mooted, would a solution of them be found; although the whole art is thereby exposed to much censure from the vulgar, who fancy that there really is no such science as medicine, since, in acute diseases, practitioners differ so much among themselves, that those things which one administers as thinking it the best that can be given, another holds to be bad; and, in this respect, they might say that the art of

medicine resembles augury, since augurs hold that the same bird (omen) if seen on the left hand is good, but if on the right bad: and in divination by the inspection of entrails you will find similar differences; but certain diviners hold the very opposite of these opinions.^[537] I say, then, that this question is a most excellent one, and allied to very many others, some of the most vital importance in the Art, for that it can contribute much to the recovery of the sick, and to the preservation of health in the case of those who are well; and that it promotes the strength of those who use gymnastic exercises, and is useful to whatever one may wish to apply it.

4. Ptisan, then, appears to me to be justly preferred before all the other preparations from grain in these diseases, and I commend those who made this choice,^[538] for the mucilage of it is smooth, consistent, pleasant, lubricant, moderately diluent, quenches thirst if this be required, and has no astringency; gives no trouble nor swells up in the bowels, for in the boiling it swells up as much as it naturally can. Those, then, who make use of ptisan in such diseases, should never for a day allow their vessels to be empty of it, if I may say so, but should use it and not intermit, unless it be necessary to stop for a time, in order to administer medicine or a clyster. And to those who are accustomed to take two meals in the day it is to be given twice, and to those accustomed to live upon a single meal it is to be given once at first, and then, if the case permit, it is to be increased and given twice to them, if they appear to stand in need of it. At first it will be proper not to give a large quantity nor very thick, but in proportion to the quantity of food which one has been accustomed to take, and so as that the veins may not be much emptied. And, with regard to the augmentation of the dose, if the disease be of a drier nature than one had supposed,^[539] one must not give more of it, but should give before the draught of ptisan, either hydromel or wine, in as great quantity as may be proper; and what is proper in each case will be afterward stated by us. But if the mouth and the passages from the lungs be in a proper state as to moisture, the quantity of the draught is to be increased, as a general rule, for an early and abundant state of moisture indicates an early crisis, but a late and deficient moisture indicates a slower crisis.^[540] And these things are as I have stated for the most part; but many other things are omitted which are important to the prognosis, as will be explained afterwards. And the more that the patient is troubled with purging, in so much greater quantity is it to be given until the crisis, and moreover until two days beyond the crisis, in such cases as it appears to take place on the fifth, seventh, or ninth day, so as to have respect both for the odd and even day; after this the draught is to be given early in the day, and the other food in place is to be given in the evening. These things are proper, for the most part, to be given to those who, from the first, have used ptisan containing its whole substance; for the pains in pleuritic affections immediately cease of their own accord whenever the patients begin to expectorate anything worth mentioning, and the purgings become much better, and empyema much more seldom takes place, than if the patients used a different regimen, and the crises are more simple, occur earlier, and the cases are less subject to relapses.

5. Ptisans are to be made of the very best barley, and are to be well boiled, more especially if you do not intend to use them strained. For, besides the other virtues of ptisan, its lubricant quality prevents the barley that is swallowed from proving injurious, for it does not stick nor remain in the region of the breast; for that which is well boiled is very lubricant, excellent for quenching thirst, of very easy digestion, and very weak, all which qualities are wanted. If, then, one do not pay proper attention to the mode of administering the ptisan, much harm may be done; for when the food is shut up in the bowels, unless one procure some evacuation speedily, before administering the draught, the pain, if present, will be exasperated; and, if not present, it will be immediately created, and the respiration will become more frequent, which does mischief, for it dries the lungs, fatigues the hypochondria, the hypogastrium, and diaphragm. And moreover if, while the pain of the side persists, and does not yield to warm fomentations, and the sputa are not brought up, but are viscid and unconcocted, unless one get the pain resolved, either by loosening the bowels; or opening a vein, whichever of these may be proper;—if to persons so circumstanced ptisan be administered, their speedy death will be the result.^[541] For these reasons, and for others of a similar kind still more, those who use unstrained ptisan die on the seventh day, or still earlier, some being seized with delirium, and others dying suffocated with orthopnoea and râles.^[542] Such persons the ancients thought *struck*, for this reason more especially, that when dead the affected side was livid, like that of a person who had been struck. The cause of this is that they die before the pain is resolved, being seized with difficulty of respiration, and by large and rapid breathing, as has been already explained, the spittle becoming thick, acid, and unconcocted, cannot be brought up, but, being retained in the bronchi of the lungs, produces râles; and, when it has come to this, death, for the most part, is inevitable; for the sputa being retained prevent the breath from being drawn in, and force it speedily out, and thus the two conspire together to aggravate the mischief; for the sputa being retained renders the respiration frequent, while the respiration being frequent thickens the sputa, and prevents them from being evacuated. These symptoms supervene, not only if ptisan be administered unseasonably, but still more if any other food or drink worse than ptisan be given.

6. For the most part, then, the results are the same, whether the patient have used the unstrained ptisan or have used the juice alone; or even only drink; and sometimes it is necessary to proceed quite differently. In general, one should do thus: if fever commences shortly after taking food, and before the bowels have been evacuated, whether with or without pain, the physician ought to withhold the draught until he thinks that the food has descended to the lower part of the belly; and if any pain be present, the patient should use oxymel, hot if it is winter, and cold if it is summer; and, if there be much thirst, he

should take hydromel and water.^[543] Then, if any pain be present, or any dangerous symptoms make their appearance, it will be proper to give the draught neither in large quantity nor thick, but after the seventh day, if the patient be strong. But if the earlier-taken food has not descended, in the case of a person who has recently swallowed food, and if he be strong and in the vigor of life, a clyster should be given, or if he be weaker, a suppository is to be administered, unless the bowels open properly of themselves. The time for administering the draught is to be particularly observed at the commencement and during the whole illness; when, then, the feet are cold, one should refrain from giving the ptisan, and more especially abstain from drink; but when the heat has descended to the feet, one may then give it; and one should look upon this season as of great consequence in all diseases, and not least in acute diseases, especially those of a febrile character, and those of a very dangerous nature. One may first use the juice, and then the ptisan, attending accurately to the rules formerly laid down.

7. When pain seizes the side, either at the commencement or at a later stage, it will not be improper to try to dissolve the pain by hot applications.^[544] Of hot applications the most powerful is hot water in a bottle, or bladder, or in a brazen vessel, or in an earthen one; but one must first apply something soft to the side, to prevent pain. A soft large sponge, squeezed out of hot water and applied, forms a good application; but it should be covered up above, for thus the heat will remain the longer, and at the same time the vapor will be prevented from being carried up to the patient's breath, unless when this is thought of use, for sometimes it is the case. And further, barley or tares may be infused and boiled in diluted vinegar, stronger than that it could be drunk, and may then be sewed into bladders and applied; and one may use bran in like manner. Salts or toasted millet in woolen bags are excellent for forming a dry fomentation, for the millet is light and soothing. A soft fomentation like this soothes pains, even such as shoot to the clavicle. Venesection, however, does not alleviate the pain unless when it extends to the clavicle. But if the pain be not dissolved by the fomentations, one ought not to foment for a length of time, for this dries the lungs and promotes suppuration; but if the pain point to the clavicle, or if there be a heaviness in the arm, or about the breast, or above the diaphragm, one should open the inner vein at the elbow, and not hesitate to abstract a large quantity, until it become much redder, or instead of being pure red, it turns livid,^[545] for both these states occur. But if the pain be below the diaphragm, and do not point to the clavicle, we must open the belly either with black hellebore^[546] or peplium,^[547] mixing the black hellebore with carrot or seseli,^[548] or cumin, or anise, or any other of the fragrant herbs; and with the peplium the juice of sulphur^[549] (asafœtida), for these substances, when mixed up together, are of a similar nature.^[550] The black hellebore acts more pleasantly and effectually than the peplium, while, on the other hand, the peplium expels wind much more effectually than the black hellebore, and both these stop the pain, and many other of the laxatives also stop it, but these two are the most efficacious that I am acquainted with. And the laxatives given in draughts are beneficial, when not very unpalatable owing to bitterness, or any other disagreeable taste, or from quantity, colour, or any apprehension. When the patient has drunk the medicine, one ought to give him to swallow but little less of the ptisan than what he had been accustomed to; but it is according to rule not to give any draughts while the medicine is under operation;^[551] but when the purging is stopped then he should take a smaller draught than what he had been accustomed to, and afterwards go on increasing it progressively, until the pain cease, provided nothing else contra-indicate. This is my rule, also, if one would use the juice of ptisan, (for I hold that it is better, on the whole, to begin with taking the decoction at once, rather than by first emptying the veins before doing so, or on the third, fourth, fifth, sixth, or seventh day, provided the disease has not previously come to a crisis in the course of this time), and similar preparations to those formerly described are to be made in those cases.

8. Such are the opinions which I entertain respecting the administering of the ptisan; and, as regards drinks, whichever of those about to be described may be administered, the same directions are generally applicable. And here I know that physicians are in the practice of doing the very reverse of what is proper, for they all wish, at the commencement of diseases, to starve their patients for two, three, or more days, and then to administer the ptisans and drinks; and perhaps it appears to them reasonable that, as a great change has taken place in the body, it should be counteracted by another great change. Now, indeed, to produce a change is no small matter, but the change must be effected well and cautiously, and after the change the administration of food must be conducted still more so. Those persons, then, would be most injured if the change is not properly managed, who used unstrained ptisans; they also would suffer who made use of the juice alone; and so also they would suffer who took merely drink, but these least of all.

9. One may derive information from the regimen of persons in good health what things are proper; for if it appear that there is a great difference whether the diet be so and so, in other respects, but more especially in the changes, how can it be otherwise in diseases, and more especially in the most acute? But it is well ascertained that even a faulty diet of food and drink steadily persevered in, is safer in the main as regards health than if one suddenly change it to another. Wherefore, in the case of persons who take two meals in the day, or of those who take a single meal, sudden changes induce suffering and weakness; and thus persons who have not been accustomed to dine, if they shall take dinner, immediately become weak, have heaviness over their whole body, and become feeble and languid, and if, in addition, they take supper, they will have acid eructations, and some will have diarrhoea whose bowels were previously dry, and not having been accustomed to be twice swelled out with food and to digest it twice a day, have been loaded beyond their wont. It is beneficial, in such cases, to counterbalance this change, for one should sleep after dinner, as if passing the night, and guard against cold in winter and heat in summer; or, if the

person cannot sleep, he may stroll about slowly, but without making stops, for a good while, take no supper, or, at all events, eat little, and only things that are not unwholesome, and still more avoid drink, and especially water. Such a person will suffer still more if he take three full meals in the day, and more still if he take more meals; and yet there are many persons who readily bear to take three full meals in the day, provided they are so accustomed. And, moreover, those who have been in the habit of eating twice a day, if they omit dinner, become feeble and powerless, averse to all work, and have heartburn; their bowels seem, as it were, to hang loose, their urine is hot and green, and the excrement is parched; in some the mouth is bitter, the eyes are hollow, the temples throb, and the extremities are cold, and the most of those who have thus missed their dinner cannot eat supper; or, if they do sup, they load their stomach, and pass a much worse night than if they had previously taken dinner. Since, then, an unwonted change of diet for half a day produces such effects upon persons in health, it appears not to be a good thing either to add or take from. If, then, he who was restricted to a single meal, contrary to usage, having his veins thus left empty during a whole day, when he supped according to custom felt heavy, it is probable that if, because he was uneasy and weak from the want of dinner, he took a larger supper than wont, he would be still more oppressed; or if, wanting food for a still greater interval, he suddenly took a meal after supper, he will feel still greater oppression. He, then, who, contrary to usage, has had his veins kept empty by want of food, will find it beneficial to counteract the bad effects during that day as follows: let him avoid cold, heat, and exertion, for he could bear all these ill; let him make his supper considerably less than usual, and not of dry food, but rather liquid; and let him take some drink, not of a watery character, nor in smaller quantity than is proportionate to the food, and on the next day he should take a small dinner, so that, by degrees, he may return to his former practice. Persons who are bilious in the stomach bear these changes worst, while those who are pituitous, upon the whole, bear the want of food best, so that they suffer the least from being restricted to one meal in the day, contrary to usage. This, then, is a sufficient proof that the greatest changes as to those things which regard our constitutions and habits are most especially concerned in the production of diseases, for it is impossible to produce unseasonably a great emptying of the vessels by abstinence, or to administer food while diseases are at their acme, or when inflammation prevails; nor, on the whole, to make a great change either one way or another with impunity.^[552]

10. One might mention many things akin to these respecting the stomach and bowels, to show how people readily bear such food as they are accustomed to, even if it is not naturally good, and drink in like manner, and how they bear unpleasantly such food as they are not accustomed to, even although not bad, and so in like manner with drink; and as to the effects of eating much flesh, contrary to usage, or garlic, or asafetida, or the stem of the plant which produces it, or things of a similar kind possessed of strong properties, one would be less surprised if such things produce pains in the bowels, but rather when one learned what trouble, swelling, flatulence, and tormina the cake (maza) will raise in the belly when eaten by a person not accustomed to it; and how much weight and distention of the bowels bread will create to a person accustomed to live upon the maza; and what thirst and sudden fullness will be occasioned by eating hot bread, owing to its desiccant and indigestible properties; and what different effects are produced by fine and coarse bread when eaten contrary to usage, or by the cake when unusually dry, moist, or viscid; and what different effects polenta produces upon those who are accustomed and those who are unaccustomed to the use of it; or drinking of wine or drinking of water, when either custom is suddenly exchanged for the other; or when, contrary to usage, diluted wine or undiluted has been suddenly drunk, for the one will create water-brash in the upper part of the intestinal canal and flatulence in the lower, while the other will give rise to throbbing of the arteries, heaviness of the head, and thirst; and white and dark-colored wine, although both strong wines, if exchanged contrary to usage, will produce very different effects upon the body, so that one need the less wonder that a sweet and strong wine, if suddenly exchanged, should have by no means the same effect.

11. Let us here briefly advert to what may be said on the opposite side; namely, that a change of diet has occurred in these cases, without any change in their body, either as to strength, so as to require an increase of food, or as to weakness, so as to require a diminution. But the strength of the patient is to be taken into consideration, and the manner of the disease, and of the constitution of the man, and the habitual regimen of the patient, not only as regards food but also drink. Yet one must much less resort to augmentation, since it is often beneficial to have recourse to abstraction, when the patient can bear it, until the disease having reached its acme and has become concocted. But in what cases this must be done will be afterwards described. One might write many other things akin to those which have been now said, but there is a better proof, for it is not akin to the matter on which my discourse has principally turned, but the subject-matter itself is a most seasonable proof. For some at the commencement of acute diseases have taken food on the same day, some on the next day; some have swallowed whatever has come in their way, and some have taken *cyceon*.^[553] Now all these things are worse than if one had observed a different regimen; and yet these mistakes, committed at that time, do much less injury than if one were to abstain entirely from food for the first two or three days, and on the fourth or fifth day were to take such food; and it would be still worse, if one were to observe total abstinence for all these days, and on the following days were to take such a diet, before the disease is concocted;^[554] for in this way death would be the consequence to most people, unless the disease were of a very mild nature. But the mistakes committed at first were not so irremediable as these, but could be much more easily repaired. This, therefore, I think a strong proof that such or such a draught need not be prescribed on the first days to those who will use the same draughts afterwards. At the bottom, therefore, they do not know, neither those using unstrained ptisans, that they are hurt by them, when

they begin to swallow them, if they abstain entirely from food for two, three, or more days; nor do those using the juice know that they are injured in swallowing them, when they do not commence with the draught seasonably. But this they guard against, and know that it does much mischief, if, before the disease be concocted, the patient swallow unstrained ptisan, when accustomed to use strained. All these things are strong proofs that physicians do not conduct the regimen of patients properly, but that in those diseases in which total abstinence from food should not be enforced on patients that will be put on the use of ptisans, they do enforce total abstinence; that in those cases in which there should be no change made from total abstinence to ptisans, they do make the change; and that, for the most part, they change from abstinence to ptisans, exactly at the time when it is often beneficial to proceed from ptisans almost to total abstinence, if the disease happen to be in the state of exacerbation. [555] And sometimes crude matters are attracted from the head, and bilious from the region near the chest, and the patients are attacked with insomnolency, so that the disease is not concocted; they become sorrowful, peevish, and delirious; there are flashes of light in their eyes, and noises in their ears; their extremities are cold, their urine unconcocted; the sputa thin, saltish, tinged with an intense colour and smell; sweats about the neck, and anxiety; respiration, interrupted in the expulsion of the air, [556] frequent and very large; expression of the eyelids dreadful; dangerous *deliquia*; tossing of the bedclothes from the breast; the hands trembling, and sometimes the lower lip agitated. These symptoms, appearing at the commencement, are indicative of strong delirium, and patients so affected generally die, or if they escape, it is with a deposit, hemorrhage from the nose, or the expectoration of thick matter, and not otherwise. Neither do I perceive that physicians are skilled in such things as these; how they ought to know such diseases as are connected with debility, and which are further weakened by abstinence from food, and those aggravated by some other irritation; those by pain, and from the acute nature of the disease, and what affections and various forms thereof our constitution and habit engender, although the knowledge or ignorance of such things brings safety or death to the patient. For it is a great mischief if to a patient debilitated by pain, and the acute nature of the disease, one administer drink, or more ptisan, or food, supposing that the debility proceeds from inanition. It is also disgraceful not to recognize a patient whose debility is connected with inanition, and to pinch him in his diet; this mistake, indeed, is attended with some danger, but much less than the other, and yet it is likely to expose one to much greater derision, for if another physician, or a private person, coming in and knowing what has happened, should give to eat or drink those things which the other had forbidden, the benefit thus done to the patient would be manifest. Such mistakes of practitioners are particularly ridiculed by mankind, for the physician or non-professional man thus coming in, seems as it were to resuscitate the dead. On this subject I will describe elsewhere the symptoms by which each of them may be recognized.

12. And the following observations are similar to those now made respecting the bowels. If the whole body rest long, contrary to usage, it does not immediately recover its strength; but if, after a protracted repose, it proceed to labor, it will clearly expose its weakness. So it is with every one part of the body, for the feet will make a similar display, and any other of the joints, if, being unaccustomed to labor, they be suddenly brought into action, after a time. The teeth and the eyes will suffer in like manner, and also every other part whatever. A couch, also, that is either softer or harder than one has been accustomed to will create uneasiness, and sleeping in the open air, contrary to usage, hardens the body. But it is sufficient merely to state examples of all these cases. If a person having received a wound in the leg, neither very serious nor very trifling, and he being neither in a condition very favorable to its healing nor the contrary, at first betakes himself to bed, in order to promote the cure, and never raises his leg, it will thus be much less disposed to inflammation, and be much sooner well, than it would have been if he had strolled about during the process of healing; but if upon the fifth or sixth day, or even earlier, he should get up and attempt to walk, he will suffer much more than if he had walked about from the commencement of the cure, and if he should suddenly make many laborious exertions, he will suffer much more than if, when the treatment was conducted otherwise, he had made the same exertions on the same days. In fine, all these things concur in proving that all great changes, either one way or another, are hurtful. Wherefore much mischief takes place in the bowels, if from a state of great inanition more food than is moderate be administered (and also in the rest of the body, if from a state of great rest it be hastily brought to greater exertion, it will be much more injured), or if from the use of much food it be changed to complete abstinence, and therefore the body in such cases requires protracted repose, and if, from a state of laborious exertion, the body suddenly falls into a state of ease and indolence, in these cases also the bowels would require continued repose from abundance of food, for otherwise it will induce pain and heaviness in the whole body.

13. The greater part of my discourse has related to changes, this way or that. For all purposes it is profitable to know these things, and more especially respecting the subject under consideration,—that in acute diseases, in which a change is made to ptisans from a state of inanition, it should be made as I direct; and then that ptisans should not be used until the disease be concocted, or some other symptom, whether of evacuation or of irritation, appear in the intestines, or in the hypochondria, such as will be described. Obstinate isomnolency impairs the digestion of the food and drink, and in other respects changes and relaxes the body, and occasions a heated state, and heaviness of the head. [557]

14. One must determine by such marks as these, when sweet, strong, and dark wine, hydromel, water and oxymel, should be given in acute diseases. [558] Wherefore the sweet affects the head less than the strong, attacks the brain less, evacuates the bowels more than the other, but induces swelling of the spleen and liver; it does not agree with bilious persons, for it causes

them to thirst; it creates flatulence in the upper part of the intestinal canal, but does not disagree with the lower part, as far as regards flatulence; and yet flatulence engendered by sweet wine is not of a transient nature, but rests for a long time in the hypochondria. And therefore it in general is less diuretic than wine which is strong and thin; but sweet wine is more expectorant than the other. But when it creates thirst, it is less expectorant in such cases than the other wine, but if it do not create thirst, it promotes expectoration better than the other. The good and bad effects of a white, strong wine, have been already frequently and fully stated in the disquisition on sweet wine; it is determined to the bladder more than the other, is diuretic and laxative, and should be very useful in such complaints; for if in other respects it be less suitable than the other, the clearing out of the bladder effected by it is beneficial to the patient, if properly administered. There are excellent examples of the beneficial and injurious effects of wine, all which were left undetermined by my predecessors. In these diseases you may use a yellow wine, and a dark austere wine for the following purposes: if there be no heaviness of the head, nor delirium, nor stoppage of the expectoration, nor retention of the urine, and if the alvine discharges be more loose and like scrapings than usual, in such cases a change from a white wine to such as I have mentioned, might be very proper. It deserves further to be known, that it will prove less injurious to all the parts above, and to the bladder, if it be of a more watery nature, but that the stronger it is, it will be the more beneficial to the bowels.

15. Hydromel, when drunk in any stage of acute disease, is less suitable to persons of a bilious temperament, and to those who have enlarged viscera, than to those of a different character; it increases thirst less than sweet wine; it softens the lungs, is moderately expectorant, and alleviates a cough; for it has some detergent quality in it, whence it lubricates the sputum.^[559] Hydromel is also moderately diuretic, unless prevented by the state of any of the viscera. And it also occasions bilious discharges downwards, sometimes of a proper character, and sometimes more intense and frothy than is suitable; but such rather occurs in persons who are bilious, and have enlarged viscera. Hydromel rather produces expectoration, and softening of the lungs, when given diluted with water.^[560] But unmixed hydromel, rather than the diluted, produces frothy evacuations, such as are unseasonably and intensely bilious, and too hot; but such an evacuation occasions other great mischiefs, for it neither extinguishes the heat in the hypochondria, but rouses it, induces inquietude, and jactitation of the limbs, and ulcerates the intestines and anus. The remedies for all these will be described afterwards. By using hydromel without ptisans, instead of any other drink, you will generally succeed in the treatment of such diseases, and fail in few cases; but in what instances it is to be given, and in what it is not to be given, and wherefore it is not to be given,—all this has been explained already, for the most part. Hydromel is generally condemned, as if it weakened the powers of those who drink it, and on that account it is supposed to accelerate death; and this opinion arose from persons who starve themselves to death, some of whom use hydromel alone for drink, as fancying that it really has this effect. But this is by no means always the case. For hydromel, if drunk alone, is much stronger than water, if it do not disorder the bowels; but in some respects it is stronger, and in some weaker, than wine that is thin, weak, and devoid of *bouquet*. There is a great difference between unmixed wine and unmixed honey, as to their nutritive powers, for if a man will drink double the quantity of pure wine, to a certain quantity of honey which is swallowed, he will find himself much stronger from the honey, provided it do not disagree with his bowels, and that his alvine evacuations from it will be much more copious. But if he shall use ptisan for a draught, and drink afterward hydromel, he will feel full, flatulent, and uncomfortable in the viscera of the hypochondrium; but if the hydromel be taken before the draught, it will not have the same injurious effects as if taken after it, but will be rather beneficial. And boiled hydromel has a much more elegant appearance than the unboiled, being clear, thin, white, and transparent, but I am unable to mention any good quality which it possesses that the other wants. For it is not sweeter than the unboiled, provided the honey be fine, and it is weaker, and occasions less copious evacuations of the bowels, neither of which effects is required from the hydromel. But one should by all means use it boiled, provided the honey be bad, impure, black, and not fragrant, for the boiling will remove the most of its bad qualities and appearances.

16. You will find the drink, called oxymel, often very useful in these complaints, for it promotes expectoration and freedom of breathing. The following are the proper occasions for administering it. When strongly acid it has no mean operation in rendering the expectoration more easy, for by bringing up the sputa, which occasion troublesome hawking, and rendering them more slippery, and, as it were, clearing the windpipe with a feather, it relieves the lungs and proves emollient to them; and when it succeeds in producing these effects it must do much good. But there are cases in which hydromel, strongly acid, does not promote expectoration, but renders it more viscid and thus does harm, and it is most apt to produce these bad effects in cases which are otherwise of a fatal character, when the patient is unable to cough or bring up the sputa. On this account, then, one ought to consider beforehand the strength of the patient, and if there be any hope, then one may give it, but if given at all in such cases it should be quite tepid, and in by no means large doses. But if slightly acrid it moistens the mouth and throat, promotes expectoration, and quenches thirst; agrees with the viscera seated in the hypochondrium, and obviates the bad effects of the honey; for the bilious quality of the honey is thereby corrected. It also promotes flatulent discharges from the bowels, and is diuretic, but it occasions watery discharges and those resembling scrapings, from the lower part of the intestine, which is sometimes a bad thing in acute diseases, more especially when the flatulence cannot be passed, but rolls backwards; and otherwise it diminishes the strength and makes the extremities cold, this is the only bad effect worth mentioning which I have known to arise from the oxymel. It may suit well to drink a little of this at night before the draught of ptisan, and when a

considerable interval of time has passed after the draught there will be nothing to prevent its being taken. But to those who are restricted entirely to drinks without draughts of ptisan, it will therefore not be proper at all times to give it, more especially from the fretting and irritation of the intestine which it occasions, (and these bad effects it will be the more apt to produce provided there be no fæces in the intestines and the patient is laboring under inanition,) and then it will weaken the powers of the hydromel. But if it appears advantageous to use a great deal of this drink during the whole course of the disease, one should add to it merely as much vinegar as can just be perceived by the taste, for thus what is prejudicial in it will do the least possible harm, and what is beneficial will do the more good. In a word, the acidity of vinegar agrees rather with those who are troubled with bitter bile, than with those patients whose bile is black; for the bitter principle is dissolved in it and turned to phlegm, by being suspended in it; whereas black bile is fermented, swells up, and is multiplied thereby: for vinegar is a melanagogue. Vinegar is more prejudicial to women than to men, for it creates pains in the uterus.

17. I have nothing further to add as to the effects of water when used as a drink in acute diseases; for it neither soothes the cough in pneumonia, nor promotes expectoration, but does less than the others in this respect, if used alone through the whole complaint. But if taken intermediate between oxymel and hydromel, in small quantity, it promotes expectoration from the change which it occasions in the qualities of these drinks, for it produces, as it were, a certain overflow. Otherwise it does not quench the thirst, for it creates bile in a bilious temperament, and is injurious to the hypochondrium; and it does the most harm, engenders most bile, and does the least good when the bowels are empty; and it increases the swelling of the spleen and liver when they are in an inflamed state; it produces a gurgling noise in the intestines and swims on the stomach; for it passes slowly downwards, as being of a coldish and indigestible nature, and neither proves laxative nor diuretic; and in this respect, too, it proves prejudicial, that it does not naturally form fæces in the intestines: and, if it be drunk while the feet are cold, its injurious effects will be greatly aggravated, in all those parts to which it may be determined. When you suspect in these diseases either strong heaviness of the head, or mental alienation, you must abstain entirely from wine, and in this case use water, or give weak, straw colored wine, entirely devoid of *bouquet*, after which a little water is to be given in addition; for thus the strength of the wine will less affect the head and the understanding: but in which cases water is mostly to be given for drink, when in large quantity, when in moderate, when cold, and when hot; all these things have either been discussed already or will be treated of at the proper time. In like manner, with respect to all the others, such as barley-water, the drinks made from green shoots, those from raisins, and the skins of grapes and wheat, and bastard saffron, and myrtles, pomegranates, and the others, when the proper time for using them is come, they will be treated of along with the disease in question, in like manner as the other compound medicines.^[561]

18. The bath is useful in many diseases, in some of them when used steadily, and in others when not so. Sometimes it must be less used than it would be otherwise, from the want of accommodation; for in few families are all the conveniences prepared, and persons who can manage them as they ought to be. And if the patient be not bathed properly, he may be thereby hurt in no inconsiderable degree, for there is required a place to cover him that is free of smoke, abundance of water, materials for frequent baths, but not very large, unless this should be required. It is better that no friction should be applied, but if so, a hot soap (*smegma*)^[562] must be used in greater abundance than is common, and an affusion of a considerable quantity of water is to be made at the same time and afterwards repeated. There must also be a short passage to the basin, and it should be of easy ingress and egress. But the person who takes the bath should be orderly and reserved in his manner, should do nothing for himself, but others should pour the water upon him and rub him, and plenty of waters, of various temperatures, should be in readiness for the *douche*, and the affusions quickly made;^[563] and sponges should be used instead of the comb (*strigil*), and the body should be anointed when not quite dry. But the head should be rubbed by the sponge until it is quite dry; the extremities should be protected from cold, as also the head and the rest of the body; and a man should not be washed immediately after he has taken a draught of ptisan or a drink; neither should he take ptisan as a drink immediately after the bath. Much will depend upon whether the patient, when in good health, was very fond of the bath, and in the custom of taking it: for such persons, especially, feel the want of it, and are benefited if they are bathed, and injured if they are not. In general it suits better with cases of pneumonia than in ardent fevers; for the bath soothes the pain in the side, chest, and back; concocts the sputa, promotes expectoration, improves the respiration, and allays lassitude; for it soothes the joints and outer skin, and is diuretic, removes heaviness of the head, and moistens the nose. Such are the benefits to be derived from the bath, if all the proper requisites be present; but if one or more of these be wanting, the bath, instead of doing good, may rather prove injurious; for every one of them may do harm if not prepared by the attendants in the proper manner. It is by no means a suitable thing in these diseases to persons whose bowels are too loose, or when they are unusually confined, and there has been no previous evacuation; neither must we bathe those who are debilitated, nor such as have nausea or vomiting, or bilious eructations; nor such as have hemorrhage from the nose, unless it be less than required at that stage of the disease, (with those stages you are acquainted:) but if the discharge be less than proper, one should use the bath, whether in order to benefit the whole body or the head alone. If then the proper requisites be at hand, and the patient be well disposed to the bath, it may be administered once every day, or if the patient be fond of the bath there will be no harm, though he should take it twice in the day. The use of the bath is much more appropriate to those who take unstrained ptisan, than to those who take only the juice of it, although even in their case it may be proper; but least of all does it suit with those who use only plain drink, although, in their case too it may be

suitable; but one must form a judgment from the rules laid down before, in which of these modes of regimen the bath will be beneficial, and in which not. Such as want some of the requisites for a proper bath, but have those symptoms which would be benefited by it, should be bathed; whereas those who want none of the proper requisites, but have certain symptoms which contra-indicate the bath, are not to be bathed.

APPENDIX TO THE WORK
ON
REGIMEN IN ACUTE DISEASES.

THE ARGUMENT.

No one can read this piece attentively without coming to the conclusion that it is not a natural continuation of the subject discussed in the preceding work, but that it is made up, in a considerable measure, of materials extracted from it. Expositions of subjects which are there given methodically are here presented in a disjointed form; and rules of practice there laid down with precision are here often delivered in a vague and indefinite shape. Still, however, it must be admitted, that the reverse is sometimes the case, and that what is presented imperfect in the former part of the work is here sometimes reproduced very much improved. It has been therefore a matter of much dispute among the critics whether this portion be the composition of Hippocrates, or whether it be altogether the work of a different hand. The most probable conjecture respecting it seems to be, that as Hippocrates in the preceding part several times announces his intention of giving a continuation of the subject, some one of his immediate disciples undertook the work which he had thus promised, and composed this treatise from fragments left by the author himself, and from materials collected from his other works. As stated by Galen in his Commentary, and as we have explained in our remarks on the "Aphorisms," in the second section of the Preliminary Discourse, it was a common practice, in ancient times, to add appendices to popular works. I can have no hesitation, then, in following the example of M. Littré, who recognizes it as an appendix to the preceding work. But I must say that I rather incline, with Galen, to think that there are many things in it which cannot have come from Hippocrates, than to hold with M. Littré that it is nearly or altogether his composition. But however that may be, it indisputably contains much interesting matter, for which we have every reason to believe that we are indebted to Hippocrates, either directly or at second hand. I shall now give a brief abstract of its contents.

He commences with some general observations on the nature and treatment of *causus*, the endemial fever of Greece. What is said on this head is much to the purpose, but incomplete. Then there is given a general rule for bleeding in diseases which certainly is well deserving of attention at the present day, when professional opinions on this point of medical practice are very much unsettled. Nowadays we have abandoned all general rules of practice, and profess to be guided solely by experience; but how variable and uncertain are its results in the present case! I myself—albeit but verging towards the decline of life—can well remember the time when a physician would have run the risk of being indicted for culpable homicide if he had ventured to bleed a patient in common fever; about twenty-five years ago venesection in fever, and in almost every disease, was the established order of the day; and now what shall I state as the general practice that has been sanctioned by the experience of the present generation? I can scarcely say, so variable has the practice in fever and in many other diseases become of late years. One thing is remarkable in the present work with regard to venesection in pneumonia and pleurisy, namely, that it is directed to be carried the length of inducing *deliquium animi*, contrary to the practice laid down in the preceding work, and to the rule which was followed by all the other ancient authorities. Another of the rules regarding bleeding here delivered is also deserving of attention, namely, that in inflammatory diseases it is improper to purge before bleeding, but that venesection should precede all other means of cure.

The section in which *cynanche* is treated of appears to me to be highly interesting and important. I think it may be a question whether the prognostic spirit of Hippocrates and his followers had not in a great measure anticipated all the results of modern diagnosis.

After this there follows some additional account of *causus*, which, although out of place, contains observations of considerable interest.

To the treatment of pleurisy and pneumonia we have already alluded, but the subject is so interesting that we cannot dismiss it with so brief a notice. In the ancient method of treating fevers and febrile affections three main objects would appear to have been kept in view: 1st, by depletions, to remove the morbid fluids from the general system, or to draw them off from a particular spot in which they had fixed; 2d, by diluents, to supply the waste of fluids occasioned by the preternatural heat of the body; and, 3d, to support the strength by a suitable supply of such nutriment as the system is then capable of receiving.

Now with regard to venesection, it will be seen in this and the preceding work that the practice is regulated by certain well-marked indications, namely, the seat of the pain, the condition of the patient, and the characters of the sputa. The purging is regulated by the state of matters below the chest, it being held as a general rule that clysters should be administered regularly every day during the first days of the fever. After purging comes the cooling drinks, such as oxymel. The administration of farinaceous food in a liquid state, that is to say, of unstrained ptisan, is to be regulated by the state of the sputa and urinary sediment, namely, when the sputa have put on a purulent appearance, and the sediment has become copious and reddish. Now this certainly seems to be a very intelligible and judicious rule for the administration of nutritious articles in febrile diseases. I need scarcely remark that at the present time there is scarcely a rule of practice in medicine which is worse defined than this respecting the administration of wine and other alimentary substances in febrile diseases. In proof of what is now stated, I

would beg leave to refer the reader to what will be admitted to be one of the best authorities in modern literature on fever, I mean to Dr. Tweedie's elaborate article on this subject, in the "Cyclopædia of Medicine." It will be seen, at vol. ii., p. 208, that the rules for the administration of wine and other articles of food are by no means well defined. A cool skin and a soft pulse, when combined with debility, are the indications upon which most stress is laid; but the pulse, as long ago it was pronounced by Celsus to be, is "res fallacissima," and of this the excellent author seems to have been sensible; for the injunctions which he gives to regulate the administration of the wine and other articles, by the effects they produce, sufficiently show that he was sensible how deficient in precision our knowledge of the subject is at present. At the same time he makes it appear that he was well aware of one important fact in the treatment of febrile diseases, which, although distinctly recognized by Hippocrates, is still frequently overlooked by ordinary practitioners, namely, that in convalescence the stomach partakes of the general debility, and is unable to digest food in any great quantity at that time.^[564] M. Littré further calls attention to another rule for the administration of wine, lately laid down by Dr. Stokes, of Dublin, which is certainly a most important one, provided it is confirmed by time and experience. It is founded on auscultation, and is to this effect; that when the impulse of the heart is abnormally weak, and when there is a diminution of the proportion between the two *bruits*, or when there is a preponderance in the sound of the second *bruit*, wine may be freely administered. Now, as I have said, this rule, if sanctioned by ample experience, is undoubtedly a most excellent one; but I may be allowed to remark, that my own observations on the heart in fever have led me to the conclusion that, as I have stated respecting the pulse, its sounds are very fallacious; and I must say that the rule of Hippocrates appears more likely to prove a certain guide in this instance. For is it not a natural view of the subject, that wine and other articles of food should be withheld while the emunctories are not in a condition to cast off the recrementitious superfluities of the system; but that when the secretions are properly established, alimentary substances may be safely administered?^[565]

There is another point connected with the regimen in acute diseases on which I have a remark or two to make—it is the administration of animal matters in a fluid state, such as beef-tea, or soups from fowls. These we see frequently administered in febrile cases by practitioners of the present day, but by the ancient authorities they would appear to have been entirely rejected. Which party is the safer guide in this case? For my own part, I have long thought that animal matters, when introduced into the system while in a febrile state, have a tendency to become putrid, and thereby to occasion an increase of the heat and general disorder.

After some defective observations on dysentery, our author treats of tetanus; but here Galen objects to the characters which he gives of the urine, and to his practice as regards the administration of wine. His views, however, are not very different from those which now prevail.

Having made some general remarks on the administration of hellebore, to which he was very partial, he proceeds to point out the bad effects resulting from any change in regimen. His views here are very similar to the observations contained in the preceding portion of the work, and in the treatise "On Ancient Medicine."

The account of dry cholera is confused and vague. By it he would seem to mean flatulent colic, or *dry belly-ache*. See Opera, ed. Littré, tom. ii., p. 388.

The paragraph on dropsy is interesting, although the views taken of the subject are incomplete. Tympanitis is recognized as a variety of dropsy. Then follow some detached observations on persons whose bowels are heated, and on the regulation of the diet, with some remarks on the different states which counter-indicate purging. At § 23 there are some practical observations on various conditions of the constitution, which it would no doubt be proper for the physician to make himself acquainted with. The contents of all the remaining paragraphs would seem to have nothing to do with the subject of this treatise.

From what is now stated the reader will readily perceive that this treatise abounds in interesting matters, which, even at the present day, may prove suggestive of important views in the theory and practice of medicine. And although the style, in the judgment of Galen, be very different from Hippocrates, and the mode of thought deficient in that precision for which he is so remarkable, the treatise is unquestionably a work of great ability, and contains what we have reason to regard as the results of his experience and meditations on many important subjects. I should have thought it quite unwarrantable, therefore, to have rejected this piece from a volume which professes to give all the genuine remains of our great author. And moreover, at the risk, perhaps, of being set down as an antique devotee, I do not hesitate to declare that in my opinion this and the preceding portion, taken together, contain more original information on the important subject to which they relate than is to be found in any medical work which has been written from the days of our author down to the present time.

I shall conclude the present Argument by giving from Cælius Aurelianus the criticisms of Soranus on the opinions of our author, as delivered in these two treatises. It is to be borne in mind that Soranus was the chief of the ancient sect of physicians called *Methodici*, which was very inimically disposed towards all the others, and more especially to Hippocrates. Though most of the strictures are evidently overstrained, it cannot fail to be interesting to the reader to have an opportunity of considering them, such as they are.

After giving an elaborate analysis of our author's views, Cælius Aurelianus proceeds as follows: "His Soranus respondens ait. In calefactionibus acres esse sales, ac necessario tumorem provocare, febremque accendere, poscam etiam constringere et stricturam passionis augere. Item milium frixum graveolens et nidorosum, atque capiti grave, maximè acutè fabricitantium esse perspicimus. Spongiis etiam erat melius quenquam in dimissione patientes partes vaporare, atque oleo calido perfundere. Est præterea improprium, ac sine ratione, tunc uti phlebotomo quoties ad superiora dolor tetenderit; prohibere autem quoties ad inferiora descenderit. Oportet ergo sub hoc argumento neque difficultate tumorum partibus inferioribus impeditos phlebotomare: neque etiam podagricos si quidem inferiora tumere videantur, sed necessariò quoties dolor ad superiora tetenderit, phlebotomiam adhibendam videmus. Siquidem sæpe pejorante ventris fluore, hoc adjutorii genus prohibetur. Neque etiam (ut ait) oportet interiorem venam dividi. Siquidem et exteriori et media divisa corpora releventur. Quippe quum e contrario interiorem prohibeant, propter magnitudinem, ne tumor augeatur. Item sanguinis mutatio iners est detractiois moderationi, sicuti de adjutoriis scribentes demonstrabimus. Sese denique idem Hippocrates impugnat in consequentibus, dicens usque ad animi defectum faciendam detractioem, quod magis vehementer est nocens: siquidem est periculosa defectio, et neque si sit temporaliter defectionis causa, sensu carens ægrotans, dolore relevatus, videbitur (quum resumptus fuerit) rursum non dolere, quum magis atque magis ejusdem passionis debilia corpora vehementius officiant. Item purgativa medicamina (quæ Græci καθαρτικά vocant) acrimoniæ causa, stomachum tumentem, atque hypezocota membranum acuunt in tumorem; et in periculum ventris effusionem provocantia, magnificam passionis ingerunt vehementiam. Nutrire etiam cibo post medicamen non oportebat. Pugnat enim purgationi faciundæ illatum cibi nutrimentum. Quippe quum medicamine corruptum, officii sui careat viribus. Mitto etiam quod ex initio acescere facile ptisanæ succus perspiciatur, confectus quippe ex ordeï succo, qui sit digestionem difficilis. Dehinc ægrotantis corpus non valet tantum sustinere nutrimentum, quantum sanitatis tempore solitum videbatur. Item mulsum ex aceto (quod oxymeli appellavit) sine discretionem accipimus. Est etiam immodica usque ad septimum diem cibi abstinentia, quam custodiendam ordinavit.^[566] Quippe cum nullus vehementiam passionis sustinere valet, nisi nutrimento quamvis parvo toleratus: et neque in declinatione passionis aliquid humanius cibo largitur, sed in iisdem sorbilibus perseverandum existimat succis. At cum fuerint sputa segniora, tunc ut existimat, erit primo æger nutriendus, quomodo necessariò declinante passione occurrunt intolerato. In cæteris relinquendum temporibus absque nutrimento ægrotantem apertissimè indicavit, quum semper plurimum utilitatis adjutorium cibi, quam cætera possunt adjutoria, largiatur. Omne etiam corpus erit unctione coequandum, et non ejus particula. Quippe cum totum cibo nutriatur, ipsa quoque unctio non exerta, anxietatem ingerit ægrotanti, quæ latentem difficultatem, atque accessionem venientem, corporis provocat incendium."

APPENDIX TO THE WORK
ON
REGIMEN IN ACUTE DISEASES.

Ardent fever (causus)^[567] takes place when the veins, being dried up in the summer season, attract acrid and bilious humors to themselves; and strong fever seizes the whole body, which experiences aches of the bones, and is in a state of lassitude and pain. It takes place most commonly from a long walk and protracted thirst, when the veins being dried up attract acrid and hot defluxions to themselves. The tongue becomes rough, dry, and very black; there are gnawing pains about the bowels; the alvine discharges are watery and yellow; there is intense thirst, insomnolency, and sometimes wandering of the mind. To a person in such a state give to drink water and as much boiled hydromel of a watery consistence as he will take; and if the mouth be bitter, it may be advantageous to administer an emetic and clyster; and if these things do not loosen the bowels, purge with the boiled milk of asses. Give nothing saltish nor acrid, for they will not be borne; and give no draughts of ptisan until the crisis be past. And the affection is resolved if there be an epistaxis, or if true critical sweats supervene with urine having white, thick, and smooth sediments, or if a deposit take place anywhere; but if it be resolved without these, there will be a relapse of the complaint, or pain in the hips and legs will ensue, with thick sputa, provided the patient be convalescent. Another species of ardent fever: belly loose, much thirst, tongue rough, dry, and saltish, retention of urine, insomnolency, extremities cold. In such a case, unless there be a flow of blood from the nose, or an abscess form about the neck, or pain in the limbs, or the patient expectorate thick sputa (these occur when the belly is constipated), or pain of the hips, or lividity of the genital organs, there is no crisis; tension of the testicle is also a critical symptom. Give attractive draughts.^[568]

2. Bleed in the acute affections, if the disease appear strong, and the patients be in the vigor of life, and if they have strength.^[569] If it be quinsy or any other of the pleuritic affections, purge with electuaries; but if the patient be weaker, or if you abstract more blood, you may administer a clyster every third day, until he be out of danger, and enjoin total abstinence if necessary.

3. Hypochondria inflamed not from retention of flatus, tension of the diaphragm, checked respiration, with dry orthopnoea, when no pus is formed, but when these complaints are connected with obstructed respiration; but more especially strong pains of the liver, heaviness of the spleen, and other phlegmasiæ and intense pains above the diaphragm, diseases connected with collections of humors,—all these diseases do not admit of resolution, if treated at first by medicine, but venesection holds the first place in conducting the treatment; then we may have recourse to a clyster, unless the disease be great and strong; but if so, purging also may be necessary; but bleeding and purging together require caution and moderation. Those who attempt to resolve inflammatory diseases at the commencement by the administration of purgative medicines, remove none of the morbid humors which produce the inflammation and tension: for the diseases while unconcocted could not yield, but they melt down those parts which are healthy and resist the disease; so when the body is debilitated the malady obtains the mastery; and when the disease has the upper hand of the body, it does not admit of a cure.^[570]

4. When a person suddenly loses his speech, in connection with obstruction of the veins,—if this happen without warning or any other strong cause, one ought to open the internal vein of the right arm, and abstract blood more or less according to the habit and age of the patient. Such cases are mostly attended with the following symptoms: redness of the face, eyes fixed, hands distended, grinding of the teeth, palpitations, jaws fixed, coldness of the extremities, retention of airs in the veins.^[571]

5. When pains precede, and there are influxes of black bile and of acrid humors, and when by their pungency the internal parts are pained, and the veins being pinched and dried become distended, and getting inflamed attract the humors running into the parts, whence the blood being vitiated, and the airs collected there not being able to find their natural passages, coldness comes on in consequence of this stasis, with vertigo, loss of speech, heaviness of the head, and convulsion, if the disease fix on the liver, the heart, or the great vein (vena cava?); whence they are seized with epilepsy or apoplexy, if the defluxions fall upon the containing parts,^[572] and if they are dried up by airs which cannot make their escape; such persons having been first fomented are to be immediately bled at the commencement, while all the peccant vapors and humors are buoyant, for then the cases more easily admit of a cure; and then supporting the strength and attending to the crisis, we may give emetics, unless the disease be alleviated; or if the bowels be not moved, we may administer a clyster and give the boiled milk of asses, to the amount of not less than twelve heminæ, or if the strength permit, to more than sixteen.

6. Quinsy takes place when a copious and viscid defluxion from the head, in the season of winter or spring, flows into the jugular veins, and when from their large size they attract a greater defluxion; and when owing to the defluxion being of a cold and viscid nature it becomes enfarcted, obstructing the passages of the respiration and of the blood, coagulates the surrounding blood, and renders it motionless and stationary, it being naturally cold and disposed to obstructions. Hence they are seized with convulsive suffocation, the tongue turning livid, assuming a rounded shape, and being bent owing to the veins which are seated below the tongue (for when an enlarged uvula, which is called *uva*, is cut, a large vein may be observed on each side). These veins, then, becoming filled, and their roots extending into the tongue, which is of a loose and spongy texture, it, owing

to its dryness receiving forcibly the juice from the veins, changes from broad and becomes round, its natural color turns to livid, from a soft consistence it grows hard, instead of being flexible it becomes inflexible, so that the patient would soon be suffocated unless speedily relieved. Bleeding, then, in the arm, and opening the sublingual veins, and purging with the electuaries, and giving warm gargles, and shaving the head, we must apply to it and the neck a cerate, and wrap them round with wool, and foment with soft sponges squeezed out of hot water; give to drink water and hydromel, not cold; and administer the juice of ptisan when, having passed the crisis, the patient is out of danger. When, in the season of summer or autumn, there is a hot and nitrous defluxion from the head (it is rendered hot and acrid by the season), being of such a nature it corrodes and ulcerates, and fills with air, and orthopnoea attended with great dryness supervenes; the fauces, when examined, do not seem swollen; the tendons on the back part of the neck are contracted, and have the appearance as if it were tetanus; the voice is lost, the breathing is small, and inspiration becomes frequent and laborious. In such persons the trachea becomes ulcerated, and the lungs engorged, from the patient's not being able to draw in the external air. In such cases, unless there be a spontaneous determination to the external parts of the neck, the symptoms become still more dreadful, and the danger more imminent, partly owing to the season, and the hot and acrid humors which cause the disease.^[573]

7. When fever seizes a person who has lately taken food, and whose bowels are loaded with faeces which have been long retained, whether it be attended with pain of the side or not, he ought to lie quiet until the food descend to the lower region of the bowels, and use oxymel for drink; but when the load descends to the loins, a clyster should be administered, or he should be purged by medicine; and when purged, he should take ptisan for food and hydromel for drink; then he may take the cerealia, and boiled fishes, and a watery wine in small quantity, at night, but during the day, a watery hydromel. When the flatus is offensive, either a suppository or clyster is to be administered; but otherwise the oxymel is to be discontinued, until the matters descend to the lower part of the bowels, and then they are to be evacuated by a clyster. But if the ardent fever (*causus*) supervene when the bowels are empty, should you still judge it proper to administer purgative medicine, it ought not to be done during the first three days, nor earlier than the fourth. When you give the medicine, use the ptisan, observing the paroxysms of the fevers, so as not to give it when the fever is setting in, but when it is ceasing, or on the decline, and as far as possible from the commencement. When the feet are cold, give neither drink nor ptisan, nor anything else of the kind, but reckon it an important rule to refrain until they become warm, and then you may administer them with advantage. For the most part, coldness of the feet is a symptom of a paroxysm of the fever coming on; and if at such a season you apply those things, you will commit the greatest possible mistake, for you will augment the disease in no small degree. But when the fever ceases, the feet, on the contrary, become hotter than the rest of the body; for when the heat leaves the feet, it is kindled up in the breast, and sends its flame up to the head. And when all the heat rushes upwards, and is exhaled at the head, it is not to be wondered at that the feet become cold, being devoid of flesh, and tendinous; and besides, they contract cold, owing to their distance from the hotter parts of the body, an accumulation of heat having taken place in the chest: and again, in like manner, when the fever is resolved and dissipated, the heat descends to the feet, and, at the same time, the head and chest become cold. Wherefore one should attend to this; that when the feet are cold, the bowels are necessarily hot, and filled with nauseous matters; the hypochondrium distended: there is jactitation of the body, owing to the internal disturbance; and aberration of the intellect, and pains; the patient is agitated, and wishes to vomit, and if he vomits bad matters he is pained; but when the heat descends to the feet, and the urine passes freely, he is every way lightened, even although he does not sweat; at this season, then, the ptisan ought to be given; it would be death to give it before.^[574]

8. When the bowels are loose during the whole course of fevers, in this case we are most especially to warm the feet, and see that they are properly treated with cerates, and wrapped in shawls, so that they may not become colder than the rest of the body; but when they are hot, no fomentation must be made to them, but care is to be taken that they do not become cold; and very little drink is to be used, either cold water or hydromel. In those cases of fever where the bowels are loose, and the mind is disordered, the greater number of patients pick the wool from their blankets, scratch their noses, answer briefly when questions are put to them, but, when left to themselves, utter nothing that is rational. Such attacks appear to me to be connected with black bile. When in these cases there is a colliquative diarrhoea, I am of opinion that we ought to give the colder and thicker ptisans, and that the drinks ought to be binding, of a vinous nature, and rather astringent. In cases of fever attended from the first with vertigo, throbbing of the head, and thin urine, you may expect the fever to be exacerbated at the crisis; neither need it excite wonder, although there be delirium. When, at the commencement, the urine is cloudy or thick, it is proper to purge gently, provided this be otherwise proper; but when the urine at first is thin, do not purge such patients, but, if thought necessary, give a clyster: such patients should be thus treated; they should be kept in a quiet state, have unguents applied to them, and be covered up properly with clothes, and they should use for drink a watery hydromel, and the juice of ptisan as a draught in the evening; clear out the bowels at first with a clyster, but give no purgative medicines to them, for, if you move the bowels strongly, the urine is not concocted, but the fever remains long, without sweats and without a crisis. Do not give draughts when the time of the crisis is at hand, if there be agitation, but only when the fever abates and is alleviated. It is proper to be guarded at the crises of other fevers, and to withhold the draughts at that season. Fevers of this description are apt to be protracted, and to have determinations, if the inferior extremities be cold, about the ears and neck, or, if these parts are not cold, to have other changes; they have epistaxis, and disorder of the bowels. But in cases of fever attended with nausea, or

distention of the hypochondria, when the patients cannot lie reclined in the same position, and the extremities are cold, the greatest care and precaution are necessary; nothing should be given to them, except oxymel diluted with water; no draught should be administered, until the fever abate and the urine be concocted; the patient should be laid in a dark apartment, and recline upon the softest couch, and he should be kept as long as possible in the same position, so as not to toss about, for this is particularly beneficial to him. Apply to the hypochondrium linseed by inunctions, taking care that he do not catch cold when the application is made; let it be in a tepid state, and boiled in water and oil. One may judge from the urine what is to take place, for if the urine be thicker, and more yellowish, so much the better; but if it be thinner, and blacker, so much the worse; ^[575] but if it undergo changes, it indicates a prolongation of the disease, and the patient, in like manner, must experience a change to the worse and the better. Irregular fevers should be let alone until they become settled, and, when they do settle, they are to be treated by a suitable diet and medicine, attending to the constitution of the patient.

9. The aspects of the sick are various; wherefore the physician should pay attention, that he may not miss observing the exciting causes, as far as they can be ascertained by reasoning, nor such symptoms as should appear on an even or odd day, but he ought to be particularly guarded in observing the odd days, as it is in them, more especially, that changes take place in patients. He should mark, particularly, the first day on which the patient became ill, considering when and whence the disease commenced, for this is of primary importance to know. When you examine the patient, inquire into all particulars; first how the head is, and if there be no headache, nor heaviness in it; then examine if the hypochondria and sides be free of pain; for if the hypochondrium be painful, swelled, and unequal, with a sense of satiety, or if there be pain in the side, and, along with the pain, either cough, tormina, or belly-ache, if any of these symptoms be present in the hypochondrium, the bowels should be opened with clysters, and the patient should drink boiled hydromel in a hot state. The physician should ascertain whether the patient be apt to faint when he is raised up, and whether his breathing be free; and examine the discharges from the bowels, whether they be very black, or of a proper color, like those of persons in good health, and ascertain whether the fever has a paroxysm every third day, and look well to such persons on those days. And should the fourth day prove like the third, the patient is in a dangerous state. ^[576] With regard to the symptoms, black stools prognosticate death; but if they resemble the discharges of a healthy person, and if such is their appearance every day, it is a favorable symptom; but when the bowels do not yield to a suppository, and when, though the respiration be natural, the patient when raised to the night-table, or even in bed, be seized with deliquium, you may expect that the patient, man or woman, who experiences these symptoms, is about to fall into a state of delirium. Attention also should be paid to the hands, for if they tremble, you may expect epistaxis; and observe the nostrils, whether the breath be drawn in equally by both; and if expiration by the nostrils be large, a convulsion is apt to take place; and should a convulsion occur to such a person, death may be anticipated, and it is well to announce it beforehand.

10. If, in a winter fever, the tongue be rough, and if there be swoonings, it is likely to be the remission of the fever. Nevertheless such a person is to be kept upon a restricted diet, with water for drink, and hydromel, and the strained juices, not trusting to the remission of the fevers, as persons having these symptoms are in danger of dying; when, therefore, you perceive these symptoms, announce this prognostic, if you shall judge proper, after making the suitable observations. When, in fevers, any dangerous symptom appears on the fifth day, when watery discharges suddenly take place from the bowels, when deliquium animi occurs, or the patient is attacked with loss of speech, convulsions, or hiccup, under such circumstances he is likely to be affected with nausea, and sweats break out under the nose and forehead, or on the back part of the neck and head, and patients with such symptoms shortly die, from stoppage of the respiration. ^[577] When, in fevers, abscesses form about the legs, and, getting into a chronic state, are not concocted while the fever persists, and if one is seized with a sense of suffocation in the throat, while the fauces are not swelled, and if it do not come to maturation, but is repressed, in such a case there is apt to be a flow of blood from the nose; if this, then, be copious, it indicates a resolution of the disease, but if not, a prolongation of the complaint; and the less the discharge, so much worse the symptoms, and the more protracted the disease; but if the other symptoms are very favorable, expect in such a case that pains will fall upon the feet; if then they attack the feet, and if those continue long in a very painful, and inflamed state, and if there be no resolution, the pains will extend by degrees to the neck, to the clavicle, shoulder, breast, or to some articulation, in which an inflammatory tumour will necessarily form. When these are reduced, if the hands are contracted, and become trembling, convulsion and delirium seize such a person; but blisters break out on the eyebrow, erythema takes place, the one eyelid being tumefied overtops the other, a hard inflammation sets in, the eye become strongly swelled, and the delirium increases much, but makes its attacks rather at night than by day. These symptoms more frequently occur on odd than on even days, but, whether on the one or the other, they are of a fatal character. Should you determine to give purgative medicines in such cases, at the commencement, you should do so before the fifth day, if there be borborygmi in the bowels, or, if not, you should omit the medicines altogether. If there be borborygmi, with bilious stools, purge moderately with scammony; but with regard to the treatment otherwise, administer as few drinks and draughts as possible, until there be some amendment, and the disease is past the fourteenth day. When loss of speech seizes a person, on the fourteenth day of a fever, there is not usually a speedy resolution, nor any removal of the disease, for this symptom indicates a protracted disease; and when it appears on that day, it will be still more prolonged. When, on the fourth day of a fever, the tongue articulates confusedly, and when there are watery and bilious discharges from the bowels, such a patient is

apt to fall into a state of delirium; the physician ought, therefore, to watch him, and attend to whatever symptoms may turn up. In the season of summer and autumn an epistaxis, suddenly occurring in acute diseases, indicates vehemence of the attack, and inflammation in the course of the veins, and on the day following, the discharge of thin urine; and if the patient be in the prime of life, and if his body be strong from exercise, and brawny, or of a melancholic temperament, or if from drinking he has trembling hands, it may be well to announce beforehand either delirium or convulsion;^[578] and if these symptoms occur on even days, so much the better; but on critical days, they are of a deadly character. If, then, a copious discharge of blood procure an issue to the fullness thereof about the nose, or what is collected about the anus, there will be an abscess, or pains in the hypochondrium, or testicles, or in the limbs; and when these are resolved, there will be a discharge of thick sputa, and of smooth, thin urine. In fever attended with singultus, give asafœtida, oxymel, and carrot, triturated together, in a draught; or galbanum in honey, and cumin in a linctus, or the juice of ptisan. Such a person cannot escape, unless critical sweats and gentle sleep supervene, and thick and acrid urine be passed, or the disease terminate in an abscess: give pine-fruit^[579] and myrrh in a linctus, and further give a very little oxymel to drink; but if they are very thirsty, some barley-water.

11. Peripneumonia, and pleuritic affections, are to be thus observed: If the fever be acute, and if there be pains on either side, or in both, and if expiration be attended with pain, if cough be present, and the sputa expectorated be of a blond or livid color, or likewise thin, frothy, and florid, or having any other character different from the common, in such a case, the physician should proceed thus: if the pain pass upward to the clavicle, or the breast, or the arm, the inner vein in the arm should be opened on the side affected, and blood abstracted according to the habit, age, and color of the patient, and the season of the year, and that largely and boldly, if the pain be acute, so as to bring on deliquium animi,^[580] and afterwards a clyster is to be given. But if the pain be below the chest, and if very intense, purge the bowels gently in such an attack of pleurisy, and during the act of purging give nothing; but after the purging give oxymel. The medicine is to be administered on the fourth day; on the first three days after the commencement, a clyster should be given, and if it does not relieve the patient, he should then be gently purged, but he is to be watched until the fever goes off, and till the seventh day; then if he appear to be free from danger, give him some unstrained ptisan, in small quantity, and thin at first, mixing it with honey. If the expectoration be easy, and the breathing free, if his sides be free of pain, and if the fever be gone, he may take the ptisan thicker, and in larger quantity, twice a day. But if he do not progress favorably, he must get less of the drink, and of the draught, which should be thin, and only given once a day, at whatever is judged to be the most favorable hour; this you will ascertain from the urine. The draught is not to be given to persons after fever, until you see that the urine and sputa are concocted, (if, indeed, after the administration of the medicine he be purged frequently, it may be necessary to give it, but it should be given in smaller quantities and thinner than usual, for from inanition he will be unable to sleep, or digest properly, or wait the crisis;) but when the melting down of crude matters has taken place, and his system has cast off what is offensive, there will then be no objection. The sputa are concocted when they resemble pus, and the urine when it has reddish sediments like tares. But there is nothing to prevent fomentation and cerates being applied for the other pains of the sides; and the legs and loins may be rubbed with hot oil, or anointed with fat; linseed, too, in the form of a cataplasm, may be applied to the hypochondrium and as far up as the breasts. When pneumonia is at its height, the case is beyond remedy if he is not purged, and it is bad if he has dyspnœa, and urine that is thin and acrid, and if sweats come out about the neck and head, for such sweats are bad, as proceeding from the suffocation, *râles*, and the violence of the disease which is obtaining the upper hand, unless there be a copious evacuation of thick urine, and the sputa be concocted; when either of these come on spontaneously, that will carry off the disease. A linctus for pneumonia: Galbanum and pine-fruit in Attic honey; and southernwood in oxymel; make a decoction of pepper and black hellebore, and give it in cases of pleurisy attended with violent pain at the commencement. It is also a good thing to boil opoponax in oxymel, and, having strained it, to give it to drink; it answers well, also, in diseases of the liver, and in severe pains proceeding from the diaphragm, and in all cases in which it is beneficial to determine to the bowels or urinary organs, when given in wine and honey; when given to act upon the bowels, it should be drunk in larger quantity, along with a watery hydromel.

12. A dysentery, when stopped, will give rise to an aposteme, or tumour, if it do not terminate in fevers with sweats, or with thick and white urine, or in a tertian fever, or the pain fix upon a varix, or the testicles, or on the hip-joints.^[581]

13. In a bilious fever, jaundice coming on with rigor before the seventh day carries off the fever, but if it occur without the fever, and not at the proper time, it is a fatal symptom.

14. When the loins are in a tetanic state, and the spirits in the veins are obstructed by melancholic humors, venesection will afford relief.^[582] But when, on the other hand, the anterior tendons are strongly contracted, and if there be sweats about the neck and face, extorted by the violent pain of the parched and dried tendons of the sacral extremity (these are very thick, sustaining the spine, and giving rise to very great ligaments, which terminate in the feet,) in such a case, unless fever and sleep come on, followed by concocted urine and critical sweats, give to drink a strong Cretan wine, and boiled barley-meal for food; anoint and rub with ointments containing wax; bathe the legs and feet in hot water, and then cover them up; and so in like manner the arms, as far as the hands, and the spine, from the neck to the sacrum, are to be wrapped in a skin smeared with wax; this must extend to the parts beyond, and intervals are to be left for applying fomentations, by means of leather bottles

filled with hot water, then, wrapping him up in a linen cloth, lay him down in bed. Do not open the bowels, unless by means of a suppository, when they have been long of being moved. If there be any remission of the disease, so far well, but otherwise, pound of the root of bryonia^[583] in fragrant wine, and that of the carrot, and give to the patient fasting early in the morning, before using the affusion, and immediately afterwards let him eat boiled barley-meal in a tepid state, and as much as he can take, and in addition let him drink, if he will, wine well diluted. If the disease yield to these means, so much the better, but, if otherwise, you must prognosticate accordingly.

15. All diseases are resolved either by the mouth, the bowels, the bladder, or some other such organ. Sweat is a common form of resolution in all these cases.^[584]

16. You should put persons on a course of hellebore who are troubled with a defluxion from the head. But do not administer hellebore to such persons as are laboring under empyema connected with abscesses, hæmoptysis, and intemperament, or any other strong cause, for it will do no good; and if anything unpleasant occur the hellebore will get the blame of it. But if the body have suddenly lost its powers, or if there be pain in the head, or obstruction of the ears and nose, or ptyalism, or heaviness of the limbs, or an extraordinary swelling of the body, you may administer the hellebore, provided these symptoms be not connected with drinking, nor with immoderate venery; nor with sorrow, vexation, nor insomnolency, for, if any of these causes exist, the treatment must have respect to it.

17. From walking arise pains of the sides, of the back, of the loins, and of the hip-joint, and disorder of the respiration has often been from the same cause, for, after excesses of wine and flatulent food, pains shoot to the loins and hips, accompanied with dysuria.^[585] Walking is the cause of such complaints, and also of coryza and hoarseness.

18. Disorders connected with regimen, for the most part, make their attack accordingly as any one has changed his habitual mode of diet.^[586] For persons who dine contrary to custom experience much swelling of the stomach, drowsiness, and fullness; and if they take supper over and above, their belly is disordered; such persons will be benefited by sleeping after taking the bath, and by walking slowly for a considerable time after sleep; if, then, the bowels be moved, he may dine and drink a small quantity of wine not much diluted; but if the bowels are not opened, he should get his body rubbed with hot oil, and, if thirsty, drink of some weak and white wine, or a sweet wine, and take repose; if he does not sleep he should repose the longer. In other respects he should observe the regimen laid down for those who have taken a debauch. With regard to the bad effects of drinks, such as are of a watery nature pass more slowly through the body, they regurgitate, as it were, and float about the hypochondria, and do not flow readily by urine; when filled up with such a drink, he should not attempt any violent exertion, requiring either strength or swiftness, but should rest as much as possible until the drink has been digested along with the food; but such drinks as are stronger or more austere, occasion palpitation in the body and throbbing in the head, and in this case the person affected will do well to sleep, and take some hot draught for which he feels disposed; for abstinence is bad in headache and the effects of a surfeit. Those who, contrary to usage, restrict themselves to one meal, feel empty and feeble, and pass hot urine in consequence of the emptiness of their vessels; they have a salt and bitter taste in the mouth; they tremble at any work they attempt; their temples throb; and they cannot digest their supper so well as if they had previously taken their dinner. Such persons should take less supper than they are wont, and a pudding of barley-meal more moist than usual instead of bread, and of potherbs the dock, or mallow, and ptisan, or beets, and along with the food they should take wine in moderation, and diluted with water; after supper they should take a short walk, until the urine descend and be passed; and they may use boiled fish.

Articles of food have generally such effects as the following:^[587] Garlic occasions flatulence and heat about the chest, heaviness of the head, and nausea, and any other habitual pain is apt to be exasperated by it; it is diuretic, which, in so far, is a good property which it possesses; but it is best to eat it when one means to drink to excess, or when intoxicated. Cheese produces flatulence and constipation, and heats the other articles of food; and it gives rise to crudities and indigestion, but it is worst of all to eat it along with drink after a full meal. Pulse of all kinds are flatulent, whether raw, boiled, or fried; least so when macerated in water, or in a green state; they should not be used except along with food prepared from the cerealia. Each of these articles, however, has bad effects peculiar to itself. The vetch, whether raw or boiled, creates flatulence and pain. The lentil is astringent, and disorders the stomach if taken with its hull. The lupine has the fewest bad effects of all these things. The stalk and the juice of silphium (*asafætida*), pass through some people's bowels very readily, but in others, not accustomed to them, they engender what is called dry cholera;^[588] this complaint is more especially produced by it if mixed with much cheese, or eaten along with beef. Melancholic diseases are most particularly exacerbated by beef, for it is of an unmanageable nature, and requires no ordinary powers of stomach to digest it; it will agree best with those who use it well boiled and pretty long kept. Goat's flesh has all the bad properties of beef; it is as indigestible, more flatulent and engenders acid eructations and cholera; such as has a fragrant smell, is firm, and sweet to the taste, is the best, when well baked and cooled; but those kinds which are disagreeable to the taste, have a bad smell, and are hard, such are particularly bad, and especially if very fresh; it is best in summer and worst in autumn. The flesh of young pigs is bad, either when it is too raw or when it is over-roasted, for it engenders bile and disorders the bowels. Of all kinds of flesh, pork is the best; it is best when neither very fat, nor, on the other

hand, very lean, and the animal had not attained the age of what is reckoned an old victim; it should be eaten without the skin, and in a coldish state.

19. In dry cholera the belly is distended with wind, there is rumbling in the bowels, pain in the sides and loins, no dejections, but, on the contrary, the bowels are constipated. In such a case you should guard against vomiting, but endeavor to get the bowels opened. As quickly as possible give a clyster of hot water with plenty of oil in it, and having rubbed the patient freely with unguents, put him into hot water, laying him down in the basin, and pouring the hot water upon him by degrees; and if, when heated in the bath, the bowels be moved, he will be freed from the complaint. To a person in such a complaint it will do good if he sleep, and drink a thin, old, and strong wine; and you should give him oil, so that he may settle, and have his bowels moved, when he will be relieved. He must abstain from all other kinds of food; but when the pain remits, give him asses' milk to drink until he is purged. But if the bowels are loose, with bilious discharges, tormina, vomitings, a feeling of suffocation, and gnawing pains, it is best to enjoin repose, and to drink hydromel, and avoid vomiting.

20. There are two kinds of dropsy, the one anasarca, which, when formed, is incurable; the other is accompanied with emphysema (tyimpanites?) and requires much good fortune to enable one to triumph over it.^[589] Laborious exertion, fomentation, and abstinence (are to be enjoined). The patient should eat dry and acrid things, for thus will he pass the more water, and his strength be kept up. If he labors under difficulty of breathing, if it is the summer season, and if he is in the prime of life, and is strong, blood should be abstracted from the arm, and then he should eat hot pieces of bread, dipped in dark wine and oil, drink very little, and labor much, and live on well-fed pork, boiled with vinegar, so that he may be able to endure hard exercises.^[590]

21. Those who have the inferior intestines hot, and who pass acrid and irregular stools of a colliquative nature, if they can bear it, should procure revulsion by vomiting with hellebore; but if not, should get a thick decoction of summer wheat in a cold state, lentil soup, bread cooked with cinders, and fish, which should be taken boiled if they have fever, but roasted if not feverish; and also dark-colored wine if free of fever; but otherwise they should take the water from medlars, myrtles, apples, services, dates, or wild vine. If there be no fever, and if there be tormina, the patient should drink hot asses' milk in small quantity at first, and gradually increase it, and linseed, and wheaten flour, and having removed the bitter part of Egyptian beans, and ground them, sprinkle on the milk and drink; and let him eat eggs half-roasted, and fine flour, and millet, and perispelt (chondrus) boiled in milk;—all these things should be eaten cold, and similar articles of food and drink should be administered.

22. The most important point of regimen to observe and be guarded about in protracted diseases, is to pay attention to the exacerbations and remissions of fevers, so as to avoid the times when food should not be given, and to know when it may be administered without danger; this last season is at the greatest possible distance from the exacerbation.

23. One should be able to recognize those who have headache from gymnastic exercises, or running, or walking or hunting, or any other unseasonable labour, or from immoderate venery; also those who are of a pale colour, or troubled with hoarseness; those who have enlarged spleen, those who are in a state of anæmia, those who are suffering from tympanites, those having dry cough and thirst, those who are flatulent, and have the course of the blood in their veins intercepted; those persons whose hypochondria, sides, and back are distended; those having torpor; those laboring under amaurosis, or having noises in their ears; those suffering from incontinence of urine or jaundice, or whose food is passed undigested; those who have discharges of blood from the nose or anus, or who have flatulence and intense pain, and who cannot retain the wind. In these cases you may do mischief, but cannot possibly do any good by purging, but may interrupt the spontaneous remissions and crises of the complaints.^[591]

24. If you think it expedient to let blood, see that the bowels be previously settled, and then bleed; enjoin abstinence, and forbid the use of wine; and complete the cure by means of a suitable regimen, and wet fomentations.^[592] But if the bowels appear to be constipated, administer a soothing clyster.

25.^[593] If you think it necessary to give medicines, you may safely purge upwards by hellebore, but none of those should be purged downwards. The most effectual mode of treatment is by the urine, sweats, and exercise; and use gentle friction so as not to harden the constitution; and if he be confined to bed let others rub him. When the pain is seated above the diaphragm, place him erect for the most part, and let him be as little reclined as possible; and when he is raised up let him be rubbed for a considerable time with plenty of hot oil. But if the pains be in the lower belly below the diaphragm, it will be useful to lie reclined and make no motion, and to such a person nothing should be administered except the friction. Those pains which are dissolved by discharges from the bowels, by urine, or moderate sweats, cease spontaneously, if they are slight, but if strong they prove troublesome; for persons so affected either die, or at least do not recover without further mischief, for they terminate in abscesses.

26. *A draught for a dropsical person.* Take three cantharides,^[594] and removing their head, feet, and wings, triturate their bodies in three cupfuls (cyathi) of water, and when the person who has drunk the draught complains of pain, let him have hot

fomentations applied. The patient should be first anointed with oil, should take the draught fasting, and eat hot bread with oil.

27.^[595] *A styptic.* Apply the juice of the fig inwardly to the vein; or having moulded biestings into a tent, introduce up the nostril, or push up some chalcitis with the finger, and press the cartilages of the nostrils together; and open the bowels with the boiled milk of asses: or having shaved the head apply cold things to it if in the summer season.

28. The sesamoides^[596] purges upwards when pounded in oxymel to the amount of a drachm and a half, and drunk; it is combined with the hellebores, to the amount of the third part, and thus it is less apt to produce suffocation.

29. *Trichiasis.* Having introduced a thread into the eye of a needle push it through the upper part of the distended eyelid, and do the same at the base of it; having stretched the threads tie a knot on them, and bind up until they drop out: and, if this be sufficient, so far well; but if otherwise, you must do the same thing again.^[597] And hemorrhoids, in like manner, you may treat by transfixing them with a needle and tying them with a very thick and large woolen thread; for thus the cure will be more certain. When you have secured them, use a septic application, and do not foment until they drop off, and always leave one behind; and when the patient recovers, let him be put upon a course of hellebore. Then let him be exercised and sweated; the friction of the gymnasium and wrestling in the morning will be proper; but he must abstain from running, drinking, and all acrid substances, except marjoram; let him take an emetic every seven days, or three times in a month; for thus will he enjoy the best bodily health. Let him take straw-colored, austere, and watery wine, and use little drink.

30. *For persons affected with empyema.* Having cut some bulbs or squill, boil in water, and when well boiled, throw this away, and having poured in more water, boil until it appear to the touch soft and well-boiled; then triturate finely and mix roasted cumin, and white sesames, and young almonds pounded in honey, form into an electuary and give; and afterwards sweet wine. In draughts, having pounded about a small acetabulum of the white poppy, moisten it with water in which summer wheat has been washed, add honey, and boil. Let him take this frequently during the day. And then taking into account what are to happen, give him supper.

31. *For dysentery.* A fourth part of a pound of cleaned beans, and twelve shoots of madder having been triturated, are to be mixed together and boiled, and given as a linctus with some fatty substance.

32. *For diseases of the eyes.* Washed spodium (tutty?) mixed with grease, and not of a thinner consistence than dough, is to be carefully triturated, and moistened with the juice of unripe raisins; and having dried in the sun, moisten until it is of the consistence of an ointment. When it becomes again dry, let it be finely levigated, anoint the eyes with it, and dust it upon the angles of the eyes.

33. *For watery eyes.* Take one drachm of ebony and nine oboli of burnt copper, rub them upon a whetstone, add three oboli of saffron; triturate all these things reduced to a fine powder, pour in an Attic hemina of sweet wine, and then place in the sun and cover up; when sufficiently digested, use it.^[598]

34. *For violent pains of the eyes.* Take of chalcitis,^[599] and of raisin, of each 1 dr., when digested for two days, strain; and pounding myrrh and saffron, and having mixed must, with these things, digest in the sun; and with this anoint the eyes when in a state of severe pain. Let it be kept in a copper vessel.

35. *Mode of distinguishing persons in an hysterical fit.* Pinch them with your fingers, and if they feel, it is hysterical; but if not, it is a convulsion.

36. *To persons in coma, (dropsy?)* give to drink meconium (*euphorbia peplus?*) to the amount of a round Attic *leciskion* (small acetabulum^[600]).

37. Of squama æris, as much as three specilla can contain, with the gluten of summer wheat: levigate, pound, form into pills, and give; it purges water downwards.

38. *A medicine for opening the bowels.* Pour upon figs the juice of spurge, in the proportion of seven to one: then put into a new vessel and lay past when properly mixed. Give before food.

39. Pounding meconium, pouring on it water, and straining, and mixing flour, and baking into a cake, with the addition of boiled honey, give in affections of the anus and in dropsy; and after eating of it, let the patient drink of a sweet watery wine, and diluted hydromel prepared from wax: or collecting meconium, lay it up for medicinal purposes.^[601]

FIRST AND THIRD BOOKS
OF
THE EPIDEMICS.

THE EPIDEMICS.

BOOK I.

THE ARGUMENT.

The ancient physicians commonly used the term Epidemic in the same sense as it is understood now, that is to say, as applying to any disease which attacks a multitude of persons in a locality at any particular period. This, as will be seen in our annotations below, is nearly the definition which Galen gives of it; and it is generally used by Hippocrates, in the first and third books of the “Epidemics,” in pretty much the same sense as it is used by our great modern authority on epidemics, Sydenham. But, although this be the strict sense in which the ancient authorities use the term, it must be borne in mind that, as applied to the whole seven books of the “Epidemics,” it must be taken in a much wider signification; for there are many things treated of in them to which the term epidemic can by no means be thus applied, such as surgical cases, fragments of anatomical descriptions, philosophical speculations, empirical remedies, general reflections on various topics, and so forth. In fact, the work entitled “The Books of Epidemics” can be viewed in no other light than as an *Adversaria*, or *Memorandum Book*, in which is collected a variety of isolated facts and detached observations, to serve as the materials for more elaborate and finished works on professional subjects. Indeed, Galen does not hesitate to give it as his opinion, that some of the most celebrated of our author’s productions, such as the “Aphorisms” and “Prognostics,” are in a great measure made up from the materials originally laid up in this capacious repertory of observations;^[602] and, with regard to the former of these works, there is no person familiarly acquainted with it but must admit the truth of Galen’s remark. But, respecting the other, although it must be obvious, upon a comparison of them, that there is a close connection between it and the “Epidemics,” there can be no doubt that, in composing the “Prognostics,” Hippocrates availed himself of other materials ready prepared for his use, in the “Prorrhethics” and “Coan Prænotions” of his predecessors, the Asclepiadæ;^[603] so that, of all his admired productions, it, perhaps, is the one which has the least pretension to any originality of matter. If it be thought strange that the term epidemics should have been applied to a work composed of such heterogeneous materials, I would remark, in explanation, that, although the subject-matters of which it consists are not all of this nature, the most valuable portion of them refers to epidemics, and it is not to be wondered at that the whole collection should have got its appellation from the most prominent subject to which it relates.

I shall now proceed to give a succinct analysis of the various subjects which are contained in the First and Third Books of the “Epidemics.”

The first book opens with a description of the leading phenomena of a certain season, which is called the First Constitution; it was southerly, coldish, rainy, clouded and misty, with some intervals of drought. The most noted diseases of spring in this constitution were *causus* and an epidemical parotitis. But the most important subject which is handled under this head, is an epidemic phthisis, of which a very interesting description is given.

The Second Constitution is described as being northerly and humid; humid ophthalmies, dysenteries, and diarrhœas are described among the prevailing diseases of the season; but the most marked affection which is said to have occurred in this constitution, is a continual fever of a serious character, which did not come to a crisis until after it had run a long course. It is described as passing off by deposits, and principally by dropsies, and an affection of the urinary organs. One cannot help being struck with the remark which Hippocrates makes, that he never knew a case prove fatal in which the strangury supervened. The directions as to the treatment he condenses into one general rule, which well deserves to be engraved in letters of gold, that “*the aim of the physician should be to do good to his patient, or, at least, to do no harm.*” The description of this constitution concludes with some general reflections on the prognostics in *causus* and phrenitis.

The Third Constitution is described as being of a very variable character; winter stormy, spring rainy, summer hot, autumn cold and dry. The ardent fevers (*or causi*) began early in the season, but did not assume a fatal character until autumn. This disease came to a crisis in four modes—by an epistaxis, by a copious flow of urine, by a deposit, or by an alvine discharge. In women, there was also sometimes a crisis by menstruation.

The Fourth Constitution is one which, by Galen and the other authorities, has been entitled the pestilential, and has attracted great attention, as being supposed to have derived its peculiar characters from the great Plague which prevailed during the Peloponnesian war, and which is described in so interesting a manner by Thucydides. Galen, not only in his Commentary, but in various other parts of his works, advocates this opinion, and it will be seen from what is stated in our annotations, that there is in reality a striking resemblance between the features of the plague, as delineated by Thucydides, and the epidemical diseases which are noticed by Hippocrates as having prevailed during this constitution. Of all the diseases here described the most remarkable is the erysipelas, which, although not of a very fatal character, was still of a formidable nature,

as it frequently terminated in gangrene. Causus, phrenitis, and anthrax are also described as being common under this constitution. The last of these being a well-known symptom of the Oriental plague, it has naturally excited a good deal of speculation to determine whether or not our author here refers to the glandular plague. See our remarks on Epidem. III.

In these books it is remarkable that phthisis is treated of as a febrile disease, and in particular as supervening upon attacks of the semitertian. There seems reason to suppose that our author means to describe a hectic fever succeeding to intermittents, which had caused organic derangement of the internal viscera, more especially of the liver and spleen. See PAULUS ÆGINETA, Book II., 32.

In the first book, fourteen cases of disease are related, and in the beginning of the third twelve, and sixteen in the end; thus making forty-two in all. It is worthy of remark, that in twenty-five of these the result was fatal. There is every reason, then, to suppose that they were selected for a purpose, but what that purpose was cannot now be easily determined. The most natural would, no doubt, have been to illustrate, by examples, the forms of the different diseases which are described as occurring during the Constitutions previously described. But there seems to be little or no reason to suppose that this is the object for which they are related. In proof of this, I may mention that there is not in the collection a single case of the epidemical erysipelas which is described as having been the prevailing disease during the fourth Constitution. Indeed it must strike everybody, who reads them carefully, as a singular feature in these cases, that the lineaments of a particular disease are seldom to be recognized, and this perhaps may be regarded as a proof of the faithfulness with which they have been copied from nature. In short, we here recognize the features of disease in the concrete, and not in the abstract. And is not this what we should expect in all true copies from Nature? How often does the candid physician find himself forced honestly to admit that he is at a loss what name to give to the combination of morbid actions which he is called upon to treat! The common herd of mankind would seem to fancy, as in Nature there are certain types of all animal and vegetable substances, and the botanist has no difficulty in classing such a plant, for example, as the *conium maculatum*; and the natural historian can readily pronounce that such a bird is the *alcedo Ispida*; that the physician, in like manner, upon examining the characteristic features of any case, should have no difficulty in pronouncing that it is *pleuritis*, for example, or *pneumonia*, or the like. But how often does it happen, that the complaint in question is an aggregate of symptoms, produced by peculiarities of constitution, and incidental circumstances, which, taken together, constitute an *ensemble* which does not well admit of being referred to any one of the general forms of disease described in our nosological systems? Now, I say the most wonderful feature in the cases related by Hippocrates, is that they are descriptive of the symptoms observed in certain diseased individuals, instead of being, what most modern cases are, symptoms drawn to correspond with certain ideal forms of disease. What, in my opinion, likewise adds very much to the value of these cases is, that (as Galen somewhere remarks in his Commentary) the author never aimed to make his Books of Epidemics a work on Therapeutics, and hence, in noting morbid phenomena, his mind is not warped by any particular hypothesis, nor by any selfish interest, in order to place some favorite mode of practice, advocated by himself, in a favorable light. May I be permitted here to remark, that the reader will be much struck with our author's admirable talent for describing the phenomena of disease as they are actually presented to us, if he will compare the case related by him in these two books with those of almost any modern authority whatever;—for example, with those related by the late Dr. James Hamilton, in his celebrated work on Purgative Medicines. In the latter, you look in vain for the strongly-marked features which present themselves in all the cases related by our author,—for a description of the condition of the hypochondriac region,—of the state of the animal heat in the extremities,—of the minute characters of the alvine and urinary discharges,—of the respiration,—of the patients' position in bed,—and many other symptoms, which are invariably noticed by Hippocrates. And what reasonable person will venture to deny, that the symptoms I have just now mentioned are most important features in every febrile disease, and that no one can be said to have a sufficient view of such a case, who does not take these into account? To confine our attention at present to only one of these symptoms,—can it ever be a matter of indifference what are the physical characters of so important an excretion as the urine? that is to say, whether the grosser particles of it, which usually fall to the bottom, be present in the urine or not? Yet in all the seventeen cases related in the modern work just now referred to, the characters of the urine are not given in a single instance. And although the object of the writer is to enforce his own peculiar views, as to the utility of purgative medicines in this disease, he scarcely ever gives the minute characters of the alvine discharges, as is uniformly the case with Hippocrates; or if they are noticed at all, it is in so confused a manner that the reader is at a loss to determine whether they are produced by the disease, or by the medicines which have been administered. For the issue of the case no obvious cause is stated, but the reader is expected to draw the conclusion that, as purgatives were freely given, and a considerable proportion of the cases did well,—(agreeably to the hackneyed rule, *post quod, ergo propter quod*,)—the purgatives brought about the fortunate result. Had the cases been fully and circumstantially detailed, it might have been found that, as in those related by Hippocrates, recovery was preceded by a critical discharge of urine, accompanied with a copious sediment; and then the more probable inference would have been, that the amendment was referable to *it*, and not to the purgative medicines which were administered. It is, I regret to say, a notable example of the want of logical training in the

education of professional men, in the present age, that inferences regarding a peculiar method of practice were allowed to be founded upon narratives of observations so defective and one-sided as those I refer to.

I cannot quit the present subject of discussion, without saying a few words in reference to what must strike the reader as a singular feature in the cases related in the books of the Epidemics; I mean the general omission of any mention of treatment. The reader will find in our annotations various remarks of Galen on this head, from which he will learn that the Great Commentator inclines to the opinion, that in all these cases the usual routine of practice was followed, but that no mention is made of medicines, unless when there was some deviation from the established rules. For example, in a certain febrile case, it is stated that the patient was bled on the eighth day, and Galen contends that venesection is noticed in this instance, merely because it was contrary to the established rule of not bleeding after the fourth day; for that if the practice had been in accordance with the general rule, it would not have been noticed at all. Now it must be admitted, that this supposition is by no means improbable, and that examples of this usage are not wanting, even in the modern literature of medicine. To give an example, which just occurs to me; in not a few of the cases of cerebral disease related by Dr. Abercrombie, in his work "On the Brain," there is no allusion whatever to remedies, although no one, who recollects the vigorous system of treatment then pursued by the profession in "Modern Athens," will doubt for a moment that they must have been applied. As this eminent authority, then, when he believed that the treatment had no perceptible effect on the course which the disease ran, thought himself warranted in omitting all mention of it, it might be supposed, in like manner, that Hippocrates may have passed over the remedies applied, from some such motive or consideration. But another reason for the absence of remedies in these Reports may be readily supposed. May not Hippocrates have been at first quite undecided what was the proper plan of treatment to be adopted in those cases, and thought it the wisest course to attempt nothing rashly, but to be for a season the quiet spectator of the course which the diseases in question were naturally disposed to run, before attempting to interfere in the struggle between morbid agents with which he was imperfectly acquainted, and their great physician, as he held Nature to be?^[604] And however much the advocates for a bold system of treating diseases may be disposed to deride this expectant method, which Asclepiades contemptuously denominated "the contemplation of death,"^[605] it does not want the sanction of a name which is second only to Hippocrates in the literature of epidemical fevers. Sydenham admits, that with all the diligence which he had applied to the study of these diseases, he was always greatly puzzled what plan of treatment to adopt at the first breaking out of a new epidemic, and that it was only "ingenti adhibita cautela intentisque animi nervis," that he could make up his mind what course of treatment to adopt in such an emergency. Need it be wondered at, then, that two thousand years earlier the modest mind of our great author should have hesitated for a time, before deciding how to act under similar circumstances? I must own, therefore, that I have long inclined to the opinion, that, distracted with the conflicting plans of treatment adopted by his contemporaries, Hippocrates at first did little or nothing in the treatment of epidemical fevers, and that it was only after a patient study of their symptoms, and many cautious trials, that he ventured to lay down those excellent rules of treatment which he has described so admirably in his work "On Regimen in Acute Diseases." This, however, is merely my individual opinion, and the reader must receive it as such.

M. Littré, in the Argument prefixed to his translation of the Epidemics, enters very fully into the discussion of the question regarding the nature of the diseases which are treated of in the course of this work. This is a task, however, which I deem it superfluous to undertake at any length, as I have stated my opinions on this subject in the Commentaries on the Second Book of PAULUS ÆGINETA, and after maturely weighing what has been elicited by subsequent inquirers, I find no cause to retract any of the opinions which are there advanced. That the causus of Hippocrates, and the other ancient authorities, was not the typhus of the more temperate parts of Europe, but a bilious fever, of the remittent type, must be quite apparent to every person at all acquainted with the medical literature of febrile diseases. M. Littré's researches lead him to exactly the same conclusion, and much deference is due to his judgment in this case, as it must be admitted that a French physician is now very favorably situated for contrasting the diseases of temperate and hot climates, owing to the familiar intercourse which at present subsists between Paris and Algiers. Of all the materials which he has collected from the observations of French physicians in Algeria, the most interesting are those which he draws from a work on Fevers, by M. Maillot. The description which is there given of "la fièvre algide," is so striking, and is so much calculated to illustrate the nature of the fevers which are treated of in this work of Hippocrates, that I shall not scruple to quote it entire.

"La fièvre algide (dit M. Maillot) n'est pas généralement, comme on le dit, la prolongation indéfinie du stade de froid; je l'ai vue rarement débiter de la sorte. Il y a même entre ces deux états un contraste frappant. Dans le premier stade des fièvres intermittentes, la sensation du froid est hors de toute proportion avec l'abaissement réel de la température de la peau, tandis que, dans la fièvre algide, le froid n'est pas perçu par le malade, alors que la peau est glacée. C'est ordinairement pendant la réaction que commencent les symptômes qui la caractérisent; souvent ils surviennent tout à coup au milieu d'une réaction qui paraissait franche. Au trouble de la circulation succède en peu d'instant et presque sans transition le ralentissement du pouls, qui devient bientôt très rare, fuit sous le doigt et disparaît; l'abaissement de la température du corps va

vite et suit la progression promptement décroissante de la circulation; les extrémités, la face, le torse, se refroidissent successivement; l'abdomen seul conserve encore quelque temps un peu de chaleur; le contact de la peau donne la sensation de froid que procure le marbre. Les lèvres sont décolorées, l'haleine froide, la voix cassée, les battemens du cœur rares, incomplets, appréciables seulement par l'auscultation; les facultés intellectuelles sont intactes, et le malade se complaît dans cet état de repos, surtout lorsqu'il succède à une fièvre violente, la physionomie est sans mobilité, l'impassibilité la plus grande est peinte sur son visage; ses traits sont morts. La marche de cette fièvre est très insidieuse; il n'est peut-être personne, dont elle n'ait surpris la vigilance, avant d'être familiarisé avec l'observation des accidens de cette nature, on prend souvent pour une très grande amélioration due aux déplétions sanguines, le calme qui succède aux accidents inflammatoires; et plus d'une fois, dans de semblables circonstances, on n'a été détrompé que par la mort soudaine du malade. Toutes les fois qu'à une réaction plus ou moins forte, on verra succéder tout à coup un ralentissement du pouls, avec pâleur de la langue et décoloration des lèvres, on ne devra hésiter à diagnostiquer une fièvre algide. La temporisation ici donne la mort, en quelques heures. Dans quelques cas très rares, j'ai cependant vu cet état algide se prolonger trois ou quatre jours. Le malade expire en conservant toutes ses facultés intellectuelles,^[606] il s'éteint comme par un arrêt de l'innervation. Lorsque la mort n'est pas le terme de cet état morbide si grave, le pouls se relève; la peau reprend sa chaleur naturelle; quelquefois alors la réaction détermine une irritation de l'encéphale ou des voies digestives; mais rarement elle est assez intense pour qu'on soit obligé de la combattre par des déplétions sanguines."^[607] I shall add a remark, which M. Littré gives on the same authority: "J'ai tenu à mentionner ici l'impression qu'éprouva M. Maillot au début de sa pratique en Algérie, et qui est si instructive; car, aller subitement de France exercer la médecine dans un pays chaud, ou lire les observations d'Hippocrate, c'est tout un: l'impression est la même, le changement de scène est aussi grand."^[608]

I cannot help remarking in this place, however, that it appears to me singular, that M. Littré should represent the febris algida as being confined to southern climates, and should speak of it as being unknown in Paris; for, at all events, there seems to be no doubt that it prevails in a more northerly region, namely, in Holland. It is thus described by the celebrated Franciscus de le Boe (*or* Sylvius), who was professor of practical medicine at Leyden about the middle of the 17th century: "Febres algidæ observantur nonnunquam, non tantum frigore præsertim, sed frigore tantum molestæ: adeo ut aliquando et frequentius levis, aliquando et rarius nullus sequatur calor. Tales, etiam semper algidas in Nosocomio academico habuimus ita manifestas, ut non tantum incipiente, atque augescente, sed etiam vigente et declinante, imò cessante paroxysmo, id est, semper tum suo, tum adstantium, tum medicorum sensu moleste ubique frigerent, nunquam teperent, minus calerent ullibi ægri. Suntque hæ algidæ graviores semper forsan quotidianæ."^[609] The febris algida is also named "rigor without heat," by the Greek authorities, and "frigus quod non calefit" by the Arabians, who, like Sylvius, as quoted above, regard it as a variety of the quotidian intermittent. See PAULUS ÆGINETA, Book II., 26.

M. Littré^[610] quotes the remark of an excellent English authority on fever, J. Johnson,^[611] that it is singular the effects of marsh effluvia should have escaped the observation of Hippocrates, more especially as the remittent and intermittent fevers, of which he treats so fully, are mostly derived from this source. Now I must say, that I am not aware of there being any passages in the works of Hippocrates where the effects of marsh effluvia in engendering such fevers are distinctly noticed; but if Hippocrates was ignorant of this fact, in the etiology of fevers, it was well known to Galen, as may be seen on reference to his very interesting work "On the Difference of Fevers."^[612] The Arabians also were familiar with the fact. See Avicenna, iv., 1, 2, 1.

In the treatise "On Aïrs," which, although not admitted by us into the list of genuine works, has considerable pretension to be so regarded, the causes of fever are treated of with great precision, and there the pestilential fevers are said to derive their origin from miasma, but whether or not under this term be included marsh effluvia, cannot be determined. But perhaps a better reason might be assigned for there being little or no allusion to malaria in the works of Hippocrates, namely, that after all, this was *not* the cause of the epidemical diseases which he describes. The following extract from a work of very high authority on fever is well deserving of consideration in this place: "A question has arisen as to whether or not the inflammatory states of fever, in warm countries, are caused by malaria, or by the other causes now instanced (excess of heat, etc.). There can be no doubt that malaria very frequently produces in the plethoric, young, and robust, who have recently arrived in a hot climate, fever of an inflammatory and continued kind; but it must also be conceded that this fever chiefly occurs, even in persons thus constituted, during the dry season, and at times and in places where the existence of malaria is doubtful, or, at least, by no means proved. It is notoriously admitted that the inflammatory states of continued fever, in both the East and West Indies, appear among those soldiers, sailors, and civilians, who have not been long in a warm country, and who have not suffered from disease since their arrival; and that they take place chiefly during the dry and warm seasons, and in situations where the usual affects of malaria are never observed. This is the result of the experience of Jackson, Annesley, Boyle, Twining, Conwell, and the other experienced practitioners in warm countries. It agrees with my own observations, and is even admitted by Dr. Fergusson, who has gone much further than any one else in assigning malaria as the cause of intertropical fevers."^[613] I may mention, moreover, that Hippocrates and his contemporaries were evidently not ignorant of the fact, that the atmosphere in the vicinity of marshes and large rivers is unwholesome to the inhabitants of warm climates. See De Diæta, ii, 2.

The following are part of the conclusions which M. Littré draws from his investigations into the nature of the fevers described by Hippocrates. I quote them as being strongly confirmatory of the opinions delivered by me in the Commentary on the Second Book of PAULUS ÆGINETA.

“Les fièvres décrites dans les *Epidémies* d’Hippocrate diffèrent de nos fièvres continués.

“Les fièvres décrites dans les *Epidémies* ont, dans leur apparence générale, une similitude très grande avec celles des pays chauds.

“La similitude n’est pas moins grande dans les détails que dans l’ensemble.

“Dans les unes comme dans les autres les hypochondres sont pour un tiers des cas, le siège d’une manifestation toute spéciale.

“Dans les unes comme dans les autres, il y a une forte tendance ou refroidissement du corps, à la sueur froide et à la lividité des extrémités.”

On almost all the other diseases treated of in these books, M. Littré’s opinions, in like manner, exactly coincide with those delivered by me in the above-mentioned work. Thus he arrives at the conclusion, that the Phrenitis and Lethargus of Hippocrates were varieties of the Causus. Compare PAULUS ÆGINETA, Book III., 6, 9. He refers them to *les fièvres pernicieuses comateuses pseudo-continués et les fièvres pernicieuses dolorantes pseudo-continués* of M. Maillot. It would appear from the extracts which he quotes from a work of M. Roux, on the Diseases of Morea, that a similar tendency to pass into phrenitis and lethargy is still observable in the land of Greece. The fevers of the East Indies also, as described by Dr. Twining,^[614] appear to partake very much of the same character. In a word, the conclusions to which a patient study of modern authorities on the subject have brought me amount to this; that the fevers described by Hippocrates in his “Epidemics,” are exactly the same as those which are now described as still prevailing in the land of Greece: that they correspond very well with those described by Cleghorn as occurring in Majorca; differ but little from those described by Pringle, Monro, and Sylvius, as happening in the Low Countries, and differ from those described by Twining, as happening in Bengal, only in a few particulars.

From the analysis of their contents given above it will readily be understood that the subject-matters of these two books are not arranged methodically. Indeed it is quite obvious from the nature of the work that the matters which are treated of in it had never been methodized by the author. Certainly then, as proposed by Desmair,^[615] it would be a much more natural arrangement to give the four Constitutions of the season first, and then to give the forty-two cases together. But the present arrangement being of old standing, no editor has thought himself warranted to depart from it.

There are two important professional subjects of which it may appear surprising that there is no mention in the “Books of the Epidemics,” I mean sphygmology and contagion. Galen repeatedly declares it as his opinion, that Hippocrates paid no attention to the characters of the arterial pulse, and that the subject was not at all studied until after his time; and as far as I can see there is no ground for calling in question this opinion of Galen. Herophilus, in fact, would appear to have been the first person that made any progress in this study. It is more remarkable that Hippocrates should omit all allusion to the other subject, more especially as the contagiousness of certain diseases would appear to have been the popular belief of his age. Thus his contemporary, Thucydides, in describing the plague, expresses himself in such terms as puts it beyond a doubt that he regarded the disease as being of a contagious nature. And another contemporary, Isocrates, makes such observations on a certain case of empyema, by which he evidently means phthisis pulmonalis, as to show that it also was regarded as being communicable.^[616] How the omission is to be accounted for I do not know, but certain it is that not the least reference to contagion, in any shape, is to be found in any of the Hippocratic treatises.

BOOK I.—OF THE EPIDEMICS.

SEC. I.—CONSTITUTION FIRST.

1. In Thasus,^[617] about the autumnal equinox, and under the Pleiades,^[618] the rains were abundant, constant, and soft, with southerly winds; the winter southerly, the northerly winds faint, droughts; on the whole, the winter having the character of spring. The spring was southerly, cool, rains small in quantity. Summer, for the most part, cloudy, no rain, the Etesian winds, rare and small, blew in an irregular manner. The whole constitution of the season being thus inclined to the southerly, and with droughts early in the spring, from the preceding opposite and northerly state, ardent fevers occurred in a few instances, and these very mild, being rarely attended with hemorrhage, and never proving fatal.^[619] Swellings appeared about the ears, in many on either side, and in the greatest number on both sides, being unaccompanied by fever so as not to confine the patient to bed; in all cases they disappeared without giving trouble, neither did any of them come to suppuration, as is common in swellings from other causes. They were of a lax, large, diffused character, without inflammation or pain, and they went away without any critical sign. They seized children, adults, and mostly those who were engaged in the exercises of the palestra and gymnasium, but seldom attacked women. Many had dry coughs without expectoration, and accompanied with hoarseness of voice. In some instances earlier, and in others later, inflammations with pain seized sometimes one of the testicles, and

sometimes both;^[620] some of these cases were accompanied with fever and some not; the greater part of these were attended with much suffering. In other respects they were free of disease, so as not to require medical assistance.^[621]

2. Early in the beginning of spring, and through the summer, and towards winter, many of those who had been long gradually declining, took to bed with symptoms of phthisis; in many cases formerly of a doubtful character the disease then became confirmed; in these the constitution inclined to the phthisical.^[622] Many, and, in fact, the most of them, died; and of those confined to bed, I do not know if a single individual survived for any considerable time; they died more suddenly than is common in such cases. But other diseases, of a protracted character, and attended with fever, were well supported, and did not prove fatal: of these we will give a description afterwards. Consumption was the most considerable of the diseases which then prevailed, and the only one which proved fatal to many persons. Most of them were affected by these diseases in the following manner: fevers accompanied with rigors, of the continual type, acute, having no complete intermissions, but of the form of the semitertians, being milder the one day, and the next having an exacerbation, and increasing in violence; constant sweats, but not diffused over the whole body; extremities very cold, and warmed with difficulty; bowels disordered, with bilious, scanty, unmixed, thin, pungent, and frequent dejections. The urine was thin, colourless, unconcocted, or thick, with a deficient sediment, not settling favorably, but casting down a crude and unseasonable sediment. Sputa small, dense, concocted, but brought up rarely and with difficulty; and in those who encountered the most violent symptoms there was no concoction at all, but they continued throughout spitting crude matters. Their fauces, in most of them, were painful from first to last, having redness with inflammation; defluxions thin, small and acrid; they were soon wasted and became worse, having no appetite for any kind of food throughout; no thirst; most persons delirious when near death. So much concerning the phthisical affections.^[623]

3. In the course of the summer and autumn many fevers of the continual type, but not violent;^[624] they attacked persons who had been long indisposed, but who were otherwise not in an uncomfortable state. In most cases the bowels were disordered in a very moderate degree, and they did not suffer thereby in any manner worth mentioning; the urine was generally well coloured, clear, thin, and after a time becoming concocted near the crisis. They had not much cough, nor was it troublesome; they were not deficient in appetite, for it was necessary to give them food, (on the whole, persons labouring under phthisis were not affected in the usual manner).^[625] They were affected with fevers, rigors, and deficient sweats, with varied and irregular paroxysms, in general not intermitting, but having exacerbations in the tertian form. The earliest crisis which occurred was about the twentieth day, in most about the fortieth, and in many about the eightieth. But there were cases in which it did not leave them thus at all, but in an irregular manner, and without any crisis; in most of these the fevers, after a brief interval, relapsed again; and from these relapses they came to a crisis in the same periods; but in many they were prolonged so that the disease was not gone at the approach of winter. Of all those which are described under this constitution, the phthisical diseases alone were of a fatal character; for in all the others the patients bore up well, and did not die of the other fevers.^[626]

SEC. II.—CONSTITUTION SECOND.

1. In Thasus, early in autumn, the winter suddenly set in rainy before the usual time, with much northerly and southerly winds. These things all continued so during the season of the Pleiades, and until their setting.^[627] The winter was northerly, the rains frequent, in torrents, and large, with snow, but with a frequent mixture of fair weather. These things were all so, but the setting in of the cold was not much out of season. After the winter solstice, and at the time when the zephyr usually begins to blow, severe winterly storms out of season, with much northerly wind, snow, continued and copious rains; the sky tempestuous and clouded; these things were protracted, and did not remit until the equinox. The spring was cold, northerly, rainy, and clouded; the summer was not very sultry, the Etesian winds blew constant, but quickly afterwards, about the rising of Arcturus, there were again many rains with north winds. The whole season being wet, cold, and northerly, people were, for the most part, healthy during winter; but early in the spring very many, indeed, the greater part, were valetudinary. At first ophthalmies set in, with rheums, pains, unconcocted discharges, small concretions, generally breaking with difficulty, in most instances they relapsed, and they did not cease until late in autumn.^[628] During summer and autumn there were dysenteric affections, attacks of tenesmus and lientery, bilious diarrhœa, with thin, copious, undigested, and acrid dejections, and sometimes with watery stools; many had copious defluxions, with pain, of a bilious, watery, slimy, purulent nature, attended with strangury, not connected with disease of the kidneys, but one complaint succeeding the other; vomitings of bile, phlegm, and undigested food, sweats, in all cases a redundancy of humors. In many instances these complaints were unattended with fever, and did not prevent the patients from walking about, but some cases were febrile, as will be described. In some all those described below occurred with pain. During autumn, and at the commencement of winter, there were phthisical complaints, continual fevers; and, in a few cases, ardent; some diurnal, others nocturnal, semitertians, true tertians, quartans, irregular fevers. All the fevers which are described attacked great numbers. The ardent fevers attacked the smallest numbers, and the patients suffered the least from them, for there were no hemorrhages, except a few and to a small amount, nor was there delirium; all the other complaints were slight; in these the crises were regular, in most instances, with the intermittents, in seventeen days; and I

know no instance of a person dying of *causus*, nor becoming phrenitic.^[629] The tertians were more numerous than the ardent fevers, and attended with more pain,^[630] but these all had four periods in regular succession from the first attack, and they had a complete crisis in seven, without a relapse in any instance. The quartans attacked many at first, in the form of regular quartans, but in no few cases a transition from other fevers and diseases into quartans took place; they were protracted, as is wont with them, indeed, more so than usual. Quotidian, nocturnal, and wandering fevers attacked many persons, some of whom continued to keep up, and others were confined to bed. In most instances these fevers were prolonged under the Pleiades and till winter. Many persons, and more especially children, had convulsions from the commencement;^[631] and they had fever, and the convulsions supervened upon the fevers; in most cases they were protracted, but free from danger, unless in those who were in a deadly state from other complaints. Those fevers which were continual in the main, and with no intermissions, but having exacerbations in the tertian form,^[632] there being remissions the one day and exacerbations the next, were the most violent of all those which occurred at that time, and the most protracted, and occurring with the greatest pains, beginning mildly, always on the whole increasing, and being exacerbated, and always turning worse, having small remissions, and after an abatement having more violent paroxysms, and growing worse, for the most part, on the critical days. Rigors, in all cases, took place in an irregular and uncertain manner, very rare and weak in them, but greater in all other fevers; frequent sweats, but most seldom in them, bringing no alleviation, but, on the contrary, doing mischief. Much cold of the extremities in them, and these were warmed with difficulty. Insomnolency, for the most part, especially in these fevers, and again a disposition to coma. The bowels, in all diseases, were disordered, and in a bad state, but worst of all in these. The urine, in most of them, was either thin and crude, yellow, and after a time with slight symptoms of concoction in a critical form, or having the proper thickness, but muddy, and neither settling nor subsiding; or having small and bad, and crude sediments; these being the worst of all. Coughs attended these fevers, but I cannot state that any harm or good ever resulted from the cough. The most of these were protracted and troublesome, went on in a very disorderly and irregular form, and, for the most part, did not end in a crisis, either in the fatal cases or in the others; for if it left some of them for a season it soon returned again. In a few instances the fever terminated with a crisis; in the earliest of these about the eightieth day, and some of these relapsed, so that most of them were not free from the fever during the winter; but the fever left most of them without a crisis, and these things happened alike to those who recovered and to those who did not. There being much want of crisis and much variety as to these diseases, the greatest and worst symptom attended the most of them, namely, a loathing of all articles of food, more especially with those who had otherwise fatal symptoms; but they were not unseasonably thirsty in such fevers. After a length of time, with much suffering and great wasting, abscesses were formed in these cases, either unusually large, so that the patients could not support them, or unusually small, so that they did no good, but soon relapsed and speedily got worse. The diseases which attacked them were in the form of dysenteries, tenesmus, lientery, and fluxes; but, in some cases, there were dropsies, with or without these complaints. Whatever attacked them violently speedily cut them off, or again, did them no good. Small rashes, and not corresponding to the violence of the disease, and quickly disappearing, or swellings occurred about the ears, which were not resolved, and brought on no crisis.^[633] In some they were determined to the joints, and especially to the hip-joint, terminating critically with a few, and quickly again increasing to its original habit. Persons died of all these diseases, but mostly of these fevers, and especially infants just weaned, and older children, until eight or ten years of age, and those before puberty. These things occurred to those affected with the complaints described above, and to many persons at first without them. The only favorable symptom, and the greatest of those which occurred, and what saved most of those who were in the greatest dangers, was the conversion of it to a strangury, and when, in addition to this, abscesses were formed.^[634] The strangury attacked, most especially, persons of the ages I have mentioned, but it also occurred in many others, both of those who were not confined to bed and those who were. There was a speedy and great change in all these cases. For the bowels, if they happened previously to have watery discharges of a bad character, became regular, they got an appetite for food, and the fevers were mild afterwards. But, with regard to the strangury itself, the symptoms were protracted and painful. Their urine was copious, thick, of various characters, red, mixed with pus, and was passed with pain. These all recovered, and I did not see a single instance of death among them.

5.^[635] With regard to the dangers of these cases, one must always attend to the seasonable concoction of all the evacuations, and to the favorable and critical abscesses. The concoctions indicate a speedy crisis and recovery of health; crude and undigested evacuations, and those which are converted into bad abscesses, indicate either want of crisis, or pains, or prolongation of the disease, or death, or relapses; which of these it is to be must be determined from other circumstances. *The physician must be able to tell the antecedents, know the present, and foretell the future—must meditate these things, and have two special objects in view with regard to diseases, namely, to do good or to do no harm. The art consists in three things—the disease, the patient, and the physician. The physician is the servant of the art, and the patient must combat the disease along with the physician.*^[636]

6. Pains about the head and neck, and heaviness of the same along with pain, occur either without fevers or in fevers. Convulsions occurring in persons attacked with frenzy, and having vomitings of verdigris-green bile, in some cases quickly prove fatal. In ardent fevers, and in those other fevers in which there is pain of the neck, heaviness of the temples, mistiness

about the eyes, and distention about the hypochondriac region, not unattended with pain, hemorrhage from the nose takes place,^[637] but those who have heaviness of the whole head, cardialgia and nausea, vomit bilious and pituitous matters; children, in such affections, are generally attacked with convulsions, and women have these and also pains of the uterus; whereas, in elder persons, and those in whom the heat is already more subdued, these cases end in paralysis, mania, and loss of sight.

THIRD CONSTITUTION.

7. In Thasus, a little before and during the season of Arcturus,^[638] there were frequent and great rains, with northerly winds. About the equinox, and till the setting of the Pleiades, there were a few southerly rains: the winter northerly and parched, cold, with great winds and snow. Great storms about the equinox, the spring northerly, dryness, rains few and cold. About the summer solstice, scanty rains, and great cold until near the season of the Dog-star.^[639] After the Dog-days, until the season of Arcturus, the summer hot, great droughts, not in intervals, but continued and severe: no rain; the Etesian winds blew; about the season of Arcturus southerly rains until the equinox.

8. In this state of things, during winter, paraplegia set in, and attacked many, and some died speedily; and otherwise the disease prevailed much in an epidemical form, but persons remained free from all other diseases.^[640] Early in the spring, ardent fevers commenced and continued through the summer until the equinox. Those then that were attacked immediately after the commencement of the spring and summer, for the most part recovered, and but few of them died. But when the autumn and the rains had set in, they were of a fatal character, and the greater part then died.^[641] When in these attacks of ardent fevers there was a proper and copious hemorrhage from the nose, they were generally saved by it, and I do not know a single person who had a proper hemorrhage who died in this constitution. Philiscus, Epaminon, and Silenus, indeed, who had a trifling epistaxis on the fourth and fifth day, died.^[642] The most of those seized with the disease had a rigor about the time of the crisis, and especially those who had no hemorrhage; these had also the rigor associated. Some were attacked with jaundice on the sixth day,^[643] but these were benefited either by an urinary purgation, or a disorder of the bowels, or a copious hemorrhage, as in the case of Heraclides, who was lodged with Aristocydes: this person, though he had the hemorrhage from the nose, the purgation by the bladder, and disorder of the bowels, experienced a favorable crisis on the twentieth day, not like the servant of Phanagoras, who had none of these symptoms, and died. The hemorrhages attacked most persons, but especially young persons and those in the prime of life, and the greater part of those who had not the hemorrhage died:^[644] elderly persons had jaundice or disorder of the bowels, such as Bion, who was lodged with Silenus. Dysenteries were epidemical during the summer, and some of those cases in which the hemorrhage occurred, terminated in dysentery, as happened to the slave of Eraton, and to Mullus, who had a copious hemorrhage, which settled down into dysentery, and they recovered. This humor was redundant in many cases, since in those who had not the hemorrhage about the crisis, but the risings about the ears disappeared, after their disappearance there was a sense of weight in the left flank extending to the extremity of the hip, and pain setting in after the crisis, with a discharge of thin urine; they began to have small hemorrhages about the twenty-fourth day, and the swelling was converted into the hemorrhage. In the case of Antiphon, the son of Critobulus, the fever ceased and came to a crisis about the fortieth day. Many women were attacked, but fewer than of the men, and there were fewer deaths among them. But most of them had difficult parturition, and after labor they were taken ill, and these most especially died, as, for example, the daughter of Telebolus died on the sixth day after delivery.^[645] Most females had the menstrual discharge during the fever, and many girls had it then for the first time: in certain individuals both the hemorrhage from the nose and the menses appeared; thus, in the case of the virgin daughter of Dætharses, the menses then took place for the first time, and she had also a copious hemorrhage from the nose, and I knew no instance of any one dying when one or other of these took place properly. But all those in the pregnant state that were attacked had abortions, as far as I observed. The urine in most cases was of the proper color, but thin, and having scanty sediments:^[646] in most the bowels were disordered with thin and bilious dejections; and many, after passing through the other crises, terminated in dysenteries, as happened to Xenophanes and Critias. The urine was watery, copious, clear, and thin; and even after the crises, when the sediment was natural, and all the other critical symptoms were favorable, as I recollect having happened to Bion, who was lodged in the house of Silenus, and Critias, who lived with Xenophanes, the slave of Areton, and the wife of Mnesistratus. But afterwards all these were attacked with dysentery. It would be worth while to inquire whether the watery urine was the cause of this.^[647] About the season of Arcturus many had the crisis on the eleventh day, and in them the regular relapses did not take place, but they became comatose about this time, especially children; but there were fewest deaths of all among them.

9. About the equinox, and until the season of the Pleiades, and at the approach of winter, many ardent fevers set in; but great numbers at that season were seized with phrenitis, and many died.^[648] a few cases also occurred during the summer. These then made their attack at the commencement of ardent fevers, which were attended with fatal symptoms; for immediately upon their setting in, there were acute fever and small rigors, insomnolency, aberration, thirst, nausea, insignificant sweats about the forehead and clavicles, but no general perspiration; they had much delirious talking, fears,

despondency, great coldness of the extremities, in the feet, but more especially in their hands: the paroxysms were on the even days; and in most cases, on the fourth day, the most violent pains set in, with sweats, generally coldish, and the extremities could not be warmed, but were livid and rather cold, and they had then no thirst; in them the urine was black, scanty, thin, and the bowels were constipated; there was an hemorrhage from the nose in no case in which these symptoms occurred, but merely a trifling epistaxis; and none of them had a relapse, but they died on the sixth day with sweats.^[649] In the phrenitic cases, all the symptoms which have been described did not occur, but in them the disease mostly came to a crisis on the eleventh day, and in some on the twentieth. In those cases in which the phrenitis did not begin immediately, but about the third or fourth day, the disease was moderate at the commencement, but assumed a violent character about the seventh day. There was a great number of diseases, and of those affected, they who died were principally infants, young persons, adults having smooth bodies, white skins, straight and black hair, dark eyes, those living recklessly and luxuriously; persons with shrill, or rough voices, who stammered and were passionate, and women more especially died from this form. In this constitution, four symptoms in particular proved salutary; either a hemorrhage from the nose, or a copious discharge by the bladder of urine, having an abundant and proper sediment, or a bilious disorder of the bowels at the proper time, or an attack of dysentery.^[650] And in many cases it happened, that the crisis did not take place by any one of the symptoms which have been mentioned, but the patient passed through most of them, and appeared to be in an uncomfortable way, and yet all who were attacked with these symptoms recovered. All the symptoms which I have described occurred also to women and girls; and whoever of them had any of these symptoms in a favorable manner, or the menses appeared abundantly, were saved thereby, and had a crisis, so that I do not know a single female who had any of these favorably that died. But the daughter of Philo, who had a copious hemorrhage from the nose, and took supper unseasonably on the seventh day, died. In those cases of acute, and more especially of ardent fevers, in which there is an involuntary discharge of tears, you may expect a hemorrhage from the nose, unless the other symptoms be of a fatal character, for in those of a bad description, they do not indicate a hemorrhage, but death. Swellings about the ears, with pain in fevers, sometimes when the fever went off critically, neither subsided nor were converted into pus; in these cases a bilious diarrhoea, or dysentery, or thick urine having a sediment, carried off the disease, as happened to Hermippus of Clazomenæ. The circumstances relating to crises, as far as we can recognize them, were so far similar and so far dissimilar. Thus two brothers became ill at the same hour (they were brothers of Epigenes, and lodged near the theatre), of these the elder had a crisis on the sixth day, and the younger on the seventh, and both had a relapse at the same hour; it then left them for five days, and from the return of the fever both had a crisis together on the seventeenth day. Most had a crisis on the sixth day; it then left them for six days, and from the relapse there was a crisis on the fifth day.^[651] But those who had a crisis on the seventh day, had an intermission for seven days; and the crisis took place on the third day after the relapse. Those who had a crisis on the sixth day, after an interval of six days were seized again on the third, and having left them for one day, the fever attacked them again on the next and came to a crisis, as happened to Evagon the son of Dætharses. Those in whom the crisis happened on the sixth day, had an intermission of seven days, and from the relapse there was a crisis on the fourth, as happened to the daughter of Aglaïdas. The greater part of those who were taken ill under this constitution of things, were affected in this manner, and I did not know a single case of recovery, in which there was not a relapse agreeably to the stated order of relapses; and all those recovered in which the relapses took place according to this form: nor did I know a single instance of those who then passed through the disease in this manner who had another relapse. In these diseases death generally happened on the sixth day, as happened to Epaminondas, Silenus, and Philiscus the son of Antagoras. Those who had parotid swellings experienced a crisis on the twentieth day, but in all these cases the disease went off without coming to a suppuration, and was turned upon the bladder. But in Cratistonax, who lived by the temple of Hercules, and in the maid servant of Scymnus the fuller, it turned to a suppuration, and they died. Those who had a crisis on the seventh day, had an intermission of nine days, and a relapse which came to a crisis on the fourth day from the return of the fever, as was the case with Pantacles, who resided close by the temple of Bacchus. Those who had a crisis on the seventh day, after an interval of six days had a relapse, from which they had a crisis on the seventh day, as happened to Phanocritus, who was lodged with Gnathon the painter. During the winter, about the winter solstices, and until the equinox, the ardent fevers and frenzies prevailed, and many died. The crisis, however, changed, and happened to the greater number on the fifth day from the commencement, left them for four days and relapsed; and after the return, there was a crisis on the fifth day, making in all fourteen days. The crisis took place thus in the case of most children, also in elder persons. Some had a crisis on the eleventh day, a relapse on the fourteenth, a complete crisis on the twentieth; but certain persons, who had a rigor about the twentieth, had a crisis on the fortieth. The greater part had a rigor along with the original crisis, and these had also a rigor about the crisis in the relapse. There were fewest cases of rigor in the spring, more in summer, still more in autumn, but by far the most in winter; then hemorrhages ceased.

SEC. III.

10. With regard to diseases, the circumstances from which we form a judgment of them are,—by attending to the general nature of all, and the peculiar nature of each individual,—to the disease, the patient, and the applications,—to the person who applies them, as that makes a difference for better or for worse,—to the whole constitution of the season, and particularly to

the state of the heavens, and the nature of each country;—to the patient's habits, regimen, and pursuits;—to his conversation, manners, taciturnity, thoughts, sleep, or absence of sleep, and sometimes his dreams, what and when they occur;—to his picking and scratching;^[652]—to his tears;—to the alvine discharges, urine, sputa, and vomitings; and to the changes of diseases from the one into the other;—to the deposits, whether of a deadly or critical character;—to the sweat, coldness, rigor, cough, sneezing, hiccup, respiration, eructation, flatulence, whether passed silently or with a noise;—to hemorrhages and hemorrhoids;—from these, and their consequences, we must form our judgment.^[653]

11. Fevers are,—the continual, some of which hold during the day and have a remission at night, and others hold during the night and have a remission during the day;^[654] semitertians, tertians, quartans, quintans, septans, nonans. The most acute, strongest, most dangerous, and fatal diseases, occur in the continual fever. The least dangerous of all, and the mildest and most protracted, is the quartan, for it is not only such from itself, but it also carries off other great diseases.^[655] In what is called the semitertian, other acute diseases are apt to occur, and it is the most fatal of all others, and moreover phthisical persons, and those laboring under other protracted diseases, are apt to be attacked by it.^[656] The nocturnal fever is not very fatal, but protracted; the diurnal is still more protracted, and in some cases passes into phthisis. The septan is protracted, but not fatal; the nonan more protracted, and not fatal. The true tertian comes quickly to a crisis, and is not fatal; but the quintan is the worst of all, for it proves fatal when it precedes an attack of phthisis, and when it supervenes on persons who are already consumptive.^[657] There are peculiar modes, and constitutions, and paroxysms, in every one of these fevers; for example,—the continual, in some cases at the very commencement, grows, as it were, and attains its full strength, and rises to its most dangerous pitch, but is diminished about and at the crisis; in others it begins gentle and suppressed, but gains ground and is exacerbated every day, and bursts forth with all its heat about and at the crisis; while in others, again, it commences mildly, increases, and is exacerbated until it reaches its acmé, and then remits until at and about the crisis.^[658] These varieties occur in every fever, and in every disease. From these observations one must regulate the regimen accordingly. There are many other important symptoms allied to these, part of which have been already noticed, and part will be described afterwards, from a consideration of which one may judge, and decide in each case, whether the disease be acute, and whether it will end in death or recovery; or whether it will be protracted, and will end in death or recovery; and in what cases food is to be given, and in what not; and when and to what amount, and what particular kind of food is to be administered.

12. Those diseases which have their paroxysms on even days have their crises on even days; and those which have their paroxysms on uneven days have their crises on uneven days. The first period of those which have the crisis on even days, is the 4th, 6th, 8th, 10th, 14th, 20th, 30th, 40th, 60th, 80th, 100th; and the first period of those which have their crises on uneven days, is the 1st, 3d, 5th, 7th, 9th, 11th, 17th, 21st, 27th, 31st. It should be known, that if the crisis take place on any other day than on those described, it indicates that there will be a relapse, which may prove fatal. But one ought to pay attention, and know in these seasons what crises will lead to recovery and what to death, or to changes for the better or the worse. Irregular fevers, quartans, quintans, septans, and nonans should be studied, in order to find out in what periods their crises take place.

13. FOURTEEN CASES OF DISEASE.^[659]

CASE I.—Philiscus, who lived by the Wall, took to bed on the first day of acute fever; he sweated; towards night was uneasy. On the second day all the symptoms were exacerbated; late in the evening had a proper stool from a small clyster; the night quiet. On the third day, early in the morning and until noon, he appeared to be free from fever; towards evening, acute fever, with sweating, thirst, tongue parched; passed black urine; night uncomfortable, no sleep; he was delirious on all subjects. On the fourth, all the symptoms exacerbated, urine black; night more comfortable, urine of a better color. On the fifth, about mid-day, had a slight trickling of pure blood from the nose; urine varied in character, having floating in it round bodies, resembling semen, and scattered, but which did not fall to the bottom; a suppository having been applied, some scanty flatulent matters were passed; night uncomfortable, little sleep, talking incoherently; extremities altogether cold, and could not be warmed; urine black; slept a little towards day; loss of speech, cold sweats; extremities livid; about the middle of the sixth day he died. The respiration throughout, like that of a person recollecting himself, was rare, and large, and spleen was swelled upon in a round tumor, the sweats cold throughout, the paroxysms on the even days.^[660]

CASE II.—Silenus lived on the Broad-way, near the house of Evalcidas. From fatigue, drinking, and unseasonable exercises, he was seized with fever. He began with having pain in the loins; he had heaviness of the head, and there was stiffness of the neck. On the first day the alvine discharges were bilious, unmixed, frothy, high colored, and copious; urine black, having a black sediment; he was thirsty, tongue dry; no sleep at night. On the second, acute fever, stools more copious, thinner, frothy; urine black, an uncomfortable night, slight delirium. On the third, all the symptoms exacerbated; an oblong distention, of a softish nature, from both sides of the hypochondrium to the navel; stools thin, and darkish; urine muddy, and darkish; no sleep at night; much talking, laughter, singing, he could not restrain himself. On the fourth, in the same state. On the fifth, stools bilious, unmixed, smooth, greasy; urine thin, and transparent; slight absence of delirium. On the sixth, slight perspiration about the head; extremities cold and livid; much tossing about; no passage from the bowels, urine suppressed,

acute fever. On the seventh, loss of speech; extremities could no longer be kept warm; no discharge of urine. On the eighth, a cold sweat all over; red rashes with sweat, of a round figure, small, like *vari*, persistent, not subsiding; by means of a slight stimulus, a copious discharge from the bowels, of a thin and undigested character, with pain; urine acrid, and passed with pain; extremities slightly heated; sleep slight, and comatose; speechless; urine thin, and transparent. On the ninth, in the same state. On the tenth, no drink taken; comatose, sleep slight; alvine discharges the same; urine abundant, and thickish; when allowed to stand, the sediment farinaceous and white; extremities again cold. On the eleventh, he died. At the commencement, and throughout, the respiration was slow and large; there was a constant throbbing in the hypochondrium; his age was about twenty.^[661]

CASE III.—Herophon was seized with an acute fever; alvine discharges at first were scanty, and attended with tenesmus; but afterwards they were passed of a thin, bilious character, and frequent; there was no sleep; urine black, and thin. On the fifth, in the morning, deafness; all the symptoms exacerbated; spleen swollen; distention of the hypochondrium; alvine discharges scanty, and black; he became delirious. On the sixth, delirious; at night, sweating, coldness; the delirium continued. On the seventh, he became cold, thirsty, was disordered in mind; at night recovered his senses; slept. On the eighth, was feverish; the spleen diminished in size; quite collected; had pain at first about the groin, on the same side as the spleen; had pains in both legs; night comfortable; urine better colored, had a scanty sediment. On the ninth, sweated; the crisis took place; fever remitted. On the fifth day afterwards, fever relapsed, spleen immediately became swollen; acute fever; deafness again. On the third day after the relapse, the spleen diminished; deafness less; legs painful; sweated during the night; crisis took place on the seventeenth day; had no disorder of the senses during the relapse.^[662]

CASE IV.—In Thasus, the wife of Philinus, having been delivered of a daughter, the lochial discharge being natural, and other matters going on mildly, on the fourteenth day after delivery was seized with fever, attended with rigor; was pained at first in the cardiac region of the stomach and right hypochondrium; pain in the genital organs; lochial discharge ceased. Upon the application of a pessary all these symptoms were alleviated; pains of the head, neck, and loins remained; no sleep; extremities cold; thirst; bowels in a hot state; stools scanty; urine thin, and colorless at first. On the sixth, towards night, senses much disordered, but again were restored. On the seventh, thirsty; the evacuations bilious, and high colored. On the eighth, had a rigor; acute fever; much spasm, with pain; talked much, incoherently; upon the application of a suppository, rose to stool, and passed copious dejections, with a bilious flux; no sleep. On the ninth, spasms. On the tenth, slightly recollected. On the eleventh, slept; had perfect recollection, but again immediately wandered; passed a large quantity of urine with spasms, (the attendants seldom putting her in mind,) it was thick, white, like urine which has been shaken after it has stood for a considerable time until it has subsided, but it had no sediment; in color and consistence, the urine resembled that of cattle, as far as I observed. About the fourteenth day, startings over the whole body; talked much; slightly collected, but presently became again delirious. About the seventeenth day became speechless, on the twentieth died.^[663]

CASE V.—The wife of Epicrates, who was lodged at the house of Archigetes, being near the term of delivery, was seized with a violent rigor, and, as was said, she did not become heated.^[664] next day the same. On the third, she was delivered of a daughter, and everything went on properly. On the day following her delivery, she was seized with acute fever, pain in the cardiac region of the stomach, and in the genital parts. Having had a suppository, was in so far relieved; pain in the head, neck, and loins; no sleep; alvine discharges scanty, bilious, thin, and unmixed; urine thin, and blackish. Towards the night of the sixth day from the time she was seized with the fever, became delirious. On the seventh, all the symptoms exacerbated; insomnolency, delirium, thirst; stools bilious, and high colored. On the eighth, had a rigor; slept more. On the ninth, the same. On the tenth, her limbs painfully affected; pain again of the cardiac region of the stomach; heaviness of the head; no delirium; slept more; bowels constipated. On the eleventh, passed urine of a better color, and having an abundant sediment; felt lighter. On the fourteenth had a rigor; acute fever. On the fifteenth, had a copious vomiting of bilious and yellow matters; sweated; fever gone; at night acute fever; urine thick, sediment white.^[665] On the seventeenth, an exacerbation; night uncomfortable; no sleep; delirium. On the eighteenth, thirsty; tongue parched; no sleep; much delirium; legs painfully affected. About the twentieth, in the morning, had a slight rigor; was comatose; slept tranquilly; had slight vomiting of bilious and black matters; towards night deafness. About the twenty-first, weight generally in the left side, with pain; slight cough; urine thick, muddy, and reddish; when allowed to stand, had no sediment; in other respects felt lighter; fever not gone; fauces painful from the commencement, and red; uvula retracted; defluxion remained acrid, pungent, and saltish throughout. About the twenty-seventh, free of fever; sediment in the urine; pain in the side. About the thirty-first, was attacked with fever, bilious diarrhœa; slight bilious vomiting on the fortieth. Had a complete crisis, and was freed from the fever on the eightieth day.^[666]

CASE VI.—Cleonactides, who was lodged above the Temple of Hercules, was seized with a fever in an irregular form; was pained in the head and left side from the commencement, and had other pains resembling those produced by fatigue: paroxysms of the fevers inconstant and irregular; occasional sweats; the paroxysms generally attacked on the critical days. About the twenty-fourth was cold in the extremities of the hands, vomitings bilious, yellow, and frequent, soon turning to a verdigris-green color; general relief. About the thirtieth, began to have hemorrhage from both nostrils, and this continued in an irregular manner until near the crisis; did not loathe food, and had no thirst throughout, nor was troubled with insomnolency;

urine thin, and not devoid of color. When about the thirtieth day, passed reddish urine, having a copious red sediment; was relieved, but afterwards the characters of the urine varied, sometimes having sediment, and sometimes not. On the sixtieth, the sediment in the urine copious, white, and smooth; all the symptoms ameliorated; intermission of the fever; urine thin, and well colored. On the seventieth, fever gone for ten days. On the eightieth had a rigor, was seized with acute fever, sweated much; a red, smooth sediment in the urine; and a perfect crisis.^[667]

CASE VII.—Meton was seized with fever; there was a painful weight in the loins. Next day, after drinking water pretty copiously, had proper evacuations from the bowels. On the third, heaviness of the head, stools thin, bilious, and reddish. On the fourth, all the symptoms exacerbated; had twice a scanty trickling of blood from the right nostril; passed an uncomfortable night; alvine discharges like those on the third day; urine darkish, had a darkish cloud floating in it, of a scattered form, which did not subside. On the fifth, a copious hemorrhage of pure blood from the left nostril; he sweated, and had a crisis. After the fever restless, and had some delirium; urine thin, and darkish; had an affusion of warm water on the head; slept and recovered his senses. In this case there was no relapse, but there were frequent hemorrhages after the crisis.^[668]

CASE VIII.—Erasinus, who lived near the Canal of Bootes, was seized with fever after supper; passed the night in an agitated state. During the first day quiet, but in pain at night. On the second, symptoms all exacerbated; at night delirious. On the third, was in a painful condition; great incoherence. On the fourth, in a most uncomfortable state; had no sound sleep at night, but dreaming and talking; then all the appearances worse, of a formidable and alarming character; fear, impatience. On the morning of the fifth, was composed, and quite coherent, but long before noon was furiously mad, so that he could not constrain himself; extremities cold, and somewhat livid; urine without sediment; died about sunset. The fever in this case was accompanied by sweats throughout; the hypochondria were in a state of meteorism, with distention and pain; the urine was black, had round substances floating in it, which did not subside; the alvine evacuations were not stopped; thirst throughout not great; much spasms with sweats about the time of death.^[669]

CASE IX.—Criton, in Thasus, while still on foot, and going about, was seized with a violent pain in the great toe; he took to bed the same day, had rigors and nausea, recovered his heat slightly, at night was delirious. On the second, swelling of the whole foot, and about the ankle erythema, with distention, and small bullæ (phlyctænæ); acute fever; he became furiously deranged; alvine discharges bilious, unmixed, and rather frequent. He died on the second day from the commencement.^[670]

CASE X.—The Clazomenian who was lodged by the Well of Phrynichides was seized with fever. He had pain in the head, neck, and loins from the beginning, and immediately afterwards deafness; no sleep, acute fever, hypochondria elevated with a swelling, but not much distention; tongue dry. On the fourth, towards night, he became delirious. On the fifth, in an uneasy state. On the sixth, all the symptoms exacerbated. About the eleventh a slight remission; from the commencement to the fourteenth day the alvine discharges thin, copious, and of the color of water, but were well supported; the bowels then became constipated. Urine throughout thin, and well colored, and had many substances scattered through it, but no sediment. About the sixteenth, urine somewhat thicker, which had a slight sediment; somewhat better, and more collected. On the seventeenth, urine again thin; swellings about both his ears, with pain; no sleep, some incoherence; legs painfully affected. On the twentieth, free of fever, had a crisis, no sweat, perfectly collected. About the twenty-seventh, violent pain of the right hip; it speedily went off. The swellings about the ears subsided, and did not suppurate, but were painful. About the thirty-first, a diarrhœa, attended with a copious discharge of watery matter, and symptoms of dysentery; passed thick urine; swellings about the ears gone. About the fortieth day, had pain in the right eye, sight dull. It went away.^[671]

CASE XI.—The wife of Dromeades having been delivered of a female child, and all other matters going on properly, on the second day after was seized with rigor and acute fever. Began to have pain about the hypochondrium on the first day; had nausea and incoherence, and for some hours afterwards had no sleep; respiration rare, large, and suddenly interrupted. On the day following that on which she had the rigor, alvine discharges proper; urine thick, white, muddy, like urine which has been shaken after standing for some time, until the sediment had fallen to the bottom; it had no sediment; she did not sleep during the night. On the third day, about noon, had a rigor, acute fever; urine the same; pain of the hypochondria, nausea, an uncomfortable night, no sleep; a coldish sweat all over, but heat quickly restored. On the fourth, slight alleviation of the symptoms about the hypochondria; heaviness of the head, with pain; somewhat comatose; slight epistaxis, tongue dry, thirst, urine thin and oily; slept a little, upon awaking was somewhat comatose; slight coldness, slept during the night, was delirious. On the morning of the sixth had a rigor, but soon recovered her heat, sweated all over; extremities cold, was delirious, respiration rare and large. Shortly afterwards spasms from the head began, and she immediately expired.^[672]

CASE XII.—A man, in a heated state, took supper, and drank more than enough; he vomited the whole during the night; acute fever, pain of the right hypochondrium, a softish inflammation from the inner part; passed an uncomfortable night; urine at the commencement thick, red, but when allowed to stand, had no sediment, tongue dry, and not very thirsty. On the fourth, acute fever, pains all over. On the fifth, urine smooth, oily, and copious; acute fever. On the sixth, in the evening, very incoherent, no sleep during the night. On the seventh, all the symptoms exacerbated; urine of the same characters; much talking, and he could not contain himself; the bowels being stimulated, passed a watery discharge with lumbrici: night equally

painful. In the morning had a rigor; acute fever, hot sweat, appeared to be free of fever; did not sleep long; after the sleep a chill, ptyalism; in the evening, great incoherence; after a little, vomited a small quantity of dark bilious matters. On the ninth, coldness, much delirium, did not sleep. On the tenth, pains in the limbs, all the symptoms exacerbated; he was delirious. On the eleventh, he died.^[673]

CASE XIII.—A woman, who lodged on the Quay, being three months gone with child, was seized with fever, and immediately began to have pains in the loins. On the third day, pain of the head and neck, extending to the clavicle, and right hand; she immediately lost the power of speech; was paralyzed in the right hand, with spasms, after the manner of paraplegia; was quite incoherent; passed an uncomfortable night; did not sleep; disorder of the bowels, attended with bilious, unmixed, and scanty stools. On the fourth, recovered the use of her tongue; spasms of the same parts, and general pains remained; swelling in the hypochondrium, accompanied with pain; did not sleep, was quite incoherent; bowels disordered, urine thin, and not of a good color. On the fifth, acute fever; pain of the hypochondrium, quite incoherent; alvine evacuations bilious; towards night had a sweat, and was freed from the fever. On the sixth, recovered her reason; was every way relieved; the pain remained about the left clavicle; was thirsty, urine thin, had no sleep. On the seventh trembling, slight coma, some incoherence, pains about the clavicle and left arm remained; in all other respects was alleviated; quite coherent. For three days remained free from fever. On the eleventh, had a relapse, with rigor and fever. About the fourteenth day, vomited pretty abundantly bilious and yellow matters, had a sweat, the fever went off, by coming to a crisis.^[674]

CASE XIV.—Melidia, who lodged near the Temple of Juno, began to feel a violent pain of the head, neck, and chest. She was straightway seized with acute fever; a slight appearance of the menses; continued pains of all these parts. On the sixth, was affected with coma, nausea, and rigor; redness about the cheeks; slight delirium. On the seventh, had a sweat; the fever intermitted, the pains remained. A relapse; little sleep; urine throughout of a good color, but thin; the alvine evacuations were thin, bilious, acrid, very scanty, black, and fetid; a white, smooth sediment in the urine; had a sweat, and experienced a perfect crisis on the eleventh day.^[675]

Though in the Argument prefixed to the First Book of the Epidemics I have given a pretty full summary of the contents both of that book and the third, I have still a few observations to make on some important points, which were not sufficiently considered on that occasion; and this I do the more readily, as it will afford me an opportunity of noticing a subject on which M. Littré has bestowed very extensive research. I allude to the origin of the Glandular Plague. As I make it a rule, in giving these my annotations, not to enter into any lengthy details, I shall now state, in a very succinct manner, the result of my inquiries. The reader is referred, for a fuller discussion of the subject, to the more ample disquisitions of M. Littré.^[676]

The opinion has been pretty generally maintained by modern authorities, that the first description which we have of the glandular plague of the East is that given by the historian Procopius, in the sixth century; and the inference drawn therefrom is that the disease was unknown until his time. This opinion is still held, to a certain extent, by Hecker, Rosenbaum, Pariset, Nauman, and others of the most distinguished scholars of the day, but it appears to be untenable after the discovery of the “Fragment” of Ruffus, published by Mai, Rome, 1831. As the passage is very important, I shall give a translation of it in this place. It is as follows: “The buboes called pestilential are most fatal and acute, especially those which are seen occurring about Libya, Egypt, and Syria, and which are mentioned by Dionysius Curtus. Dioscorides and Posidonius make much mention of them in the plague which occurred in their time in Libya; they say it was accompanied by acute fever, pain, and prostration of the whole body, delirium, and the appearance of large and hard buboes, which did not suppurate, not only in the accustomed parts, but also in the groins and armpits.” The only thing which detracts from the value of this paragraph is the difficulty of determining exactly who the authorities are which are referred to in it. Of Dionysius Curtus nothing is known; indeed it is more than probable, that there is some mistake in this name. There are several medical authors of the name of Dioscorides and Posidonius, and it is difficult to determine to which of them reference is here made. Still, however, there seems to be no reason for questioning the authenticity of the passage. Ruffus, I may add, is generally admitted to have flourished in the reign of Trajan.^[677]

To this important document let me join an interesting extract from Galen’s work “On Fevers.” Galen, *treating professedly of Pestilential Fevers*, which he maintains are all connected with a tendency to putridity, expresses himself as follows: “Moreover, as Hippocrates says, all fevers from buboes are bad, with the exception of ephemerals; although the bubo is also of the class of phlegmons. And I agree in so far with what is said of putrefaction, for this is the cause of the fever in inflammations, and not as Erasistratus supposed.^[678] But yet there are certain fevers from buboes of the class of ephemerals, as certain others proclaim them to be; diseases difficult to cure, which derive their origin from an inflammation, an ulcer, an abscess, or some other such affection in a viscus. But the ephemeral fevers from buboes differ from those connected with putrefactions, either in a certain viscus, or in the hollow and very large vessels, that in those from buboes, which always impart their heat to the surrounding parts, the heat is communicated to the heart, and the putrefactive fume does not reach it, but remains circumscribed in the seat of the bubo, and the heat reaching the heart solely by a change in the connecting parts, in like manner as in those exposed to excessive heat and fatigue, the diffusion of the heat takes place from the parts first warmed to the source of vitality; but in a putrefaction about the viscera and large vessels, a fume, as it were, from the putrefying humors reaches the cavities of the heart, etc.”^[679] From these two passages alone, without taking into account several others of less importance, which might be gathered from other medical authorities,^[680] it must be quite obvious that the glandular plague was known, at all events, in the second century of the Christian era. Moreover it is equally clear, that Galen did not look upon it as a new disease, but considered that it was noticed in the works of Hippocrates. To my mind, then, there can be no doubt that the pestilence which prevailed during the Peloponnesian war partook of the nature of the glandular plague. What has tended to create doubts on this subject, in the minds of many learned men, is the omission of any distinct mention of buboes in the graphic description of it given by Thucydides. But it should always be taken into account that Thucydides was not a professional man, and therefore there is a strong presumption that his acquaintance with the disease, even although, as he states, he himself had experienced an attack of it, must have been altogether of a general nature. Indeed Galen, both in the treatise from which I have quoted above and in many other parts of his works, does not hesitate to declare, that the historian describes the disease as a common, that is to say, a non-professional man, whereas Hippocrates gives its characters as a physician. It is also to be borne in mind, that the description of it given by Thucydides applies to it only at its outbreak in the city of Athens, and it is a well-known characteristic of pestilential epidemics that they change very much during their progress. This character of them was well illustrated in the Plague of Aleppo, so admirably described by Dr. Russel; for although the glandular form of the disease prevailed in a large number of cases, a considerable proportion of them were unaffected with buboes. Indeed it appears to me to be too much the practice for the profession, as well as the public, to imagine to themselves a certain type or ideal of every disease, and when they do not recognize the exact characters which they fancy it should present, they immediately set down such cases as constituting an entirely different disease. This is an error that is constantly committed, and one which I believe to be at the bottom of the discordant opinions which prevail among professional men, on

the subject of the glandular plague. It would be well for the physician to bear in mind how many varieties of symptoms the fever designated as Typhus puts on,—some with the rash reckoned peculiar to this fever, and some without it,—some with petechiæ, and miliary eruptions, and others without them; and many other complications of symptoms, which are sometimes present and sometimes not.

With regard to the hypothesis lately advanced by Mr. Theod. Krause,^[681] and in so far countenanced by M. Littré, that the plague of Athens was an epidemical variola, I must say that I can see no probability in this supposition; for that a disease so strongly marked as smallpox should have prevailed in ancient times, and yet not be distinctly noticed by the Greek and Roman writers on medicine, I cannot conceive, more especially when we call to recollection the very accurate descriptions which they have left us of other cutaneous diseases, by no means attended with symptoms of so obvious a nature. Indeed it appears to me most wonderful, that such an opinion should have been entertained by any person at all acquainted with the Arabic writers on medicine, who described most distinctly both the plague and the smallpox. Not to lose ourselves amidst a host of authorities, I would refer the reader, in particular, to Avicenna, iv., 1, 4, where the two diseases are treated of most distinctly, so that I cannot entertain a doubt that the Arabian physicians considered them to be essentially different.

In a considerable number of the cases reported in this book, there are affixed to them in the original certain characters, the interpretation of which the reader will find given in the translation. It will be necessary, then, to give the reader some account of the origin of these characters, regarding which our sole authority is Galen, who, in his Commentaries on this book, enters on the question in his usual elegant and attractive style. He admits that he derived his information principally from Zeuxis, one of his predecessors in the office of commenting upon the works of Hippocrates. (See § 2, of the Preliminary Discourse.) It appears that Ptolemy Philadelphus was so zealous in his search for books to adorn his library, in Alexandria, that he gave instructions to the masters of ships going on distant voyages to collect all the books they could procure, and bring them back with them; that he ordered copies to be taken of books brought to him in this way, and kept the originals, but returned the copies, along with large sums of money, in certain cases, to those who had lent them to him; and that the works so obtained were preserved in a separate department of the library, with the inscription, “The Books of the Ships.” Among these was found a copy of the Third Book of the Epidemics, with the inscription, “One of the Books of the Ships, according to the *redacteur* Memnon of Sida.” Others say, that the term “*redacteur*” was wanting, and that the book bore simply the inscription of “Memnon;” and that the servants of the king inscribed the names of all the seamen who had brought these books, when they were installed on the shelves of the library. This, it would seem, was not done immediately after their arrival in Alexandria, but that at first they were collected together in certain houses. Memnon, the librarian, then, is generally supposed to have surreptitiously introduced the characters into one of the copies, in order that he might raise himself into importance by interpreting them. But whether or not this ruse was actually perpetrated by Memnon, the general belief of the commentators was, that Hippocrates himself had nothing to do with them. In fact, Zeno would appear to have been the only commentator who held them to be genuine, and ascribed the introduction of them to our author. The opinion thus advanced by Zeno led him into a violent controversy with the two Apollonii, namely, the Empiric and Biblas, who strenuously maintained that the characters were an interpolation executed by Memnon. This came to be the settled opinion of the commentators, and among others of Galen, who, although he gives a key to the interpretation of the characters, maintains, on all occasions, that they are of no authority, and had in fact been forged by Memnon.

The following is the key which Galen gives to the interpretation of the characters: α, signifies ἀποφορὰν, *abortion*, or ἀπώλειαν, *loss*; γ, signifies γονοειδὲς οὖρον, *urine resembling semen*; δ, punctuated below, thus, δ̣, signifies ἰδρῶτα, *sweat*, and διάρροϊαν, *diarrhœa*, and διαφόρησιν, *perspiration*, or in fact any other *evacuation* which it is wished to express; ε, signifies ἐποχὴν, *retention*, or ἔδραν, *seat*; ζ, signifies ζητήμα, *the object of research*; θ, signifies θάνατον, *death*; ι, signifies ἰδρῶτα, *sweat*; κ, signifies κρίσιν, *crisis*, or κοιλιακὴν διάθεσιν; μ, signifies μανίαν, *madness*, or μήτραν, *the womb*; ν, signifies νεότητα, *youth*, or νέκρωσιν, *mortification*; ξ, signifies ξανθὴν χολήν, *yellow bile*, or ξένον τι καὶ σπάνιον, *something strange and rare*, or ξυσμὸν, *irritation*, or ξηρότητα, *dryness*; ο, signifies ὀδύνας, *pains*, or οὔρον, *urine* (but some think that it is only when it has a ù above it that it signifies urine); π signifies πλῆθος, *abundance*, or πτύελον, *sputum*, or πυρὸν (πυρρὸν?), *yellow*, or πυρετὸν, *fever*, or πνεύμονος τάθος, *affection of the lungs*; π, with a character ι in its middle (π̣ or π̣̣), signifies πιθανόν, *probable*; ρ, signifies ῥύσιν, *flux*, or ρίγος, *chill*; φ, signifies φρενίτιν, *phrensy*; ζ, signifies σπασμὸν, *convulsion*, or στομαχοῦ ἢ στόματός κάκωσιν, *illness of the stomach or mouth*; τ, signifies τόκον, *accouchement*; υ, signifies ὑγίαν, *health*, or ὑποχόνδριον, *hypochondrium*; χ, signifies χολήν, *bile*, or χολῶδες, *bilious*; ψ, signifies ψύξιν, *congealing*; ω, signifies ὠμότητα, *crudity*. See Galeni Opera, t. v., p. 412, ed. Basil.; and Littré's Hippocrates, t. iii., p. 33.

According to this key, the characters at the end of the first case are thus explained by Galen: they are π̣ ΠΙΟΥΜΥ. Here, then, π̣ signifies πιθανόν, *it is probable*, Π, πλῆθος, *that an abundance*, ου, οὔρων, *of urine*; Μ, on the 40th day; Υ, ὑγίαν, *brought health*. It is more fully expressed thus by Galen: πιθανόν εἰναδιὰ τὸ πλῆθος τῶν ἐκριθέντων οὔρων αὐτὸ λυρθῆναι τὸ νοσημα καὶ ὑγῇ γενέσθαι τὸν ἄνθρωπον ἐν τῇ τεσσαρακοστῇ τῶν ἡμερῶν, *that is to say, “it is probable that, owing to the copious discharge of urine, the disease was resolved, and the patient became well on the fortieth day.”*

CASE I.—Pythion, who lived by the Temple of the Earth, on the first day, trembling commencing from his hands; acute fever, delirium. On the second, all the symptoms were exacerbated. On the third, the same. On the fourth alvine discharges scanty, unmixed, and bilious. On the fifth, all the symptoms were exacerbated, the tremors remained; little sleep, the bowels constipated. On the sixth sputa mixed, reddish. On the seventh, mouth drawn aside. On the eighth, all the symptoms were exacerbated; the tremblings were again constant; urine, from the beginning to the eighth day, thin, and devoid of color; substances floating in it, cloudy. On the tenth he sweated; sputa somewhat digested, had a crisis; urine thinnish about the crisis; but after the crisis, on the fortieth day, an abscess about the anus, which passed off by a strangury.

Explanation of the characters. It is probable that the great discharge of urine brought about the resolution of the disease, and the cure of the patient on the fortieth day.^[682]

CASE II.—Hermocrates, who lived by the New Wall,^[683] was seized with fever. He began to have pain in the head and loins; an empty distention of the hypochondrium; the tongue at first was parched; deafness at the commencement; there was no sleep; not very thirsty; urine thick and red, when allowed to stand it did not subside; alvine discharge very dry, and not scanty. On the fifth, urine thin, had substances floating in it which did not fall to the bottom;^[684] at night he was delirious. On the sixth, had jaundice;^[685] all the symptoms were exacerbated; had no recollection. On the seventh, in an uncomfortable state; urine thin, as formerly; on the following days the same. About the eleventh day, all the symptoms appeared to be lightened. Coma set in; urine thicker, reddish, thin substances below, had no sediment; by degrees he became collected. On the fourteenth, fever gone; had no sweat; slept, quite collected; urine of the same characters. About the seventeenth, had a relapse, became hot. On the following days, acute fever, urine thin, was delirious. Again, on the twentieth, had a crisis; free of fever; had no sweat; no appetite through the whole time; was perfectly collected; could not speak, tongue dry, without thirst; deep sleep. About the twenty-fourth day he became heated; bowels loose, with a thin, watery discharge; on the following days acute fever, tongue parched. On the twenty-seventh he died. In this patient deafness continued throughout;^[686] the urine either thick and red, without sediment, or thin, devoid of color, and having substances floating in it; he could taste nothing.

Explanation of the characters. It is probable that it was the suppression of the discharges from the bowels which occasioned death on the twenty-seventh day.

CASE III.—The man who was lodged in the Garden of Dealces:^[687] had heaviness of the head and pain in the right temple for a considerable time, from some accidental cause, was seized with fever, and took to bed. On the second, there was a trickling of pure blood from the left nostril, but the alvine discharges were proper, urine thin, mixed, having small substances floating in it, like coarse barley meal, or semen. On the third, acute fever; stools black, thin, frothy, a livid sediment in the dejections; slight coma; uneasiness at the times he had to get up; sediment in the urine livid, and somewhat viscid. On the fourth, slight vomiting of bilious, yellow matters, and, after a short interval, of the color of verdigris; a few drops of pure blood ran from the left nostril; stools the same; urine the same; sweated about the head and clavicles; spleen enlarged, pain of the thigh on the same side; loose swelling of the right hypochondrium; at night had no sleep, slight delirium. On the sixth, stools black, fatty, viscid, fetid; slept, more collected. On the seventh, tongue dry, thirsty, did not sleep; was somewhat delirious; urine thin, not of a good color. On the eighth, stools black, scanty, and compact; slept, became collected; not very thirsty. On the ninth had a rigor, acute fever, sweated, a chill, was delirious, strabismus of the right eye, tongue dry, thirsty, without sleep.^[688] On the tenth, much the same. On the eleventh, became quite collected; free from fever, slept, urine thin about the crisis. The two following days without fever; it returned on the fourteenth, then immediately insomnolency and complete delirium. On the fifteenth, urine muddy, like that which has been shaken after the sediment has fallen to the bottom; acute fever, quite delirious, did not sleep; knees and legs painful; after a suppository, had alvine dejection of a black color. On the sixteenth, urine thin, had a cloudy eneoema, was delirious. On the seventeenth, in the morning, extremities cold, was covered up with the bedclothes, acute fever, general sweat, felt relieved, more collected; not free of fever, thirsty, vomited yellow bile, in small quantities; formed fæces passed from the bowels, but soon afterwards black, scanty, and thin; urine thin, not well colored. On the eighteenth, not collected, comatose. On the nineteenth, in the same state. On the twentieth, slept; quite collected, sweated, free from fever, not thirsty, but the urine thin. On the twenty-first, slight delirium; somewhat thirsty, pain of the hypochondrium, and throbbing about the navel throughout. On the twenty-fourth, sediment in the urine, quite collected. Twenty-seventh, pain of the right hip joint; urine thin and bad, a sediment; all the other symptoms milder. About the twenty-ninth, pain of the right eye; urine thin. Fortieth, dejections pituitous, white, rather frequent; sweated abundantly all over; had a complete crisis.^[689]

Explanation of the characters. It is probable that, by means of the stools, the urine, and the sweat, this patient was cured in forty days.

CASE IV.—In Thasus, Philistes had headache of long continuance, and sometimes was confined to bed, with a tendency to deep sleep; having been seized with continual fevers from drinking, the pain was exacerbated; during the night he, at first, became hot. On the first day, he vomited some bilious matters, at first yellow, but afterwards of a verdigris-green color, and in greater quantity; formed fæces passed from the bowels; passed the night uncomfortably. On the second, deafness, acute fever; retraction of the right hypochondrium; urine thin, transparent, had some small substances like semen floating in it; delirium ferox about mid-day. On the third, in an uncomfortable state. On the fourth, convulsions; all the symptoms exacerbated. On the fifth, early in the morning, died.

Explanation of the characters. It is probable that the death of the patient on the fifth day is to be attributed to a phrenitis, with unfavorable evacuations.^[690]

CASE V.—Charion, who was lodged at the house of Demænetus, contracted a fever from drinking. Immediately he had a painful heaviness of the head; did not sleep; bowels disordered, with thin and somewhat bilious discharges. On the third day, acute fever; trembling of the head, but especially of the lower lip; after a little time a rigor, convulsions; he was quite delirious; passed the night uncomfortably. On the fourth, quiet, slept little, talked incoherently. On the fifth, in pain; all the symptoms exacerbated; delirium; passed the night uncomfortably; did not sleep. On the sixth, in the same state. On the seventh had a rigor, acute fever, sweated all over his body; had a crisis. Throughout the alvine discharges were bilious, scanty, and unmixed; urine thin, well colored, having cloudy substances floating in it. About the eighth day, passed urine of a better color, having a white scanty sediment; was collected, free from fever for a season. On the ninth it relapsed. About the fourteenth, acute fever. On the sixteenth, vomited pretty frequently yellow, bilious matters. On the seventeenth had a rigor, acute fever, sweated, free of fever; had a crisis; urine, after the relapse and the crisis, well colored, having a sediment; neither was he delirious in the relapse. On the eighteenth, became a little heated; some thirst, urine thin, with cloudy substances floating in it; slight wandering in his mind. About the nineteenth, free of fever, had a pain in his neck; a sediment in the urine. Had a complete crisis on the twentieth.

Explanation of the characters. It is probable that the patient was cured in twenty days, by the abundance of bilious stools and urine.^[691]

CASE VI.—The daughter of Euryanax, a maid, was taken ill of fever. She was free of thirst throughout, but had no relish for food. Alvine discharges small, urine thin, scanty, not well colored. In the beginning of the fever, had a pain about the nates. On the sixth day, was free of fever, did not sweat, had a crisis; the complaint about the nates came to a small suppuration, and burst at the crisis. After the crisis, on the seventh day, had a rigor, became slightly heated, sweated. On the eighth day after the rigor, had an inconsiderable rigor; the extremities cold ever after. About the tenth day, after a sweat which came on, she became delirious, and again immediately afterwards was collected; these symptoms were said to have been brought on by eating grapes. After an intermission of the twelfth day, she again talked much incoherently; her bowels disordered with bilious, scanty, unmixed, thin, acrid discharges; she required to get frequently up. She died on the seventh day after the return of the delirium. At the commencement of the disease she had pain in the throat, and it was red throughout; uvula retracted, defluxions abundant, thin, acrid; coughed, but had no concocted sputa; during the whole time loathed all kinds of food, nor had the least desire of anything; had no thirst, nor drank anything worth mentioning; was silent, and never spoke a word; despondency; had no hopes of herself. She had a congenital tendency to phthisis.^[692]

CASE VII.—The woman affected with quinsy, who lodged in the house of Aristion: her complaint began in the tongue; speech inarticulate; tongue red and parched. On the first day, felt chilly, and afterwards became heated. On the third day, a rigor, acute fever; a reddish and hard swelling on both sides of the neck and chest, extremities cold and livid; respiration elevated; the drink returned by the nose; she could not swallow; alvine and urinary discharges suppressed. On the fourth, all the symptoms were exacerbated. On the sixth she died of the quinsy.

Explanation of the characters. It is probable that the cause of death on the sixth day was the suppression of the discharges.^[693]

CASE VIII.—The young man who was lodged by the Liars' Market was seized with fever from fatigue, labor, and running out of season. On the first day, the bowels disordered, with bilious, thin, and copious dejections; urine thin and blackish; had no sleep; was thirsty. On the second all the symptoms were exacerbated; dejections more copious and unseasonable; he had no sleep; disorder of the intellect; slight sweat. On the third day, restless, thirst, nausea, much tossing about, bewilderment, delirium; extremities livid and cold; softish distention of the hypochondrium on both sides. On the fourth, did not sleep; still worse. On the seventh he died. He was about twenty years of age.

Explanation of the characters. It is probable that the cause of his death on the seventh day was the unseasonable practices mentioned above. An acute affection.^[694]

CASE IX.—The woman who lodged at the house of Tisamenas had a troublesome attack of iliac passion; much vomiting; could not keep her drink; pains about the hypochondria, and pains also in the lower part of the belly; constant tormina; not thirsty; became hot; extremities cold throughout, with nausea and insomnolency; urine scanty and thin; dejections undigested, thin, scanty. Nothing could do her any good. She died.^[695]

CASE X.—A woman of those who lodged with Pantimides, from a miscarriage, was taken ill of fever. On the first day, tongue dry, thirst, nausea, insomnolency, belly disordered, with thin, copious, undigested dejections. On the second day, had a rigor, acute fever; alvine discharges copious; had no sleep. On the third, pains greater. On the fourth, delirious. On the seventh she died. Belly throughout loose, with copious, thin, undigested evacuations; urine scanty, thin. An ardent fever.^[696]

CASE XI.—Another woman, after a miscarriage about the fifth month, the wife of Ocetes, was seized with fever. At first had sometimes coma and sometimes insomnolency; pain of the loins; heaviness of the head. On the second, the bowels were disordered, with scanty, thin, and at first unmixed dejections. On the third, more copious, and worse; at night did not sleep. On the fourth was delirious; frights, despondency; strabismus of the right eye; a faint cold sweat about the head; extremities cold. On the fifth day, all the symptoms were exacerbated; talked much incoherently, and again immediately became collected; had no thirst; labored under insomnolency; alvine dejections copious, and unseasonable throughout; urine scanty, thin, darkish; extremities cold, somewhat livid. On the sixth day, in the same state. On the seventh she died. Phrenitis.^[697]

CASE XII.—A woman who lodged near the Liars' Market, having then brought forth a son in a first and difficult labor, was seized with fever. Immediately on the commencement had thirst, nausea, and cardialgia; tongue dry; bowels disordered, with thin and scanty dejections; had no sleep. On the second, had slight rigor, acute fever; a faint cold sweat about the head. On the third, painfully affected; evacuations from the bowels undigested, thin, and copious. On the fourth, had a rigor; all the symptoms exacerbated; insomnolency. On the fifth, in a painful state. On the sixth, in the same state; discharges from the bowels liquid and copious. On the seventh, had a rigor, fever acute; much thirst; much tossing about; towards evening a cold sweat over all; extremities cold; could no longer be kept warm; and again at night had a rigor; extremities could not be warmed; she did not sleep; was slightly delirious, and again speedily collected. On the eighth, about mid-day, she became warm, was thirsty, comatose, had nausea; vomited small quantities of yellowish bile; restless at night, did not sleep; passed frequently large quantities of urine without consciousness. On the ninth, all the symptoms gave way; comatose, towards evening slight rigors; small vomitings of bile. On the tenth, rigor; exacerbation of the fever, did not sleep at all; in the morning passed much urine having a sediment; extremities recovered their heat. On the eleventh, vomited bile of a verdigris-green color; not long after had a rigor, and again the extremities cold; towards evening a rigor, a cold sweat, much vomiting; passed a painful night. On the twelfth, had copious black and fetid vomitings; much hiccup, painful thirst. On the thirteenth, vomitings black, fetid, and copious; rigor about mid-day, loss of speech. On the fourteenth, some blood ran from her nose, she died. In this case the bowels were loose throughout; with rigors; her age about seventeen. An ardent fever.^[698]

SECTION III.—CONSTITUTION 2.^[699]

The year was southerly, rainy; no winds throughout.^[700] Droughts having prevailed during the previous seasons of the year, the south winds towards the rising of Arcturus were attended with much rain. Autumn gloomy and cloudy, with copious rains. Winter southerly, damp, and soft. But long after the solstice, and near the equinox, much winterly weather out of season; and when now close to the equinox, northerly, and winterly weather for no long time. The spring again southerly, calm, much rain throughout until the dog-days. Summer fine and hot; great suffocating heats. The Etesian winds blew small and irregular; again, about the season of Arcturus, much rains with north winds. The year being southerly, damp, and soft towards winter, all were healthy, except those affected with phthisis, of whom we shall write afterwards.

3. Early in spring, along with the prevailing cold, there were many cases of erysipelas, some from a manifest cause, and some not.^[701] They were of a malignant nature, and proved fatal to many; many had sore-throat and loss of speech. There were many cases of ardent fever, phrensy, aphthous affections of the mouth,^[702] tumors on the genital organs; of ophthalmia, anthrax,^[703] disorder of the bowels, anorexia, with thirst and without it; of disordered urine, large in quantity, and bad in quality; of persons affected with coma for a long time, and then falling into a state of insomnolency. There were many cases of failure of crisis, and many of unfavorable crisis; many of dropsy and of phthisis. Such were the diseases then epidemic.^[704] There were patients affected with every one of the species which have been mentioned, and many died. The symptoms in each of these cases were as follows:

4. In many cases erysipelas, from some obvious cause, such as an accident, and sometimes from even a very small wound, broke out all over the body, especially, in persons about sixty years of age, about the head, if such an accident was neglected in the slightest degree; and this happened in some who were under treatment; great inflammation took place, and the erysipelas quickly spread all over.^[705] In the most of them the abscesses ended in suppurations, and there were great fallings off (sloughing) of the flesh, tendons, and bones; and the defluxion which seated in the part was not like pus, but a sort of

putrefaction, and the running was large and of various characters. Those cases in which any of these things happened about the head were accompanied with falling off of the hairs of the head and chin, the bones were laid bare and separated, and there were excessive runnings; and these symptoms happened in fevers and without fevers. But these things were more formidable in appearance than dangerous; for when the concoction in these cases turned to a suppuration, most of them recovered; but when the inflammation and erysipelas disappeared, and when no abscess was formed, a great number of these died.^[706] In like manner, the same things happened to whatever part of the body the disease wandered, for in many cases both forearm and arm dropped off; and in those cases in which it fell upon the sides, the parts there, either before or behind, got into a bad state; and in some cases the whole femur and bones of the leg and whole foot were laid bare. But of all such cases, the most formidable were those which took place about the pubes and genital organs.^[707] Such was the nature of these cases when attended with sores, and proceeding from an external cause; but the same things occurred in fevers, before fevers, and after fevers. But those cases in which an abscess was formed, and turned to a suppuration, or a seasonable diarrhoea or discharge of good urine took place, were relieved thereby: but those cases in which none of these symptoms occurred, but they disappeared without a crisis, proved fatal. The greater number of these erysipelatous cases took place in the spring, but were prolonged through the summer and during autumn.

5. In certain cases there was much disorder, and tumors about the fauces, and inflammations of the tongue, and abscesses about the teeth. And many were attacked with impairment or loss of speech,^[708] at first, those in the commencement of phthisis, but also persons in ardent fever and in phrenitis.

6. The cases of ardent fever and phrenitis occurred early in spring after the cold set in, and great numbers were taken ill at that time, and these cases were attended with acute and fatal symptoms. The constitution of the ardent fevers which then occurred was as follows: at the commencement they were affected with coma, nausea, and rigors; fever not acute, not much thirst, nor delirium, slight epistaxis,^[709] the paroxysms for the most part on even days; and, about the time of the paroxysms, forgetfulness, loss of strength and of speech, the extremities, that is to say, the hands and feet, at all times, but more especially about the time of the paroxysms, were colder than natural; they slowly and imperfectly became warmed, and again recovered their recollection and speech.^[710] They were constantly affected either with coma, in which they got no sleep, or with insomnolency, attend with pains;^[711] most had disorders of the bowels, attended with undigested, thin, and copious evacuations; urine copious, thin, having nothing critical nor favorable about it; neither was there any other critical appearance in persons affected thus; for neither was there any proper hemorrhage, nor any other of the accustomed evacuations, to prove a crisis. They died, as it happened, in an irregular manner, mostly about the crisis, but in some instances after having lost their speech for a long time, and having had copious sweats. These were the symptoms which marked the fatal cases of ardent fever; similar symptoms occurred in the phrenitic cases; but these were particularly free from thirst, and none of these had wild delirium^[712] as in other cases, but they died oppressed by a bad tendency to sleep, and stupor.

7. But there were also other fevers, as will be described. Many had their mouths affected with aphthous ulcerations. There were also many defluxions about the genital parts, and ulcerations, boils (phymata), externally and internally, about the groins.^[713] Watery ophthalmies of a chronic character, with pains; fungous excrescences of the eyelids, externally and internally, called fici, which destroyed the sight of many persons.^[714] There were fungous growths, in many other instances, on ulcers, especially on those seated on the genital organs. There were many attacks of carbuncle (anthrax) through the summer, and other affections, which are called “the putrefaction” (*seps*); also large ecthymata,^[715] and large tetter (herpetes) in many instances.

8. And many and serious complaints attacked many persons in the region of the belly. In the first place, tenesmus, accompanied with pain, attacked many, but more especially children, and all who had not attained to puberty; and the most of these died. There were many cases of lientery and of dysentery; but these were not attended with much pain.^[716] The evacuations were bilious, and fatty, and thin, and watery; in many instances the disease terminated in this way, with and without fever; there were painful tormina and volvuli of a malignant kind; copious evacuations of the contents of the guts, and yet much remained behind; and the passages did not carry off the pains, but yielded with difficulty to the means administered; for in most cases purgings were hurtful to those affected in this manner; many died speedily, but in many others they held out longer. In a word, all died, both those who had acute attacks and those who had chronic, most especially from affections of the belly, for it was the belly which carried them all off.

9. All persons had an aversion to food in all the afore-mentioned complaints to a degree such as I never met with before,^[717] and persons in these complaints most especially, and those recovering from them, and in all other diseases of a mortal nature. Some were troubled with thirst, and some not; and both in febrile complaints and in others no one drank unseasonably or disobeyed injunctions.

10. The urine in many cases was not in proportion to the drink administered, but greatly in excess; and the badness of the urine voided was great, for it had not the proper thickness, nor concoction, nor purged properly; for in many cases purgings by

the bladder indicate favorably, but in the greatest number they indicated a melting of the body, disorder of the bowels, pains, and a want of crisis.^[718]

11. Persons laboring under phrenitis and causus were particularly disposed to coma; but also in all other great diseases which occurred along with fever. In the main, most cases were attended either by heavy coma, or by short and light sleep.

12. And many other forms of fevers were then epidemic, of tertian, of quartan, of nocturnal,^[719] of continual, of chronic, of erratic, of fevers attended with nausea, and of irregular fevers. All these were attended with much disorder, for the bowels in most cases were disordered, accompanied with rigors, sweats not of a critical character, and with the state of the urine as described. In most instances the disease was protracted, for neither did the deposits which took place prove critical as in other cases; for in all complaints and in all cases there was difficulty of crisis, want of crisis, and protraction of the disease, but most especially in these. A few had the crisis about the eightieth day, but in most instances it (the disease?) left them irregularly. A few of them died of dropsy without being confined to bed. And in many other diseases people were troubled with swelling, but more especially in phthisical cases.

13. The greatest and most dangerous disease, and the one that proved fatal to the greatest number, was the consumption.^[720] With many persons it commenced during the winter, and of these some were confined to bed, and others bore up on foot; the most of those died early in spring who were confined to bed; of the others, the cough left not a single person, but it became milder through the summer; during the autumn, all these were confined to bed, and many of them died, but in the greater number of cases the disease was long protracted. Most of these were suddenly attacked with these diseases, having frequent rigors, often continual and acute fevers; unseasonable, copious, and cold sweats throughout; great coldness, from which they had great difficulty in being restored to heat; the bowels variously constipated, and again immediately in a loose state, but towards the termination in all cases with violent looseness of the bowels; a determination downwards of all matters collected about the lungs; urine excessive, and not good; troublesome melting. The coughs throughout were frequent, and sputa copious, digested, and liquid, but not brought up with much pain; and even when they had some slight pain, in all cases the purging of the matters about the lungs went on mildly. The fauces were not very irritable, nor were they troubled with any saltish humors; but there were viscid, white, liquid, frothy, and copious defluxions from the head. But by far the greatest mischief attending these and the other complaints, was the aversion to food, as has been described. For neither had they any relish for drink along with their food, but continued without thirst. There was heaviness of the body, disposition to coma, in most cases swelling, which ended in dropsy; they had rigors, and were delirious towards death.

14. The form of body peculiarly subject to phthisical complaints was the smooth, the whitish, that resembling the lentil; the reddish, the blue-eyed, the leucophlegmatic,^[721] and that with the scapulæ having the appearance of wings: and women in like manner,^[722] with regard to the melancholic and subsanguineous, phrenitic and dysenteric affections principally attacked them. Tenesmus troubled young persons of a phlegmatic temperament. Chronic diarrhœa, acrid and viscid discharges from the bowels, attacked those who were troubled with bitter bile.

15. To all those which have been described, the season of spring was most inimical, and proved fatal to the greatest numbers: the summer was the most favorable to them, and the fewest died then; in autumn, and under the Pleiades, again there died great numbers. It appears to me, according to the reason of things, that the coming on of summer should have done good in these cases; for winter coming on cures the diseases of summer, and summer coming on removes the diseases of winter. And yet the summer in question was not of itself well constituted, for it became suddenly hot, southerly, and calm; but, notwithstanding, it proved beneficial by producing a change on the other constitution.

16. I look upon it as being a great part of the art to be able to judge properly of that which has been written. For he that knows and makes a proper use of these things, would appear to me not likely to commit any great mistake in the art. He ought to learn accurately the constitution of every one of the seasons, and of the diseases; whatever that is common in each constitution and disease is good, and whatever is bad; whatever disease will be protracted and end in death, and whatever will be protracted and end in recovery; which disease of an acute nature will end in death, and which in recovery. From these it is easy to know the order of the critical days, and prognosticate from them accordingly. And to a person who is skilled in these things, it is easy to know to whom, when, and how aliment ought to be administered.^[723]

SEC. 17. SIXTEEN CASES.^[724]

CASE I.—In Thasus, the Parian who lodged above the Temple of Diana was seized with an acute fever, at first of a continual and ardent type; thirsty, inclined to be comatose at first, and afterwards troubled with insomnolency; bowels disordered at the beginning, urine thin. On the sixth day, passed oily urine, was delirious. On the seventh, all the symptoms were exacerbated; had no sleep, but the urine of the same characters, and the understanding disordered; alvine dejections bilious and fatty. On the eighth, a slight epistaxis; small vomiting of verdigris-green matters; slept a little. On the ninth, in the same state. On the tenth, all the symptoms gave way. On the eleventh, he sweated, but not over the whole body; he became

cold, but immediately recovered his heat again. On the fourteenth, acute fever; discharges bilious, thin, and copious; substances floating in the urine; he became incoherent. On the seventeenth, in a painful state, for he had no sleep, and the fever was more intense. On the twentieth, sweated all over; apyrexia, dejections bilious; aversion to food, comatose. On the twenty-fourth, had a relapse. On the thirty-fourth, apyrexia; bowels not confined; and he again recovered his heat. Fortieth, apyrexia, bowels confined for no long time, aversion to food; had again slight symptoms of fever, and throughout in an irregular form; apyrexia at times, and at others not; for if the fever intermitted, and was alleviated for a little, it immediately relapsed again; he used much and improper food; sleep bad; about the time of the relapse he was delirious; passed thick urine at that time, but troubled, and of bad characters; bowels at first confined, and again loose; slight fevers of a continual type; discharges copious and thin. On the hundred and twentieth day he died. In this patient the bowels were constantly from the first either loose, with bilious, liquid, and copious dejection, or constipated with hot and undigested fæces; the urine throughout bad; for the most part coma, or insomnolency with pain; continued aversion to food. Ardent fever.

Explanation of the characters. It is probable that the weakness produced by the fever, the phrenitis, and affection of the hypochondrium caused death on the hundred and twentieth day.^[725]

CASE II.—In Thasus, the woman who lodged near the Cold Water, on the third day after delivery of a daughter, the lochial discharge not taking place, was seized with acute fever, accompanied with rigors. But a considerable time before delivery she was feverish, confined to bed, and loathed her food. After the rigor which took place, continual and acute fevers, with rigors. On the eighth and following days, was very incoherent, and immediately afterwards became collected; bowels disordered, with copious, thin, watery, and bilious stools; no thirst. On the eleventh was collected, but disposed to coma; urine copious, thin, and black; no sleep. On the twentieth, slight chills, and immediately afterwards was warm; slight incoherence; no sleep; with regard to the bowels, in the same condition; urine watery, and copious. On the twenty-seventh, free from fever; bowels constipated; not long afterwards violent pain of the right hip-joint for a considerable time; fevers afterwards supervened; urine watery. On the fortieth, complaints about the hip-joint better; continued coughs, with copious, watery sputa; bowels constipated; aversion to food; urine the same; fever not leaving her entirely, but having paroxysms in an irregular form, sometimes present, sometimes not. On the sixtieth, the coughs left her without a crisis, for no concoction of the sputa took place, nor any of the usual abscesses; jaw on the right side convulsively retracted; comatose, was again incoherent, and immediately became collected; utter aversion to food; the jaw became relaxed; alvine discharges small, and bilious; fever more acute, affected with rigors; on the following days lost her speech, and again became collected, and talked. On the eightieth she died. In this case the urine throughout was black, thin, and watery; coma supervened; there was aversion to food, despondency, and insomnolency; irritability, restlessness; she was of a melancholic turn of mind.

Explanation of the characters. It is probable that the suppression of the lochial discharge caused death on the eightieth day.^[726]

CASE III.—In Thasus, Pythion, who was lodged above the Temple of Hercules, from labor, fatigue, and neglected diet, was seized with strong rigor and acute fever; tongue dry, thirsty, and bilious; had no sleep; urine darkish, eneorema floating on the top of the urine, did not subside. On the second day, about noon, coldness of the extremities, especially about the hands and head; loss of speech and of articulation; breathing short for a considerable time; recovered his heat; thirst; passed the night quietly; slight sweats about the head. On the third, passed the day in a composed state; in the evening, about sunset, slight chills; nausea, agitation; passed the night in a painful state; had no sleep; small stools of compact fæces passed from the bowels. On the fourth, in the morning, composed; about noon all the symptoms became exacerbated; coldness, loss of speech, and of articulation; became worse; recovered his heat after a time; passed black urine, having substances floating in it; the night quiet; slept. On the fifth, seemed to be lightened, but a painful weight about the belly; thirsty, passed the night in a painful state. On the sixth, in the morning, in a quiet state; in the evening the pains greater; had a paroxysm; in the evening the bowels properly opened by a small clyster; slept at night. On the seventh, during the day, in a state of nausea, somewhat disturbed; passed urine of the appearance of oil; at night, much agitation, was incoherent, did not sleep. On the eighth, in the morning, slept a little; but immediately coldness, loss of speech, respiration small and weak; but in the evening recovered his heat again; was delirious, but towards day was somewhat lightened; stools small, bilious, and unmixed. On the ninth, affected with coma, and with nausea when roused; not very thirsty; about sunset he became restless and incoherent; passed a bad night. On the tenth, in the morning, had become speechless; great coldness; acute fever; much perspiration; he died. His sufferings were on the even days.

Explanation of the characters. It is probable that the excessive sweats caused death on the tenth day.^[727]

CASE IV.—The patient affected with phrenitis, having taken to bed on the first day, vomited largely of verdigris-green and thin matters; fever, accompanied with rigors, copious and continued sweats all over; heaviness of the head and neck, with pain; urine thin, substances floating in the urine small, scattered, did not subside; had copious dejections from the bowels; very delirious; no sleep. On the second, in the morning, loss of speech; acute fever; he sweated, fever did not leave him; palpitations over the whole body, at night, convulsions. On the third, all the symptoms exacerbated; he died.

Explanation of the characters. It is probable that the sweats and convulsions caused death.^[728]

CASE V.—In Larissa, a man, who was bald, suddenly was seized with pain in the right thigh; none of the things which were administered did him any good. On the first day, fever acute, of the ardent type, not agitated, but the pains persisted. On the second, the pains in the thigh abated, but the fever increased; somewhat tossed about; did not sleep; extremities cold; passed a large quantity of urine, not of a good character. On the third, the pain of the thigh ceased; derangement of the intellect, confusion, and much tossing about. On the fourth, about noon, he died. An acute disease.^[729]

CASE VI.—In Abdera, Pericles was seized with a fever of the acute, continual type, with pain; much thirst, nausea, could not retain his drink; somewhat swelled about the spleen, with heaviness of the head. On the first day, had hemorrhage from the left nostril, but still the fever became more violent; passed much muddy, white urine, which when allowed to stand did not subside. On the second day, all the symptoms were exacerbated, yet the urine was thick, and more inclined to have a sediment; the nausea less; he slept. On the third, fever was milder; abundance of urine, which was concocted, and had a copious sediment; passed a quiet night. On the fourth, had a copious and warm sweat all over about noon; was free of fever, had a crisis, no relapse. An acute affection.^[730]

CASE VII.—In Abdera, the young woman who was lodged in the Sacred Walk was seized with an ardent fever. She was thirsty, and could not sleep; had menstruation for the first time. On the sixth, much nausea, flushing, was chilly, and tossed about. On the seventh, in the same state; urine thin, but of a good color; no disturbance about the bowels. On the eighth, deafness, acute fever, insomnolency, nausea, rigors, became collected; urine the same. On the ninth, in the same state, and also on the following days; thus the deafness persisted. On the fourteenth, disorder of the intellect; the fever abated. On the seventeenth, a copious hemorrhage from the nose; the deafness slightly better; and on the following days, nausea, deafness, and incoherence. On the twentieth, pain of the feet; deafness and delirium left her; a small hemorrhage from the nose; sweat, apyrexia. On the twenty-fourth, the fever returned, deafness again; pain of the feet remained; incoherence. On the twenty-seventh, had a copious sweat, apyrexia; the deafness left her; the pain of her feet partly remained; in other respects had a complete crisis.

Explanation of the characters. It is probable that the restoration of health on the twentieth day was the result of the evacuation of urine.^[731]

CASE VIII.—In Abdera, Anaxion, who was lodged near the Thracian Gates, was seized with an acute fever; continued pain of the right side; dry cough, without expectoration during the first days, thirst, insomnolency; urine well colored, copious, and thin. On the sixth, delirious; no relief from the warm applications. On the seventh, in a painful state, for the fever increased, while the pains did not abate, and the cough was troublesome, and attended with dyspnea. On the eighth, I opened a vein at the elbow, and much blood, of a proper character, flowed; the pains were abated, but the dry coughs continued. On the eleventh, the fever diminished; slight sweats about the head; coughs, with more liquid sputa; he was relieved. On the twentieth, sweat, apyrexia; but after the crisis he was thirsty, and the expectorations were not good. On the twenty-seventh the fever relapsed; he coughed, and brought up much concocted sputa: sediment in the urine copious and white; he became free of thirst, and the respiration was good. On the thirty-fourth, sweated all over, apyrexia, general crisis.

Explanation of the characters. It is probable that the evacuation of the sputa brought about the recovery on the thirty-fourth day.^[732]

CASE IX.—In Abdera, Heropythus, while still on foot, had pain in the head, and not long afterwards he took to bed; he lived near the High Street. Was seized with acute fever of the ardent type; vomitings at first of much bilious matter; thirst; great restlessness; urine thin, black, substances sometimes floating high in it, and sometimes not; passed the night in a painful state; paroxysms of the fever diversified, and for the most part irregular. About the fourteenth day, deafness; the fever increased; urine the same. On the twentieth and following days, much delirium. On the thirtieth, copious hemorrhage from the nose, and became more collected; deafness continued, but less; the fever diminished; on the following days, frequent hemorrhages, at short intervals. About the sixtieth, the hemorrhages ceased, but violent pain of the hip-joint, and increase of fever. Not long afterwards, pains of all the inferior parts; it then became a rule, that either the fever and deafness increased, or, if these abated and were lightened, the pains of the inferior parts were increased. About the eightieth day, all the complaints gave way, without leaving any behind; for the urine was of a good color, and had a copious sediment, while the delirium became less. About the hundredth day, disorder of the bowels, with copious and bilious evacuations, and these continued for a considerable time, and again assumed the dysenteric form with pain; but relief of all the other complaints. On the whole, the fevers went off, and the deafness ceased. On the hundred and twentieth day, had a complete crisis. Ardent fever.

Explanation of the characters. It is probable that the bilious discharge brought about the recovery on the hundred and twentieth day.^[733]

CASE X.—In Abdera, Nicodemus was seized with fever from venery and drinking. At the commencement he was troubled with nausea and cardialgia; thirsty, tongue was parched; urine thin and dark. On the second day, the fever exacerbated; he was

troubled with rigors and nausea; had no sleep; vomited yellow bile; urine the same; passed a quiet night, and slept. On the third, a general remission; amelioration; but about sunset felt again somewhat uncomfortable; passed an uneasy night. On the fourth, rigor, much fever, general pains; urine thin, with substances floating in it; again a quiet night. On the fifth, all the symptoms remained, but there was an amelioration. On the sixth, some general pains; substances floating in the urine; very incoherent. On the seventh, better. On the eighth, all the other symptoms abated. On the tenth, and following days, there were pains, but all less; in this case throughout, the paroxysms and pains were greater on the even days. On the twentieth, the urine white and thick, but when allowed to stand had no sediment; much sweat; seemed to be free from fever; but again in the evening he became hot, with the same pains, rigor, thirst, slightly incoherent. On the twenty-fourth, urine copious, white, with an abundant sediment; a copious and warm sweat all over; apyrexia; the fever came to its crisis.

Explanation of the characters. It is probable that the cure was owing to the bilious evacuations and the sweats.^[734]

CASE XI.—In Thasus, a woman, of a melancholic turn of mind, from some accidental cause of sorrow, while still going about, became affected with loss of sleep, aversion to food, and had thirst and nausea. She lived near the Pylades, upon the Plain. On the first, at the commencement of night, frights, much talking; despondency, slight fever; in the morning, frequent spasms, and when they ceased, she was incoherent and talked obscurely; pains frequent, great, and continued. On the second, in the same state; had no sleep; fever more acute. On the third, the spasms left her; but coma, and disposition to sleep, and again awake, started up, and could not contain herself; much incoherence; acute fever; on that night a copious sweat all over; apyrexia, slept, quite collected; had a crisis. About the third day, the urine black, thin, substances floating in it generally round, did not fall to the bottom; about the crisis a copious menstruation.^[735]

CASE XII.—In Larissa,^[736] a young unmarried woman was seized with a fever of the acute and ardent type; insomnolency, thirst; tongue sooty and dry; urine of a good color, but thin. On the second, in an uneasy state, did not sleep. On the third, alvine discharges copious, watery, and greenish, and on the following days passed such with relief. On the fourth, passed a small quantity of thin urine, having substances floating towards its surface, which did not subside; was delirious towards night. On the sixth, a great hemorrhage from the nose; a chill, with a copious and hot sweat all over; apyrexia, had a crisis. In the fever, and when it had passed the crisis, the menses took place for the first time, for she was a young woman. Throughout she was oppressed with nausea, and rigors; redness of the face; pain of the eyes; heaviness of the head; she had no relapse, but the fever came to a crisis. The pains were on the even days.^[737]

CASE XIII.—Apollonius, in Abdera, bore up (under the fever?) for some time, without betaking himself to bed. His viscera were enlarged, and for a considerable time there was a constant pain about the liver, and then he became affected with jaundice; he was flatulent, and of a whitish complexion. Having eaten beef, and drunk unseasonably, he became a little heated at first, and betook himself to bed, and having used large quantities of milk, that of goats and sheep, and both boiled and raw, with a bad diet otherwise, great mischief was occasioned by all these things; for the fever was exacerbated, and of the food taken scarcely any portion worth mentioning was passed from the bowels; the urine was thin and scanty; no sleep; troublesome meteorism; much thirst; disposition to coma; painful swelling of the right hypochondrium; extremities altogether coldish; slight incoherence, forgetfulness of everything he said; he was beside himself. About the fourteenth day after he betook himself to bed, had a rigor, became heated, and was seized with furious delirium; loud cries, much talking, again composed, and then coma came on; afterwards the bowels disordered, with copious, bilious, unmixed, and undigested stools; urine black, scanty, and thin; much restlessness; alvine evacuations of varied characters, either black, scanty, and verdigris-green, or fatty, undigested, and acrid; and at times the dejections resembled milk. About the twenty-fourth, enjoyed a calm; other matters in the same state; became somewhat collected; remembered nothing that had happened since he was confined to bed; immediately afterwards became delirious; every symptom rapidly getting worse. About the thirtieth, acute fever; stools copious and thin; was delirious; extremities cold; loss of speech. On the thirty-fourth he died. In this case, as far as I saw, the bowels were disordered; urine thin and black; disposition to coma; insomnolency; extremities cold; delirious throughout. Phrenitis.^[738]

CASE XIV.—In Cyzicus,^[739] a woman who had brought forth twin daughters, after a difficult labor, and in whom the lochial discharge was insufficient, at first was seized with an acute fever, attended with chills; heaviness of the head and neck, with pain; insomnolency from the commencement; she was silent, sullen, and disobedient; urine thin, and devoid of color; thirst, nausea for the most part; bowels irregularly disordered, and again constipated. On the sixth, towards night, talked much incoherently; had no sleep. About the eleventh day was seized with wild delirium, and again became collected; urine black, thin, and again deficient, and of an oily appearance; copious, thin, and disordered evacuations from the bowels. On the fourteenth, frequent convulsions; extremities cold; not in anywise collected; suppression of urine. On the sixteenth loss of speech. On the seventeenth, she died. Phrenitis.

Explanation of the characters. It is probable that death was caused, on the seventeenth day, by the affection of the brain consequent upon her accouchement.^[740]

CASE XV.—In Thasus, the wife of Dealces, who was lodged upon the Plain, from sorrow was seized with an acute fever, attended with chills. From first to last she wrapped herself up in her bedclothes; still silent, she fumbled, picked, bored, and gathered hairs (from them); tears, and again laughter; no sleep; bowels irritable, but passed nothing; when directed, drank a little; urine thin and scanty; to the touch of the hand the fever was slight; coldness of the extremities. On the ninth, talked much incoherently, and again became composed and silent. On the fourteenth, breathing rare, large, at intervals; and again hurried respiration. On the sixteenth, looseness of the bowels from a stimulant clyster; afterwards she passed her drink, nor could retain anything, for she was completely insensible; skin parched and tense. On the twentieth, much talk, and again became composed; loss of speech; respiration hurried. On the twenty-first she died. Her respiration throughout was rare and large; she was totally insensible; always wrapped up in her bedclothes; either much talk, or complete silence throughout. Phrenitis.^[741]

CASE XVI.—In Melibœa,^[742] a young man having become heated by drinking and much venery, was confined to bed; he was affected with rigors and nausea; insomnolency and absence of thirst. On the first day much fæces passed from the bowels along with a copious flux; and on the following days he passed many watery stools of a green color; urine thin, scanty, and deficient in color; respiration rare, large, at long intervals; softish distention of the hypochondrium, of an oblong form, on both sides; continued palpitation in the epigastric region throughout; passed urine of an oily appearance. On the tenth, he had calm delirium, for he was naturally of an orderly and quiet disposition; skin parched and tense; dejections either copious and thin, or bilious and fatty. On the fourteenth, all the symptoms were exacerbated; he became delirious, and talked much incoherently. On the twentieth, wild delirium, jactitation, passed no urine; small drinks were retained. On the twenty-fourth he died. Phrenitis.^[743]

ON INJURIES OF THE HEAD.

THE ARGUMENT.

This treatise opens with a description of the bones of the head, which, although in most respects pretty accurate, is remarkable for containing an account of particular configurations of the cranium, and of certain varieties in the arrangement of the sutures, which it has puzzled modern authorities in anatomy to explain, otherwise than upon the supposition that the writer must have been but imperfectly acquainted with the subject. But as the work otherwise bears evidence that our author must have examined the bones of the head very carefully, and moreover, as in all his works he displays a wonderfully minute acquaintance with osteology, (to say nothing of the historical tradition, mentioned by Pausanias, that he was possessed of a skeleton, which at his death he bequeathed to the Temple of Apollo, at Delphi,) it seems incredible that he should have committed most glaring blunders in describing the prominent features of a part to which it is clear that he had paid very great attention. Moreover, the reputation of Hippocrates for accuracy stood so high, that an eminent authority does not hesitate to declare of him, that he was a man who knew not how to deceive or be deceived.^[744] An easy way of getting rid of the difficulty would no doubt be, to adopt the conjecture advanced by Scaliger,^[745] and in part approved of by Riolanus,^[746] that the treatise had suffered much in early times, from the interpolations of ignorant transcribers; or to hold, with M. Malgaigne, that the whole work is to be condemned as spurious. But it would be a dangerous practice in ancient criticism, to reject as spurious a work which has such unexceptionable evidence in its favor, although it may contain matter which appears to us derogatory to the reputation of its author, and it will be admitted, by any competent judge who examines the arguments by Scaliger, that the proofs which he brings forward of great interpolations in this treatise, are generally of a very fanciful nature.

On a point so obscure, and which has puzzled so many eminent scholars, it is to be feared that I shall not be able to throw much additional light, but as, consistently with my general plan, I cannot well avoid stating some opinion on the question I shall endeavor to elucidate it in so far by giving in the first place a brief sketch of the information supplied by all the other ancient authorities who have touched upon this subject. I shall begin, then, with Aristotle, the contemporary of our author, who, in his work "On the History of Animals," gives the following very inaccurate description of the sutures of the human skull: "The female cranium has one circular suture, but men generally three, which unite in one point. But a male skull has been seen not having a suture."^[747] Celsus describes the sutures in the following terms: "Ex ceteris, quo suturæ pauciores sunt, eo capitis valetudo commodior est. Neque enim certus eorum numerus est, sicut ne locus quidem. Ferè tamen duæ, super aures, tempora a superiori parte discernunt; tertia ad aures, occipitium a summo capite deducit; quarta, ab eodem vertice per medium caput ad frontem procedit; eaque modo sub imo capillo desinit, modo frontem ipsam secans inter supercilia finitur." (viii., 1.) "Nam neque utique certa sedes, supra posui, suturarum est." (viii., 4.) Pliny gives the following description of the head, which it is impossible not to recognize as having been borrowed from our author: "Vertices bini hominum tantum aliquibus. Capitis ossa plana, tenuia, sine medullis, serratis pectinatim structa compagibus."^[748] Of Ruffus Ephesius I may just mention, that his descriptions of the human body are in general remarkable for their correctness, which is not to be wondered at, as he would appear to have followed, in general, Erasistratus and the other authorities belonging to the great Alexandrian period in anatomy; and that he has described very accurately all the sutures of the human cranium, but says not a word of the different configurations of the head, as here given by our author.^[749] We now come to Galen, who gives a very lengthy description of the various forms of the head, in nearly the same terms as our author, and after alluding to the uses of the sutures, the principal of which he holds to be to permit transpiration from the brain, he proceeds thus to describe the distribution of the sutures: "That there is one which runs straight along the middle of the head, (the sagittal?) and two transverse, (the coronal and lambdoid?) has been stated previously, and need not require many words in this place. For, the head being like an oblong sphere, one was justly made to extend straight through its middle from behind forwards, and two transverse sutures meet it, and the form of the three sutures is like the letter H. For the whole head being more elongated in this case than usual, and, as it were, compressed towards the ears, it was equitable that the number of the sutures should be unequal as to length and breadth, otherwise Nature would undeservedly have been named just, by Hippocrates, in thus giving equal gifts to the unequal. But it is not the case; for being most just, she formed the strongest suture which extends along the length of the head single, being thus proportionate to the width of the parts on both sides of it; namely, on the right and on the left; but she formed the transverse double in number, the one behind, as formerly said, called the lambdoid, and the other before, called the coronal, so that the bone of the head between these two sutures might be equal to those in the middle, on each side (the parietal bones?). The sutures of the head, in that configuration which is acuminate,^[750] furnish a very great example of the justness of Nature. For there are three principal figures of the head: the one entirely opposed to the natural configuration already described, when the head loses both its protuberances, that behind and the other before, and is equal on all hands, and like a true sphere; and two others, the one form having no protuberance in front, and the other none in the occiput. The sutures of the spherical head are like the letter χ , two only in number, and intersecting one another; the one extending transverse from the one ear to the other, and the other extending straight through the middle of the vertex to the middle of the forehead. For, as

when one part of the head is excessive, being longer than the other, it was just that the longer form should have more sutures, so, when both are alike, Nature bestowed an equal number on both. But in the head which wants the protuberance at the occiput, the straight and the coronal sutures remain, but the lambdoid is wanting (it being near to the protuberance that is wanting), so that the figure of the two resembles the letter T; as also when the protuberance of the head in front is wanting, the coronal at the same time is wanting, but there remains the one running lengthways and joining the lambdoid, and this form of construction is made to resemble the letter T. A fourth species of acuminate (sugar-loaf) head might be imagined, but which does not occur, with the head more prominent at the two ears than in front and behind." He goes on to state the reasons why there is no such construction of the head as this, and concludes as follows: "Wherefore Hippocrates described four configurations, and the sutures of each, in the manner we have now said that they exist, being justly distributed to each configuration by Nature as to position and number."^[751] The description of the bones and sutures of the head, given in the Latin work "De Ossibus," generally attributed to Galen, is to the same effect. The same number of distinct configurations of the head, and the same characters as regards the sutures, is also given by Avicenna, who professedly copies from Galen. (I., i., 5, 3.)

When examined together, these descriptions certainly must be admitted to have the appearance of being all derived from one original, namely, from our author, in this place; and taken literally, there can be no doubt that their meaning amounts to this: that the number of the sutures varies with the form of the head; that when there are protuberances both before and behind, the head in its upper part has two transverse sutures, namely, the coronal and the lambdoid, and one longitudinal, namely, the sagittal; that if the anterior protuberance be wanting, the coronal is wanting, and, if the posterior, the lambdoid. Now I need scarcely remark, that modern anatomists do not recognize such varieties in the configuration of the head nor in the numbers of the sutures, and that it is very rare indeed for either the coronal or the lambdoid suture to be found wanting. To all appearance, then, Galen was mistaken, and it only appears remarkable that, with all his knowledge of anatomy, theoretical and practical, and considering the opportunities which he must have possessed of examining human skeletons in Alexandria, he should have failed to observe and describe the bones of the cranium for himself.

Before stating my own conjectures on this question, it may be interesting to examine the solution of it attempted by authorities who lived about the period when the original study of human anatomy was revived in modern times. In the first place, then, I may mention that Ambrose Paré, who, I need scarcely say, was possessed of no mean talent for original observation, in treating of fractures of the head, adopts exactly the description given by Hippocrates; thus he describes "the bunches of the head" in nearly the same terms as our author, and adds, that such "bunches change the figure and site of the sutures," and that "there be some skulls that want the foremost suture, and other some the hind, and sometimes none of the true sutures, but only the false, *or* spurious, remain."^[752] Nay, it cannot but appear remarkable, that Vesalius, the great antagonist of Galen and of the ancient authorities in general, in the present instance does not venture to call in question their opinion, but gives a description of the different forms of the head, and the varieties of the sutures, which scarcely at all differs from that given by Hippocrates.^[753] It is singular, also, that certain other authorities, who were much more disposed to show a leaning to antiquity, such as Columbus, Eustachius, Fallopius, and Riolanus, should, in the present instance, have manifested a more independent spirit in challenging the authority of Hippocrates, though, at the same time, they show a disposition to find out some mode of bringing him clear off. Thus, for example, Riolanus is compelled to admit that there is no such variety in the forms and numbers of the sutures as Hippocrates describes; but he attempts to free him from error, by suggesting that the cases in which Hippocrates found them wanting must have been those of old men.^[754] He also quotes some very extraordinary instances, in which something approaching the varieties described by our author had been remarked.^[755] Fallopius does not hesitate, in his great anatomical work, to express the surprise he felt that all the authorities should have assented to the descriptions of the protuberances and sutures of the head given by Hippocrates; for that he, after having examined large heaps of crania in the Musea of Ferrara and Florence, had not found that they agreed with the descriptions given by Hippocrates; that he had seen crania without a suture, and yet not wanting in the protuberances; and in like manner, that he had seen the coronal suture obliterated, and yet the skull possessed its anterior prominence, and the lambdoid wanting, although the posterior protuberance was as usual. Altogether, then, in this work he modestly ventures to impugn the authority of Hippocrates.^[756] In his work entitled "Expositio in Librum Galeni de Ossibus," he adopts the same views, and there declares that he had never seen the sutures obliterated except from old age. But, in his work entitled "Expositio in Lib. Hippocrat. de Vulneribus Capitis," he gives two suppositions, which he had devised in order to defend the authority of Hippocrates: first, that Hippocrates did not give these varieties of form as real, but as hypothetical; and second, that he merely described them as being the vulgar opinion, without pledging himself to the correctness of the description. These, as far as I am aware, are the only defences which have ever been set up for our author in this matter, and it must be admitted that they are not very satisfactory. I shall now present the reader with the conjectural explanation which has occurred to myself. I have imagined that what Hippocrates meant was to express himself to the following effect: when the forehead is remarkably prominent, and, at the same time, there is a great depression behind, the cranium, if looked upon from above, will show the coronal suture running across the fore part of the head, and the sagittal through its middle, while the lambdoid will be inconspicuous, from being below the level of the coronal.

The two together, then, would form some resemblance to the letter T. When, on the other hand, the forehead is low, that is to say, wants its normal development, and the occiput is unusually prominent, the lambdoid suture joins the sagittal, so as to present some appearance of the same letter reversed. But in a square-built head, where the frontal and occipital regions have protuberances equally developed, the coronal and lambdoid sutures run nearly parallel to one another, and are joined in the middle by the sagittal, in which case the three sutures may be imagined to present some resemblance to the Greek letter H. When there is no protuberance either before or behind, and the sagittal suture passes through the middle of the bone down to the nasal process, the coronal suture intersects it, so as to give them something like the shape of the Greek letter χ .^[757] I offer this explanation, however, merely as a conjecture, and wish the reader to judge of it accordingly.

I now proceed to give an analysis of the contents of this treatise, and to attempt to form a correct estimate of their value.

Injuries of the cranial bones are divided by our author into five orders, as follows: 1, simple fractures, *or* fissures of various kinds and sizes (§ 4); 2, contusion, without fracture or depression (§ 5); 3, fractures attended with depression (§ 6); 4, the *hedra*, that is to say, the indentation or cut in the outer table of the bone, and not necessarily attended either with fracture or contusion (§ 7); 5, the counter-fissure, *or fracture par contre-coup* fracture and the severe contusion, require the operation of trepanning; whereas neither the *hedra* (*or* simple cut) nor the depressed fracture require it, and the counter-fissure does not admit it, owing to the obscurity of the symptoms with which it is attended (§ 9).

In the first place, the surgeon is to ascertain the nature and situation of the wound, by a careful investigation of all the circumstances of the case, but so as to avoid the use of the sound, if possible (§§ 9, 10).

Next are described the various kind of injury which the different sorts of weapons are most likely to inflict, and from the consideration of them the surgeon is to form an estimate of the probable nature of the accident (§ 11).

The characters of the *hedra*, *or* superficial injury of the cranium, and the difficulty of forming a correct estimate of it, when complicated by the presence of a suture, are strongly insisted upon (§ 12).

The principles upon which the treatment of injuries situated in different parts of the head should be treated, are carefully defined and stated. Great, and as now would be thought, superfluous directions are given, for ascertaining whether or not a fissure exists in the bone. The treatment, as far as applications go, is to be mild and desiccant. When a fracture cannot be made to disappear by scraping, the trepan is to be applied (§§ 13, 14).

The dangers which the bone incurs of becoming affected from the soft parts, are strongly insisted upon, and applications of a drying nature are prescribed (§ 15).

The condition of a piece of bone which is going to exfoliate is correctly and strikingly described (§ 16).

The treatment of depression is laid down, and the danger of applying the trepan in this case is strongly insisted upon (§ 17).

The peculiarities in the case of children are pointed out. Under certain circumstances, when there is contusion combined with the fracture, he admits of perforating the skull with a small trepan (§ 18).

When, after a severe injury, symptoms of irritation and inflammation appear to be coming on, the surgeon is to lose no time in proceeding to the operation. Some correct observations are made on the consequences of injuries of the head on other parts of the body (§ 19).

The treatment of erysipelatous inflammation is distinctly laid down (§20).

The operation of trepanning the skull is circumstantially described, and an interesting description is given of a mode of doing the operation peculiar to our author^[758] (§ 21).

This, then, as far as I know, is the first exposition ever made of a highly important subject in surgery, upon which professional men are still greatly divided in opinion. I cannot, then, resist the temptation to offer some remarks on the views of practice here recommended, and to institute a comparison between them and certain methods of treatment which have been in vogue of late years.

I can scarcely doubt but it will be generally admitted that the exposition of the subject here given is remarkably lucid, that our author's divisions of it are strongly marked, and his rules of practice, whether correct or not, distinctly laid down. At all events, it will not be affirmed that there is any confusion in his ideas, or that his principles of treatment are not properly defined. After all that has been written on injuries of the head, it would be difficult to point to any better arrangement of them than that of our author, into five orders: 1st, simple fractures without depression; 2d, contusions without fracture or depression; 3d, depression with fracture; 4th, simple incisions without fracture; 5th, fractures *par contre-coup*.

As regards the operation of trepanning the skull, then, our author's rule of practice is sufficiently well defined: we are to operate in the first two of these cases, that is to say, in simple fractures and contusions, but not in the last three, that is to say, in fracture with depression, in simple incisions in the skull, and in the counter-fissure. To begin, then, with the examination of

those cases in which the operation is proscribed: it is not to be had recourse to in the counter-fissure, because, from the nature of it, there is generally no rule by which its existence can be positively ascertained, and therefore the case is to be given up as hopeless.

In the simple incision of the bone, that is to say, in the slash or indentation, when the effects of the injury are not transmitted to the brain, it must be obvious that all instrumental interference must be strongly contraindicated.^[759]

At first sight it will appear remarkable to a surgeon, who approaches the subject with views exclusively modern, that our author should have interdicted the use of instruments in that class of injuries in which one would be inclined to suppose that they are most clearly indicated, namely, in a fracture of considerable extent, attended with depression of part of the bone from its natural level. Several questions present themselves here to be solved. Is the operation generally required? Has it been successful when it has been had recourse to? When it is to be performed, should it be done immediately, or not until the bad effects of the injury have manifested themselves?

With regard, then, to the necessity of the operation for depressed fractures, the most discordant opinions have prevailed in modern times, and even within a very recent period. Not to go farther back than Pott, it is well known that he established it as the general rule of practice, that in every case of fracture with depression, the skull should be perforated, and the depressed portion of the bone either raised to its level, or entirely removed. But since his time a great change of opinion has taken place on this subject, and of late it has become the general rule of practice (if rule can be predicated, where opinions are so vague and indeterminate) not to interfere, even in cases of depression, unless urgent symptoms have supervened. The late Mr. Abernethy took the lead in questioning the propriety of the rule laid down by Pott; and with the view of demonstrating that the operation may be often dispensed with in fractures complicated with depression, and in order, as he says, “to counteract in some degree the bias which long-accustomed modes of thinking and acting are apt to impress on the minds of practitioners,” he relates the histories of five cases of fracture with depression, which, in the space of twelve months, occurred under his own eyes in St. Bartholomew’s Hospital, and which all terminated favorably, although no operation was performed. These cases, supported by the authority of so great a name as Mr. Abernethy, made a deep impression on the profession, especially in this country, so that it became the established rule of practice in British surgery never to interfere in cases of fracture, unless with the view of removing urgent symptoms. See Cooper’s Surgical Dictionary, edit. 1825, and the previous edition. The old Hippocratic rule in regard to the trepan, when it is at all to be applied, namely, that of applying it as a preventive of bad consequences, was altogether eschewed, and it was held to be perfectly unwarrantable to perforate the skull, except with the intention of removing substances which were creating irritation and pressure of the brain. This practice, I say, was sanctioned by all the best army and hospital surgeons, from about the beginning of the present century, down to a very recent period. What, then, it will be asked, have been the results? Has experience confirmed the safety of this rule of practice, or has it not? To enable us to solve these queries, we have most elaborate and trustworthy statistics, published a few years ago by Dr. Laurie of Glasgow, which deserve to be seriously studied by every surgeon who may be called upon to discharge the duties of his profession in such cases. I cannot find room for long extracts from these valuable papers, but may be allowed to state a few of the more important results which are to be deduced from Dr. Laurie’s interesting investigation. Coming then at once to the point, it deserves to be remarked that Dr. Laurie’s ample experience has led him to reject decidedly the rule of practice, which, as I have stated, was established by Mr. Abernethy, about forty years ago, namely, that, in cases of depression, the symptoms of compression should be our guide to the employment of the trephine. He adds, “however well this rule may sound, when delivered *ex cathedra*, it will be found of very little practical utility, for this reason, that if we limit interference to cases exhibiting symptoms of compression, we had much better not interfere at all, inasmuch as such cases prove almost invariably fatal. Such, at least, has been the experience of the Glasgow hospitals; for out of fifty-six cases operated upon, including, in point of time, a period little short of fifty years, there does not appear in our records a single unequivocal instance of profound insensibility, in which the mere operation of trepanning removed the coma and paralysis, or in any way conduced to the recovery of the patient. We wish to be clearly understood as speaking of the trephine used in reference to the state of the bone in cases of profound insensibility, not employed to remove extravasated blood. Nor does the cause of our want of success appear at all obscure. We believe that in practice the cases of urgent compression dependent on depressed bone alone are very few indeed; we are well aware that many such are on record, we do not presume to impugn their accuracy, we merely affirm that the records of the Glasgow Infirmary do not add to the number.” He thus states his views with regard to the principles by which the application of the trephine should be regulated. “From what we have said, it will appear that we coincide with these who, in using the trephine, in cases of compound fracture of the skull, look more to the state of the bone than to the general symptoms, and *who employ it more as a preventive of inflammation and its consequences, than as a cure for urgent symptoms, the immediate result of the accident.*” He goes on to state that “the details we have given are by no means in favor of the trephine. Of fifty-six cases operated upon, eleven recovered, and forty-five died. We feel assured that this affords too favorable a view of the actual results.”^[760]

From the extracts now given, it will readily be seen that this very able authority has rejected entirely the rule of practice established by Mr. Abernethy, and that, in so far, he has reverted to the principle upon which the use of the instruments in simple fractures of the skull was regulated by Hippocrates, namely, as a preventive of the bad consequences of fracture on the brain, rather than with the view of relieving them when established. It will further be seen that, in whatever way applied, the use of perforating instruments in the case of depressed fractures is attended with so unsatisfactory results, that it may be doubted if any other operation in surgery, recognized as legitimate, be equally fatal.^[761] Less than one fifth of the patients operated upon recovered. In fact, he very candidly admits “that it would not have been greatly to the disadvantage of the patients admitted into the Glasgow Infirmary, if the trephine had never found its way within its walls.” He further, in conclusion, adverts to the well-known fact that Desault, in the end, completely abandoned the operation, and that Mr. Lawrence states, “as far as the experience of this Hospital (St. Bartholomew’s) goes, he can cite very few instances in which the life of the patient had been saved by the operation of trephining.”^[762]

Altogether, then, it will be allowed to no very questionable whether, in general, the Hippocratic treatment, in cases of fracture with depression, would not be fully as successful as the modern practice of perforating the skull. Moreover, it is by no means well ascertained, as generally assumed by superficial observers of facts in medical practice, that depressed fractures are more dangerous than other injuries of the skull attended with less formidable appearances. Indeed, recent experience has shown, in confirmation of the opinion advanced by our author, that extensive fractures, with great depression, are frequently not followed by any very dangerous train of consequences. (See Thomson’s “Observations made in the Military Hospitals of Belgium,” pp. 59, 60; Hennen’s “Military Surgery,” p. 287; Cooper’s “Lectures,” xiii.; Mr. Guthrie’s “Lectures on Injuries of the Head,” p. 56.) All these, in substance, coincide with Mr. Guthrie, who mentions with approbation that “it has been stated from the earliest antiquity, that the greater the fracture, the less the concussion of the brain.” I may mention further, that I myself, in the course of my own experience, have known many instances in which fractures with considerable depression were not followed, either immediately or afterwards, by any bad consequences; while, on the other hand, I have known cases in which simple contusion of the bone, without fracture or extravasation, and without even very urgent symptoms of concussion at first, have proved fatal in the course of a day or two. Now, in such circumstances, Hippocrates would have operated by either perforating the skull at once, down to the meninx, and removing a piece of it, or by sawing it nearly through, and leaving the piece of bone to exfoliate. It will be asked here, what object can he have had in view by this procedure? This he has nowhere distinctly defined; but, judging from the whole tenor of this treatise, and that of his commentator, Galen, I can have no doubt in my mind that what he wished to accomplish was to loosen the bones of the head, and give greater room to the brain, which he conceived to be in a state of congestion and swelling brought on by the vibration, or *trémoussement*, communicated directly to the brain by the contusion. It is, in fact, an opinion which Hippocrates repeatedly inculcates, not only with regard to the brain, but also respecting injuries of the chest and joints, that severe contusions are, in general, more dangerous than fractures, the effects of the vibration in the former case being more violent than in the latter.^[763] Believing, then, that, in contusions, the internal structure of the brain is extensively injured, and that irritation, with hypertrophy, are the consequences, he advocated instrumental interference, in order as I have stated, to give more room to the brain, and relieve it from its state of compression.^[764] This, no doubt, was the rationale of his practice also in simple fractures, not attended with depression, that is to say, his object in perforating the skull was to remove tension, and furnish an outlet to the collection within, whether of a liquid or a gaseous nature.

There can be no doubt that our author also had it in view, by perforating the skull, to afford an issue to extravasated blood and other matters collected within the cranium. This clearly appears from what is stated in section 18, and the same rule of practice is distinctly described by Celsus in the following terms: “Raro, sed aliquando tamen evenit, ut os quidem totum integrum maneat, intus vero ex ictu vena aliqua in cerebri membrana rupta aliquid sanguinis mittat; isque ibi concretus magnos dolores moveat, et oculos quibusdam obcæcet.... Sed ferè contra id dolor est, et, eo loco cute incisa, pallidum os reperitur: ideoque id os quoque excidendum est.” (viii., 4.) It is quite certain, then, that one of the objects for which our author recommended trepanning, was to give issue to extravasated blood on the surface of the skull. This naturally leads me to compare the results of modern experience in the treatment of cases of contusion, with or without extravasation of blood.

All the earlier of our modern authorities on surgery, such as Theodoric, Pet. c. Largelata, Ambrose Paré, Wiseman, and Fallopius, distinctly held that contusions of the skull, even when not complicated with a fracture, are often of so formidable a nature as to require the use of perforating instruments. The same views are strenuously advocated by Pott, who has described the effects of contusion in very elegant and impressive language. See page 42; ed. Lond. 1780. The upshot is, that one of the consequences of a severe contusion of the bone frequently is separation of the pericranium, “which is almost always followed by a separation between the cranium and the dura mater; a circumstance extremely well worth attending to in fissures and undepressed fractures of the skull, because it is from this circumstance principally that the bad symptoms and the hazard in such cases arise.” (p. 50.)^[765] After insisting, in very strong terms, on the danger attending severe contusions of the skull, he proceeds to lay down the rules of treatment, which, in a word, are comprehended in the two following intentions:—first, to prevent bad consequences by having recourse, at first, to depletion; and, second, to procure the discharge of matter collected

under the cranium, which can be answered only by the perforation of it. He agrees with Archigenes that the operation is generally too long deferred, and that the sooner it is performed the better. Still, however, it is to be borne in mind that even Potts does not make it a general rule to operate at first, *before* the bad symptoms have come on, that is to say, during the first three days, and that he rather appears to have followed Celsus, who alludes to the method of Hippocrates, and describes his rule of practice in the following terms:

—“In omni vero fissio fractoque osse, protinus antiquiores medici ad ferramenta veniebant, quibus id exciderent. Sed multo melius est ante emplastra experiri, etc.... Si vero sub prima curatione febris intenditur, ... magni dolores sunt, cibique super hæc fastidium increseit; tum demum ad manum scalpumque veniendum est.” (viii., 4.) Pott then, it appears, follows the rule of Celsus, and does not operate until unpleasant effects have developed themselves;^[766] but, at the same time, he candidly admits that, although the course now described be all that our art is capable of doing in these melancholy cases, he wishes he could say that it was frequently successful. He then goes on to relate several cases: first, of simple contusion without a wound; second, of contusion with a wound; and, third, of contusion with extravasation. In all these classes of cases he operated with very equivocal results; but then it is to be borne in mind, that, as I have said, he operated, like Celsus, after the bad effects had come on, and not, like Hippocrates, at first, in order to prevent them. Even with all these discouraging results, he continued to adhere to this rule of treatment, which, under the sanction of his name, became the established practice of the profession. The late Mr. Abernethy, who took the lead in innovating upon Pott’s rules for the application of the trephine, did not venture to make any material change in this case when he supposed that there was any considerable extravasation of blood; and he delivered it as a test whereby we might judge whether or not a great vessel had been ruptured within the skull, to examine whether or no the bone bled, having generally found, as, indeed, had been clearly laid down by Celsus, that in these cases the bone does not bleed. The rule of practice, then, to operate in order to remove the coagula of blood and matters which form between the skull and the dura mater, was sanctioned by Sir Charles Bell and Sir Astley Cooper; but they, like Mr. Abernethy, generally condemn interference when the fluids are situated below the membrane. On this subject Mr. Guthrie remarks:—“The operation of incising the dura mater, to admit of the discharge of blood or matter from beneath, and even of puncturing the brain, is much more commonly performed in France than in Great Britain, where it is very rarely had recourse to, *and which may be an error*,” etc. (p. 125.)

After thirty years’ further experience, this practice has been tested by the recent statistics of Dr. Laurie, and the results, as stated by him, are very discouraging. In the Hospital of Glasgow, it was found in practice that there was no certain symptom whereby it could be determined at what part of the head the blood had been effused, nor, when discovered, could it, in general, be removed by trephining the skull. The results, in short, were the following: “We have thus thirty-nine cases in which extravasation existed as the principal lesion, or as an important complication, in only one of which extravasation existed as the principal lesion, or as an important complication; in only one of which could an operation have saved the patient; and of the seventeen cases operated upon, *not one* recovered after, or was benefited by, the removal of the coagula.”

Such, then, are the results of modern experience, as far as they are at present ascertained, in the use of the trephine for the treatment of contusion, and undepressed fracture, complicated with the effusion either of blood or of matter, from the days of Pott down to the present time. The reader, however, should bear in remembrance that the practice, of which the results have been shown to be so unsatisfactory, is not that of Hippocrates, but of Celsus; for, in the present instance, even Dr. Laurie repudiates the idea of operating “for the purpose of relieving the evil consequences *which may follow* concussion of the brain,” and holds distinctly in this case that one is not warranted in even entertaining the idea of operating, unless—“first, when the puffy tumor indicates the spot which probably has sustained the greatest amount of injury; second, such an inflamed and suppurating condition of the injured soft parts as renders it more than probable that the corresponding portion of the dura mater is in a similarly diseased condition; third, inflammatory fever, preceded or followed by rigors, and symptoms of compression.” From what has been stated, then, it must appear evident that the recent statistics furnish no test whatever of the results of the practice laid down by Hippocrates, which was founded upon an entirely different principle, namely, the preventive.

But, however anxious I may feel to prosecute further this comparison of the results of ancient and of modern experience on this highly interesting subject, my necessary limits compel me to bring this discussion to a close. Before doing so, however, I shall briefly state the inferences which I think may be drawn from a careful study of all the principle authorities who have written on injuries of the head from Hippocrates down to the present time:

1. All the serious injuries of the skull may be arranged conveniently under the classes of contusions, simple fractures and fractures with depressions.
2. Hippocrates recommended the operation of perforating the cranium, in cases of simple fractures and contusions, whenever he apprehended that these would be followed by serious consequences, such as inflammation, extravasation of blood, and the effusion of matter.
3. Hippocrates operated in these cases during the first three days, before any serious symptoms had come on, but Celsus rejected this rule, and postponed the operation until after these effects had been developed.

4. The objects which Hippocrates had in view by perforating the skull, either entirely through or nearly so, would appear to have been to slacken the tightness of the skull, and procure the evacuation of extravasated blood lying within it.

5. The object for which Celsus opened the skull would appear to have been solely to remove bodies which were creating irritation in the brain.

6. All the ancient authorities looked upon contusions and simple fractures as being very formidable injuries, which generally produce congestion in the brain, with inflammation and effusion.

7. In modern times, at least within the last hundred years, the trephine has never been applied in cases of contusion and simple fracture, upon the principle of the operation acting as a preventive of subsequent mischief, but only with the object of relieving effusion when it was supposed to have taken place within the cranium, that is to say, upon the plan recommended by Celsus.

8. The most contradictory accounts are given by modern authorities, especially by the French surgeons of the eighteenth century, as to the different results in cases of this description, when let alone, and when treated upon the Celsian principle; and the recent statistics of the operation are extremely unfavorable.

9. Hippocrates regarded fractures accompanied with depression and a considerable separation of the bones as being generally less dangerous than severe contusions and simple fractures, as in the former case the brain is usually less hurt by the vibration of the shock which inflicted the injury, and there is an outlet to any noxious matters which may get congested in the brain.

10. Hippocrates, as a general rule, did not operate in cases of depression, not even in cases of comminuted fracture, but in the latter case left the pieces of bone to separate gradually by suppuration.

11. Celsus, on the other hand, approved of removing spiculæ at once, of raising the depressed corner of a fractured bone, by sawing off the superincumbent part, and even of perforating the adjoining bone, and, in certain instances, of removing the whole of the depressed portion.

12. Pott laid it down as a general rule of practice, to operate with the trephine in all cases of fracture accompanied with any considerable degree of depression, and this formed the established practice in this country, until the late Mr. Abernethy, about forty years ago, introduced the rule of not interfering in such cases until urgent symptoms had come on.

13. Of late years a further innovation has taken place in this rule of practice in cases of depressed fracture, the operation being had recourse to by Dr. Laurie and others, on the principle of preventing the bad effects likely to result from the injury.

14. On whatever principle applied, the statistics of large hospitals exhibit the results of the operation in a most unfavorable light, insomuch that many of the most able and experienced surgeons of the day hesitate whether, as a general rule, the operation ought not to be abandoned altogether.

Finally, a careful study of the whole literature of the subject, from Hippocrates down to the present time, leads to the conclusion that what constitutes the great difficulty in the treatment of injuries of the head is, that the operation, to be successful, would require to be performed early, and rather with a view of preventing serious consequences, than of removing them after they have come on; and that these can seldom be estimated so correctly as could be wished, since they frequently bear no proportion to the apparent magnitude of the mischief which the cranium has sustained.^[767]

As the reader may find some difficulty in apprehending correctly the nature of the instruments and other apparatus used by the ancients in surgical operations, I have subjoined drawings of them, taken principally from the works of Vidus Vidius and Andreas à Cruce, who both lived at a time when these instruments must have been sufficiently common in the cabinets of learned physicians, so that there is every presumption that the figures which they give are sufficiently correct. The manner in which they were used will readily be comprehended from their shapes, assisted by the following lucid description of the ancient process of trepanning the skull, given by Mr. Pott. "If the piece of bone intended to be removed was larger than could be comprehended within the modiolus (*trephine?*) then in use, and which was a very defective instrument in many respects, the operation was thus performed by means of *terebræ*. The piece intended to be taken away was surrounded with perforations made at small distances from each other, and then either the scalper *excisorius* or the scalper *lenticulatus* was introduced, and, by means of repeated strokes with a heavy mallet, was driven through all the interspaces between each perforation. By these means the portion of bone so surrounded was removed, and the *dura mater* was laid bare."^[768] That the modiolus of Celsus was a small circular saw with a pivot, exactly like the modern trephine, seems quite obvious from his own description of it; and that the instrument called by our author *terebra serrata* (πίων χαρακτός) was identical with it, cannot admit of any doubt. See Foës, Œc. Hipp. in voce *πίων*.

Before concluding, I must also say a few words on one important point connected with the constitutional treatment, which the modern reader may at first sight be surprised to find no mention made of in this treatise—I mean the use of venesection in the treatment of injuries of the head. Now certainly it does not appear that Hippocrates regarded bleeding as necessarily

forming a portion of the system of treatment in injuries of the bones of the head any more than in those of other bones. But, although these were his views, it can be as little doubted, by any one who is acquainted with his general views of practice, that he bled whenever the abstraction of blood was indicated, either to produce evacuation or revulsion. We know, for example, that in pains of the back part of the head he opened the temporal vessels,^[769] and that in all inflammations and febrile diseases he abstracted blood freely, nay, perhaps, *ad deliquium animi*.^[770] And that Hippocrates enforced the depletory system of treatment in injuries of the head, when pain and inflammatory fever supervened, is quite obvious, from its having been the system pursued in such cases by all subsequent authorities, who looked up to him as their great guide in practice. See PAULUS ÆGINETA, Book VI., 90, Syd. Soc. Edit. I may mention further, as a proof that I am not straining a point in the present instance, in order, as might be supposed, to bring my author clear off in a case where he would appear to have been in fault, that Ambrose Paré, who is a great advocate for depletion in the treatment of fractures of the skull, is at great pains to show that he has Hippocrates on his side in support of this practice.^[771] But while it is maintained that our author did not omit venesection when properly indicated, I did not mean to say that he or any of the ancient authorities carried the abstraction of the blood to the extent practiced by Pott, or the members of the Royal Academy of Surgery in France, nor as was done by the army and hospital surgeons of this country during the late war.^[772] Whether or not this was a defect in ancient practice I shall not take it upon me to offer an opinion. Suffice it to say, that there is undoubted evidence that in injuries of the head the ancient surgeon, as is *naively* recommended by Avicenna, “bled his patient when he stood in need of being bled;”^[773] that is to say, according to special indications, and not in obedience to any general rule.^[774]

There is another point of practice in injuries of the head to which it is proper that I should draw attention—I mean cold applications. Now it is beyond a doubt that the application of cold in diseases of the brain is pointedly condemned by Hippocrates, and that he used hot applications instead;^[775] and, moreover, that most of the ancient authorities adhered to his rule on this point. At the same time it would appear, that in extreme cases certain of them did not scruple to apply ice to the shaved head.^[776] I shall only remark further, that in this case, as in diseases of the eyes, perhaps the safest rule is, to be guided very much by the feelings and habits of the patient.

[The Plates referred to will be found at the end of the work.]

ON INJURIES OF THE HEAD

1. Men’s heads are by no means all like to one another, nor are the sutures of the head of all men constructed in the same form. Thus, whoever has a prominence in the anterior part of the head (by prominence is meant the round protuberant part of the bone which projects beyond the rest of it), in him the sutures of the head take the form of the Greek letter *tau*, T; for the head has the shorter line running transverse before the prominence, while the other line runs through the middle of the head, all the way to the neck.^[777] But whoever has the prominence in the back part of the head, in him the sutures are constructed in quite the opposite form to the former; for in this case the shorter line runs in front of the prominence, while the longer runs through the middle all along to the forehead.^[778] But whoever has a prominence of the head both before and behind, in him the sutures resemble the Greek letter *êta* H; for the long lines of the letter run transverse before each prominence while the short one runs through the middle and terminates in the long lines.^[779] But whoever has no prominence on either part he has the sutures of the head resembling the Greek letter *chi*; for the one line comes transverse to the temple while the other passes along the middle of the head.^[780] The bone at the middle of the head is double, the hardest and most compact part being the upper portion, where it is connected with the skin, and the lowest, where it is connected with the meninx (*dura mater*); and from the uppermost and lowermost parts the bone gradually becomes softer and less compact, till you come to the *diploe*.^[781] The *diploe* is the most porous, the softest, and most cavernous part. But the whole bone of the head, with the exception of a small portion of the uppermost and lowermost portions of it, is like a sponge; and the bone has in it many juicy substances, like caruncles; and if one will rub them with the fingers, some blood will issue from them.^[782] There are also in the bone certain very slender and hollow vessels full of blood. So it is with regard to hardness, softness, and porosity.

2. In respect to thickness and thinness; the thinnest and weakest part of the whole head is the part about the bregma; and the bone there has the smallest and thinnest covering of flesh upon it, and the largest proportion of brain is situated in that region of the head. And hence it happens that from similar or even smaller wounds and instruments, when a person is wounded to the same or a less degree, the bone of the head there is more contused, fractured, and depressed; and that injuries there are more deadly and more difficult to cure; and it is more difficult to save one’s life in injuries there than in any other part of the head; that from having sustained a similar or even a less wound a man will die, and that, too, in a shorter space of time than from a wound in any other part of the head. For the brain about the bregma feels more quickly and strongly any mischief that may occur to the flesh or the bone; for the brain about the bregma is in largest quantity, and is covered by the thinnest bone and the least flesh. Of the other portions, the weakest is that about the temples; for it is the conjunction of the lower jaw with the cranium, and there is motion there up and down as at a joint; and the organ of hearing is near it; and further, a hollow and important vein runs along the temple. But the whole bone of the head behind the vertex and the ear is stronger than the whole

anterior part, and the bone itself has a larger and deeper covering of flesh upon it. And hence it follows, that when exposed to the same or even greater injuries from instruments of the same or greater size, the bone is less liable to be fractured and depressed than elsewhere; and that in a fatal accident the patient will live longer when the wound is in the posterior part of the head than when elsewhere; and that pus takes longer time to form and penetrate through the bone to the brain, owing to the thickness of the bone; and moreover, as there is less brain in that part of the head, more persons who are wounded in the back part of the head escape than of those who are wounded in the anterior part.^[783] And in fatal cases, a man will survive longer in winter than in summer, whatever be the part of the head in which the wound is situated.

3. As to the *hædræ* (dints or marks?) of sharp and light weapons, when they take place in the bone without fissure, contusion, or depression inwards (and these take place equally in the anterior and posterior part of the head), death, when it does occur, does not properly result from them. A suture appearing in a wound, when the bone is laid bare, on whatever part of the head the wound may have been inflicted, is the weakest point of the head to resist a blow or a weapon, when the weapon happens to be impinged into the suture itself; but more especially when this occurs in the bregma at the weakest part of the head, and the sutures happen to be situated near the wound, and the weapon has hit the sutures themselves.^[784]

4. The bone in the head is liable to be wounded in the following modes, and there are many varieties in each of these modes of fracture: When a wounded bone breaks, in the bone comprehending the fissure, contusion necessarily takes place where the bone is broken; for an instrument that breaks the bone occasions a contusion thereof more or less, both at the fracture and in the parts of the bone surrounding the fracture.^[785] This is the first mode. But there are all possible varieties of fissures; for some of them are fine, and so very fine that they cannot be discovered, either immediately after the injury, or during the period in which it would be of use to the patient if this could be ascertained. And some of these fissures are thicker and wider, certain of them being very wide. And some of them extend to a greater, and some to a smaller, distance. And some are more straight, nay, completely straight; and some are more curved, and that in a remarkable degree. And some are deep, so as to extend downwards and through the whole bone; and some are less so, and do not penetrate through the whole bone.

5. But a bone may be contused, and yet remain in its natural condition without any fracture in it; this is the second mode. And there are many varieties of contusion; for they occur to a greater and less degree, and to a greater depth, so as sometimes to extend through the whole bone; or to a less depth, so as not to extend through the whole bone; and to a greater and smaller length and breadth. But it is not possible to recognize any of these varieties by the sight, so as to determine their form and extent; neither, indeed, is it visible to the eyes when any mischief of this kind takes place, and immediately after the injury, whether or not the bone has been actually bruised, as is likewise the case with certain fractures at a distance from the seat of injury.^[786]

6. And the bone being fractured, is sometimes depressed inwards from its natural level along with the fractures, otherwise there would be no depression; for the depressed portion being fractured and broken off, is pushed inwards, while the rest of the bone remains in its natural position; and in this manner a fracture is combined with the depression.^[787] This is the third mode. There are many varieties of depression, for it may comprehend a greater and a smaller extent of bone, and may either be to a greater depth, or less so, and more superficial.^[788]

7. When a *hedra*, or dint of a weapon, takes place in a bone, there may be a fracture combined with it; and provided there be a fracture, contusion must necessarily be joined, to a greater or less extent, in the seat of the dint and fracture, and in the bone which comprehends them.^[789] This is the fourth mode. And there may be a *hedra*, or indentation of the bone, along with contusion of the surrounding bone, but without any fracture either in the *hedra* or in the contusion inflicted by the weapon. But the indentation of a weapon takes place in a bone, and is called *hedra*, when the bone remaining in its natural state, the weapon which struck against the bone leaves its impression on the part which it struck. In each of these modes there are many varieties, with regard to the contusion and fracture, if both these be combined with the *hedra*, or if contusion alone, as it has been already stated that there are many varieties of contusion and fracture. And the *hedra*, or dint, of itself may be longer and shorter, crooked, straight, and circular; and there are many varieties of this mode, according to the shape of the weapon; and they may be more or less deep, and narrower or broader, and extremely broad. When a part is cleft, the cleft or notch which occurs in the bone, to whatever length or breadth, is a *hedra*, if the other bones comprehending the cleft remain in their natural position, and be not driven inwards; for in this case it would be a depression, and no longer a *hedra*.^[790]

8. A bone may be injured in a different part of the head from that on which the person has received the wound, and the bone has been laid bare. This is the fifth mode. And for this misfortune, when it occurs, there is no remedy; for when this mischief takes place, there is no means of ascertaining by any examination whether or not it has occurred, or on what part of the head.^[791]

9. Of these modes of fracture, the following require trepanning: the contusion, whether the bone be laid bare or not; and the fissure, whether apparent or not. And if, when an indentation (*hedra*) by a weapon takes place in a bone it be attended with fracture and contusion, and even if contusion alone, without fracture, be combined with the indentation, it requires trepanning.

A bone depressed from its natural position rarely requires trepanning; and those which are most pressed and broken require trepanning the least; neither does an indentation (*hedra*) without fracture and contusion require trepanning; nor does a notch, provided it is large and wide; for a notch and a *hedra* are the same.^[792]

10. In the first place, one must examine the wounded person, in what part of the head the wound is situated, whether in the stronger or weaker parts; and ascertain respecting the hairs about the wound, whether they have been cut off by the instrument, and have gone into the wound; and if so, one should declare that the bone runs the risk of being denuded of flesh, and of having sustained some injury from the weapon. These things one should say from a distant inspection, and before laying a hand on the man;^[793] but on a close examination one should endeavor to ascertain clearly whether the bone be denuded of flesh or not; and if the denuded bone be visible to the eyes, this will be enough; but otherwise an examination must be made with the sound. And if you find the bone denuded of the flesh, and not safe from the wound, you must first ascertain the state of the bone, and the extent of the mischief, and what assistance it stands in need of. One should also inquire of the wounded person how and in what way he sustained the injury; and if it be not apparent whether the bone has sustained an injury or not, it will be still more necessary, provided the bone be denuded, to make inquiry how the wound occurred, and in what manner; for when contusions and fractures exist in the bone, but are not apparent, we must ascertain, in the first place from the patient's answers, whether or not the bone has sustained any such injuries, and then find out the nature of the case by word and deed, with the exception of sounding. For sounding does not discover to us whether the bone has sustained any of these injuries or not; but sounding discovers to us an indentation inflicted by a weapon, and whether a bone be depressed from its natural position, and whether the bone be strongly fractured; all which may also be ascertained visibly with the eyes.^[794]

11. And a bone sustains fractures, either so fine as to escape the sight, or such as are apparent, and contusions which are not apparent, and depression from its natural position, especially when one person is intentionally wounded by another, or when, whether intentionally or not, a blow or stroke is received from an elevated place, and if the instrument in the hand, whether used in throwing or striking, be of a powerful nature, and if a stronger person wound a weaker. Of those who are wounded in the parts about the bone, or in the bone itself, by a fall, he who falls from a very high place upon a very hard and blunt object is in most danger of sustaining a fracture and contusion of the bone, and of having it depressed from its natural position; whereas he that falls upon more level ground, and upon a softer object, is likely to suffer less injury in the bone, or it may not be injured at all. Of those instruments which, falling upon the head, wound the parts about the bone, or the bone itself, that which falls from a very high place, and the least on a level with the person struck, and which is at the same time very hard, very blunt, and very heavy, and which is the least light, sharp, and soft, such an instrument would occasion a fracture and contusion of the bone. And there is most danger that the bone may sustain these injuries, under such circumstances, when the wound is direct and perpendicular to the bone, whether struck from the hand or from a throw, or when any object falls upon the person, or when he is wounded by falling, or in whatever way the bone sustains a direct wound from this instrument. Those weapons which graze the bone obliquely are less apt to fracture, contuse, or depress the bone, even when the bone is denuded of flesh; for in some of those wounds thus inflicted the bone is not laid bare of the flesh. Those instruments more especially produce fractures in the bone, whether apparent or not, and contusions, and inward depression of the bone, which are rounded, globular, smooth on all sides, blunt, heavy, and hard; and such weapons bruise, compress, and pound the flesh; and the wounds inflicted by such instruments, whether obliquely or circularly, are round, and are more disposed to suppurate, and to have a discharge, and take longer time to become clean; for the flesh which has been bruised and pounded must necessarily suppurate and slough away. But weapons of an oblong form, being, for the most part, slender, sharp, and light, penetrate the flesh rather than bruise it, and the bone in like manner; and such an instrument may occasion a *hedra* and a cut (for a *hedra* and a cut are same thing); but weapons of this description do not produce contusions, nor fractures, nor depressions inwardly. And in addition to the appearances in the bone, which you can detect by the sight, you should make inquiry as to all these particulars (for they are symptoms of a greater or less injury), whether the wounded person was stunned, and whether darkness was diffused over his eyes, and whether he had vertigo, and fell to the ground.^[795]

12. When the bone happens to be denuded of flesh by the weapon, and when the wound occurs upon the sutures, it is difficult to distinguish the indentation (*hedra*) of a weapon which is clearly recognized in other parts of the bone, whether it exist or not, and especially if the *hedra* be seated in the sutures themselves. For the suture being rougher than the rest of the bone occasions confusion, and it is not clear which is the suture, and which the mark inflicted by the instrument, unless the latter (*hedra*) be large. Fracture also for the most part is combined with the indentation when it occurs in the sutures; and this fracture is more difficult to discern when the bone is broken, on this account, that if there be a fracture, it is situated for the most part in the suture. For the bone is liable to be broken and slackened there, owing to the natural weakness of the bone there, and to its porosity, and from the suture being readily ruptured and slackened: but the other bones which surround the suture remain unbroken, because they are stronger than the suture.^[796] For the fracture which occurs at the suture is also a slackening of the suture, and it is not easy to detect whether the bone be broken and slackened by the indentation of a weapon occurring in the suture, or from a contusion of the bone at the sutures; but it is still more difficult to detect a fracture connected with contusion. For the sutures, having the appearance of fissures, elude the discernment and sight of the physician, as being

rougher than the rest of the bone, unless the bone be strongly cut and slackened, (for a cut and a *hedra* are the same thing.)^[797] But it is necessary, if the wound has occurred at the sutures, and the weapon has impinged on the bone or the parts about it, to pay attention and find out what injury the bone has sustained. For a person wounded to the same, or a much smaller, extent, and by weapons of the same size and quality, and even much less, will sustain a much greater injury, provided he has received the blow at the sutures, than if it was elsewhere. And many of these require trepanning, but you must not apply the trepan to the sutures themselves, but on the adjoining bone.^[798]

13. And with regard to the cure of wounds in the head, and the mode of detecting injuries in the bone which are not apparent, the following is my opinion:—In a wound of the head, you must not apply anything liquid, not even wine, but as little as possible, nor a cataplasm, nor conduct the treatment with tents, nor apply a bandage to an ulcer on the head, unless it be situated on the forehead, in the part which is bare of hairs, or about the eyebrow and eye, for wounds occurring there require cataplasms and bandages more than upon any other part of the head.^[799] For the rest of the head surrounds the whole forehead, and the wounds wherever situated become inflamed and swelled, owing to an influx of blood from the surrounding parts.^[800] And neither must you apply cataplasms and bandages to the forehead at all times; but when the inflammation is stopped and the swelling has subsided, you must give up the cataplasms and bandages. A wound in any other part of the head must not be treated with tents, bandages, or cataplasms, unless it also requires incision. You must perform incision on wounds situated on the head and forehead, whenever the bone is denuded of flesh, and appears to have sustained some injury from the blow, but the wound has not sufficient length and breadth for the inspection of the bone, so that it may be seen whether it has received any mischief from the blow, and of what nature the injury is, and to what extent the flesh has been contused, and whether the bone has sustained any injury, or whether it be uninjured by the blow, and has suffered no mischief; and with regard to the treatment, what the wound, and the flesh, and the injury of the bone stand in need of. Ulcers of this description stand in need of incision; and, if the bone be denuded of the flesh, and if it be hollow, and extend far obliquely, we cut up the cavity wherever the medicine cannot penetrate readily, whatever medicine it may be; and wounds which are more inclined to be circular and hollow, and for the most part others of the like shape, are cut up by making a double incision in the circle lengthways, according to the figure of the man, so as to make the wound of a long form. Incisions may be practiced with impunity on other parts of the head, with the exception of the temple and the parts above it, where there is a vein that runs across the temple, in which region an incision is not to be made. For convulsions seize on a person who has been thus treated; and if the incision be on the left temple, the convulsions seize on the right side; and if the incision be on the right side, the convulsions take place on the left side.^[801]

14. When, then, you lay open a wound in the head on account of the bones having been denuded of the flesh, as wishing to ascertain whether or not the bone has received an injury from the blow, you must make an incision proportionate to the size of the wound, and as much as shall be judged necessary. And in making the incision you must separate the flesh from the bone where it is united to the membrane (*pericranium*?) and to the bone, and then fill the whole wound with a tent, which will expand the wound very wide next day with as little pain as possible; and along with the tents apply a cataplasm, consisting of a mass (*maza*) of fine flour pounded in vinegar, or boiled so as to render it as glutinous as possible.^[802] On the next day, when you remove the tent, having examined the bone to see what injury it has sustained, if the wound in the bone be not right seen by you, nor can you discover what mischief the bone itself has sustained, but the instrument seems to have penetrated to the bone so as to have injured it, you must scrape the bone with a raspatory to a depth and length proportionate to the suture of the patient, and again in a transverse direction, for the sake of the fractures which are not seen, and of the contusions which are not discovered, as not being accompanied with depression of the bone from its natural position. For the scraping discovers the mischief, if the injuries in the bone be not otherwise manifest. And if you perceive an indentation (*hedra*) left in the bone by the blow, you must scrape the dint itself and the surrounding bones, lest, as often happens, there should be a fracture and contusion, or a contusion alone, combined with the dint, and escape observation. And when you scrape the bone with the raspatory, and it appears that the wound in the bone requires the operation, you must not postpone it for three days, but do it during this period, more especially if the weather be hot, and you have had the management of the treatment from the commencement. If you suspect that the bone is broken or contused, or has sustained both these injuries, having formed your judgment from the severity of the wound, and from the information of the patient, as that the person who inflicted the wound, provided it was done by another person, was remarkably strong, and that the weapon by which he was wounded was of a dangerous description, and then that the man had been seized with vertigo, dimness of vision, and stupor, and fell to the ground,—under these circumstances, if you cannot discover whether the bone be broken, contused, or both the one and the other, nor can see the truth of the matter, you must dissolve the jet-black ointment,^[803] and fill the wound with it when thus dissolved, and apply a linen rag smeared with oil, and then a cataplasm of the *maza* with a bandage; and on the next day, having cleaned out the wound, scrape the bone with the raspatory. And if the bone is not sound, but fractured and contused, the rest of it which is scraped will be white; but the fracture and contusion, having imbibed the preparation, will appear black, while the rest of the bone is white. And you must again scrape more deeply the fracture where it appears black; and, if you thus remove the fissure, and cause it to disappear, you may conclude that there has been a contusion of the bone to a greater or less

extent, which has occasioned the fracture that has disappeared under the raspatory; but it is less dangerous, and a matter of less consequence, when the fissure has been effaced. But if the fracture extend deep, and do not seem likely to disappear when scraped, such an accident requires trepanning. But having performed this operation, you must apply the other treatment to the wound.

15. You must be upon your guard lest the bone sustain any injury from the fleshy parts if not properly treated. When the bone has been sawed and otherwise denuded, whether it be actually sound, or only appears to be so, but has sustained some injury from the blow, there may be danger of its suppurating (although it would not otherwise have done so), if the flesh which surrounds the bone be ill cured, and become inflamed and strangled; for it gets into a febrile state, and becomes much inflamed.^[804] For the bone acquires heat and inflammation from the surrounding flesh, along with irritation and throbbing, and the other mischiefs which are in the flesh itself, and from these it gets into a state of suppuration. It is a bad thing for the flesh (*granulations?*) in an ulcer to be moist and mouldy, and to require a long time to become clean. But the wound should be made to suppurate as quickly as possible; for, thus the parts surrounding the wound would be the least disposed to inflammation, and would become the soonest clean; for the flesh which has been chopped and bruised by the blow, must necessarily suppurate and slough away. But when cleaned the wound must be dried, for thus the wound will most speedily become whole, when flesh devoid of humors grows up, and thus there will be no fungous flesh in the sore. The same thing applies to the membrane which surrounds the brain: for when, by sawing the bone, and removing it from the meninx, you lay the latter bare, you must make it clean and dry as quickly as possible, lest being in a moist state for a considerable time, it become soaked therewith and swelled; for when these things occur, there is danger of its mortifying.^[805]

16. A piece of bone that must separate from the rest of the bone, in consequence of a wound in the head, either from the indentation (*hedra*) of a blow in the bone, or from the bone being otherwise denuded for a long time, separates mostly by becoming exsanguous. For the bone becomes dried up and loses its blood by time and a multiplicity of medicines which are used; and the separation will take place most quickly, if one having cleaned the wound as quickly as possible will next dry it, and the piece of bone, whether larger or smaller. For a piece of bone which is quickly dried and converted, as it were, into a shell, is most readily separated from the rest of the bone which retains its blood and vitality; for, the part having become exsanguous and dry, more readily drops off from that which retains its blood and is alive.^[806]

17. Such pieces of bone as are depressed from their natural position, either being broken off or chopped off to a considerable extent, are attended with less danger, provided the membrane be safe; and bones which are broken by numerous and broader fractures are still less dangerous and more easily extracted.^[807] And you must not trepan any of them, nor run any risks in attempting to extract the pieces of bone, until they rise up of their own accord, upon the subsidence of the swelling. They rise up when the flesh (*granulations*) grows below, and it grows from the diploe of the bone, and from the sound portion, provided the upper table alone be in a state of necrosis. And the flesh will shoot up and grow below the more quickly, and the pieces of bone ascend, if one will get the wound to suppurate and make it clean as quickly as possible. And when both the tables of the bone are driven in upon the membrane, I mean the upper and lower, the wound, if treated in the same way, will very soon get well, and the depressed bones will quickly rise up.^[808]

18. The bones of children are thinner and softer, for this reason, that they contain more blood [than those of adults]; and they are porous and spongy, and neither dense nor hard. And when wounded to a similar or inferior degree by weapons of the same or even of an inferior power, the bone of a young person more readily and quickly suppurates, and that in less time than the bone of an older person; and in accidents, which are to prove fatal, the younger person will die sooner than the elder. But if the bone is laid bare of flesh, one must attend and try to find out, what even is not obvious to the sight, and discover whether the bone be broken and contused, or only contused; and if, when there is an indentation in the bone, whether contusion, or fracture, or both be joined to it; and if the bone has sustained any of these injuries, we must give issue to the blood by perforating the bone with a small trepan, observing the greatest precautions, for the bone of young persons is thinner and more superficial than that of elder persons.^[809]

19. When a person has sustained a mortal wound on the head, which cannot be cured, nor his life preserved, you may form an opinion of his approaching dissolution, and foretell what is to happen from the following symptoms which such a person experiences.^[810] When a bone is broken, or cleft, or contused, or otherwise injured, and when by mistake it has not been discovered, and neither the raspatory nor trepan has been applied as required, but the case has been neglected as if the bone were sound, fever will generally come on before the fourteenth day if in winter, and in summer the fever usually seizes after seven days. And when this happens, the wound loses its color, and the inflammation dies in it; and it becomes glutinous, and appears like a pickle, being of a tawny and somewhat livid color; and the bone then begins to sphacelate, and turns black where it was white before, and at last becomes pale and blanched. But when suppuration is fairly established in it, small blisters form on the tongue and he dies delirious. And, for the most part, convulsions seize the other side of the body; for, if the wound be situated on the left side, the convulsions will seize the right side of the body; or if the wound be on the right side of the head, the convulsion attacks the left side of the body.^[811] And some become apoplectic. And thus they die before the end of

seven days, if in summer; and before fourteen, if in winter. And these symptoms indicate, in the same manner, whether the wound be older or more recent. But if you perceive that fever is coming on, and that any of these symptoms accompany it, you must not put off, but having sawed the bone to the membrane (*meninx*), or scraped it with a raspator, (and it is then easily sawed or scraped,) you must apply the other treatment as may seem proper, attention being paid to circumstances.^[812]

20. When in any wound of the head, whether the man has been trepanned or not, but the bone has been laid bare, a red and erysipelatous swelling supervenes in the face, and in both eyes, or in either of them, and if the swelling be painful to the touch, and if fever and rigor come on, and if the wound look well, whether as regards the flesh or the bone, and if the parts surrounding the wound be well, except the swelling in the face, and if the swelling be not connected with any error in the regimen, you must purge the bowels in such a case with a medicine which will evacuate bile; and when thus purged the fever goes off, the swelling subsides, and the patient gets well. In giving the medicine you must pay attention to the strength of the patient.^[813]

21. With regard to trepanning, when there is a necessity for it, the following particulars should be known. If you have had the management of the case from the first, you must not at once saw the bone down to the meninx; for it is not proper that the membrane should be laid bare and exposed to injuries for a length of time, as in the end it may become fungous. And there is another danger if you saw the bone down to the meninx and remove it at once, lest in the act of sawing you should wound the meninx. But in trepanning, when only a very little of the bone remains to be sawed through, and the bone can be moved, you must desist from sawing, and leave the bone to fall out of itself.^[814] For to a bone not sawed through, and where a portion is left of the sawing, no mischief can happen; for the portion now left is sufficiently thin. In other respects you must conduct the treatment as may appear suitable to the wound. And in trepanning you must frequently remove the trepan, on account of the heat in the bone, and plunge it in cold water. For the trepan being heated by running round, and heating and drying the bone, burns it and makes a larger piece of bone around the sawing to drop off, than would otherwise do. And if you wish to saw at once down to the membrane, and then remove the bone, you must also, in like manner, frequently take out the trepan and dip it in cold water. But if you have not charge of the treatment from the first, but undertake it from another after a time, you must saw the bone at once down to the meninx with a serrated trepan,^[815] and in doing so must frequently take out the trepan and examine with a sound (specillum), and otherwise along the tract of the instrument.^[816] For the bone is much sooner sawn through, provided there be matter below it and in it, and it often happens that the bone is more superficial,^[817] especially if the wound is situated in that part of the head where the bone is rather thinner than in other parts. But you must take care where you apply the trepan, and see that you do so only where it appears to be particularly thick, and having fixed the instrument there, that you frequently make examinations and endeavor by moving the bone to bring it up. Having removed it, you must apply the other suitable remedies to the wound. And if, when you have the management of the treatment from the first, you wish to saw through the bone at once, and remove it from the membrane, you must, in like manner, examine the tract of the instrument frequently with the sound, and see that it is fixed on the thickest part of the bone, and endeavor to remove the bone by moving it about. But if you use a perforator (*trepan?*), you must not penetrate to the membrane, if you operate on a case which you have had the charge of from the first, but must leave a thin scale of bone, as described in the process of sawing.

END OF VOL. I.